Supplementary Materials for the paper "Multilevel Main Effects Matrix Factor Model"

A Iterative Projection for Estimating $\mathbf{Q}_m$ and $\mathbf{J}_m$

For finding the common components in Step 3 in Section 3.3, we assume that the directions that align more to the strongest factors of $\mathbf{Q}_m$ and $\mathbf{J}_m$ are available for all $m \in [M]$, namely $\mathbf{v}_a^{(m)}$ and $\mathbf{v}_b^{(m)}$ for $m \in [M]$ in (3.8) and (3.9), which are of course not the case in reality. In the following, we present an iterative algorithm for finding $\mathbf{v}_a^{(m)}$ and $\mathbf{v}_b^{(m)}$, and more accurate estimators of $\mathbf{Q}_m$ and $\mathbf{J}_m$ in Step 3 in Section 3.3.

Algorithm A.1 Algorithm for choosing good projection directions for estimating $\mathbf{Q}_m$ and $\mathbf{J}_m$

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1. Preset K_0 (recommend $K_0 = 20$). Let $\{S_{m,k}\}_{m\in[M],k\in[K_0]}$ be a sequence (in k) of subsets of $[q_m]$ for each $m \in [M]$, which chooses elements in $[q_m]$ uniformly, with $|S_{m,k}| = \lfloor q_m/2 \rfloor$. Denote by $\mathbf{1}_{d,S}$ a vector of ones and zeros of length d, where the positions in $S \subseteq [d]$ are 1 and 0 otherwise.

2. Let $m = 1$, and let $k_n \in [K_0]$ for each $n \neq m$. Define

$$\mathbf{X}_{Q,m,k|k_n} := T^{-2}(M-1)^{-1} \sum_{n\neq m;n=1}^{M} \Big\{ \sum_{t=1}^{T} \breve{\mathbf{Y}}_t^{(m)} \mathbf{1}_{q_m,S_{m,k}} \mathbf{1}'_{q_n,S_{n,k_n}} \breve{\mathbf{Y}}_t^{(n)'} \Big\}^{\otimes 2}.$$

   Find $\widetilde{k}_m$ such that $\mathbf{X}_{Q,m,\widetilde{k}_m|k_n}$ has the largest eigenvalue. Take $\mathbf{v}_b^{(m)} = \mathbf{1}_{q_m,S_{m,\widetilde{k}_m}}$.

3. For $m \in [M]$ larger than 1, repeat Step 2, replacing k_n by $\widetilde{k}_n$ found in previous steps in the definition of $\mathbf{X}_{Q,m,k|k_n}$.

4. With $\mathbf{v}_b^{(m)} = \mathbf{1}_{q_m,S_{m,\widetilde{k}_m}}$ for each $m \in [M]$, take $\widehat{\mathbf{Q}}_m$ as the first k_r eigenvectors of $\mathbf{X}_{Q,m}$ defined in (3.8).

5. Set $\mathbf{v}_a^{(m)} = \widehat{\mathbf{Q}}_{m,\cdot 1}$ for each $m \in [M]$.

   Fix $m = 1$. Replace $\mathbf{v}_a^{(m)} = \widehat{\mathbf{Q}}_{m,\cdot k}$ and find the eigenvalue of $\mathbf{X}_{J,m}$ defined in (3.9), $k \in [k_r]$. Set $\mathbf{v}_a^{(m)} = \widehat{\mathbf{Q}}_{m,\cdot k}$ so that $\mathbf{X}_{J,m}$ has the largest eigenvalue. Repeat for $m = 2, \ldots, M$, resulting in a new set of $\mathbf{v}_a^{(m)}$ for $m \in [M]$.

   Estimate $\widehat{\mathbf{J}}_m$ as the first k_c eigenvectors of $\mathbf{X}_{J,m}$.

6. Set $\mathbf{v}_b^{(m)} = \widehat{\mathbf{J}}_{m,\cdot 1}$ for each $m \in [M]$.

Fix $m = 1$. Replace $\mathbf{v}_b^{(m)} = \widehat{\mathbf{J}}_{m,\cdot k}$ and find the eigenvalue of $\mathbf{X}_{Q,m}$ defined in (3.8), $k \in [k_c]$. Set $\mathbf{v}_b^{(m)} = \widehat{\mathbf{J}}_{m,\cdot k}$ so that $\mathbf{X}_{Q,m}$ has the largest eigenvalue. Repeat for $m = 2, \ldots, M$, resulting in a new set of $\mathbf{v}_b^{(m)}$ for $m \in [M]$.

Estimate $\widehat{\mathbf{Q}}_m$ as the first k_r eigenvectors of $\mathbf{X}_{Q,m}$.

7. Iterate Step 5 and 6 until some kind of convergence is reached.

One major difference between Algorithm A.1 and the iterative algorithm in Zhang et al. (2025) is that we only use the directions that align more to the strongest factors for iterative projection in Steps 5 and 6 above. This is to avoid projection using weak factors when it is likely that in real data such weak factors exist, which can numerically lead to less accurate estimators if used as projection directions. Our final two-step aligned estimators $\widetilde{\mathbf{Q}}_m$ and $\widetilde{\mathbf{J}}_m$ in step 3 in Section 3.3 are constructed based on $\widehat{\mathbf{Q}}_m$ and $\widehat{\mathbf{J}}_m$ and indeed perform better than the augmented estimators from Zhang et al. (2025). See Table 3 in Section 6 for more details.

B Alternative Procedures for Global Common Component Estimation

The estimation procedure in Step 3 of Section 3.3 uses the nearest Kronecker product approximation, described in Section 6 of Loan (2000), implemented through a matrix rearrangement and singular value decomposition, for both the alignment procedure and the re-estimation procedure. For completeness, we describe two alternative iterative procedures. Subsection B.1 presents a two-sided Procrustes alternative for the alignment step, while Subsection B.2 presents an alternating least-squares alternative. These two alternatives concern separate stages of the estimation procedure and may be applied independently.

B.1 Two-sided Procrustes alignment

To align the coordinate systems of the $\widehat{\mathbf{Q}}_m$ and $\widehat{\mathbf{J}}_m$ across $m \in [M]$, we first choose a reference group $m_0 \in [M]$. For each $m \neq m_0$, we find orthogonal alignment matrices $\widehat{\mathbf{O}}_{Q,m}$ and $\widehat{\mathbf{O}}_{J,m}$ by the two-sided Procrustes criterion, namely

$$(\widehat{\mathbf{O}}_{Q,m}, \widehat{\mathbf{O}}_{J,m}) = \underset{\mathbf{O}_Q \in \mathbb{O}(k_r), \mathbf{O}_J \in \mathbb{O}(k_c)}{\arg\min} \sum_{t=1}^{T} \left\| \widehat{\mathbf{Q}}'_{m_0} \breve{\mathbf{Y}}_{G,t}^{(m_0)} \widehat{\mathbf{J}}_{m_0} - \mathbf{O}'_Q \widehat{\mathbf{Q}}'_m \breve{\mathbf{Y}}_{G,t}^{(m)} \widehat{\mathbf{J}}_m \mathbf{O}_J \right\|_F^2, \tag{B.1}$$

where $\mathbb{O}(a)$ denotes the set of $a \times a$ orthogonal matrices. To this end, given $\mathbf{O}_{J,m}$, define

$$\mathbf{\Psi}_{Q,m} := \sum_{t=1}^{T} \widehat{\mathbf{Q}}_m' \check{\mathbf{Y}}_{G,t}^{(m)} \widehat{\mathbf{J}}_m \mathbf{O}_{J,m} \widehat{\mathbf{J}}_{m_0}' \check{\mathbf{Y}}_{G,t}^{(m_0)'} \widehat{\mathbf{Q}}_{m_0}.$$

The corresponding Procrustes update is

$$\mathbf{O}_{Q,m} = \mathbf{U}_{Q,m} \mathbf{V}_{Q,m}', \tag{B.2}$$

where $\mathbf{U}_{Q,m}$ and $\mathbf{V}_{Q,m}$ are the left- and right-singular vectors of $\mathbf{\Psi}_{Q,m}$. Similarly, given $\mathbf{O}_{Q,m}$, define

$$\mathbf{\Psi}_{J,m} := \sum_{t=1}^{T} \widehat{\mathbf{J}}_m' \check{\mathbf{Y}}_{G,t}^{(m)'} \widehat{\mathbf{Q}}_m \mathbf{O}_{Q,m} \widehat{\mathbf{Q}}_{m_0}' \check{\mathbf{Y}}_{G,t}^{(m_0)} \widehat{\mathbf{J}}_{m_0},$$

and its corresponding Procrustes update is

$$\mathbf{O}_{J,m} = \mathbf{U}_{J,m} \mathbf{V}_{J,m}', \tag{B.3}$$

where $\mathbf{U}_{J,m}$ and $\mathbf{V}_{J,m}$ are the left- and right-singular vectors of $\mathbf{\Psi}_{J,m}$. Starting from $\mathbf{O}_{Q,m} = \mathbf{I}_{k_r}$ and $\mathbf{O}_{J,m} = \mathbf{I}_{k_c}$, we iterate (B.2) and (B.3) to obtain $\widehat{\mathbf{O}}_{Q,m}$ and $\widehat{\mathbf{O}}_{J,m}$. We then replace $\widehat{\mathbf{Q}}_m$ and $\widehat{\mathbf{J}}_m$ with $\widehat{\mathbf{Q}}_m \widehat{\mathbf{O}}_{Q,m}$ and $\widehat{\mathbf{J}}_m \widehat{\mathbf{O}}_{J,m}$ respectively for $m \neq m_0$. After the alignment, the global common factor $\mathbf{G}_t$ is estimated by least square as

$$\widehat{\mathbf{G}}_t := M^{-1} \sum_{m=1}^{M} \widehat{\mathbf{Q}}_m' (\check{\mathbf{Y}}_t^{(m)} - \widehat{\mathbf{R}}_m \widehat{\mathbf{F}}_t^{(m)} \widehat{\mathbf{C}}_m') \widehat{\mathbf{J}}_m. \tag{B.4}$$

B.2 Alternating least-squares refitting

For $m \in [M]$, we re-estimate $\mathbf{Q}_m$ and $\mathbf{J}_m$ by minimizing $\sum_{t=1}^{T} \|\check{\mathbf{Y}}_t^{(m)} - \widehat{\mathbf{R}}_m \widehat{\mathbf{F}}_t^{(m)} \widehat{\mathbf{C}}_m' - \mathbf{Q}_m \widehat{\mathbf{G}}_t \mathbf{J}_m'\|_F^2$ over $\mathbf{Q}_m$ and $\mathbf{J}_m$. To this end, given $\mathbf{J}_m$, we have

$$\widetilde{\mathbf{Q}}_m = \Big(\sum_{t=1}^{T} (\check{\mathbf{Y}}_t^{(m)} - \widehat{\mathbf{R}}_m \widehat{\mathbf{F}}_t^{(m)} \widehat{\mathbf{C}}_m') \mathbf{J}_m \widehat{\mathbf{G}}_t' \Big) \Big(\sum_{t=1}^{T} \widehat{\mathbf{G}}_t \mathbf{J}_m' \mathbf{J}_m \widehat{\mathbf{G}}_t' \Big)^{-1}, \tag{B.5}$$

and given $\mathbf{Q}_m$, we have

$$\widetilde{\mathbf{J}}_m = \Big(\sum_{t=1}^{T} (\check{\mathbf{Y}}_t^{(m)} - \widehat{\mathbf{R}}_m \widehat{\mathbf{F}}_t^{(m)} \widehat{\mathbf{C}}_m')' \mathbf{Q}_m \widehat{\mathbf{G}}_t \Big) \Big(\sum_{t=1}^{T} \widehat{\mathbf{G}}_t' \mathbf{Q}_m' \mathbf{Q}_m \widehat{\mathbf{G}}_t \Big)^{-1}. \tag{B.6}$$

Using estimators $\check{\mathbf{Q}}_m$ and $\check{\mathbf{J}}_m$, we can iterate (B.5) and (B.6) to obtain the final estimators $\widetilde{\mathbf{Q}}_m$ and $\widetilde{\mathbf{J}}_m$ for $\mathbf{Q}_m$ and $\mathbf{J}_m$, respectively. This is in parallel to the algorithm in Section 2.2 of Cai et al. (2025).

Table B.1: Common component estimation accuracy using Procrustes alignment and ALS refitting. All entries are Monte Carlo means multiplied by 10^2, with standard deviations in parentheses.

Settings	IIa		IIb		IIc		IId	
	$T = 100$	$T = 400$	$T = 100$	$T = 400$	$T = 100$	$T = 400$	$T = 100$	$T = 400$
$\mathcal{D}(\mathbf{Q}_m, \widehat{\mathbf{Q}}_m)$	7.98(3.09)	3.90(1.45)	9.84(5.71)	4.52(1.77)	9.21(3.03)	4.46(1.41)	5.75(2.21)	2.73(1.05)
$\mathcal{D}(\mathbf{Q}_m, \widetilde{\mathbf{Q}}_m)$	1.67(0.42)	0.75(0.15)	2.09(4.45)	0.76(0.20)	2.85(0.46)	1.35(0.17)	1.06(0.22)	0.48(0.07)
$\mathcal{D}(\mathbf{J}_m, \widehat{\mathbf{J}}_m)$	8.22(3.15)	3.93(1.57)	9.69(5.65)	4.51(1.74)	9.27(3.16)	4.24(1.25)	8.24(3.12)	3.73(1.32)
$\mathcal{D}(\mathbf{J}_m, \widetilde{\mathbf{J}}_m)$	1.42(0.20)	0.66(0.07)	1.66(4.15)	0.62(0.07)	2.69(0.34)	1.28(0.13)	1.31(0.15)	0.61(0.05)
$\mathcal{D}(\mathbf{R}_m, \widehat{\mathbf{R}}_m)$	2.53(0.64)	1.01(0.18)	3.28(4.47)	1.05(0.22)	4.23(0.79)	2.42(0.55)	2.23(0.72)	0.76(0.15)
$\mathcal{D}(\mathbf{C}_m, \widehat{\mathbf{C}}_m)$	2.50(0.66)	1.01(0.16)	3.33(4.54)	1.04(0.22)	4.23(0.90)	2.41(0.56)	2.09(0.48)	0.84(0.10)
$\text{MSE}(\widetilde{\mathbf{Q}}_m \widehat{\mathbf{G}}_t \widetilde{\mathbf{J}}'_m)$	0.52(2.44)	0.06(0.64)	1.23(9.15)	0.03(0.00)	0.58(2.27)	0.15(0.57)	0.30(1.87)	0.02(0.00)
$\text{MSE}(\widehat{\mathbf{R}}_m \widehat{\mathbf{F}}_t^{(m)} \widehat{\mathbf{C}}'_m)$	0.25(0.11)	0.13(0.03)	0.82(10.24)	0.12(0.03)	0.71(0.20)	0.48(0.10)	0.14(0.06)	0.06(0.01)

C Proof of Theorems

Proof of Theorem 1. The grand mean $c_t^{(m)}$ (as defined in (3.2)) is identified when we pre- and post-multiply by $\mathbf{1}'_{p_m}$ and $\mathbf{1}_{q_m}$ respectively on both sides of model (3.1) under (IC1). Then for $\{\mathbf{a}_t^{(m)}\}$ and $\{\mathbf{b}_t^{(m)}\}$ we can post-multiply by $\mathbf{1}_{q_m}$ and pre-multiply by $\mathbf{1}'_{p_m}$ on both sides of (3.1) respectively. The remaining common component is then identified as well.

For global and local effects, since $c_t^{(m)} = \mu_t + \mu_t^{(m)}$ is identified, we have $\mu_t = M^{-1}\sum_{m=1}^M c_t^{(m)}$ using (IC2), and hence both $\{\mu_t\}$ and $\{\mu_t^{(m)}\}$ are identified. Next, we in fact identify the row and column main effects at group m respectively as

$$\mathbf{a}_t^{(m)} = q_m^{-1}\mathbf{M}_{p_m}(\mathbf{Y}_t^{(m)} - \mathbf{E}_t^{(m)})\mathbf{1}_{q_m}, \quad \mathbf{b}_t^{(m)} = p_m^{-1}\mathbf{M}_{q_m}(\mathbf{Y}_t^{(m)} - \mathbf{E}_t^{(m)})'\mathbf{1}_{p_m}. \tag{C.1}$$

Hence for $m \neq n$, using (3.2) and (IC2), we have

$$\begin{aligned}
\mathbf{A}_{1m}\mathbb{E}\Big(T^{-1}\sum_{t=1}^T \boldsymbol{\alpha}_t^{\otimes 2}\Big)\mathbf{A}'_{1n} &= \mathbb{E}\Big(T^{-1}\sum_{t=1}^T \mathbf{a}_t^{(m)}\mathbf{a}_t^{(n)'}\Big) \\
&= q_m^{-1}q_n^{-1}\mathbf{M}_{p_m}\mathbb{E}\Big\{T^{-1}\sum_{t=1}^T(\mathbf{Y}_t^{(m)} - \mathbf{E}_t^{(m)})\mathbf{1}_{q_m}\mathbf{1}'_{q_n}(\mathbf{Y}_t^{(n)} - \mathbf{E}_t^{(n)})'\Big\}\mathbf{M}_{p_n} \\
&= T^{-1}q_m^{-1}q_n^{-1}\mathbf{M}_{p_m}\sum_{t=1}^T \mathbb{E}\{\mathbf{Y}_t^{(m)}\mathbf{1}_{q_m}\mathbf{1}'_{q_n}\mathbf{Y}_t^{(n)'}\}\mathbf{M}_{p_n}, \qquad \text{(C.2)} \\
\mathbf{B}_{1m}\mathbb{E}\Big(T^{-1}\sum_{t=1}^T \boldsymbol{\beta}_t^{\otimes 2}\Big)\mathbf{B}'_{1n} &= \mathbb{E}\Big(T^{-1}\sum_{t=1}^T \mathbf{b}_t^{(m)}\mathbf{b}_t^{(n)'}\Big) \\
&= p_m^{-1}p_n^{-1}\mathbf{M}_{q_m}\mathbb{E}\Big\{T^{-1}\sum_{t=1}^T(\mathbf{Y}_t^{(m)} - \mathbf{E}_t^{(m)})'\mathbf{1}_{p_m}\mathbf{1}'_{p_n}(\mathbf{Y}_t^{(n)} - \mathbf{E}_t^{(n)})\Big\}\mathbf{M}_{q_n} \\
&= T^{-1}p_m^{-1}p_n^{-1}\mathbf{M}_{q_m}\sum_{t=1}^T \mathbb{E}\{\mathbf{Y}_t^{(m)'}\mathbf{1}_{p_m}\mathbf{1}'_{p_n}\mathbf{Y}_t^{(n)}\}\mathbf{M}_{q_n}.
\end{aligned}$$

If we define

$$\begin{aligned}
\boldsymbol{\Omega}_{A_1,m} &:= (M-1)^{-1}\sum_{n=1;n\neq m}^M \Big\{T^{-1}q_m^{-1}q_n^{-1}\mathbf{M}_{p_m}\sum_{t=1}^T \mathbb{E}\{\mathbf{Y}_t^{(m)}\mathbf{1}_{q_m}\mathbf{1}'_{q_n}\mathbf{Y}_t^{(n)'}\}\mathbf{M}_{p_n}\Big\}^{\otimes 2}, \\
\boldsymbol{\Omega}_{B_1,m} &:= (M-1)^{-1}\sum_{n=1;n\neq m}^M \Big\{T^{-1}p_m^{-1}p_n^{-1}\mathbf{M}_{q_m}\sum_{t=1}^T \mathbb{E}\{\mathbf{Y}_t^{(m)'}\mathbf{1}_{p_m}\mathbf{1}'_{p_n}\mathbf{Y}_t^{(n)}\}\mathbf{M}_{q_n}\Big\}^{\otimes 2},
\end{aligned}$$

then a version of $\mathbf{A}_{1m}$ (resp. $\mathbf{B}_{1m}$) can be identified as the first r_1 eigenvectors of $\boldsymbol{\Omega}_{A_1,m}$ (resp. first ℓ_1 eigenvectors of $\boldsymbol{\Omega}_{B_1,m}$).

As for $\mathbf{A}_{2m}$, since $\mathbf{A}_{1m}^{\perp}$ have orthonormal columns spanning the orthogonal complement of the column

space of $\mathbf{A}_{1m}$, from (3.2), we have

$$\mathbf{A}_{1m}^{\perp'}\mathbf{a}_t^{(m)} = \mathbf{A}_{1m}^{\perp'}\mathbf{A}_{2m}\boldsymbol{\alpha}_t^{(m)}.$$

Hence

$$T^{-1}\sum_{t=1}^{T}\mathbf{a}_t^{(m)}\mathbf{a}_t^{(m)'}\mathbf{A}_{1m}^{\perp} = \mathbf{A}_{1m}\Big(T^{-1}\sum_{t=1}^{T}\boldsymbol{\alpha}_t\boldsymbol{\alpha}_t^{(m)'}\Big)\mathbf{A}_{2m}'\mathbf{A}_{1m}^{\perp} + \mathbf{A}_{2m}\Big(T^{-1}\sum_{t=1}^{T}\boldsymbol{\alpha}_t^{(m)}\boldsymbol{\alpha}_t^{(m)'}\Big)\mathbf{A}_{2m}'\mathbf{A}_{1m}^{\perp}.$$

Hence (IC2) gives

$$\Big\{T^{-1}\sum_{t=1}^{T}(\mathbf{a}_t^{(m)\otimes 2})\mathbf{A}_{1m}^{\perp}\Big\}^{\otimes 2} = \mathbf{A}_{2m}\Big\{T^{-1}\sum_{t=1}^{T}(\boldsymbol{\alpha}_t^{(m)\otimes 2})\mathbf{A}_{2m}'\mathbf{A}_{1m}^{\perp}\Big\}^{\otimes 2}\mathbf{A}_{2m}'. \tag{C.3}$$

Therefore, we can take the first $r_2^{(m)}$ eigenvectors of $\Big\{T^{-1}\sum_{t=1}^{T}\mathbf{a}_t^{(m)\otimes 2}\mathbf{A}_{1m}^{\perp}\Big\}^{\otimes 2}$ as a version of $\mathbf{A}_{2m}$. Similarly, we can take the first $\ell_2^{(m)}$ eigenvectors of $\Big\{T^{-1}\sum_{t=1}^{T}\mathbf{b}_t^{(m)\otimes 2}\mathbf{B}_{1m}^{\perp}\Big\}^{\otimes 2}$ as a version of $\mathbf{B}_{2m}$.

To identify $\boldsymbol{\alpha}_t^{(m)}$, since $\mathbf{A}_{1m}^{\perp'}\mathbf{A}_{2m}\boldsymbol{\alpha}_t^{(m)} = q_m^{-1}\mathbf{A}_{1m}^{\perp'}\mathbf{M}_{p_m}(\mathbf{Y}_t^{(m)} - \mathbf{E}_t^{(m)})\mathbf{1}_{q_m}$, we have

$$\boldsymbol{\alpha}_t^{(m)} = q_m^{-1}(\mathbf{A}_{2m}'\mathbf{A}_{1m}^{\perp}\mathbf{A}_{1m}^{\perp'}\mathbf{A}_{2m})^{-1}\mathbf{A}_{2m}'\mathbf{A}_{1m}^{\perp}\mathbf{A}_{1m}^{\perp'}\mathbf{M}_{p_m}(\mathbf{Y}_t^{(m)} - \mathbf{E}_t^{(m)})\mathbf{1}_{q_m}.$$

Similarly,

$$\boldsymbol{\beta}_t^{(m)} = p_m^{-1}(\mathbf{B}_{2m}'\mathbf{B}_{1m}^{\perp}\mathbf{B}_{1m}^{\perp'}\mathbf{B}_{2m})^{-1}\mathbf{B}_{2m}'\mathbf{B}_{1m}^{\perp}\mathbf{B}_{1m}^{\perp'}\mathbf{M}_{q_m}(\mathbf{Y}_t^{(m)} - \mathbf{E}_t^{(m)})'\mathbf{1}_{p_m}.$$

We can then identify $\boldsymbol{\alpha}_t$ and $\boldsymbol{\beta}_t$ as

$$\begin{aligned}
\boldsymbol{\alpha}_t &= (\mathbf{A}_{1m}'\mathbf{A}_{1m})^{-1}\mathbf{A}_{1m}'\{q_m^{-1}\mathbf{M}_{p_m}(\mathbf{Y}_t^{(m)} - \mathbf{E}_t^{(m)})\mathbf{1}_{q_m} - \mathbf{A}_{2m}\boldsymbol{\alpha}_t^{(m)}\},\\
\boldsymbol{\beta}_t &= (\mathbf{B}_{1m}'\mathbf{B}_{1m})^{-1}\mathbf{B}_{1m}'\{p_m^{-1}\mathbf{M}_{q_m}(\mathbf{Y}_t^{(m)} - \mathbf{E}_t^{(m)})'\mathbf{1}_{p_m} - \mathbf{B}_{2m}\boldsymbol{\beta}_t^{(m)}\}.
\end{aligned}$$

(A refinement procedure can be carried out to estimate $\boldsymbol{\alpha}_t$ and $\boldsymbol{\beta}_t$ as an average over all $m \in [M]$, with $\mathbf{A}_{1m}$ and $\mathbf{B}_{1m}$ re-estimated using least squares estimation. This is indeed integrated in our estimation procedure in Section 3.3.)

Define

$$\widetilde{\mathbf{Y}}_t^{(m)} := \mathbf{Y}_t^{(m)} - \mathbf{E}_t^{(m)} - c_t\mathbf{1}_{p_m}\mathbf{1}_{q_m}' - \mathbf{a}_t^{(m)}\mathbf{1}_{q_m}' - \mathbf{1}_{p_m}\mathbf{b}_t^{(m)'}.$$

If we know $\mathbf{v}_1^{(m)}$ as a vector aligning more to the direction of the strongest factors in $\mathbf{Q}_m$, and $\mathbf{v}_2^{(m)}$ as such a vector for $\mathbf{J}_m$ (practically this can be done using similar pre-averaging ideas in Chen and Lam (2024), which is also integrated in our estimation procedure), then since for $m \neq n$, by (IC2),

$$\mathbb{E}(\widetilde{\mathbf{Y}}_t^{(m)}\mathbf{v}_2^{(m)}\mathbf{v}_2^{(n)'}\widetilde{\mathbf{Y}}_t^{(n)'}) = \mathbf{Q}_m\mathbb{E}(\mathbf{G}_t\mathbf{J}_m'\mathbf{v}_2^{(m)}\mathbf{v}_2^{(n)'}\mathbf{J}_n'\mathbf{G}_t')\mathbf{Q}_n'$$

$$
\begin{aligned}
&= \mathbf{Q}_m \mathbb{E}\Big\{ \mathbf{vec}(\mathbf{G}_t \mathbf{J}_m' \mathbf{v}_2^{(m)}) \mathbf{vec}(\mathbf{G}_t \mathbf{J}_n \mathbf{v}_2^{(n)})' \Big\} \mathbf{Q}_n' \\
&= \mathbf{Q}_m \mathbb{E}\Big\{ (\mathbf{v}_2^{(m)'} \mathbf{J}_m \otimes \mathbf{I}_{k_r}) \mathbf{vec}(\mathbf{G}_t) \mathbf{vec}'(\mathbf{G}_t) (\mathbf{J}_m' \mathbf{v}_2^{(m)} \otimes \mathbf{I}_{k_r}) \Big\} \mathbf{Q}_n' \\
&= \mathbf{Q}_m (\mathbf{v}_2^{(m)'} \mathbf{J}_m \otimes \mathbf{I}_{k_r}) \mathbb{E}(\mathbf{vec}(\mathbf{G}_t))^{\otimes 2} (\mathbf{J}_n' \mathbf{v}_2^{(n)} \otimes \mathbf{I}_{k_r}) \mathbf{Q}_n', \qquad \text{(C.4)}
\end{aligned}
$$

which is of rank k_r through having $\mathbb{E}(\mathbf{vec}(\mathbf{G}_t))^{\otimes 2}$ of full rank by (IC3), we can identify $\mathbf{Q}_m$ as the first k_r eigenvectors of

$$
\sum_{n=1; n \neq m}^{M} \{ \mathbb{E}(\widetilde{\mathbf{Y}}_t^{(m)} \mathbf{v}_2^{(m)} \mathbf{v}_2^{(n)'} \widetilde{\mathbf{Y}}_t^{(n)'}) \}^{\otimes 2}.
$$

Similarly, we can identify $\mathbf{J}_m$ as the first k_c eigenvectors of

$$
\sum_{n=1; n \neq m}^{M} \{ \mathbb{E}(\widetilde{\mathbf{Y}}_t^{(m)'} \mathbf{v}_1^{(m)} \mathbf{v}_1^{(n)'} \widetilde{\mathbf{Y}}_t^{(n)}) \}^{\otimes 2}.
$$

We can also identify $\mathbf{R}_m$ as the first $k_r^{(m)}$ eigenvectors of

$$
\mathbb{E}(\widetilde{\mathbf{Y}}_t^{(m)} \mathbf{J}_m^{\perp})^{\otimes 2} = \mathbf{R}_m \mathbb{E}[\mathbf{F}_t^{(m)} \mathbf{C}_m' \mathbf{J}_m^{\perp} \mathbf{J}_m^{\perp'} \mathbf{C}_m \mathbf{F}_t^{(m)'}] \mathbf{R}_m',
$$

which is of full rank since $\mathbb{E}(\mathbf{vec}(\mathbf{F}_t^{(m)}))^{\otimes 2}$ and $\mathbf{J}_m^{\perp'} \mathbf{C}_m$ have full rank by (IC3). Similarly, $\mathbf{C}_m$ can be identified as the first $k_c^{(m)}$ eigenvectors of

$$
\mathbb{E}(\widetilde{\mathbf{Y}}_t^{(m)'} \mathbf{Q}_m^{\perp})^{\otimes 2} = \mathbf{C}_m \mathbb{E}[\mathbf{F}_t^{(m)'} \mathbf{R}_m' \mathbf{Q}_m^{\perp} \mathbf{Q}_m^{\perp'} \mathbf{R}_m \mathbf{F}_t^{(m)}] \mathbf{C}_m'.
$$

Then $\mathbf{F}_t^{(m)}$ can be identified as

$$
\mathbf{F}_t^{(m)} = (\mathbf{R}_m' \mathbf{Q}_m^{\perp} \mathbf{Q}_m^{\perp'} \mathbf{R}_m)^{-1} \mathbf{R}_m' \mathbf{Q}_m^{\perp} \mathbf{Q}_m^{\perp'} \widetilde{\mathbf{Y}}_t^{(m)} \mathbf{C}_m.
$$

Finally $\mathbf{G}_t$ can be identified as

$$
\mathbf{G}_t = \mathbf{Q}_m' (\widetilde{\mathbf{Y}}_t^{(m)} - \mathbf{R}_m \mathbf{F}_t^{(m)} \mathbf{C}_m') \mathbf{J}_m.
$$

Again, we can consider the average of the above over all $m \in [M]$ as an estimator of $\mathbf{G}_t$, and re-estimate $\mathbf{Q}_m$ and $\mathbf{J}_m$ using least square method, which is indeed integrated into our estimation procedure. □

Proof of Theorem 2. We first derive the convergence rate for grand mean and its corresponding global and local components. By left- and right-multiplying $\mathbf{1}_{p_m}'$ and $\mathbf{1}_{q_m}$ on (3.1), we have

$$
\mathbf{1}_{p_m}' \mathbf{Y}_t^{(m)} \mathbf{1}_{q_m} = p_m q_m c_t^{(m)} + \mathbf{1}_{p_m}' \mathbf{E}_t^{(m)} \mathbf{1}_{q_m}. \qquad \text{(C.5)}
$$

Thus,

$$\mathbb{E}(\widehat{c}_t^{(m)} - c_t^{(m)})^2 = (p_m q_m)^{-2}\mathbb{E}(\mathbf{1}'_{p_m}\mathbf{E}_t^{(m)}\mathbf{1}_{q_m})^2 = (p_m q_m)^{-2}\mathbb{E}\big(\sum_{i=1}^{p_m}\sum_{j=1}^{q_m} E_{t,ij}^{(m)}\big)^2$$
$$= (p_m q_m)^{-2}\sum_{i=1}^{p_m}\sum_{l=1}^{p_m}\sum_{j=1}^{q_m}\sum_{h=1}^{q_m}\mathbb{E}(E_{t,ij}^{(m)}E_{t,lh}^{(m)}) = O(p_m^{-1}q_m^{-1}), \tag{C.6}$$

where we apply Lemma 1 in the last equality. Hence $|\widehat{c}_t^{(m)} - c_t^{(m)}| = O_P\big((p_m q_m)^{-1/2}\big)$. Taking sum on both sides of (C.5) w.r.t. m, we then have $\mu_t = M^{-1}\sum_{m=1}^M (p_m q_m)^{-1}\mathbf{1}'_{p_m}(\mathbf{Y}_t^{(m)} - \mathbf{E}_t^{(m)})\mathbf{1}_{q_m}$. Therefore, since $\widehat{\mu}_t = M^{-1}\sum_{m=1}^M \widehat{c}_t^{(m)}$, we have

$$(\widehat{\mu}_t - \mu_t)^2 \lesssim \big(\sum_{m=1}^M (p_m q_m)^{-1}\mathbf{1}'_{p_m}\mathbf{E}_t^{(m)}\mathbf{1}_{q_m}\big)^2 \lesssim \sum_{m=1}^M \big((p_m q_m)^{-1}\mathbf{1}'_{p_m}\mathbf{E}_t^{(m)}\mathbf{1}_{q_m}\big)^2$$
$$= \sum_{m=1}^M \big(\widehat{c}_t^{(m)} - c_t^{(m)}\big)^2 = \sum_{m=1}^M O_P\big((p_m q_m)^{-1}\big) = O_P\Big(\max_{m\in[M]}(p_m q_m)^{-1}\Big).$$

Furthermore, as $\widehat{\mu}_t^{(m)} = \widehat{c}_t^{(m)} - \widehat{\mu}_t$ and $\mu_t^{(m)} = c_t^{(m)} - \mu_t$, for each $m \in [M]$,

$$\big(\widehat{\mu}_t^{(m)} - \mu_t^{(m)}\big)^2 \lesssim \big(\widehat{c}_t^{(m)} - c_t^{(m)}\big)^2 + \big(\widehat{\mu}_t - \mu_t\big)^2 = O_P\Big(\max_{n\in[M]}(p_n q_n)^{-1}\Big),$$

which is the stated rate.

By (C.1), the definitions of $\widehat{\mathbf{a}}_t^{(m)}$ and $\widehat{\mathbf{b}}_t^{(m)}$, and (IC1), we have

$$\big\|\widehat{\mathbf{a}}_t^{(m)} - \mathbf{a}_t^{(m)}\big\|^2 = q_m^{-2}\big\|\mathbf{M}_{p_m}\mathbf{E}_t^{(m)}\mathbf{1}_{q_m}\big\|^2 = q_m^{-2}\big\|\mathbf{E}_t^{(m)}\mathbf{1}_{q_m} - p_m^{-1}\mathbf{1}_{p_m}\mathbf{1}'_{p_m}\mathbf{E}_t^{(m)}\mathbf{1}_{q_m}\big\|^2.$$

For the first part, by Lemma 1,

$$\mathbb{E}\big\|\mathbf{E}_t^{(m)}\mathbf{1}_{q_m}\big\|^2 = \sum_{i=1}^{p_m}\sum_{j=1}^{q_m}\sum_{h=1}^{q_m}\mathbb{E}(E_{t,ij}^{(m)}E_{t,ih}^{(m)}) \le \sum_{i=1}^{p_m}\sum_{j=1}^{q_m}\sum_{k=1}^{p_m}\sum_{h=1}^{q_m}|\mathbb{E}(E_{t,ij}^{(m)}E_{t,kh}^{(m)})| = O(p_m q_m).$$

And for the second part, write

$$\mathbb{E}\big\|p_m^{-1}\mathbf{1}_{p_m}\mathbf{1}'_{p_m}\mathbf{E}_t^{(m)}\mathbf{1}_{q_m}\big\|^2 = p_m^{-2}\mathbb{E}(\mathbf{1}'_{q_m}\mathbf{E}_t^{(m)'}\mathbf{1}_{p_m}\mathbf{1}'_{p_m}\mathbf{1}_{p_m}\mathbf{1}'_{p_m}\mathbf{E}_t^{(m)}\mathbf{1}_{q_m})$$
$$= p_m^{-1}\mathbb{E}(\mathbf{1}'_{p_m}\mathbf{E}_t^{(m)}\mathbf{1}_{q_m})^2 = O(q_m),$$

where the last equality follows by (C.6). By triangle inequality, combining the two bounds, we obtain

$\mathbb{E}\|\widehat{\mathbf{a}}_t^{(m)} - \mathbf{a}_t^{(m)}\|^2 = O(p_m q_m^{-1})$. Therefore, by Markov's inequality, we have

$$p_m^{-1}\|\widehat{\mathbf{a}}_t^{(m)} - \mathbf{a}_t^{(m)}\|^2 = O_P(q_m^{-1}), \tag{C.7}$$

$$q_m^{-1}\|\widehat{\mathbf{b}}_t^{(m)} - \mathbf{b}_t^{(m)}\|^2 = O_P(p_m^{-1}), \tag{C.8}$$

where column effect $\widehat{\mathbf{b}}_t^{(m)}$ follows similarly. This ends the proof. □

For the remaining main effect proofs and lemmas, we use the notation $\gamma_{r,\max}^{(m)} := \gamma_r \vee \gamma_r^{(m)}$ and $\gamma_{r,\min}^{(m)} := \gamma_r \wedge \gamma_r^{(m)}$, and define $\gamma_{c,\max}^{(m)}$ and $\gamma_{c,\min}^{(m)}$ analogously. These quantities are related to the notation in (R1) by

$$p_m^{\gamma_{r,\max}^{(m)} - \gamma_r} = 1 \vee \nu_{r,m}, \qquad p_m^{\gamma_r - \gamma_{r,\min}^{(m)}} = 1 \vee \frac{1}{\nu_{r,m}}, \qquad p_m^{\gamma_{r,\max}^{(m)} - \gamma_{r,\min}^{(m)}} = \nu_{r,m} \vee \frac{1}{\nu_{r,m}},$$

$$p_m^{\gamma_{r,\max}^{(m)} - \gamma_r^{(m)}} = \frac{1}{1 \wedge \nu_{r,m}}, \qquad \frac{p_m^{1-\gamma_{r,\min}^{(m)}}}{q_m} = \frac{p_m^{1-\gamma_r}}{q_m(1 \wedge \nu_{r,m})}.$$

The column-side identities follow similarly.

Proof of Theorem 3. For $m \neq n \in [M]$, define

$$\mathbf{S}_{A_1,mn} := T^{-1} q_m^{-1} q_n^{-1} \mathbf{M}_{p_m} \sum_{t=1}^{T} \mathbb{E}\left(\mathbf{Y}_t^{(m)} \mathbf{1}_{q_m} \mathbf{1}_{q_n}' \mathbf{Y}_t^{(n)'}\right) \mathbf{M}_{p_n}, \tag{C.9}$$

$$\widehat{\mathbf{S}}_{A_1,mn} := T^{-1} q_m^{-1} q_n^{-1} \mathbf{M}_{p_m} \sum_{t=1}^{T} \left(\mathbf{Y}_t^{(m)} \mathbf{1}_{q_m} \mathbf{1}_{q_n}' \mathbf{Y}_t^{(n)'}\right) \mathbf{M}_{p_n}. \tag{C.10}$$

The corresponding population matrix and its estimator are

$$\mathbf{\Omega}_{A_1,m} = (M-1)^{-1} \sum_{n=1, n\neq m}^{M} \mathbf{S}_{A_1,mn}^{\otimes 2}, \qquad \widehat{\mathbf{\Omega}}_{A_1,m} = (M-1)^{-1} \sum_{n=1, n\neq m}^{M} \widehat{\mathbf{S}}_{A_1,mn}^{\otimes 2}.$$

By (C.2), we have $\mathbf{S}_{A_1,mn} = T^{-1} \mathbf{A}_{1m} \left(\sum_{t=1}^{T} \mathbb{E}(\boldsymbol{\alpha}_t \boldsymbol{\alpha}_t')\right) \mathbf{A}_{1n}'$. Hence,

$$\mathbf{\Omega}_{A_1,m} = T^{-2}(M-1)^{-1} \mathbf{A}_{1m} \left(\sum_{t=1}^{T} \mathbb{E}(\boldsymbol{\alpha}_t \boldsymbol{\alpha}_t')\right) \left(\sum_{n\neq m} \mathbf{A}_{1n}' \mathbf{A}_{1n}\right) \left(\sum_{t=1}^{T} \mathbb{E}(\boldsymbol{\alpha}_t \boldsymbol{\alpha}_t')\right) \mathbf{A}_{1m}'. \tag{C.11}$$

Define $\omega_{A_1,m} := p_m^{\gamma_r} \sum_{n\neq m} p_n^{\gamma_r}$. Since (C.11) implies that the column space of $\mathbf{\Omega}_{A_1,m}$ is contained in that of $\mathbf{A}_{1m}$, with $\mathbf{P}_{A_1,m} := \breve{\mathbf{A}}_{1m} \breve{\mathbf{A}}_{1m}'$, we have

$$(\mathbf{I}_{p_m} - \mathbf{P}_{A_1,m}) \mathbf{\Omega}_{A_1,m} = \mathbf{0}. \tag{C.12}$$

Since $\widehat{\mathbf{A}}_{1m}$ consists of the leading r_1 eigenvectors of $\widehat{\mathbf{\Omega}}_{A_1,m}$,

$$\widehat{\mathbf{\Omega}}_{A_1,m}\widehat{\mathbf{A}}_{1m} = \widehat{\mathbf{A}}_{1m}\widehat{\mathbf{D}}_{A_1,m}, \qquad \widehat{\mathbf{D}}_{A_1,m} := \operatorname{diag}\big(\lambda_1(\widehat{\mathbf{\Omega}}_{A_1,m}), \ldots, \lambda_{r_1}(\widehat{\mathbf{\Omega}}_{A_1,m})\big). \tag{C.13}$$

Define $\mathbf{\Delta}_{A_1,mn} := \widehat{\mathbf{S}}_{A_1,mn} - \mathbf{S}_{A_1,mn}$. Then,

$$\widehat{\mathbf{\Omega}}_{A_1,m} - \mathbf{\Omega}_{A_1,m} = (M-1)^{-1} \sum_{n\neq m} \left(\mathbf{\Delta}_{A_1,mn}\mathbf{S}'_{A_1,mn} + \mathbf{S}_{A_1,mn}\mathbf{\Delta}'_{A_1,mn} + \mathbf{\Delta}_{A_1,mn}\mathbf{\Delta}'_{A_1,mn}\right). \tag{C.14}$$

Left-multiplying (C.14) by $\mathbf{I}_{p_m} - \mathbf{P}_{A_1,m}$ and using $(\mathbf{I}_{p_m} - \mathbf{P}_{A_1,m})\mathbf{S}_{A_1,mn} = \mathbf{0}$ gives

$$\begin{aligned}
&(\mathbf{I}_{p_m} - \mathbf{P}_{A_1,m})(\widehat{\mathbf{\Omega}}_{A_1,m} - \mathbf{\Omega}_{A_1,m}) \\
&= (M-1)^{-1} \sum_{n\neq m} \left\{(\mathbf{I}_{p_m} - \mathbf{P}_{A_1,m})\mathbf{\Delta}_{A_1,mn}\mathbf{S}'_{A_1,mn} + (\mathbf{I}_{p_m} - \mathbf{P}_{A_1,m})\mathbf{\Delta}_{A_1,mn}\mathbf{\Delta}'_{A_1,mn}\right\}.
\end{aligned}$$

By (D.43) and (D.44) in Lemma 5, $\left\|\mathbf{S}_{A_1,mn}\right\|_F^2 + \left\|\mathbf{\Delta}_{A_1,mn}\right\|_F^2 = O_P(p_m^{\gamma_r} p_n^{\gamma_r})$. Hence, (D.45) gives

$$\begin{aligned}
&\left\|(\mathbf{I}_{p_m} - \mathbf{P}_{A_1,m})(\widehat{\mathbf{\Omega}}_{A_1,m} - \mathbf{\Omega}_{A_1,m})\right\|_F^2 \\
&= O_P\Big(R(T,d_{r,\star}) p_m^{\gamma_r^{(m)}+\gamma_r} \sum_{n\neq m} p_n^{\gamma_r^{(n)}+\gamma_r} + T^{-1} p_m^{\gamma_r+1} q_m^{-1} \sum_{n\neq m} p_n^{\gamma_{r,\max}^{(n)}+\gamma_r}\Big) \\
&= O_P\left(\omega_{A_1,m}^2\Big(R(T,d_{r,\star}) p_m^{\gamma_r^{(m)}-\gamma_r} \max_{n\in[M]\setminus\{m\}} p_n^{\gamma_r^{(n)}-\gamma_r} + T^{-1} p_m^{1-\gamma_r} q_m^{-1} \max_{n\in[M]\setminus\{m\}} p_n^{\gamma_{r,\max}^{(n)}-\gamma_r}\Big)\right).
\end{aligned} \tag{C.15}$$

Define $\mathbf{H}_{A_1,m} := \check{\mathbf{A}}'_{1m}\widehat{\mathbf{A}}_{1m}$. Since $\check{\mathbf{A}}_{1m}$ and $\widehat{\mathbf{A}}_{1m}$ both have orthonormal columns, Lemma 4 gives

$$\widehat{\mathbf{A}}_{1m} - \check{\mathbf{A}}_{1m}\mathbf{H}_{A_1,m} = (\mathbf{I}_{p_m} - \mathbf{P}_{A_1,m})\widehat{\mathbf{A}}_{1m}. \tag{C.16}$$

Left-multiplying (C.13) by $\mathbf{I}_{p_m} - \mathbf{P}_{A_1,m}$, and using (C.12) and (C.16), gives

$$\left(\widehat{\mathbf{A}}_{1m} - \check{\mathbf{A}}_{1m}\mathbf{H}_{A_1,m}\right)\widehat{\mathbf{D}}_{A_1,m} = (\mathbf{I}_{p_m} - \mathbf{P}_{A_1,m})(\widehat{\mathbf{\Omega}}_{A_1,m} - \mathbf{\Omega}_{A_1,m})\widehat{\mathbf{A}}_{1m}. \tag{C.17}$$

Combining (C.15), (C.17), and Lemma 6, together with $\left\|\widehat{\mathbf{A}}_{1m}\right\| = 1$, we obtain

$$\begin{aligned}
\left\|\widehat{\mathbf{A}}_{1m} - \check{\mathbf{A}}_{1m}\mathbf{H}_{A_1,m}\right\|_F^2 &\le \left\|(\mathbf{I}_{p_m} - \mathbf{P}_{A_1,m})(\widehat{\mathbf{\Omega}}_{A_1,m} - \mathbf{\Omega}_{A_1,m})\right\|_F^2 \left\|\widehat{\mathbf{A}}_{1m}\right\|^2 \left\|\widehat{\mathbf{D}}_{A_1,m}^{-1}\right\|_F^2 \\
&= O_P\Big(R(T,d_{r,\star}) p_m^{\gamma_r^{(m)}-\gamma_r} \max_{n\in[M]\setminus\{m\}} p_n^{\gamma_r^{(n)}-\gamma_r} + T^{-1} p_m^{1-\gamma_r} q_m^{-1} \max_{n\in[M]\setminus\{m\}} p_n^{\gamma_{r,\max}^{(n)}-\gamma_r}\Big).
\end{aligned}$$

Using $p_n^{\gamma_{r,\max}^{(n)}-\gamma_r} = 1 \vee \nu_{r,n}$, this is exactly the row-side rate stated in Theorem 3. Both terms above are $o(1)$ under (R1), so that $\left\|\widehat{\mathbf{A}}_{1m} - \check{\mathbf{A}}_{1m}\mathbf{H}_{A_1,m}\right\|_F = o_P(1)$. Hence, Lemma 4 gives $\mathbf{H}'_{A_1,m}\mathbf{H}_{A_1,m} = \mathbf{I}_{r_1} + o_P(1)$,

$\mathbf{H}_{A_1,m}\mathbf{H}'_{A_1,m} = \mathbf{I}_{r_1} + o_P(1)$, and $\mathbf{H}_{A_1,m}$ is invertible with probability tending to one. Moreover, defining $\widehat{\mathbf{P}}_{A_1,m} := \widehat{\mathbf{A}}_{1m}\widehat{\mathbf{A}}'_{1m}$, we have

$$\big\|\widehat{\mathbf{P}}_{A_1,m} - \mathbf{P}_{A_1,m}\big\|_F^2 = 2\big\|\widehat{\mathbf{A}}_{1m} - \breve{\mathbf{A}}_{1m}\mathbf{H}_{A_1,m}\big\|_F^2. \tag{C.18}$$

The proof for the global column-loading estimator $\widehat{\mathbf{B}}_{1m}$ follows similarly, with $\mathbf{H}_{B_1,m} := \breve{\mathbf{B}}'_{1m}\widehat{\mathbf{B}}_{1m}$. This ends the proof. □

Proof of Theorem 4. Recall that

$$\mathbf{S}_{A_2,m} := T^{-1}\sum_{t=1}^{T}\mathbf{a}_t^{(m)}\mathbf{a}_t^{(m)'}\mathbf{P}^{\perp}_{A_1,m}, \qquad \widehat{\mathbf{S}}_{A_2,m} := T^{-1}\sum_{t=1}^{T}\widehat{\mathbf{a}}_t^{(m)}\widehat{\mathbf{a}}_t^{(m)'}\widehat{\mathbf{P}}^{\perp}_{A_1,m}, \tag{C.19}$$

where $\mathbf{P}^{\perp}_{A_1,m} := \mathbf{I}_{p_m} - \mathbf{P}_{A_1,m}$ and $\widehat{\mathbf{P}}^{\perp}_{A_1,m} := \mathbf{I}_{p_m} - \widehat{\mathbf{P}}_{A_1,m}$. Since $\widehat{\mathbf{A}}^{\perp}_{1m}\widehat{\mathbf{A}}^{\perp\prime}_{1m} = \widehat{\mathbf{P}}^{\perp}_{A_1,m}$, the sample matrix in the estimation procedure satisfies

$$\widehat{\mathbf{\Omega}}_{A_2,m} = \left(T^{-1}\sum_{t=1}^{T}\widehat{\mathbf{a}}_t^{(m)}\widehat{\mathbf{a}}_t^{(m)'}\right)\widehat{\mathbf{P}}^{\perp}_{A_1,m}\left(T^{-1}\sum_{t=1}^{T}\widehat{\mathbf{a}}_t^{(m)}\widehat{\mathbf{a}}_t^{(m)'}\right) = \widehat{\mathbf{S}}_{A_2,m}\widehat{\mathbf{S}}'_{A_2,m}. \tag{C.20}$$

Define its population counterpart as

$$\mathbf{\Omega}_{A_2,m} := \left(T^{-1}\sum_{t=1}^{T}\mathbf{a}_t^{(m)}\mathbf{a}_t^{(m)'}\right)\mathbf{P}^{\perp}_{A_1,m}\left(T^{-1}\sum_{t=1}^{T}\mathbf{a}_t^{(m)}\mathbf{a}_t^{(m)'}\right) = \mathbf{S}_{A_2,m}\mathbf{S}'_{A_2,m}. \tag{C.21}$$

By (IC2),

$$\mathbf{S}_{A_2,m} = \mathbf{A}_{2m}\left(T^{-1}\sum_{t=1}^{T}\boldsymbol{\alpha}_t^{(m)}\boldsymbol{\alpha}_t^{(m)'}\right)\mathbf{A}'_{2m}\mathbf{P}^{\perp}_{A_1,m}.$$

Hence, with $\mathbf{P}_{A_2,m} := \breve{\mathbf{A}}_{2m}\breve{\mathbf{A}}'_{2m}$,

$$(\mathbf{I}_{p_m} - \mathbf{P}_{A_2,m})\mathbf{S}_{A_2,m} = \mathbf{0}, \qquad (\mathbf{I}_{p_m} - \mathbf{P}_{A_2,m})\mathbf{\Omega}_{A_2,m} = \mathbf{0}. \tag{C.22}$$

Since $\widehat{\mathbf{A}}_{2m}$ consists of the leading $r_2^{(m)}$ eigenvectors of $\widehat{\mathbf{\Omega}}_{A_2,m}$,

$$\widehat{\mathbf{\Omega}}_{A_2,m}\widehat{\mathbf{A}}_{2m} = \widehat{\mathbf{A}}_{2m}\widehat{\mathbf{D}}_{A_2,m}, \qquad \widehat{\mathbf{D}}_{A_2,m} := \mathrm{diag}\big(\lambda_1(\widehat{\mathbf{\Omega}}_{A_2,m}), \ldots, \lambda_{r_2^{(m)}}(\widehat{\mathbf{\Omega}}_{A_2,m})\big). \tag{C.23}$$

Define $\mathbf{H}_{A_2,m} := \breve{\mathbf{A}}'_{2m}\widehat{\mathbf{A}}_{2m}$. Since $\breve{\mathbf{A}}_{2m}$ and $\widehat{\mathbf{A}}_{2m}$ both have orthonormal columns, Lemma 4 gives

$$\widehat{\mathbf{A}}_{2m} - \breve{\mathbf{A}}_{2m}\mathbf{H}_{A_2,m} = (\mathbf{I}_{p_m} - \mathbf{P}_{A_2,m})\widehat{\mathbf{A}}_{2m}. \tag{C.24}$$

Left-multiplying (C.23) by $\mathbf{I}_{p_m} - \mathbf{P}_{A_2,m}$, and using (C.22) and (C.24), gives

$$\left(\widehat{\mathbf{A}}_{2m} - \breve{\mathbf{A}}_{2m}\mathbf{H}_{A_2,m}\right)\widehat{\mathbf{D}}_{A_2,m} = (\mathbf{I}_{p_m} - \mathbf{P}_{A_2,m})(\widehat{\mathbf{\Omega}}_{A_2,m} - \mathbf{\Omega}_{A_2,m})\widehat{\mathbf{A}}_{2m}. \tag{C.25}$$

Let $\mathbf{\Delta}_{A_2,m} := \widehat{\mathbf{S}}_{A_2,m} - \mathbf{S}_{A_2,m}$. Since

$$\widehat{\mathbf{\Omega}}_{A_2,m} - \mathbf{\Omega}_{A_2,m} = \mathbf{\Delta}_{A_2,m}\mathbf{S}'_{A_2,m} + \mathbf{S}_{A_2,m}\mathbf{\Delta}'_{A_2,m} + \mathbf{\Delta}_{A_2,m}\mathbf{\Delta}'_{A_2,m},$$

using $(\mathbf{I}_{p_m} - \mathbf{P}_{A_2,m})\mathbf{S}_{A_2,m} = \mathbf{0}$ gives

$$\left\|(\mathbf{I}_{p_m} - \mathbf{P}_{A_2,m})(\widehat{\mathbf{\Omega}}_{A_2,m} - \mathbf{\Omega}_{A_2,m})\right\|_F^2 \lesssim \left\|(\mathbf{I}_{p_m} - \mathbf{P}_{A_2,m})\mathbf{\Delta}_{A_2,m}\right\|_F^2 \left(\left\|\mathbf{S}_{A_2,m}\right\|_F^2 + \left\|\mathbf{\Delta}_{A_2,m}\right\|_F^2\right). \tag{C.26}$$

By (D.56) and (D.57) in Lemma 7, $\left\|\mathbf{S}_{A_2,m}\right\|_F^2 + \left\|\mathbf{\Delta}_{A_2,m}\right\|_F^2 = O_P(p_m^{2\gamma_r^{(m)}})$. Combining (C.26), (D.58), (C.25), and Lemma 8, together with $\left\|\widehat{\mathbf{A}}_{2m}\right\| = 1$, we obtain

$$\begin{aligned}
&\left\|\widehat{\mathbf{A}}_{2m} - \breve{\mathbf{A}}_{2m}\mathbf{H}_{A_2,m}\right\|_F^2 \le \left\|(\mathbf{I}_{p_m} - \mathbf{P}_{A_2,m})(\widehat{\mathbf{\Omega}}_{A_2,m} - \mathbf{\Omega}_{A_2,m})\right\|_F^2 \left\|\widehat{\mathbf{A}}_{2m}\right\|^2 \left\|\widehat{\mathbf{D}}_{A_2,m}^{-1}\right\|_F^2 \\
&= O_P\Big(R(T,d_{r,\star})p_m^{\gamma_r-\gamma_r^{(m)}} \max_{n\in[M]\setminus\{m\}} p_n^{\gamma_r^{(n)}-\gamma_r} + T^{-1}p_m^{1+\gamma_r-2\gamma_r^{(m)}} q_m^{-1} \max_{n\in[M]} p_n^{\gamma_{r,\max}^{(n)}-\gamma_r} + p_m^{1-2\gamma_r^{(m)}} q_m^{-2}\Big).
\end{aligned}$$

Together with the definition of $\nu_{r,m}$, this proves the row-side loading rate in Theorem 4. All three terms above are $o(1)$ under (R1), so that $\left\|\widehat{\mathbf{A}}_{2m} - \breve{\mathbf{A}}_{2m}\mathbf{H}_{A_2,m}\right\|_F = o_P(1)$. Hence, Lemma 4 gives $\mathbf{H}'_{A_2,m}\mathbf{H}_{A_2,m} = \mathbf{I}_{r_2^{(m)}} + o_P(1)$, $\mathbf{H}_{A_2,m}\mathbf{H}'_{A_2,m} = \mathbf{I}_{r_2^{(m)}} + o_P(1)$, and $\mathbf{H}_{A_2,m}$ is invertible with probability tending to one. Moreover, defining $\widehat{\mathbf{P}}_{A_2,m} := \widehat{\mathbf{A}}_{2m}\widehat{\mathbf{A}}'_{2m}$, we have

$$\left\|\widehat{\mathbf{P}}_{A_2,m} - \mathbf{P}_{A_2,m}\right\|_F^2 = 2\left\|\widehat{\mathbf{A}}_{2m} - \breve{\mathbf{A}}_{2m}\mathbf{H}_{A_2,m}\right\|_F^2. \tag{C.27}$$

The proof for the local column-loading estimator $\widehat{\mathbf{B}}_{2m}$ follows similarly, with $\mathbf{H}_{B_2,m} := \breve{\mathbf{B}}'_{2m}\widehat{\mathbf{B}}_{2m}$.

We next prove the local main effect factor series. For $m \in [M]$, define

$$\mathcal{L}_{\alpha,m} := \mathbf{H}_{A_2,m}^{-1}(\breve{\mathbf{A}}'_{2m}\mathbf{P}^{\perp}_{A_1,m}\breve{\mathbf{A}}_{2m})^{-1}\breve{\mathbf{A}}'_{2m}\mathbf{P}^{\perp}_{A_1,m}, \qquad \widehat{\mathcal{L}}_{\alpha,m} := (\widehat{\mathbf{A}}'_{2m}\widehat{\mathbf{P}}^{\perp}_{A_1,m}\widehat{\mathbf{A}}_{2m})^{-1}\widehat{\mathbf{A}}'_{2m}\widehat{\mathbf{P}}^{\perp}_{A_1,m}.$$

By (S1) and the asymptotic orthogonality of $\mathbf{H}_{A_2,m}$, we have $\left\|\mathcal{L}_{\alpha,m}\right\| = O_P(1)$. Moreover, $\mathcal{L}_{\alpha,m}\mathbf{A}_{1m} = \mathbf{0}$, $\mathcal{L}_{\alpha,m}\breve{\mathbf{A}}_{2m} = \mathbf{H}_{A_2,m}^{-1}$, $\widehat{\mathcal{L}}_{\alpha,m}\widehat{\mathbf{A}}_{2m} = \mathbf{I}_{r_2^{(m)}}$, and $\widehat{\mathcal{L}}_{\alpha,m}\widehat{\mathbf{P}}_{A_1,m} = \mathbf{0}$.

We first control $\widehat{\mathcal{L}}_{\alpha,m} - \mathcal{L}_{\alpha,m}$. Note that

$$\widehat{\mathbf{A}}'_{2m}\widehat{\mathbf{P}}^{\perp}_{A_1,m}\widehat{\mathbf{A}}_{2m} - \mathbf{H}'_{A_2,m}\breve{\mathbf{A}}'_{2m}\mathbf{P}^{\perp}_{A_1,m}\breve{\mathbf{A}}_{2m}\mathbf{H}_{A_2,m}$$

$$= (\widehat{\mathbf{A}}_{2m} - \breve{\mathbf{A}}_{2m}\mathbf{H}_{A_2,m})'\widehat{\mathbf{P}}^{\perp}_{A_1,m}\widehat{\mathbf{A}}_{2m} + \mathbf{H}'_{A_2,m}\breve{\mathbf{A}}'_{2m}(\widehat{\mathbf{P}}^{\perp}_{A_1,m} - \mathbf{P}^{\perp}_{A_1,m})\widehat{\mathbf{A}}_{2m}$$
$$+ \mathbf{H}'_{A_2,m}\breve{\mathbf{A}}'_{2m}\mathbf{P}^{\perp}_{A_1,m}(\widehat{\mathbf{A}}_{2m} - \breve{\mathbf{A}}_{2m}\mathbf{H}_{A_2,m}).$$

Since $\breve{\mathbf{A}}_{2m}$ and $\widehat{\mathbf{A}}_{2m}$ have orthonormal columns, $\|\mathbf{H}_{A_2,m}\| = O_P(1)$, and $\|\widehat{\mathbf{P}}^{\perp}_{A_1,m} - \mathbf{P}^{\perp}_{A_1,m}\| = \|\widehat{\mathbf{P}}_{A_1,m} - \mathbf{P}_{A_1,m}\|$, it follows that

$$\begin{aligned}&\|\widehat{\mathbf{A}}'_{2m}\widehat{\mathbf{P}}^{\perp}_{A_1,m}\widehat{\mathbf{A}}_{2m} - \mathbf{H}'_{A_2,m}\breve{\mathbf{A}}'_{2m}\mathbf{P}^{\perp}_{A_1,m}\breve{\mathbf{A}}_{2m}\mathbf{H}_{A_2,m}\| \\ &= O_P\Big(\|\widehat{\mathbf{A}}_{2m} - \breve{\mathbf{A}}_{2m}\mathbf{H}_{A_2,m}\|_F + \|\widehat{\mathbf{P}}_{A_1,m} - \mathbf{P}_{A_1,m}\|\Big). \end{aligned} \tag{C.28}$$

By (S1), the eigenvalues of $\breve{\mathbf{A}}'_{2m}\mathbf{P}^{\perp}_{A_1,m}\breve{\mathbf{A}}_{2m}$ are bounded away from zero. Together with the asymptotic orthogonality of $\mathbf{H}_{A_2,m}$, this implies $\lambda_{r_2^{(m)}}\big(\mathbf{H}'_{A_2,m}\breve{\mathbf{A}}'_{2m}\mathbf{P}^{\perp}_{A_1,m}\breve{\mathbf{A}}_{2m}\mathbf{H}_{A_2,m}\big) \gtrsim_P 1$. Since the right-hand side of (C.28) is $o_P(1)$, Weyl's inequality further gives

$$\|(\widehat{\mathbf{A}}'_{2m}\widehat{\mathbf{P}}^{\perp}_{A_1,m}\widehat{\mathbf{A}}_{2m})^{-1}\| = O_P(1), \qquad \|(\mathbf{H}'_{A_2,m}\breve{\mathbf{A}}'_{2m}\mathbf{P}^{\perp}_{A_1,m}\breve{\mathbf{A}}_{2m}\mathbf{H}_{A_2,m})^{-1}\| = O_P(1).$$

Therefore, using $\mathbf{A}^{-1} - \mathbf{B}^{-1} = \mathbf{A}^{-1}(\mathbf{B} - \mathbf{A})\mathbf{B}^{-1}$,

$$\begin{aligned}&\|(\widehat{\mathbf{A}}'_{2m}\widehat{\mathbf{P}}^{\perp}_{A_1,m}\widehat{\mathbf{A}}_{2m})^{-1} - (\mathbf{H}'_{A_2,m}\breve{\mathbf{A}}'_{2m}\mathbf{P}^{\perp}_{A_1,m}\breve{\mathbf{A}}_{2m}\mathbf{H}_{A_2,m})^{-1}\| \\ &= O_P\Big(\|\widehat{\mathbf{A}}_{2m} - \breve{\mathbf{A}}_{2m}\mathbf{H}_{A_2,m}\|_F + \|\widehat{\mathbf{P}}_{A_1,m} - \mathbf{P}_{A_1,m}\|\Big). \end{aligned} \tag{C.29}$$

Moreover,

$$\widehat{\mathbf{A}}'_{2m}\widehat{\mathbf{P}}^{\perp}_{A_1,m} - \mathbf{H}'_{A_2,m}\breve{\mathbf{A}}'_{2m}\mathbf{P}^{\perp}_{A_1,m} = (\widehat{\mathbf{A}}_{2m} - \breve{\mathbf{A}}_{2m}\mathbf{H}_{A_2,m})'\widehat{\mathbf{P}}^{\perp}_{A_1,m} + \mathbf{H}'_{A_2,m}\breve{\mathbf{A}}'_{2m}(\widehat{\mathbf{P}}^{\perp}_{A_1,m} - \mathbf{P}^{\perp}_{A_1,m}),$$

and hence

$$\|\widehat{\mathbf{A}}'_{2m}\widehat{\mathbf{P}}^{\perp}_{A_1,m} - \mathbf{H}'_{A_2,m}\breve{\mathbf{A}}'_{2m}\mathbf{P}^{\perp}_{A_1,m}\| = O_P\Big(\|\widehat{\mathbf{A}}_{2m} - \breve{\mathbf{A}}_{2m}\mathbf{H}_{A_2,m}\|_F + \|\widehat{\mathbf{P}}_{A_1,m} - \mathbf{P}_{A_1,m}\|\Big). \tag{C.30}$$

Since $\mathcal{L}_{\alpha,m} = (\mathbf{H}'_{A_2,m}\breve{\mathbf{A}}'_{2m}\mathbf{P}^{\perp}_{A_1,m}\breve{\mathbf{A}}_{2m}\mathbf{H}_{A_2,m})^{-1}\mathbf{H}'_{A_2,m}\breve{\mathbf{A}}'_{2m}\mathbf{P}^{\perp}_{A_1,m}$, combining (C.29) and (C.30) yields

$$\|\widehat{\mathcal{L}}_{\alpha,m} - \mathcal{L}_{\alpha,m}\| = O_P\Big(\|\widehat{\mathbf{A}}_{2m} - \breve{\mathbf{A}}_{2m}\mathbf{H}_{A_2,m}\|_F + \|\widehat{\mathbf{P}}_{A_1,m} - \mathbf{P}_{A_1,m}\|\Big).$$

Thus, by Lemma 4,

$$\|\widehat{\mathcal{L}}_{\alpha,m} - \mathcal{L}_{\alpha,m}\|^2 = O_P\Big(\|\widehat{\mathbf{A}}_{1m} - \breve{\mathbf{A}}_{1m}\mathbf{H}_{A_1,m}\|_F^2 + \|\widehat{\mathbf{A}}_{2m} - \breve{\mathbf{A}}_{2m}\mathbf{H}_{A_2,m}\|_F^2\Big). \tag{C.31}$$

In particular, $\|\widehat{\mathcal{L}}_{\alpha,m}\| = O_P(1)$.

By the definition of $\widehat{\boldsymbol{\alpha}}_t^{(m)}$ and $\widehat{\mathbf{a}}_t^{(m)} - \mathbf{a}_t^{(m)} = q_m^{-1}\mathbf{M}_{p_m}\mathbf{E}_t^{(m)}\mathbf{1}_{q_m}$,

$$\begin{aligned}\widehat{\boldsymbol{\alpha}}_t^{(m)} - \mathbf{H}_{A_2,m}^{-1}\breve{\boldsymbol{\alpha}}_t^{(m)} &= q_m^{-1}\mathcal{L}_{\alpha,m}\mathbf{M}_{p_m}\mathbf{E}_t^{(m)}\mathbf{1}_{q_m} + q_m^{-1}(\widehat{\mathcal{L}}_{\alpha,m} - \mathcal{L}_{\alpha,m})\mathbf{M}_{p_m}\mathbf{E}_t^{(m)}\mathbf{1}_{q_m} \\ &\quad + \widehat{\mathcal{L}}_{\alpha,m}(\mathbf{P}_{A_1,m} - \widehat{\mathbf{P}}_{A_1,m})\mathbf{A}_{1m}\boldsymbol{\alpha}_t - \widehat{\mathcal{L}}_{\alpha,m}(\widehat{\mathbf{A}}_{2m} - \breve{\mathbf{A}}_{2m}\mathbf{H}_{A_2,m})\mathbf{H}_{A_2,m}^{-1}\breve{\boldsymbol{\alpha}}_t^{(m)} \\ &=: \mathcal{I}_{\alpha,1}^{(m)} + \mathcal{I}_{\alpha,2}^{(m)} + \mathcal{I}_{\alpha,3}^{(m)} + \mathcal{I}_{\alpha,4}^{(m)}, \end{aligned} \tag{C.32}$$

where the first equality uses $\mathcal{L}_{\alpha,m}\mathbf{A}_{1m} = \mathbf{0}$, $\widehat{\mathcal{L}}_{\alpha,m}\widehat{\mathbf{P}}_{A_1,m} = \mathbf{0}$, $\mathcal{L}_{\alpha,m}\breve{\mathbf{A}}_{2m} = \mathbf{H}_{A_2,m}^{-1}$ and $\widehat{\mathcal{L}}_{\alpha,m}\widehat{\mathbf{A}}_{2m} = \mathbf{I}_{r_2^{(m)}}$.

For the first term in (C.32), note that the matrix $(\breve{\mathbf{A}}_{2m}'\mathbf{P}_{A_1,m}^{\perp}\breve{\mathbf{A}}_{2m})^{-1}\breve{\mathbf{A}}_{2m}'\mathbf{P}_{A_1,m}^{\perp}$ has bounded Frobenius norm by (S1). Since $\|\mathbf{H}_{A_2,m}^{-1}\| = O_P(1)$, Lemma 1 gives $\|\mathcal{I}_{\alpha,1}^{(m)}\|^2 = O_P(q_m^{-1})$. Moreover, Lemma 1, (C.31), (ME1) and (ME2) yield respectively

$$\begin{aligned}\|\mathcal{I}_{\alpha,2}^{(m)}\|^2 &= O_P\left(p_m q_m^{-1}\left(\|\widehat{\mathbf{A}}_{1m} - \breve{\mathbf{A}}_{1m}\mathbf{H}_{A_1,m}\|_F^2 + \|\widehat{\mathbf{A}}_{2m} - \breve{\mathbf{A}}_{2m}\mathbf{H}_{A_2,m}\|_F^2\right)\right), \\ \|\mathcal{I}_{\alpha,3}^{(m)}\|^2 &= O_P\Big(p_m^{\gamma_r}\|\widehat{\mathbf{A}}_{1m} - \breve{\mathbf{A}}_{1m}\mathbf{H}_{A_1,m}\|_F^2\Big), \qquad \|\mathcal{I}_{\alpha,4}^{(m)}\|^2 = O_P\Big(p_m^{\gamma_r^{(m)}}\|\widehat{\mathbf{A}}_{2m} - \breve{\mathbf{A}}_{2m}\mathbf{H}_{A_2,m}\|_F^2\Big).\end{aligned}$$

Consequently,

$$\begin{aligned}p_m^{-\gamma_r^{(m)}}\|\widehat{\boldsymbol{\alpha}}_t^{(m)} - \mathbf{H}_{A_2,m}^{-1}\breve{\boldsymbol{\alpha}}_t^{(m)}\|^2 = O_P\Big(&p_m^{-\gamma_r^{(m)}}q_m^{-1} + (p_m^{1-\gamma_r^{(m)}}q_m^{-1} + p_m^{\gamma_r - \gamma_r^{(m)}})\|\widehat{\mathbf{A}}_{1m} - \breve{\mathbf{A}}_{1m}\mathbf{H}_{A_1,m}\|_F^2 \\ &+ (p_m^{1-\gamma_r^{(m)}}q_m^{-1} + 1)\|\widehat{\mathbf{A}}_{2m} - \breve{\mathbf{A}}_{2m}\mathbf{H}_{A_2,m}\|_F^2\Big).\end{aligned}$$

Substituting the rates from Theorems 3 and 4 obtained above, and using $1 + p_m^{\gamma_r - \gamma_r^{(m)}} \lesssim p_m^{\gamma_{r,\max}^{(m)} - \gamma_r^{(m)}}$, and noting that $p_m^{1-2\gamma_r^{(m)}}q_m^{-2} = o(p_m^{-\gamma_r^{(m)}}q_m^{-1})$ under (R1), gives

$$\begin{aligned}p_m^{-\gamma_r^{(m)}}\|\widehat{\boldsymbol{\alpha}}_t^{(m)} - \mathbf{H}_{A_2,m}^{-1}\breve{\boldsymbol{\alpha}}_t^{(m)}\|^2 = O_P\Big(&p_m^{-\gamma_r^{(m)}}q_m^{-1} + R(T,d_{r,\star})p_m^{\gamma_{r,\max}^{(m)} - \gamma_r^{(m)}}\max_{n\in[M]\setminus\{m\}}p_n^{\gamma_r^{(n)} - \gamma_r} \\ &+ T^{-1}p_m^{1+\gamma_{r,\max}^{(m)} - 2\gamma_r^{(m)}}q_m^{-1}\max_{n\in[M]}p_n^{\gamma_{r,\max}^{(n)} - \gamma_r}\Big).\end{aligned} \tag{C.33}$$

Since $p_m^{\gamma_{r,\max}^{(m)} - \gamma_r^{(m)}} = (1 \wedge \nu_{r,m})^{-1}$ and $p_n^{\gamma_{r,\max}^{(n)} - \gamma_r} = 1 \vee \nu_{r,n}$, this is the row-side factor rate stated in Theorem 4. The column-side rate follows by symmetry. This ends the proof. □

Proof of Theorem 5. We prove the row-side results; the column-side results follow by symmetry. Let $\mathbf{A}_\alpha := (\boldsymbol{\alpha}_1, \ldots, \boldsymbol{\alpha}_T)$, $\boldsymbol{\Gamma}_{\alpha,T} := T^{-1}\mathbf{A}_\alpha\mathbf{A}_\alpha'$, $\breve{\mathbf{A}}_\alpha := \boldsymbol{\Gamma}_{\alpha,T}^{-1/2}\mathbf{A}_\alpha = (\breve{\boldsymbol{\alpha}}_1, \ldots, \breve{\boldsymbol{\alpha}}_T)$. Also write $\widehat{\mathbf{A}}_\alpha := (\widehat{\boldsymbol{\alpha}}_1, \ldots, \widehat{\boldsymbol{\alpha}}_T)$ and define $\mathbf{P}_\alpha := T^{-1}\breve{\mathbf{A}}_\alpha'\breve{\mathbf{A}}_\alpha$. By Lemma 2, (S1), and Weyl's inequality, $\lambda_{r_1}(\boldsymbol{\Gamma}_{\alpha,T}) \asymp_P \lambda_1(\boldsymbol{\Gamma}_{\alpha,T}) \asymp_P 1$. Hence $\|\boldsymbol{\Gamma}_{\alpha,T}^{-1/2}\| = O_P(1)$ and $\|\breve{\boldsymbol{\alpha}}_t\| = O_P(1)$ for each $t \in [T]$.

For each $m \in [M]$, define $\mathbf{Z}_{\alpha,m} := \widehat{\mathbf{A}}_{1m}'\mathbf{A}_{1m}\mathbf{A}_\alpha$ and $\boldsymbol{\Delta}_{\alpha,m} := \widehat{\mathbf{Z}}_{\alpha,m} - \mathbf{Z}_{\alpha,m}$, and let $\boldsymbol{\delta}_{\alpha,m,t}$ denote the

tth column of $\boldsymbol{\Delta}_{\alpha,m}$. Since $\widehat{\mathbf{A}}'_{1m}\mathbf{A}_{1m} = \mathbf{H}'_{A_1,m}(\mathbf{A}'_{1m}\mathbf{A}_{1m})^{1/2}$, Theorem 3 and (ME1) imply that $\widehat{\mathbf{A}}'_{1m}\mathbf{A}_{1m}$ is invertible with probability tending to one and has squared singular values of order $p_m^{\gamma_r}$. Thus,

$$\mathbf{P}_{\mathbf{Z}_{\alpha,m}} = \mathbf{A}'_\alpha(\mathbf{A}_\alpha\mathbf{A}'_\alpha)^{-1}\mathbf{A}_\alpha = \mathbf{P}_\alpha, \qquad \sigma^2_{r_1}(\mathbf{Z}_{\alpha,m}) \asymp_P Tp_m^{\gamma_r}. \tag{C.34}$$

Write

$$\widehat{\mathbf{y}}_t^{(m)} - \mathbf{A}_{1m}\boldsymbol{\alpha}_t = q_m^{-1}\mathbf{M}_{p_m}\mathbf{E}_t^{(m)}\mathbf{1}_{q_m} - (\widehat{\mathbf{A}}_{2m}\widehat{\boldsymbol{\alpha}}_t^{(m)} - \mathbf{A}_{2m}\boldsymbol{\alpha}_t^{(m)}),$$
$$\boldsymbol{\delta}_{\alpha,m,t} = \widehat{\mathbf{A}}'_{1m}\left(q_m^{-1}\mathbf{M}_{p_m}\mathbf{E}_t^{(m)}\mathbf{1}_{q_m} - (\widehat{\mathbf{A}}_{2m}\widehat{\boldsymbol{\alpha}}_t^{(m)} - \mathbf{A}_{2m}\boldsymbol{\alpha}_t^{(m)})\right).$$

Using $\widehat{\mathbf{A}}_{2m}\widehat{\boldsymbol{\alpha}}_t^{(m)} - \mathbf{A}_{2m}\boldsymbol{\alpha}_t^{(m)} = \widehat{\mathbf{A}}_{2m}\big(\widehat{\boldsymbol{\alpha}}_t^{(m)} - \mathbf{H}_{A_2,m}^{-1}\breve{\boldsymbol{\alpha}}_t^{(m)}\big) + (\widehat{\mathbf{A}}_{2m} - \breve{\mathbf{A}}_{2m}\mathbf{H}_{A_2,m})\mathbf{H}_{A_2,m}^{-1}\breve{\boldsymbol{\alpha}}_t^{(m)}$, the loading and series rates in Theorem 4, the asymptotic orthogonality of $\mathbf{H}_{A_2,m}$, and $\big\|\breve{\boldsymbol{\alpha}}_t^{(m)}\big\|^2 = O_P(p_m^{\gamma_r^{(m)}})$ give

$$\big\|\widehat{\mathbf{A}}_{2m}\widehat{\boldsymbol{\alpha}}_t^{(m)} - \mathbf{A}_{2m}\boldsymbol{\alpha}_t^{(m)}\big\|^2 = O_P\Big(q_m^{-1} + R(T,d_{r,\star})p_m^{\gamma_{r,\max}^{(m)}} \max_{n\in[M]\setminus\{m\}} p_n^{\gamma_r^{(n)}-\gamma_r}$$
$$+ T^{-1}p_m^{1+\gamma_{r,\max}^{(m)}-\gamma_r^{(m)}} q_m^{-1} \max_{n\in[M]} p_n^{\gamma_{r,\max}^{(n)}-\gamma_r}\Big). \tag{C.35}$$

The same rate holds for $T^{-1}\sum_{t=1}^T \big\|\widehat{\mathbf{A}}_{2m}\widehat{\boldsymbol{\alpha}}_t^{(m)} - \mathbf{A}_{2m}\boldsymbol{\alpha}_t^{(m)}\big\|^2$, Indeed, the same argument for $\mathcal{I}_{\alpha,1}^{(m)}$ gives $T^{-1}\sum_t \big\|\mathcal{I}_{\alpha,1}^{(m)}\big\|^2 = O_P(q_m^{-1})$. Applying the remaining bounds in (C.32), together with

$$T^{-1}\sum_t \big\|q_m^{-1}\mathbf{M}_{p_m}\mathbf{E}_t^{(m)}\mathbf{1}_{q_m}\big\|^2 = O_P(p_m q_m^{-1}),$$
$$T^{-1}\sum_t \big\|\mathbf{A}_{1m}\boldsymbol{\alpha}_t\big\|^2 = O_P(p_m^{\gamma_r}), \qquad T^{-1}\sum_t \big\|\breve{\boldsymbol{\alpha}}_t^{(m)}\big\|^2 = O_P(p_m^{\gamma_r^{(m)}}),$$

gives the stated time-average reconstruction rate.

Next, since

$$q_m^{-1}\widehat{\mathbf{A}}'_{1m}\mathbf{M}_{p_m}\mathbf{E}_t^{(m)}\mathbf{1}_{q_m} = q_m^{-1}\mathbf{H}'_{A_1,m}\breve{\mathbf{A}}'_{1m}\mathbf{M}_{p_m}\mathbf{E}_t^{(m)}\mathbf{1}_{q_m} + q_m^{-1}(\widehat{\mathbf{A}}_{1m} - \breve{\mathbf{A}}_{1m}\mathbf{H}_{A_1,m})'\mathbf{M}_{p_m}\mathbf{E}_t^{(m)}\mathbf{1}_{q_m},$$

Lemma 1 gives $\big\|q_m^{-1}\breve{\mathbf{A}}'_{1m}\mathbf{M}_{p_m}\mathbf{E}_t^{(m)}\mathbf{1}_{q_m}\big\|^2 = O_P(q_m^{-1})$, with the same rate after averaging over t. Using Theorem 3 and (R1), this gives

$$\big\|q_m^{-1}\widehat{\mathbf{A}}'_{1m}\mathbf{M}_{p_m}\mathbf{E}_t^{(m)}\mathbf{1}_{q_m}\big\|^2, \quad T^{-1}\sum_{t=1}^T \big\|q_m^{-1}\widehat{\mathbf{A}}'_{1m}\mathbf{M}_{p_m}\mathbf{E}_t^{(m)}\mathbf{1}_{q_m}\big\|^2$$
$$= O_P\Big(q_m^{-1} + R(T,d_{r,\star})p_m^{\gamma_r^{(m)}} \max_{n\in[M]\setminus\{m\}} p_n^{\gamma_r^{(n)}-\gamma_r} + T^{-1}p_m q_m^{-1} \max_{n\in[M]\setminus\{m\}} p_n^{\gamma_{r,\max}^{(n)}-\gamma_r}\Big),$$

which is bounded by the right-hand side of (C.35). Since $\|\widehat{\mathbf{A}}_{1m}\| = 1$, we therefore obtain

$$\|\boldsymbol{\delta}_{\alpha,m,t}\|^2, \quad T^{-1}\|\boldsymbol{\Delta}_{\alpha,m}\|_F^2 = O_P\Big(q_m^{-1} + R(T,d_{r,\star})p_m^{\gamma_{r,\max}^{(m)}} \max_{n\in[M]\setminus\{m\}} p_n^{\gamma_r^{(n)}-\gamma_r} + T^{-1}p_m^{1+\gamma_{r,\max}^{(m)}-\gamma_r^{(m)}} q_m^{-1} \max_{n\in[M]} p_n^{\gamma_{r,\max}^{(n)}-\gamma_r}\Big). \tag{C.36}$$

Define $\widehat{\mathbf{P}}_{\alpha,m} := \widehat{\mathbf{Z}}'_{\alpha,m}(\widehat{\mathbf{Z}}_{\alpha,m}\widehat{\mathbf{Z}}'_{\alpha,m})^{-1}\widehat{\mathbf{Z}}_{\alpha,m}$. By (C.34), (C.36), and (R1), $\|\boldsymbol{\Delta}_{\alpha,m}\|_F^2 = o_P(Tp_m^{\gamma_r})$. Hence Weyl's inequality gives $\sigma_{r_1}^2(\widehat{\mathbf{Z}}_{\alpha,m}) \asymp_P Tp_m^{\gamma_r}$, so that $\widehat{\mathbf{Z}}_{\alpha,m}\widehat{\mathbf{Z}}'_{\alpha,m}$ is invertible with probability tending to one. Since $\mathbf{P}_\alpha^\perp \mathbf{Z}'_{\alpha,m} = \mathbf{0}$,

$$\mathbf{P}_\alpha^\perp \widehat{\mathbf{P}}_{\alpha,m} = \mathbf{P}_\alpha^\perp \boldsymbol{\Delta}'_{\alpha,m}(\widehat{\mathbf{Z}}_{\alpha,m}\widehat{\mathbf{Z}}'_{\alpha,m})^{-1}\widehat{\mathbf{Z}}_{\alpha,m}. \tag{C.37}$$

Using $\|(\widehat{\mathbf{Z}}_{\alpha,m}\widehat{\mathbf{Z}}'_{\alpha,m})^{-1}\widehat{\mathbf{Z}}_{\alpha,m}\| = \sigma_{r_1}^{-1}(\widehat{\mathbf{Z}}_{\alpha,m})$ and $\|\widehat{\mathbf{P}}_{\alpha,m} - \mathbf{P}_\alpha\|_F^2 = 2\|\mathbf{P}_\alpha^\perp \widehat{\mathbf{P}}_{\alpha,m}\|_F^2$, we obtain

$$\|\widehat{\mathbf{P}}_{\alpha,m} - \mathbf{P}_\alpha\|_F^2 = O_P\Big(p_m^{-\gamma_r}q_m^{-1} + R(T,d_{r,\star})p_m^{\gamma_{r,\max}^{(m)}-\gamma_r} \max_{n\in[M]\setminus\{m\}} p_n^{\gamma_r^{(n)}-\gamma_r} + T^{-1}p_m^{1-\gamma_{r,\min}^{(m)}} q_m^{-1} \max_{n\in[M]} p_n^{\gamma_{r,\max}^{(n)}-\gamma_r}\Big). \tag{C.38}$$

All terms on the right-hand side are $o(1)$ under (R1).

For the rowwise bound, let $\mathbf{U}_\alpha := T^{-1/2}\breve{\mathbf{A}}'_\alpha$, so that $\mathbf{P}_\alpha = \mathbf{U}_\alpha\mathbf{U}'_\alpha$ and $\|\mathbf{e}_t'\mathbf{U}_\alpha\|^2 = T^{-1}\|\breve{\boldsymbol{\alpha}}_t\|^2 = O_P(T^{-1})$. Hence

$$\|\mathbf{e}_t'\mathbf{P}_\alpha^\perp \boldsymbol{\Delta}'_{\alpha,m}\|^2 \lesssim \|\boldsymbol{\delta}_{\alpha,m,t}\|^2 + T^{-1}\|\breve{\boldsymbol{\alpha}}_t\|^2\|\boldsymbol{\Delta}_{\alpha,m}\|_F^2.$$

Combining this with (C.37) and (C.36) yields

$$T\|\mathbf{e}_t'\mathbf{P}_\alpha^\perp(\widehat{\mathbf{P}}_{\alpha,m} - \mathbf{P}_\alpha)\|^2 = O_P\Big(p_m^{-\gamma_r}q_m^{-1} + R(T,d_{r,\star})p_m^{\gamma_{r,\max}^{(m)}-\gamma_r} \max_{n\in[M]\setminus\{m\}} p_n^{\gamma_r^{(n)}-\gamma_r} + T^{-1}p_m^{1-\gamma_{r,\min}^{(m)}} q_m^{-1} \max_{n\in[M]} p_n^{\gamma_{r,\max}^{(n)}-\gamma_r}\Big). \tag{C.39}$$

Now let $\widehat{\mathbf{K}}_\alpha := M^{-1}\sum_{m=1}^M \widehat{\mathbf{P}}_{\alpha,m}$ and $\widehat{\mathbf{U}}_\alpha := T^{-1/2}\widehat{\mathbf{A}}'_\alpha$, and define $\mathbf{H}_\alpha := \mathbf{U}'_\alpha\widehat{\mathbf{U}}_\alpha = T^{-1}\breve{\mathbf{A}}_\alpha\widehat{\mathbf{A}}'_\alpha$. By construction, $\widehat{\mathbf{K}}_\alpha\widehat{\mathbf{U}}_\alpha = \widehat{\mathbf{U}}_\alpha\widehat{\mathbf{D}}_\alpha$, where $\widehat{\mathbf{D}}_\alpha$ contains the leading r_1 eigenvalues of $\widehat{\mathbf{K}}_\alpha$. Since M is fixed, (C.38) gives $\|\widehat{\mathbf{K}}_\alpha - \mathbf{P}_\alpha\|_F = o_P(1)$. The nonzero eigenvalues of $\mathbf{P}_\alpha$ are all equal to one, so $\lambda_{r_1}(\widehat{\mathbf{K}}_\alpha) \asymp_P 1$ and $\|\widehat{\mathbf{D}}_\alpha^{-1}\| = O_P(1)$.

By Lemma 4, $\widehat{\mathbf{U}}_\alpha - \mathbf{U}_\alpha\mathbf{H}_\alpha = \mathbf{P}_\alpha^\perp\widehat{\mathbf{U}}_\alpha$. Left-multiplying the preceding eigenequation by $\mathbf{P}_\alpha^\perp$ therefore

gives

$$(\widehat{\mathbf{U}}_\alpha - \mathbf{U}_\alpha \mathbf{H}_\alpha)\widehat{\mathbf{D}}_\alpha = \mathbf{P}_\alpha^\perp (\widehat{\mathbf{K}}_\alpha - \mathbf{P}_\alpha)\widehat{\mathbf{U}}_\alpha. \tag{C.40}$$

Equations (C.38) and (C.40) imply $\|\widehat{\mathbf{U}}_\alpha - \mathbf{U}_\alpha \mathbf{H}_\alpha\|_F = o_P(1)$. Hence Lemma 4 shows that $\mathbf{H}_\alpha$ is asymptotically orthogonal.

For each $t \in [T]$, left-multiplying (C.40) by $\mathbf{e}_t'$ and using (C.39) gives

$$\|\widehat{\boldsymbol{\alpha}}_t - \mathbf{H}_\alpha' \breve{\boldsymbol{\alpha}}_t\|^2 = O_P\Bigg(\max_{m\in[M]} \Big\{ p_m^{-\gamma_r} q_m^{-1} + R(T, d_{r,\star}) p_m^{\gamma_{r,\max}^{(m)} - \gamma_r} \max_{n \in [M]\setminus\{m\}} p_n^{\gamma_r^{(n)} - \gamma_r} + T^{-1} p_m^{1-\gamma_{r,\min}^{(m)}} q_m^{-1} \max_{n\in[M]} p_n^{\gamma_{r,\max}^{(n)} - \gamma_r} \Big\} \Bigg).$$

Moreover, $\mathbf{H}_\alpha' \mathbf{H}_\alpha = \mathbf{I} - (\widehat{\mathbf{U}}_\alpha - \mathbf{U}_\alpha \mathbf{H}_\alpha)'(\widehat{\mathbf{U}}_\alpha - \mathbf{U}_\alpha \mathbf{H}_\alpha)$. Since $\mathbf{H}_\alpha^{-1} = (\mathbf{H}_\alpha' \mathbf{H}_\alpha)^{-1} \mathbf{H}_\alpha'$ and $\mathbf{H}_\alpha$ is asymptotically orthogonal, $\|\mathbf{H}_\alpha^{-1} - \mathbf{H}_\alpha'\| = O_P(\|\widehat{\mathbf{U}}_\alpha - \mathbf{U}_\alpha \mathbf{H}_\alpha\|_F^2)$. Together with $\|\breve{\boldsymbol{\alpha}}_t\| = O_P(1)$, replacing $\mathbf{H}_\alpha'$ by $\mathbf{H}_\alpha^{-1}$ contributes only the square of the preceding bound and is therefore negligible under (R1). Therefore,

$$\|\widehat{\boldsymbol{\alpha}}_t - \mathbf{H}_\alpha^{-1} \breve{\boldsymbol{\alpha}}_t\|^2 = O_P\Bigg(\max_{m\in[M]} \Big\{ p_m^{-\gamma_r} q_m^{-1} + R(T, d_{r,\star}) p_m^{\gamma_{r,\max}^{(m)} - \gamma_r} \max_{n \in [M]\setminus\{m\}} p_n^{\gamma_r^{(n)} - \gamma_r} + T^{-1} p_m^{1-\gamma_{r,\min}^{(m)}} q_m^{-1} \max_{n\in[M]} p_n^{\gamma_{r,\max}^{(n)} - \gamma_r} \Big\} \Bigg),$$

which proves the stated rate for the global row main effect series.

We next consider the refitted global loading $\widetilde{\mathbf{A}}_{1m}$. Let $\widehat{\mathbf{Y}}_m := (\widehat{\mathbf{y}}_1^{(m)}, \ldots, \widehat{\mathbf{y}}_T^{(m)})$ and $\mathbf{Y}_m := \mathbf{A}_{1m}\mathbf{A}_\alpha$. By (3.7) and the definitions of $\breve{\breve{\mathbf{A}}}_{1m}$ and $\mathbf{H}_\alpha$,

$$\widetilde{\mathbf{A}}_{1m} - \breve{\breve{\mathbf{A}}}_{1m}\mathbf{H}_\alpha = T^{-1}(\widehat{\mathbf{Y}}_m - \mathbf{Y}_m)\widehat{\mathbf{A}}_\alpha'. \tag{C.41}$$

Define $\boldsymbol{\Pi}_{2|1,m} := \breve{\mathbf{A}}_{2m}(\breve{\mathbf{A}}_{2m}' \mathbf{P}_{A_1,m}^\perp \breve{\mathbf{A}}_{2m})^{-1} \breve{\mathbf{A}}_{2m}' \mathbf{P}_{A_1,m}^\perp$, and define $\widehat{\boldsymbol{\Pi}}_{2|1,m}$ analogously using $\widehat{\mathbf{A}}_{2m}$ and $\widehat{\mathbf{P}}_{A_1,m}^\perp$. By (S1) and Theorem 4, $\|\widehat{\boldsymbol{\Pi}}_{2|1,m}\| = O_P(1)$. Since $\widehat{\mathbf{A}}_{2m}\widehat{\boldsymbol{\alpha}}_t^{(m)} = \widehat{\boldsymbol{\Pi}}_{2|1,m}\widehat{\mathbf{a}}_t^{(m)}$ and $\mathbf{A}_{2m}\boldsymbol{\alpha}_t^{(m)} = \boldsymbol{\Pi}_{2|1,m}\mathbf{a}_t^{(m)}$, we have

$$\widehat{\mathbf{y}}_t^{(m)} - \mathbf{A}_{1m}\boldsymbol{\alpha}_t = q_m^{-1}(\mathbf{I}_{p_m} - \widehat{\boldsymbol{\Pi}}_{2|1,m})\mathbf{M}_{p_m}\mathbf{E}_t^{(m)}\mathbf{1}_{q_m} - (\widehat{\boldsymbol{\Pi}}_{2|1,m} - \boldsymbol{\Pi}_{2|1,m})\mathbf{a}_t^{(m)}.$$

By (IC2), $T^{-1}\sum_t \mathbf{a}_t^{(m)}\breve{\boldsymbol{\alpha}}_t' = \mathbf{A}_{1m}\boldsymbol{\Gamma}_{\alpha,T}^{1/2} = \breve{\breve{\mathbf{A}}}_{1m}$ and $\boldsymbol{\Pi}_{2|1,m}\breve{\breve{\mathbf{A}}}_{1m} = \mathbf{0}$. Consequently,

$$T^{-1}(\widehat{\mathbf{Y}}_m - \mathbf{Y}_m)\breve{\mathbf{A}}_\alpha' = (\mathbf{I}_{p_m} - \widehat{\boldsymbol{\Pi}}_{2|1,m})\Big(T^{-1}\sum_{t=1}^T q_m^{-1}\mathbf{M}_{p_m}\mathbf{E}_t^{(m)}\mathbf{1}_{q_m}\breve{\boldsymbol{\alpha}}_t'\Big) - \widehat{\boldsymbol{\Pi}}_{2|1,m}\breve{\breve{\mathbf{A}}}_{1m}.$$

Since $T^{-1}\sum_t q_m^{-1}\mathbf{M}_{p_m}\mathbf{E}_t^{(m)}\mathbf{1}_{q_m}\breve{\boldsymbol{\alpha}}_t' = q_m^{-1}\mathbf{M}_{p_m}(T^{-1}\sum_t \mathbf{E}_t^{(m)}\mathbf{1}_{q_m}\boldsymbol{\alpha}_t')\boldsymbol{\Gamma}_{\alpha,T}^{-1/2}$, (D.33) in Lemma 3 gives its squared Frobenius norm as $O_P(T^{-1}p_m q_m^{-1})$. Also $\mathbf{P}_{A_1,m}^{\perp}\breve{\breve{\mathbf{A}}}_{1m} = \mathbf{0}$, and hence

$$\big\|\widehat{\boldsymbol{\Pi}}_{2|1,m}\breve{\breve{\mathbf{A}}}_{1m}\big\|_F^2 \lesssim_P \big\|(\widehat{\mathbf{P}}_{A_1,m} - \mathbf{P}_{A_1,m})\breve{\breve{\mathbf{A}}}_{1m}\big\|_F^2 = O_P\Big(p_m^{\gamma_r}\big\|\widehat{\mathbf{A}}_{1m} - \breve{\mathbf{A}}_{1m}\mathbf{H}_{A_1,m}\big\|_F^2\Big).$$

Substituting Theorem 3 gives

$$\begin{aligned}\big\|T^{-1}(\widehat{\mathbf{Y}}_m - \mathbf{Y}_m)\breve{\mathbf{A}}_\alpha'\big\|_F^2 = O_P\Big(&R(T,d_{r,\star})p_m^{\gamma_r^{(m)}}\max_{n\in[M]\setminus\{m\}}p_n^{\gamma_r^{(n)}-\gamma_r}\\ &+ T^{-1}p_m q_m^{-1}\max_{n\in[M]\setminus\{m\}}p_n^{\gamma_{r,\max}^{(n)}-\gamma_r}\Big).\end{aligned} \tag{C.42}$$

Furthermore, (C.35) and $T^{-1}\sum_t \big\|q_m^{-1}\mathbf{M}_{p_m}\mathbf{E}_t^{(m)}\mathbf{1}_{q_m}\big\|^2 = O_P(p_m q_m^{-1})$ give

$$T^{-1}\big\|\widehat{\mathbf{Y}}_m - \mathbf{Y}_m\big\|_F^2 = O_P\Big(p_m q_m^{-1} + R(T,d_{r,\star})p_m^{\gamma_{r,\max}^{(m)}}\max_{n\in[M]\setminus\{m\}}p_n^{\gamma_r^{(n)}-\gamma_r}\Big), \tag{C.43}$$

where the remaining term in (C.35) is absorbed into $p_m q_m^{-1}$ under (R1).

Finally, since $\mathbf{P}_\alpha\widehat{\mathbf{A}}_\alpha' = \breve{\mathbf{A}}_\alpha'\mathbf{H}_\alpha$, (C.41) gives

$$\widetilde{\mathbf{A}}_{1m} - \breve{\breve{\mathbf{A}}}_{1m}\mathbf{H}_\alpha = \big(T^{-1}(\widehat{\mathbf{Y}}_m - \mathbf{Y}_m)\breve{\mathbf{A}}_\alpha'\big)\mathbf{H}_\alpha + T^{-1}(\widehat{\mathbf{Y}}_m - \mathbf{Y}_m)\mathbf{P}_\alpha^{\perp}\widehat{\mathbf{A}}_\alpha'.$$

The first term is controlled by (C.42). For the second term, since $\mathbf{P}_\alpha^{\perp}\widehat{\mathbf{A}}_\alpha' = \sqrt{T}\mathbf{P}_\alpha^{\perp}\widehat{\mathbf{U}}_\alpha$,

$$\big\|T^{-1}(\widehat{\mathbf{Y}}_m - \mathbf{Y}_m)\mathbf{P}_\alpha^{\perp}\widehat{\mathbf{A}}_\alpha'\big\|_F^2 \le \big\{T^{-1}\big\|\widehat{\mathbf{Y}}_m - \mathbf{Y}_m\big\|_F^2\big\}\big\|\mathbf{P}_\alpha^{\perp}\widehat{\mathbf{U}}_\alpha\big\|_F^2.$$

By (C.38) and (C.40), $\big\|\mathbf{P}_\alpha^{\perp}\widehat{\mathbf{U}}_\alpha\big\|_F^2$ satisfies the same bound obtained above for the global row main effect series. Combining this with (C.42) and (C.43), and dividing the resulting bound by $p_m^{\gamma_r}$, gives

$$p_m^{-\gamma_r}\big\|\widetilde{\mathbf{A}}_{1m} - \breve{\breve{\mathbf{A}}}_{1m}\mathbf{H}_\alpha\big\|_F^2 = O_P\Big(\eta_{A_1,m} + \Big\{\frac{p_m^{1-\gamma_r}}{q_m} + R(T,d_{r,\star})(1\vee\nu_{r,m})\max_{n\in[M]\setminus\{m\}}\nu_{r,n}\Big\}\eta_\alpha\Big),$$

which proves Theorem 5. The column-side result follows by symmetry.

Expanding $\eta_{A_1,m}$ and η_α gives the equivalent explicit rate:

$$\begin{aligned}p_m^{-\gamma_r}\big\|\widetilde{\mathbf{A}}_{1m} - \breve{\breve{\mathbf{A}}}_{1m}\mathbf{H}_\alpha\big\|_F^2 = O_P\Big(&R(T,d_{r,\star})p_m^{\gamma_r^{(m)}-\gamma_r}\max_{n\in[M]\setminus\{m\}}p_n^{\gamma_r^{(n)}-\gamma_r}\\ &+ T^{-1}p_m^{1-\gamma_r}q_m^{-1}\max_{n\in[M]\setminus\{m\}}p_n^{\gamma_{r,\max}^{(n)}-\gamma_r} + \Big\{p_m^{1-\gamma_r}q_m^{-1} + R(T,d_{r,\star})p_m^{\gamma_{r,\max}^{(m)}-\gamma_r}\max_{n\in[M]\setminus\{m\}}p_n^{\gamma_r^{(n)}-\gamma_r}\Big\}\\ &\cdot\max_{n\in[M]}\Big\{p_n^{-\gamma_r}q_n^{-1} + R(T,d_{r,\star})p_n^{\gamma_{r,\max}^{(n)}-\gamma_r}\max_{s\in[M]\setminus\{n\}}p_s^{\gamma_r^{(s)}-\gamma_r} + T^{-1}p_n^{1-\gamma_{r,\min}^{(n)}}q_n^{-1}\max_{s\in[M]}p_s^{\gamma_{r,\max}^{(s)}-\gamma_r}\Big\}\Big),\end{aligned}$$

$$q_m^{-\gamma_c}\big\|\widetilde{\mathbf{B}}_{1m} - \breve{\breve{\mathbf{B}}}_{1m}\mathbf{H}_\beta\big\|_F^2 = O_P\Big(R(T,d_{c,\star})q_m^{\gamma_c^{(m)}-\gamma_c} \max_{n\in[M]\setminus\{m\}} q_n^{\gamma_c^{(n)}-\gamma_c}$$
$$+ T^{-1}q_m^{1-\gamma_c}p_m^{-1} \max_{n\in[M]\setminus\{m\}} q_n^{\gamma_{c,\max}^{(n)}-\gamma_c} + \Big\{q_m^{1-\gamma_c}p_m^{-1} + R(T,d_{c,\star})q_m^{\gamma_{c,\max}^{(m)}-\gamma_c} \max_{n\in[M]\setminus\{m\}} q_n^{\gamma_c^{(n)}-\gamma_c}\Big\}$$
$$\cdot \max_{n\in[M]}\Big\{q_n^{-\gamma_c}p_n^{-1} + R(T,d_{c,\star})q_n^{\gamma_{c,\max}^{(n)}-\gamma_c} \max_{s\in[M]\setminus\{n\}} q_s^{\gamma_c^{(s)}-\gamma_c} + T^{-1}q_n^{1-\gamma_{c,\min}^{(n)}}p_n^{-1} \max_{s\in[M]} q_s^{\gamma_{c,\max}^{(s)}-\gamma_c}\Big\}\Big),$$

where the column-side results follow by symmetry. This completes the proof. □

Proof of Corollary 6. The result follows by substituting $\gamma_r = \gamma_c = \gamma_r^{(m)} = \gamma_c^{(m)} = 1$ and $R(T,d_{r,\star}) = R(T,d_{c,\star}) = T^{-1}$ into Theorems 3–5. □

Proof of Theorem 7. For the local factor loading estimators, it suffices to show the results for $\widehat{\mathbf{C}}_m$, and analogous arguments follow for $\widehat{\mathbf{R}}_m$. To this end, we adapt the decomposition and arguments used in the proof of Lemma 9, with the roles of rows and columns interchanged appropriately. Recall from (D.74) that $\check{\mathbf{Y}}_t^{(m)} = \mathbf{Q}_m \mathbf{G}_t \mathbf{J}_m' + \mathbf{R}_m \mathbf{F}_t^{(m)} \mathbf{C}_m' + \mathbf{M}_{p_m} \mathbf{E}_t^{(m)} \mathbf{M}_{q_m}$. Hence

$$\begin{aligned}\check{\mathbf{Y}}_t^{(m)\prime} \widehat{\mathbf{Q}}_m^{\perp} &= \mathbf{J}_m \mathbf{G}_t' \mathbf{Q}_m' \widehat{\mathbf{Q}}_m^{\perp} + \mathbf{C}_m \mathbf{F}_t^{(m)\prime} \mathbf{R}_m' \widehat{\mathbf{Q}}_m^{\perp} + \mathbf{M}_{q_m} \mathbf{E}_t^{(m)\prime} \mathbf{M}_{p_m} \widehat{\mathbf{Q}}_m^{\perp} \\ &=: \mathcal{A}_t + \mathcal{B}_t + \mathcal{C}_t. \end{aligned} \tag{C.44}$$

With (C.44), we may expand $\mathbf{X}_{C,m}$ as

$$\begin{aligned}\mathbf{X}_{C,m} &= \frac{1}{T} \sum_{t=1}^{T} \big(\check{\mathbf{Y}}_t^{(m)\prime} \widehat{\mathbf{Q}}_m^{\perp}\big)\big(\check{\mathbf{Y}}_t^{(m)\prime} \widehat{\mathbf{Q}}_m^{\perp}\big)' \\ &= \frac{1}{T} \sum_{t=1}^{T} \mathcal{B}_t \mathcal{B}_t' + \frac{1}{T} \sum_{t=1}^{T} \mathcal{B}_t \mathcal{C}_t' + \frac{1}{T} \sum_{t=1}^{T} \mathcal{B}_t \mathcal{A}_t' + \frac{1}{T} \sum_{t=1}^{T} \mathcal{C}_t \mathcal{B}_t' + \frac{1}{T} \sum_{t=1}^{T} \mathcal{C}_t \mathcal{C}_t' \\ &\quad + \frac{1}{T} \sum_{t=1}^{T} \mathcal{C}_t \mathcal{A}_t' + \frac{1}{T} \sum_{t=1}^{T} \mathcal{A}_t \mathcal{B}_t' + \frac{1}{T} \sum_{t=1}^{T} \mathcal{A}_t \mathcal{C}_t' + \frac{1}{T} \sum_{t=1}^{T} \mathcal{A}_t \mathcal{A}_t' =: \sum_{i=1}^{9} \mathcal{J}_i. \end{aligned} \tag{C.45}$$

For convenience, define the matrices $\mathbf{P}_{Q,m}^{\perp} := \mathbf{Q}_m^{\perp} \mathbf{Q}_m^{\perp\prime}$ and $\widehat{\mathbf{P}}_{Q,m}^{\perp} := \widehat{\mathbf{Q}}_m^{\perp} \widehat{\mathbf{Q}}_m^{\perp\prime}$. Then we can write $\mathcal{B}_t \mathcal{B}_t' = \mathbf{C}_m \mathbf{F}_t^{(m)\prime} \mathbf{R}_m' \widehat{\mathbf{P}}_{Q,m}^{\perp} \mathbf{R}_m \mathbf{F}_t^{(m)} \mathbf{C}_m'$. Further define

$$\mathcal{B}_t^0 := \mathbf{C}_m \mathbf{F}_t^{(m)\prime} \mathbf{R}_m' \mathbf{Q}_m^{\perp}, \quad \mathcal{J}_1^0 := \frac{1}{T} \sum_{t=1}^{T} \mathcal{B}_t^0 \mathcal{B}_t^{0\prime}. \tag{C.46}$$

Then by Lemmas 4, 9, and following Lemma S5 in Cen (2026), we have

$$\begin{aligned}\big\|\widehat{\mathbf{P}}_{Q,m}^{\perp} - \mathbf{P}_{Q,m}^{\perp}\big\|_F^2 &= \big\|\widehat{\mathbf{P}}_{Q,m} - \mathbf{P}_{Q,m}\big\|_F^2 \\ &= O_P\left\{ R(T, d_{\mathrm{fac},\star})\, p_m^{2(\delta_{q,1}^{(m)} - \delta_{q,k_r}^{(m)})} \left(\frac{\max_{n \in [M]\setminus\{m\}} s_{J,n}^2 p_n^{\delta_{q,1}^{(n)}}}{\max_{n \in [M]\setminus\{m\}} s_{J,n}^2 p_n^{\delta_{q,k_r}^{(n)}}} \right)^2 \right\}. \end{aligned} \tag{C.47}$$

Consider $\mathcal{J}_1$. Note that $\mathcal{J}_1 - \mathcal{J}_1^0 = T^{-1} \sum_{t=1}^{T} \mathbf{C}_m \mathbf{F}_t^{(m)\prime} \mathbf{R}_m' (\widehat{\mathbf{P}}_{Q,m}^{\perp} - \mathbf{P}_{Q,m}^{\perp}) \mathbf{R}_m \mathbf{F}_t^{(m)} \mathbf{C}_m'$, we have

$$\begin{aligned}\big\|\mathcal{J}_1 - \mathcal{J}_1^0\big\|_F^2 &\le \frac{1}{T} \sum_{t=1}^{T} \Big\| \mathbf{C}_m \mathbf{F}_t^{(m)\prime} \mathbf{R}_m' (\widehat{\mathbf{P}}_{Q,m}^{\perp} - \mathbf{P}_{Q,m}^{\perp}) \mathbf{R}_m \mathbf{F}_t^{(m)} \mathbf{C}_m' \Big\|_F^2 \\ &\lesssim \big\|\mathbf{C}_m\big\|^4 \big\|\mathbf{R}_m\big\|^4 \big\|\widehat{\mathbf{P}}_{Q,m}^{\perp} - \mathbf{P}_{Q,m}^{\perp}\big\|_F^2 \cdot \frac{1}{T} \sum_{t=1}^{T} \big\|\mathbf{F}_t^{(m)}\big\|_F^4, \end{aligned} \tag{C.48}$$

where the last inequality used the sub-multiplicativity and the fact that the Frobenius norm of a product is bounded by products of spectral and Frobenius norms. By Assumption (F1), $\mathbf{F}_t^{(m)}$ has uniformly bounded

fourth moments, so $T^{-1}\sum_{t=1}^{T}\|\mathbf{F}_t^{(m)}\|_F^4 = O_P(1)$. Combining (C.48), (C.47), and Assumption (L1), we thus have

$$\big\|\mathcal{J}_1 - \mathcal{J}_1^0\big\|_F^2 \lesssim_P q_m^{2\delta_{c,1}^{(m)}} p_m^{2\delta_{r,1}^{(m)}} \left\{ R(T, d_{\mathrm{fac},\star})\, p_m^{2(\delta_{q,1}^{(m)} - \delta_{q,k_r}^{(m)})} \left(\frac{\max_{n\in[M]\setminus\{m\}} s_{J,n}^2 p_n^{\delta_{q,1}^{(n)}}}{\max_{n\in[M]\setminus\{m\}} s_{J,n}^2 p_n^{\delta_{q,k_r}^{(n)}}} \right)^2 \right\}. \tag{C.49}$$

Next, by Assumptions (L1) and (S1), the nonzero eigenvalues of $\mathcal{J}_1^0$ satisfy that, for any $i \in [k_c^{(m)}]$,

$$\lambda_i\big(\mathcal{J}_1^0\big) \asymp q_m^{\delta_{c,i}^{(m)}} p_m^{\delta_{r,1}^{(m)}}, \tag{C.50}$$

which follows analogous arguments as for (D.97) in the proof of Lemma 9.

It remains to bound $\mathcal{J}_i$, $i = 2, \ldots, 9$. Note that these terms all involve at least one of $\mathcal{A}_t$, $\mathcal{C}_t$, or the projection perturbation (cf. (C.47)). We bound their squared Frobenius norms separately. To this end, for $\mathcal{J}_2$ and $\mathcal{J}_4$, by Assumptions (E2), (F1), and the same calculation as in (D.86), we have

$$\big\|\mathcal{J}_2\big\|_F^2,\ \big\|\mathcal{J}_4\big\|_F^2 = \left\| \frac{1}{T}\sum_{t=1}^{T} \mathcal{B}_t\mathcal{C}_t' \right\|_F^2 \lesssim_P \frac{1}{T} q_m^{\delta_{c,1}^{(m)}} p_m^{\delta_{r,1}^{(m)}} q_m p_m = \frac{1}{T} q_m^{1+\delta_{c,1}^{(m)}} p_m^{1+\delta_{r,1}^{(m)}}. \tag{C.51}$$

For $\mathcal{J}_5$, using Lemma 3 and the proof of (D.35), it holds that

$$\big\|\mathcal{J}_5\big\|_F^2 \lesssim_P q_m^{2-2\delta_{c,1}^{(m)}} + \frac{1}{T} q_m^2 p_m. \tag{C.52}$$

For $\mathcal{J}_6$ and $\mathcal{J}_8$, it suffices to consider the former and the latter follows in similar arguments. Recall that $\mathcal{A}_t = \mathbf{J}_m \mathbf{G}_t' \mathbf{Q}_m' \widehat{\mathbf{Q}}_m^{\perp}$ from (C.44). Note that

$$\begin{aligned} \big\|\mathbf{Q}_m' \widehat{\mathbf{Q}}_m^{\perp}\big\|_F &= \big\|\mathbf{Q}_m' (\widehat{\mathbf{Q}}_m^{\perp} \widehat{\mathbf{Q}}_m^{\perp\prime} - \mathbf{Q}_m^{\perp} \mathbf{Q}_m^{\perp\prime}) \widehat{\mathbf{Q}}_m^{\perp}\big\| \\ &\le \big\|\mathbf{Q}_m\big\| \cdot \big\|\widehat{\mathbf{P}}_{Q,m}^{\perp} - \mathbf{P}_{Q,m}^{\perp}\big\|_F \cdot \big\|\widehat{\mathbf{Q}}_m^{\perp}\big\| = \big\|\mathbf{Q}_m\big\| \cdot \big\|\widehat{\mathbf{P}}_{Q,m}^{\perp} - \mathbf{P}_{Q,m}^{\perp}\big\|_F, \end{aligned} \tag{C.53}$$

Also, with $f(\mathbf{X}) = \|\mathbf{X}\|_F^2$ being convex, we have $\big\|T^{-1}\sum_{t=1}^{T}\mathbf{X}_t\big\|_F^2 = f\big(T^{-1}\sum_{t=1}^{T}\mathbf{X}_t\big) \le T^{-1}\sum_{t=1}^{T} f(\mathbf{X}_t) = T^{-1}\sum_{t=1}^{T}\|\mathbf{X}_t\|_F^2$. Altogether, by Assumption (F1), and the sub-multiplicativity of the Frobenius norm, we arrive at

$$\begin{aligned} \big\|\mathcal{J}_6\big\|_F^2 &= \left\| \frac{1}{T}\sum_{t=1}^{T} \mathcal{C}_t\mathcal{A}_t' \right\|_F^2 \le \frac{1}{T}\sum_{t=1}^{T} \big\|\mathcal{C}_t\big\|_F^2 \big\|\mathcal{A}_t\big\|_F^2 \\ &\le \frac{1}{T}\sum_{t=1}^{T} \big\|\mathcal{C}_t\big\|_F^2 \cdot \big\|\mathbf{J}_m\big\|^2 \cdot \big\|\mathbf{G}_t\big\|_F^2 \cdot \big\|\mathbf{Q}_m\big\|_F^2 \cdot \big\|\widehat{\mathbf{P}}_{Q,m}^{\perp} - \mathbf{P}_{Q,m}^{\perp}\big\|_F^2 \end{aligned}$$

$$\lesssim_P q_m^{1+\delta_{j,1}^{(m)}} p_m^{1+\delta_{q,1}^{(m)}} \left\{ R(T, d_{\mathrm{fac},\star})\, p_m^{2(\delta_{q,1}^{(m)} - \delta_{q,k_r}^{(m)})} \left(\frac{\max_{n\in[M]\setminus\{m\}} s_{J,n}^2 p_n^{\delta_{q,1}^{(n)}}}{\max_{n\in[M]\setminus\{m\}} s_{J,n}^2 p_n^{\delta_{q,k_r}^{(n)}}} \right)^2 \right\}, \tag{C.54}$$

where the second inequality used (C.47), (C.53), and $\|\mathcal{C}_t\|_F^2 = O_P(p_m q_m)$ which follows from similar arguments for (D.81). Next, $\mathcal{J}_3$ and $\mathcal{J}_7$ can be bounded analogously by the above arguments, with $\mathcal{C}_t$ replaced by $\mathcal{B}_t$ where $\|\mathcal{B}_t\|_F^2 = O_P(q_m^{\delta_{c,1}^{(m)}} p_m^{\delta_{r,1}^{(m)}})$, thus giving

$$\left\|\mathcal{J}_3\right\|_F^2,\ \left\|\mathcal{J}_7\right\|_F^2 \lesssim_P q_m^{\delta_{c,1}^{(m)}+\delta_{j,1}^{(m)}} p_m^{\delta_{r,1}^{(m)}+\delta_{q,1}^{(m)}} \left\{ R(T, d_{\mathrm{fac},\star})\, p_m^{2(\delta_{q,1}^{(m)} - \delta_{q,k_r}^{(m)})} \left(\frac{\max_{n\in[M]\setminus\{m\}} s_{J,n}^2 p_n^{\delta_{q,1}^{(n)}}}{\max_{n\in[M]\setminus\{m\}} s_{J,n}^2 p_n^{\delta_{q,k_r}^{(n)}}} \right)^2 \right\}. \tag{C.55}$$

For $\mathcal{J}_9$, using Assumption (F1), the bound of $\|\mathcal{A}_t\|_F^2$ as in (C.53) and (C.54), we have

$$\left\|\mathcal{J}_9\right\|_F^2 \le \frac{1}{T}\sum_{t=1}^{T} \left\|\mathcal{A}_t\right\|_F^4 \lesssim_P q_m^{2\delta_{j,1}^{(m)}} p_m^{2\delta_{q,1}^{(m)}} \left\{ R(T, d_{\mathrm{fac},\star})\, p_m^{2(\delta_{q,1}^{(m)} - \delta_{q,k_r}^{(m)})} \left(\frac{\max_{n\in[M]\setminus\{m\}} s_{J,n}^2 p_n^{\delta_{q,1}^{(n)}}}{\max_{n\in[M]\setminus\{m\}} s_{J,n}^2 p_n^{\delta_{q,k_r}^{(n)}}} \right)^2 \right\}^2. \tag{C.56}$$

Finally, recall (C.46). Then combining (C.49), (C.51), (C.52), (C.54), (C.55), and (C.56) give the rate on $\left\|\mathbf{X}_{C,m} - \mathcal{J}_1^0\right\|_F^2$, which, together with (C.50) and applying the Davis–Kahan theorem (Yu et al., 2015, Theorem 2), complete the proof for $\left\|\widehat{\mathbf{C}}_m - \check{\mathbf{C}}_m \mathbf{H}_{C,m}\right\|_F^2$.

It remains to show the results for the local factor estimators. To this end, define

$$\mathcal{L}_{R,m} := \left(\check{\mathbf{R}}_m' \mathbf{P}_{Q,m}^{\perp} \check{\mathbf{R}}_m\right)^{-1} \check{\mathbf{R}}_m' \mathbf{P}_{Q,m}^{\perp}, \quad \widehat{\mathcal{L}}_{R,m} := \left(\widehat{\mathbf{R}}_m' \widehat{\mathbf{P}}_{Q,m}^{\perp} \widehat{\mathbf{R}}_m\right)^{-1} \widehat{\mathbf{R}}_m' \widehat{\mathbf{P}}_{Q,m}^{\perp}.$$

By Assumptions (L1) and (S1), the matrix $\check{\mathbf{R}}_m' \mathbf{P}_{Q,m}^{\perp} \check{\mathbf{R}}_m$ is positive definite with eigenvalues bounded away from zero, and it holds that

$$\left\|\mathcal{L}_{R,m}\right\|,\ \left\|\widehat{\mathcal{L}}_{R,m}\right\| = O_P\left(p_m^{-\delta_{r,1}^{(m)}/2}\right), \quad \mathcal{L}_{R,m}\check{\mathbf{R}}_m = \mathbf{I}, \quad \mathcal{L}_{R,m}\mathbf{Q}_m = \mathbf{0}. \tag{C.57}$$

Then by the definition of $\widehat{\mathbf{F}}_t^{(m)}$ in Step 3 of Section 3.3 and (D.74), we may decompose

$$\begin{aligned} \widehat{\mathbf{F}}_t^{(m)} &= \widehat{\mathcal{L}}_{R,m} \check{\mathbf{Y}}_t^{(m)} \widehat{\mathbf{C}}_m \\ &= \widehat{\mathcal{L}}_{R,m} \mathbf{Q}_m \mathbf{G}_t \mathbf{J}_m' \widehat{\mathbf{C}}_m + \widehat{\mathcal{L}}_{R,m} \check{\mathbf{R}}_m \check{\mathbf{F}}_t^{(m)} \check{\mathbf{C}}_m' \widehat{\mathbf{C}}_m + \widehat{\mathcal{L}}_{R,m} \mathbf{M}_{p_m} \mathbf{E}_t^{(m)} \mathbf{M}_{q_m} \widehat{\mathbf{C}}_m =: \mathcal{K}_1 + \mathcal{K}_2 + \mathcal{K}_3. \end{aligned} \tag{C.58}$$

For $\mathcal{K}_1$, by (C.57), (C.53), and Assumption (F1), we have

$$\begin{aligned}\big\|\mathcal{K}_1\big\|_F^2 &\le \big\|\widehat{\mathcal{L}}_{R,m}\big\|^2\big\|\mathbf{Q}_m'\widehat{\mathbf{P}}_{Q,m}^{\perp}\big\|_F^2\big\|\mathbf{G}_t\big\|_F^2\big\|\mathbf{J}_m\big\|^2\big\|\widehat{\mathbf{C}}_m\big\|^2 \\ &\lesssim_P p_m^{\delta_{q,1}^{(m)}-\delta_{r,1}^{(m)}} q_m^{\delta_{j,1}^{(m)}}\eta_{Q,m}\big\|\mathbf{G}_t\big\|_F^2 \lesssim_P p_m^{\delta_{q,1}^{(m)}-\delta_{r,1}^{(m)}} q_m^{\delta_{j,1}^{(m)}}\eta_{Q,m},\end{aligned} \tag{C.59}$$

with $\eta_{Q,m}$ defined in the statement of Theorem 7.

Consider $\mathcal{K}_2$ next. Note that we can write

$$\begin{aligned}\mathcal{K}_2 &= \big(\mathbf{H}_{R,m}^{-1}+\widehat{\mathcal{L}}_{R,m}\check{\mathbf{R}}_m-\mathbf{H}_{R,m}^{-1}\big)\check{\mathbf{F}}_t^{(m)}\big(\mathbf{H}_{C,m}+\check{\mathbf{C}}_m'\widehat{\mathbf{C}}_m-\mathbf{H}_{C,m}\big) \\ &= \mathbf{H}_{R,m}^{-1}\check{\mathbf{F}}_t^{(m)}\mathbf{H}_{C,m}+(\widehat{\mathcal{L}}_{R,m}\check{\mathbf{R}}_m-\mathbf{H}_{R,m}^{-1})\check{\mathbf{F}}_t^{(m)}\mathbf{H}_{C,m} \\ &\quad+\mathbf{H}_{R,m}^{-1}\check{\mathbf{F}}_t^{(m)}(\check{\mathbf{C}}_m'\widehat{\mathbf{C}}_m-\mathbf{H}_{C,m})+(\widehat{\mathcal{L}}_{R,m}\check{\mathbf{R}}_m-\mathbf{H}_{R,m}^{-1})\check{\mathbf{F}}_t^{(m)}(\check{\mathbf{C}}_m'\widehat{\mathbf{C}}_m-\mathbf{H}_{C,m}).\end{aligned} \tag{C.60}$$

First, by the rates of convergence of the local loading matrices, we have

$$\begin{aligned}&\big\|\widehat{\mathcal{L}}_{R,m}\check{\mathbf{R}}_m-\mathbf{H}_{R,m}^{-1}\big\|_F^2=\big\|\big(\widehat{\mathbf{R}}_m'\widehat{\mathbf{P}}_{Q,m}^{\perp}\widehat{\mathbf{R}}_m\big)^{-1}\widehat{\mathbf{R}}_m'\widehat{\mathbf{P}}_{Q,m}^{\perp}\check{\mathbf{R}}_m-\mathbf{H}_{R,m}^{-1}\big\|_F^2 \\ &=\big\|\big(\widehat{\mathbf{R}}_m'\widehat{\mathbf{P}}_{Q,m}^{\perp}\widehat{\mathbf{R}}_m\big)^{-1}\widehat{\mathbf{R}}_m'\widehat{\mathbf{P}}_{Q,m}^{\perp}\check{\mathbf{R}}_m\mathbf{H}_{R,m}-\mathbf{I}\big\|_F^2 \\ &=\big\|\big(\widehat{\mathbf{R}}_m'\widehat{\mathbf{P}}_{Q,m}^{\perp}\widehat{\mathbf{R}}_m\big)^{-1}\widehat{\mathbf{R}}_m'\widehat{\mathbf{P}}_{Q,m}^{\perp}(\check{\mathbf{R}}_m\mathbf{H}_{R,m}-\widehat{\mathbf{R}}_m)+\big(\widehat{\mathbf{R}}_m'\widehat{\mathbf{P}}_{Q,m}^{\perp}\widehat{\mathbf{R}}_m\big)^{-1}\widehat{\mathbf{R}}_m'\widehat{\mathbf{P}}_{Q,m}^{\perp}\widehat{\mathbf{R}}_m-\mathbf{I}\big\|_F^2 \\ &\le\big\|\big(\widehat{\mathbf{R}}_m'\widehat{\mathbf{P}}_{Q,m}^{\perp}\widehat{\mathbf{R}}_m\big)^{-1}\widehat{\mathbf{R}}_m'\widehat{\mathbf{P}}_{Q,m}^{\perp}\big\|^2\cdot\big\|\check{\mathbf{R}}_m\mathbf{H}_{R,m}-\widehat{\mathbf{R}}_m\big\|_F^2\lesssim_P\Delta_{R,m},\end{aligned} \tag{C.61}$$

where $\Delta_{R,m}$ is defined in the statement of Theorem 7. Similarly, we have

$$\begin{aligned}&\big\|\check{\mathbf{C}}_m'\widehat{\mathbf{C}}_m-\mathbf{H}_{C,m}\big\|_F^2=\big\|\check{\mathbf{C}}_m'(\widehat{\mathbf{C}}_m-\check{\mathbf{C}}_m\mathbf{H}_{C,m})+\check{\mathbf{C}}_m'\check{\mathbf{C}}_m\mathbf{H}_{C,m}-\mathbf{H}_{C,m}\big\|_F^2 \\ &=\big\|\check{\mathbf{C}}_m'(\widehat{\mathbf{C}}_m-\check{\mathbf{C}}_m\mathbf{H}_{C,m})+(\mathbf{C}_m'\mathbf{C}_m)^{-1/2}\mathbf{C}_m'\mathbf{C}_m(\mathbf{C}_m'\mathbf{C}_m)^{-1/2}\mathbf{H}_{C,m}-\mathbf{H}_{C,m}\big\|_F^2\lesssim_P\Delta_{C,m},.\end{aligned} \tag{C.62}$$

Also, recall the definition $\check{\mathbf{F}}_t^{(m)}=\big(\mathbf{R}_m'\mathbf{R}_m\big)^{1/2}\mathbf{F}_t^{(m)}\big(\mathbf{C}_m'\mathbf{C}_m\big)^{1/2}$, and it is straightforward from Assumptions (F1) and (L1) that $\big\|\check{\mathbf{F}}_t^{(m)}\big\|_F^2=O_P\big(p_m^{\delta_{r,1}^{(m)}}q_m^{\delta_{c,1}^{(m)}}\big)$, which, combined with (C.60), (C.61), (C.62), give

$$\Big\|\mathcal{K}_2-\mathbf{H}_{R,m}^{-1}\check{\mathbf{F}}_t^{(m)}\mathbf{H}_{C,m}\Big\|_F^2\lesssim_P p_m^{\delta_{r,1}^{(m)}}q_m^{\delta_{c,1}^{(m)}}\big(\Delta_{R,m}+\Delta_{C,m}\big). \tag{C.63}$$

Finally, for $\mathcal{K}_3$, by (C.57) and the same argument as for (D.81),

$$\begin{aligned}&\big\|\mathcal{K}_3\big\|_F^2=\Big\|\widehat{\mathcal{L}}_{R,m}\mathbf{M}_{p_m}\mathbf{E}_t^{(m)}\mathbf{M}_{q_m}\widehat{\mathbf{C}}_m\Big\|_F^2 \\ &\le\big\|\widehat{\mathcal{L}}_{R,m}\big\|^2\cdot\Big\|\mathbf{M}_{p_m}\mathbf{E}_t^{(m)}\mathbf{M}_{q_m}\{\check{\mathbf{C}}_m\mathbf{H}_{C,m}+(\widehat{\mathbf{C}}_m-\check{\mathbf{C}}_m\mathbf{H}_{C,m})\}\Big\|_F^2 \\ &\lesssim\big\|\widehat{\mathcal{L}}_{R,m}\big\|^2\cdot\Big\|\mathbf{M}_{p_m}\mathbf{E}_t^{(m)}\mathbf{M}_{q_m}\check{\mathbf{C}}_m\mathbf{H}_{C,m}\Big\|_F^2+\big\|\widehat{\mathcal{L}}_{R,m}\big\|^2\cdot\Big\|\mathbf{M}_{p_m}\mathbf{E}_t^{(m)}\mathbf{M}_{q_m}(\widehat{\mathbf{C}}_m-\check{\mathbf{C}}_m\mathbf{H}_{C,m})\Big\|_F^2\end{aligned}$$

$$\lesssim_P p_m^{1-\delta_{r,1}^{(m)}} + p_m^{1-\delta_{r,1}^{(m)}} q_m \Delta_{C,m}. \tag{C.64}$$

By combining (C.58), (C.59), (C.63), and (C.64), we conclude that

$$\frac{1}{p_m^{\delta_{r,1}^{(m)}} q_m^{\delta_{c,1}^{(m)}}} \left\| \widehat{\mathbf{F}}_t^{(m)} - \mathbf{H}_{R,m}^{-1} \breve{\mathbf{F}}_t^{(m)} \mathbf{H}_{C,m} \right\|_F^2 \lesssim_P \Delta_{R,m} + \Delta_{C,m} + p_m^{1-2\delta_{r,1}^{(m)}} q_m^{1-\delta_{c,1}^{(m)}} \Delta_{C,m} + p_m^{1-2\delta_{r,1}^{(m)}} q_m^{-\delta_{c,1}^{(m)}},$$

where the rate in (C.59), after scaled by $p_m^{\delta_{r,1}^{(m)}} q_m^{\delta_{c,1}^{(m)}}$, is omitted as it is dominated by $\Delta_{C,m}$, particularly:

$$q_m^{-2\delta_{c,k_c^{(m)}}^{(m)}} p_m^{-2\delta_{r,1}^{(m)}} \left(q_m^{2\delta_{c,1}^{(m)}} p_m^{2\delta_{r,1}^{(m)}} + q_m^{1+\delta_{j,1}^{(m)}} p_m^{1+\delta_{q,1}^{(m)}} \right) \eta_{Q,m}.$$

Noting that all these arguments hold if we alternatively work on $\widehat{\mathbf{F}}_t^{(m)\prime}$, with the resulting rates of convergence symmetric (i.e., with row and column dimensions swapped), and taking the minimum of these two completes the proof of this theorem. □

Proof of Theorem 8. Recall $\breve{\mathbf{G}}_{t,m}$ from Lemma 10 and we can write

$$\begin{aligned}
&\breve{\mathbf{G}}_{t,m} = \breve{\mathbf{Q}}_m' \mathbf{Q}_m \mathbf{G}_t \mathbf{J}_m' \breve{\mathbf{J}}_m \\
&= (\mathbf{Q}_m' \mathbf{Q}_m)^{-1/2} \mathbf{Q}_m' \mathbf{Q}_m \mathbf{G}_t \mathbf{J}_m' \mathbf{J}_m (\mathbf{J}_m' \mathbf{J}_m)^{-1/2} = (\mathbf{Q}_m' \mathbf{Q}_m)^{1/2} \mathbf{G}_t (\mathbf{J}_m' \mathbf{J}_m)^{1/2},
\end{aligned}$$

so that for the reference group m_0, we have $\breve{\mathbf{G}}_{t,m_0} = \breve{\mathbf{G}}_t$, where $\breve{\mathbf{G}}_t = \left(\mathbf{Q}_{m_0}' \mathbf{Q}_{m_0}\right)^{1/2} \mathbf{G}_t \left(\mathbf{J}_{m_0}' \mathbf{J}_{m_0}\right)^{1/2}$ as defined in the statement of Theorem 8.

For each group $m \in [M]$ and $t \in [T]$, define the population quantities

$$\begin{aligned}
\mathbf{\Xi}_{Q,m}^* &:= \mathbf{H}_{Q,m_0}' \left(\mathbf{Q}_{m_0}' \mathbf{Q}_{m_0}\right)^{1/2} \left(\mathbf{Q}_m' \mathbf{Q}_m\right)^{-1/2} \mathbf{H}_{Q,m}, \\
\mathbf{\Xi}_{J,m}^* &:= \mathbf{H}_{J,m_0}' \left(\mathbf{J}_{m_0}' \mathbf{J}_{m_0}\right)^{1/2} \left(\mathbf{J}_m' \mathbf{J}_m\right)^{-1/2} \mathbf{H}_{J,m}, \\
\mathbf{G}_{t,m}^* &:= \mathbf{H}_{Q,m}' \left(\mathbf{Q}_m' \mathbf{Q}_m\right)^{1/2} \mathbf{G}_t \left(\mathbf{J}_m' \mathbf{J}_m\right)^{1/2} \mathbf{H}_{J,m}.
\end{aligned}$$

First, note that for all $t \in [T]$,

$$\mathbf{vec}\left(\mathbf{G}_{t,m_0}^*\right) = \left(\mathbf{\Xi}_{J,m}^* \otimes \mathbf{\Xi}_{Q,m}^*\right) \mathbf{vec}\left(\mathbf{G}_{t,m}^*\right) + o_P\left(p_m^{\delta_{q,1}^{(m)}/2} q_m^{\delta_{j,1}^{(m)}/2}\right). \tag{C.65}$$

As $\mathbf{H}_{Q,m}$ and $\mathbf{H}_{J,m}$ are asymptotically orthogonal by Lemmas 9 and 12, it also holds that

$$\begin{aligned}
\mathbf{\Xi}_{Q,m}^* \mathbf{H}_{Q,m}' \left(\mathbf{Q}_m' \mathbf{Q}_m\right)^{1/2} &= \mathbf{H}_{Q,m_0}' \left(\mathbf{Q}_{m_0}' \mathbf{Q}_{m_0}\right)^{1/2} \left(\mathbf{Q}_m' \mathbf{Q}_m\right)^{-1/2} \mathbf{H}_{Q,m} \mathbf{H}_{Q,m}' \left(\mathbf{Q}_m' \mathbf{Q}_m\right)^{1/2} \\
&= \mathbf{H}_{Q,m_0}' \left(\mathbf{Q}_{m_0}' \mathbf{Q}_{m_0}\right)^{1/2} + O_P\left(\eta_{Q,m}^{1/2} p_{m_0}^{\delta_{q,1}^{(m_0)}/2}\right),
\end{aligned}$$

$$\begin{aligned}\boldsymbol{\Xi}^*_{J,m}\mathbf{H}'_{J,m}\big(\mathbf{J}'_m\mathbf{J}_m\big)^{1/2} &= \mathbf{H}'_{J,m_0}\big(\mathbf{J}'_{m_0}\mathbf{J}_{m_0}\big)^{1/2}\big(\mathbf{J}'_m\mathbf{J}_m\big)^{-1/2}\mathbf{H}_{J,m}\mathbf{H}'_{J,m}\big(\mathbf{J}'_m\mathbf{J}_m\big)^{1/2}\\ &= \mathbf{H}'_{J,m_0}\big(\mathbf{J}'_{m_0}\mathbf{J}_{m_0}\big)^{1/2} + O_P\Big(\eta_{J,m}^{1/2}q_{m_0}^{\delta_{j,1}^{(m_0)}/2}\Big).\end{aligned}$$

so that together with $\lambda_{\min}\big(\mathbf{Q}'_{m_0}\mathbf{Q}_{m_0}\big)\lambda_{\min}\big(\mathbf{J}'_{m_0}\mathbf{J}_{m_0}\big) = O\big\{\lambda_{\min}\big(\mathbf{Q}'_m\mathbf{Q}_m\big)\lambda_{\min}\big(\mathbf{J}'_m\mathbf{J}_m\big)\big\}$ from the statement of Theorem 8, we arrive at

$$\begin{aligned}&\boldsymbol{\Xi}^*_{Q,m}\mathbf{G}^*_{t,m}\boldsymbol{\Xi}^{*\prime}_{J,m} = \boldsymbol{\Xi}^*_{Q,m}\mathbf{H}'_{Q,m}\big(\mathbf{Q}'_m\mathbf{Q}_m\big)^{1/2}\mathbf{G}_t\big(\mathbf{J}'_m\mathbf{J}_m\big)^{1/2}\mathbf{H}_{J,m}\boldsymbol{\Xi}^{*\prime}_{J,m}\\ &= \mathbf{H}'_{Q,m_0}\big(\mathbf{Q}'_{m_0}\mathbf{Q}_{m_0}\big)^{1/2}\mathbf{G}_t\big(\mathbf{J}'_{m_0}\mathbf{J}_{m_0}\big)^{1/2}\mathbf{H}_{J,m_0} + O_P\big(\eta_{Q,m}^{1/2}\eta_{J,m}^{1/2}p_m^{\delta_{q,1}^{(m)}/2}q_m^{\delta_{j,1}^{(m)}/2}\big)\\ &= \mathbf{G}^*_{t,m_0} + O_P\big(\eta_{Q,m}^{1/2}\eta_{J,m}^{1/2}p_m^{\delta_{q,1}^{(m)}/2}q_m^{\delta_{j,1}^{(m)}/2}\big) = \mathbf{H}'_{Q,m_0}\breve{\mathbf{G}}_t\mathbf{H}_{J,m_0} + O_P\big(\eta_{Q,m}^{1/2}\eta_{J,m}^{1/2}p_m^{\delta_{q,1}^{(m)}/2}q_m^{\delta_{j,1}^{(m)}/2}\big). \qquad \text{(C.66)}\end{aligned}$$

Also by Lemma 10, we have

$$\widehat{\mathbf{G}}_{t,m} = \mathbf{G}^*_{t,m} + O_P(\Delta_{G,m}^{1/2}) = \mathbf{H}'_{Q,m}\big(\mathbf{Q}'_m\mathbf{Q}_m\big)^{1/2}\mathbf{G}_t\big(\mathbf{J}'_m\mathbf{J}_m\big)^{1/2}\mathbf{H}_{J,m} + O_P(\Delta_{G,m}^{1/2}), \qquad \text{(C.67)}$$

where $O_P(\cdot)$ is in terms of Frobenius norm, and $\Delta_{G,m}$ is defined in the statement of Theorem 8 such that

$$\Delta_{G,m} = \min(p_m, q_m) + p_m^{\delta_{r,1}^{(m)}}q_m^{\delta_{c,1}^{(m)}}\big(\Delta_{R,m} + \Delta_{C,m} + \Delta_{F,m}\big) + \big(\eta_{Q,m} + \eta_{J,m}\big)p_m q_m.$$

By the construction of $\widehat{\mathbf{Z}}_{G,m}$ in Step 3 of Section 3.3 and with $\widehat{\mathbf{G}}_{t,m}$ defined in Lemma 10, note that

$$\widehat{\mathbf{Z}}_{G,m} = \Big(\mathbf{vec}(\widehat{\mathbf{Q}}'_m\check{\mathbf{Y}}^{(m)}_{G,1}\widehat{\mathbf{J}}_m), \ldots, \mathbf{vec}(\widehat{\mathbf{Q}}'_m\check{\mathbf{Y}}^{(m)}_{G,T}\widehat{\mathbf{J}}_m)\Big) = \Big(\mathbf{vec}(\widehat{\mathbf{G}}_{1,m}), \ldots, \mathbf{vec}(\widehat{\mathbf{G}}_{T,m})\Big).$$

Then we may define the population version of $\widehat{\mathbf{Z}}_{G,m}$ as $\mathbf{Z}^*_{G,m} := \Big(\mathbf{vec}(\mathbf{G}^*_{1,m}), \ldots, \mathbf{vec}(\mathbf{G}^*_{T,m})\Big)$, so that by (C.67), we have

$$\big\|\widehat{\mathbf{Z}}_{G,m} - \mathbf{Z}^*_{G,m}\big\|_F^2 \lesssim_P T\cdot\Delta_{G,m}. \qquad \text{(C.68)}$$

Next, recall the matrix $\widehat{\boldsymbol{\Xi}}_m = \widehat{\mathbf{Z}}_{G,m_0}\widehat{\mathbf{Z}}'_{G,m}(\widehat{\mathbf{Z}}_{G,m}\widehat{\mathbf{Z}}'_{G,m})^{-1}$ for the two-sided alignment in Step 3 of Section 3.3. For any group $m \neq m_0$, the population version of $\widehat{\boldsymbol{\Xi}}_m$ is $\boldsymbol{\Xi}^*_m := \mathbf{Z}^*_{G,m_0}\mathbf{Z}^{*\prime}_{G,m}\big(\mathbf{Z}^*_{G,m}\mathbf{Z}^{*\prime}_{G,m}\big)^{-1}$. With (C.65), we can write

$$\mathbf{Z}^*_{G,m_0} = \big(\mathbf{vec}(\mathbf{G}^*_{1,m_0}), \ldots, \mathbf{vec}(\mathbf{G}^*_{T,m_0})\big) = \big(\boldsymbol{\Xi}^*_{J,m}\otimes\boldsymbol{\Xi}^*_{Q,m}\big)\mathbf{Z}^*_{G,m} + o_P\Big(\big\|\mathbf{Z}^*_{G,m}\big\|_F\Big),$$

which implies $\boldsymbol{\Xi}^*_m = \mathbf{Z}^*_{G,m_0}\mathbf{Z}^{*\prime}_{G,m}\big(\mathbf{Z}^*_{G,m}\mathbf{Z}^{*\prime}_{G,m}\big)^{-1} = \boldsymbol{\Xi}^*_{J,m}\otimes\boldsymbol{\Xi}^*_{Q,m} + o_P(1)$ given Assumption (S1). Meanwhile,

also by Assumption (S1), we have

$$\left\|\left(\mathbf{Z}^*_{G,m}\mathbf{Z}^{*\prime}_{G,m}\right)^{-1}\right\| = O_P\left(\frac{1}{TS_m}\right), \tag{C.69}$$

where S_m is defined in the main text. Hence we may expand

$$\begin{aligned}\widehat{\mathbf{\Xi}}_m &= \left\{\mathbf{Z}^*_{G,m_0} + (\widehat{\mathbf{Z}}_{G,m_0} - \mathbf{Z}^*_{G,m_0})\right\}\left\{\mathbf{Z}^*_{G,m} + (\widehat{\mathbf{Z}}_{G,m} - \mathbf{Z}^*_{G,m})\right\}' \\ &\quad \cdot \left[\left\{\mathbf{Z}^*_{G,m} + (\widehat{\mathbf{Z}}_{G,m} - \mathbf{Z}^*_{G,m})\right\}\left\{\mathbf{Z}^*_{G,m} + (\widehat{\mathbf{Z}}_{G,m} - \mathbf{Z}^*_{G,m})\right\}'\right]^{-1} \\ &=: \mathbf{Z}^*_{G,m_0}\mathbf{Z}^{*\prime}_{G,m}\left(\mathbf{Z}^*_{G,m}\mathbf{Z}^{*\prime}_{G,m}\right)^{-1} + \mathcal{R}_{\Xi,m},\end{aligned} \tag{C.70}$$

where $\mathcal{R}_{\Xi,m}$ denotes all remaining terms in the decomposition and would involve at least one $(\widehat{\mathbf{Z}}_{G,m} - \mathbf{Z}^*_{G,m})$ or $(\widehat{\mathbf{Z}}_{G,m_0} - \mathbf{Z}^*_{G,m_0})$. As detailed, we bound all terms in $\mathcal{R}_{\Xi,m}$. To this end, by (C.68) and (C.69), we first have

$$\begin{aligned}&\left\|(\widehat{\mathbf{Z}}_{G,m_0} - \mathbf{Z}^*_{G,m_0})\mathbf{Z}^{*\prime}_{G,m}\left(\mathbf{Z}^*_{G,m}\mathbf{Z}^{*\prime}_{G,m}\right)^{-1}\right\|_F^2 \lesssim_P \left\|\widehat{\mathbf{Z}}_{G,m_0} - \mathbf{Z}^*_{G,m_0}\right\|_F^2 \cdot \left\|\mathbf{Z}^{*\prime}_{G,m}\right\|_F^2 \cdot \left\|\left(\mathbf{Z}^*_{G,m}\mathbf{Z}^{*\prime}_{G,m}\right)^{-1}\right\|^2 \\ &= O_P\left(T \cdot \Delta_{G,m_0}\right) \cdot O_P\left(TS_m\right) \cdot O_P\left((TS_m)^{-2}\right) = O_P\left(\frac{\Delta_{G,m_0}}{S_m}\right),\end{aligned} \tag{C.71}$$

where $\left\|\mathbf{Z}^{*\prime}_{G,m}\right\|_F^2 = O_P(TS_m)$ used Assumption (F1). Similarly,

$$\begin{aligned}&\left\|\mathbf{Z}^*_{G,m_0}(\widehat{\mathbf{Z}}_{G,m} - \mathbf{Z}^*_{G,m})'\left(\mathbf{Z}^*_{G,m}\mathbf{Z}^{*\prime}_{G,m}\right)^{-1}\right\|_F^2 \lesssim_P \left\|\mathbf{Z}^*_{G,m_0}\right\|^2 \cdot \left\|\widehat{\mathbf{Z}}_{G,m} - \mathbf{Z}^*_{G,m}\right\|_F^2 \cdot \left\|\left(\mathbf{Z}^*_{G,m}\mathbf{Z}^{*\prime}_{G,m}\right)^{-1}\right\|^2 \\ &= O_P\left(TS_{m_0}\right) \cdot O_P\left(T \cdot \Delta_{G,m}\right) \cdot O_P\left((TS_m)^{-2}\right) = O_P\left(\frac{\Delta_{G,m}S_{m_0}}{S_m^2}\right).\end{aligned} \tag{C.72}$$

To bound the cross term in $\mathcal{R}_{\Xi,m}$ involving both $\widehat{\mathbf{Z}}_{G,m_0} - \mathbf{Z}^*_{G,m_0}$ and $\widehat{\mathbf{Z}}_{G,m} - \mathbf{Z}^*_{G,m}$, let $\mathbf{E}_{G,m} := \widehat{\mathbf{Z}}_{G,m} - \mathbf{Z}^*_{G,m}$. By the condition $\Delta_{G,m} = o\{\lambda_{\min}(\mathbf{Q}'_m\mathbf{Q}_m)\lambda_{\min}(\mathbf{J}'_m\mathbf{J}_m)\}$ in the statement of Theorem 8, we have $\Delta_{G,m} = o(S_m)$. Then the cross term satisfies

$$\begin{aligned}&\left\|\mathbf{E}_{G,m_0}\mathbf{E}'_{G,m}\left[\left(\mathbf{Z}^*_{G,m} + \mathbf{E}_{G,m}\right)\left(\mathbf{Z}^*_{G,m} + \mathbf{E}_{G,m}\right)'\right]^{-1}\right\|_F^2 \\ &\lesssim_P \left\|\mathbf{E}_{G,m_0}\right\|_F^2\left\|\mathbf{E}_{G,m}\right\|_F^2\left\|\left[\left(\mathbf{Z}^*_{G,m} + \mathbf{E}_{G,m}\right)\left(\mathbf{Z}^*_{G,m} + \mathbf{E}_{G,m}\right)'\right]^{-1}\right\|^2 \\ &\lesssim_P \left(T\Delta_{G,m_0}\right)\left(T\Delta_{G,m}\right)\left(T^{-2}S_m^{-2}\right) = \frac{\Delta_{G,m_0}\Delta_{G,m}}{S_m^2},\end{aligned} \tag{C.73}$$

where we used $\left\|\left[\left(\mathbf{Z}^*_{G,m} + \mathbf{E}_{G,m}\right)\left(\mathbf{Z}^*_{G,m} + \mathbf{E}_{G,m}\right)'\right]^{-1}\right\|^2 \lesssim_P T^{-2}S_m^{-2}$, which follows from applying Weyl's inequality on $\lambda_{\min}(A_m) = O_P(TS_m)$ (as in (C.74) below), combined with (C.75) and $\Delta_{G,m} = o(S_m)$.

Finally, we bound

$$\mathbf{Z}^*_{G,m_0}\mathbf{Z}^{*\prime}_{G,m}\Big\{\big(\mathbf{Z}^*_{G,m}\mathbf{Z}^{*\prime}_{G,m}\big)^{-1}-\big(\mathbf{Z}^*_{G,m}\mathbf{Z}^{*\prime}_{G,m}+B_m\big)^{-1}\Big\}=\mathbf{Z}^*_{G,m_0}\mathbf{Z}^{*\prime}_{G,m}\Big\{A_m^{-1}-\big(A_m+B_m\big)^{-1}\Big\}, \quad \text{(C.74)}$$

where $A_m := \mathbf{Z}^*_{G,m}\mathbf{Z}^{*\prime}_{G,m}$ and $B_m := \mathbf{E}_{G,m}\mathbf{Z}^{*\prime}_{G,m}+\mathbf{Z}^*_{G,m}\mathbf{E}'_{G,m}+\mathbf{E}_{G,m}\mathbf{E}'_{G,m}$. Note that $\lambda_{\min}\big(A_m\big) = O_P(TS_m)$ by Assumption (S1) and the definition of S_m. Also, by the condition $\Delta_{G,m}=o(S_m)$ from the statement of Theorem 8 and (C.68), we have

$$\big\|B_m\big\| \lesssim_P \big\|\mathbf{E}_{G,m}\big\|_F\big\|\mathbf{Z}^*_{G,m}\big\|+\big\|\mathbf{E}_{G,m}\big\|_F^2 = O_P\Big(\sqrt{T\Delta_{G,m}}\cdot\sqrt{TS_m}\Big)+O_P\big(T\Delta_{G,m}\big)=O_P\Big(T\sqrt{\Delta_{G,m}S_m}\Big), \quad \text{(C.75)}$$

where we used $\big\|\mathbf{Z}^*_{G,m}\big\|^2 \asymp_P TS_m$ under Assumptions (S1) and (F1).

Using the matrix identity $\mathbf{A}^{-1}-(\mathbf{A}+\mathbf{B})^{-1}=\mathbf{A}^{-1}\mathbf{B}(\mathbf{A}+\mathbf{B})^{-1}$ for any matrices $\mathbf{A},\mathbf{B}$, we obtain

$$\begin{aligned}&\Big\|A_m^{-1}-\big(A_m+B_m\big)^{-1}\Big\| \le \big\|A_m^{-1}\big\|\big\|B_m\big\|\Big\|\big(A_m+B_m\big)^{-1}\Big\|\\ &= O_P\Big(T^{-1}S_m^{-1}T\sqrt{\Delta_{G,m}S_m}T^{-1}S_m^{-1}\Big)=O_P\Big(T^{-1}S_m^{-3/2}\Delta_{G,m}^{1/2}\Big).\end{aligned}$$

Hence the corresponding term in $\mathcal{R}_{\Xi,m}$ from (C.74) is bounded by

$$\begin{aligned}&\Big\|\mathbf{Z}^*_{G,m_0}\mathbf{Z}^{*\prime}_{G,m}\Big[A_m^{-1}-\big(A_m+B_m\big)^{-1}\Big]\Big\|_F^2 \lesssim \big\|\mathbf{Z}^*_{G,m_0}\big\|^2\big\|\mathbf{Z}^*_{G,m}\big\|^2\Big\|A_m^{-1}-\big(A_m+B_m\big)^{-1}\Big\|^2\\ &= O_P\Big(TS_{m_0}TS_mT^{-2}S_m^{-3}\Delta_{G,m}\Big)=O_P\left(\frac{S_{m_0}}{S_m^2}\Delta_{G,m}\right)=O_P\left(\frac{\Delta_{G,m}}{S_m}\right), \quad \text{(C.76)}\end{aligned}$$

where the last equality used $\lambda_{\min}\big(\mathbf{Q}'_{m_0}\mathbf{Q}_{m_0}\big)\lambda_{\min}\big(\mathbf{J}'_{m_0}\mathbf{J}_{m_0}\big) = O\big\{\lambda_{\min}\big(\mathbf{Q}'_{m}\mathbf{Q}_{m}\big)\lambda_{\min}\big(\mathbf{J}'_{m}\mathbf{J}_{m}\big)\big\}$ so that $S_{m_0}=O(S_m)$ from the statement of Theorem 8.

Combining (C.71), (C.72), (C.73), and (C.76), we conclude $\|\mathcal{R}_{\Xi,m}\|_F^2 = O_P(\Delta_{G,m_0}/S_m+\Delta_{G,m}/S_m)$, and thus

$$\big\|\widehat{\boldsymbol{\Xi}}_m-\boldsymbol{\Xi}^*_m\big\|_F^2 \lesssim_P \max_{m'\in[M]}\frac{\Delta_{G,m'}}{S_{m'}}. \quad \text{(C.77)}$$

Next, recall from Step 3 of Section 3.3 the definition of the notation $\mathcal{R}(\mathbf{W})$ for a block matrix $\mathbf{W}=(\mathbf{W}_{ij})_{i\in[k_c],j\in[k_c]}$, with each $\mathbf{W}_{ij}$ of size $k_r\times k_r$, such that

$$\mathcal{R}(\mathbf{W})=(\mathbf{vec}(\mathbf{W}_{11}),\ldots,\mathbf{vec}(\mathbf{W}_{k_c1}),\ldots,\mathbf{vec}(\mathbf{W}_{1k_c}),\ldots,\mathbf{vec}(\mathbf{W}_{k_ck_c})).$$

Note that $\mathcal{R}(\cdot)$ is a fixed permutation of the entries of the matrix input and hence preserves the Frobenius

norm. Thus, $\big\|\mathcal{R}(\widehat{\mathbf{\Xi}}_m) - \mathcal{R}(\mathbf{\Xi}^*_m)\big\|_F = \big\|\widehat{\mathbf{\Xi}}_m - \mathbf{\Xi}^*_m\big\|_F$, and hence by (C.77),

$$\big\|\mathcal{R}(\widehat{\mathbf{\Xi}}_m) - \mathcal{R}(\mathbf{\Xi}^*_m)\big\|_F^2 = O_P\bigg(\max_{m' \in [M]} \frac{\Delta_{G,m'}}{S_{m'}}\bigg). \tag{C.78}$$

By construction in Step 3 of Section 3.3, the matrix $\widehat{\sigma}_{\Xi,m}\widehat{\mathbf{u}}_{\Xi,m}\widehat{\mathbf{v}}'_{\Xi,m}$ is the rank-1 singular value decomposition of $\mathcal{R}(\widehat{\mathbf{\Xi}}_m)$. Then let $\sigma^*_{\Xi,m}\mathbf{u}^*_{\Xi,m}\mathbf{v}^{*\prime}_{\Xi,m}$ be the rank-1 singular value decomposition of $\mathcal{R}(\mathbf{\Xi}^*_m)$. From Lemma 11, we have

$$\mathcal{R}\big(\widehat{\mathbf{\Xi}}_{J,m} \otimes \widehat{\mathbf{\Xi}}_{Q,m}\big) = \widehat{\sigma}_{\Xi,m}\widehat{\mathbf{u}}_{\Xi,m}\widehat{\mathbf{v}}'_{\Xi,m}.$$

Recall previously that $\mathbf{\Xi}^*_m = \mathbf{\Xi}^*_{J,m} \otimes \mathbf{\Xi}^*_{Q,m}$, and by analogous arguments in the proof of Lemma 11, we also have

$$\sigma^*_{\Xi,m}\mathbf{u}^*_{\Xi,m}\mathbf{v}^{*\prime}_{\Xi,m} = \mathcal{R}\big(\mathbf{\Xi}^*_{J,m} \otimes \mathbf{\Xi}^*_{Q,m}\big) = \mathbf{vec}\big(\mathbf{\Xi}^*_{Q,m}\big)\mathbf{vec}\big(\mathbf{\Xi}^*_{J,m}\big)',$$

which implies

$$\sigma^*_{\Xi,m} = \big\|\mathbf{\Xi}^*_{Q,m}\big\|_F \cdot \big\|\mathbf{\Xi}^*_{J,m}\big\|_F = O_P(1), \tag{C.79}$$

where we used the definitions of $\mathbf{\Xi}^*_{Q,m}$ and $\mathbf{\Xi}^*_{J,m}$ at the beginning of this proof, together with the rate condition $\lambda_{\min}\big(\mathbf{Q}'_{m_0}\mathbf{Q}_{m_0}\big)\lambda_{\min}\big(\mathbf{J}'_{m_0}\mathbf{J}_{m_0}\big) = O\big\{\lambda_{\min}\big(\mathbf{Q}'_{m}\mathbf{Q}_{m}\big)\lambda_{\min}\big(\mathbf{J}'_{m}\mathbf{J}_{m}\big)\big\}$ in the statement of Theorem 8. Altogether, using the fact that $\mathcal{R}(\cdot)$ preserves the Frobenius norm (as in (C.78)), we have

$$\begin{aligned}
&\Big\|\widehat{\mathbf{\Xi}}_{J,m} \otimes \widehat{\mathbf{\Xi}}_{Q,m} - \mathbf{\Xi}^*_{J,m} \otimes \mathbf{\Xi}^*_{Q,m}\Big\|_F^2 = \Big\|\mathcal{R}(\widehat{\mathbf{\Xi}}_{J,m} \otimes \widehat{\mathbf{\Xi}}_{Q,m}) - \mathcal{R}(\mathbf{\Xi}^*_{J,m} \otimes \mathbf{\Xi}^*_{Q,m})\Big\|_F^2 \\
&= \Big\|\widehat{\sigma}_{\Xi,m}\widehat{\mathbf{u}}_{\Xi,m}\widehat{\mathbf{v}}'_{\Xi,m} - \sigma^*_{\Xi,m}\mathbf{u}^*_{\Xi,m}\mathbf{v}^{*\prime}_{\Xi,m}\Big\|_F^2 \\
&\lesssim \big|\widehat{\sigma}_{\Xi,m} - \sigma^*_{\Xi,m}\big|^2 \cdot \big\|\widehat{\mathbf{u}}_{\Xi,m}\widehat{\mathbf{v}}'_{\Xi,m}\big\|_F^2 + (\sigma^*_{\Xi,m})^2\big\|\widehat{\mathbf{u}}_{\Xi,m}\widehat{\mathbf{v}}'_{\Xi,m} - \mathbf{u}^*_{\Xi,m}\mathbf{v}^{*\prime}_{\Xi,m}\big\|_F^2 \\
&\lesssim_P \big\|\mathcal{R}(\widehat{\mathbf{\Xi}}_m) - \mathcal{R}(\mathbf{\Xi}^*_m)\big\|_F^2 = \big\|\widehat{\mathbf{\Xi}}_m - \mathbf{\Xi}^*_m\big\|_F^2 = O_P\bigg(\max_{m' \in [M]} \frac{\Delta_{G,m'}}{S_{m'}}\bigg),
\end{aligned} \tag{C.80}$$

where the last line used Weyl's inequality and the Davis–Kahan theorem (Yu et al., 2015, Theorem 2), in addition to (C.78) and (C.79).

By the construction of the global factor estimator $\widehat{\mathbf{G}}_t$ in Step 3 of Section 3.3, we have

$$\Big\|\widehat{\mathbf{G}}_t - \mathbf{H}'_{Q,m_0}\breve{\mathbf{G}}_t\mathbf{H}_{J,m_0}\Big\|_F^2 \lesssim \max_{m \in [M]} \Big\|\widehat{\mathbf{\Xi}}_{Q,m}\widehat{\mathbf{G}}_{t,m}\widehat{\mathbf{\Xi}}'_{J,m} - \mathbf{H}'_{Q,m_0}\breve{\mathbf{G}}_t\mathbf{H}_{J,m_0}\Big\|_F^2. \tag{C.81}$$

For each fixed m, we decompose

$$\widehat{\mathbf{\Xi}}_{Q,m}\widehat{\mathbf{G}}_{t,m}\widehat{\mathbf{\Xi}}'_{J,m} - \mathbf{H}'_{Q,m_0}\breve{\mathbf{G}}_t\mathbf{H}_{J,m_0} = \widehat{\mathbf{\Xi}}_{Q,m}\big(\widehat{\mathbf{G}}_{t,m} - \mathbf{G}^*_{t,m}\big)\widehat{\mathbf{\Xi}}'_{J,m} + \big(\widehat{\mathbf{\Xi}}_{Q,m}\mathbf{G}^*_{t,m}\widehat{\mathbf{\Xi}}'_{J,m} - \mathbf{H}'_{Q,m_0}\breve{\mathbf{G}}_t\mathbf{H}_{J,m_0}\big)$$

$$=: \mathcal{I}_{\Xi,G,1} + \mathcal{I}_{\Xi,G,2}. \tag{C.82}$$

For $\mathcal{I}_{\Xi,G,1}$, by (C.79) and (C.67), we have $\|\widehat{\mathbf{\Xi}}_{Q,m}\|, \|\widehat{\mathbf{\Xi}}_{J,m}\| = O_P(1)$, and $\|\widehat{\mathbf{G}}_{t,m} - \mathbf{G}^*_{t,m}\|_F^2 = O_P(\Delta_{G,m})$, so that

$$\|\mathcal{I}_{\Xi,G,1}\|_F^2 \lesssim_P \Delta_{G,m}. \tag{C.83}$$

For $\mathcal{I}_{\Xi,G,2}$, we can write

$$\begin{aligned}
&\mathbf{vec}(\mathcal{I}_{\Xi,G,2}) = (\widehat{\mathbf{\Xi}}_{J,m} \otimes \widehat{\mathbf{\Xi}}_{Q,m})\mathbf{vec}(\mathbf{G}^*_{t,m}) - (\mathbf{H}'_{J,m_0} \otimes \mathbf{H}'_{Q,m_0})\mathbf{vec}(\check{\mathbf{G}}_t) \\
&= \Big[(\widehat{\mathbf{\Xi}}_{J,m} \otimes \widehat{\mathbf{\Xi}}_{Q,m}) - (\mathbf{\Xi}^*_{J,m} \otimes \mathbf{\Xi}^*_{Q,m})\Big]\mathbf{vec}(\mathbf{G}^*_{t,m}) \\
&\quad + \Big[(\mathbf{\Xi}^*_{J,m} \otimes \mathbf{\Xi}^*_{Q,m})\mathbf{vec}(\mathbf{G}^*_{t,m}) - (\mathbf{H}'_{J,m_0} \otimes \mathbf{H}'_{Q,m_0})\mathbf{vec}(\check{\mathbf{G}}_t)\Big] =: \mathcal{I}_{\Xi,G,21} + \mathcal{I}_{\Xi,G,22}.
\end{aligned} \tag{C.84}$$

In particular, for $\mathcal{I}_{\Xi,G,21}$, by (C.80) and $\|\mathbf{G}^*_{t,m}\|_F^2 = O_P(S_m)$, we have

$$\|\mathcal{I}_{\Xi,G,21}\|_F^2 \lesssim_P \left\|\widehat{\mathbf{\Xi}}_{J,m} \otimes \widehat{\mathbf{\Xi}}_{Q,m} - \mathbf{\Xi}^*_{J,m} \otimes \mathbf{\Xi}^*_{Q,m}\right\|_F^2 \cdot \|\mathbf{G}^*_{t,m}\|_F^2 \lesssim_P \max_{m' \in [M]} \frac{\Delta_{G,m'}}{S_{m'}} S_m. \tag{C.85}$$

Also, by (C.66),

$$\|\mathcal{I}_{\Xi,G,22}\|_F^2 = O_P\Big(\eta_{Q,m}\eta_{J,m} p_m^{\delta^{(m)}_{q,1}} q_m^{\delta^{(m)}_{j,1}}\Big). \tag{C.86}$$

Combining (C.81), (C.82), (C.84), (C.83), (C.85), and (C.86), we obtain

$$\left\|\widehat{\mathbf{G}}_t - \mathbf{H}'_{Q,m_0}\check{\mathbf{G}}_t\mathbf{H}_{J,m_0}\right\|_F^2 \lesssim_P \max_{m \in [M]} \Delta_{G,m} + \max_{m' \in [M]} \frac{\Delta_{G,m'}}{S_{m'}} \max_{m \in [M]} S_m + \max_{m \in [M]} \eta_{Q,m}\eta_{J,m} p_m^{\delta^{(m)}_{q,1}} q_m^{\delta^{(m)}_{j,1}},$$

which thus concludes the proof for $\|\widehat{\mathbf{G}}_t - \mathbf{H}'_{Q,m_0}\check{\mathbf{G}}_t\mathbf{H}_{J,m_0}\|_F^2$, by noting that the last term above is dominated by the first term.

It remains to derive the rate for the re-estimated global loading matrices $\widetilde{\mathbf{Q}}_m$ and $\widetilde{\mathbf{J}}_m$ in Step 3 of Section 3.3. First, as $\mathbf{H}_{Q,m_0}$ and $\mathbf{H}_{J,m_0}$ are asymptotically orthogonal by Lemma 12, the population version of the re-estimated Kronecker product matrix is

$$\mathbf{K}^0_m := \Big\{\mathbf{J}_m\big(\mathbf{J}'_{m_0}\mathbf{J}_{m_0}\big)^{-1/2}\mathbf{H}_{J,m_0}\Big\} \otimes \Big\{\mathbf{Q}_m\big(\mathbf{Q}'_{m_0}\mathbf{Q}_{m_0}\big)^{-1/2}\mathbf{H}_{Q,m_0}\Big\}. \tag{C.87}$$

For each $m \in [M]$, define the matrices

$$\mathbf{Y}_m := \Big(\mathbf{vec}\big(\check{\mathbf{Y}}_{G,1}^{(m)}\big), \ldots, \mathbf{vec}\big(\check{\mathbf{Y}}_{G,T}^{(m)}\big)\Big), \quad \mathbf{G}^* := \Big(\mathbf{vec}\big(\mathbf{H}'_{Q,m_0}\check{\mathbf{G}}_1\mathbf{H}_{J,m_0}\big), \ldots, \mathbf{vec}\big(\mathbf{H}'_{Q,m_0}\check{\mathbf{G}}_T\mathbf{H}_{J,m_0}\big)\Big),$$
$$\widehat{\mathbf{G}} := \Big(\mathbf{vec}\big(\widehat{\mathbf{G}}_1\big), \ldots, \mathbf{vec}\big(\widehat{\mathbf{G}}_T\big)\Big), \quad \mathbf{U}_m := \mathbf{Y}_m - \mathbf{K}_m^0\mathbf{G}^*,$$

so that $\mathbf{U}_m$ satisfies

$$\mathbf{U}_m = \mathbf{Y}_m - \mathbf{K}_m^0\mathbf{G}^* = \big(\mathbf{Y}_m - \mathbf{Y}_m^0\big) + \big(\mathbf{Y}_m^0 - \mathbf{K}_m^0\mathbf{G}^*\big), \tag{C.88}$$

where $\mathbf{Y}_m^0$ denotes the matrix whose columns are $\mathbf{vec}(\mathbf{Q}_m\mathbf{G}_t\mathbf{J}'_m)$, $t \in [T]$. Consider the first term in (C.88), which has columns $\mathbf{vec}\big(\check{\mathbf{Y}}_{G,t}^{(m)} - \mathbf{Q}_m\mathbf{G}_t\mathbf{J}'_m\big)$. By our already constructed result in Theorem 8 and with the notation Δ_G in the theorem's statement, we have $\big\|\widehat{\mathbf{G}} - \mathbf{G}^*\big\|_F^2 = O_P(T\Delta_G)$. For each $m \in [M]$, by Theorem 7 and Lemma 3, we have

$$\begin{aligned}\frac{1}{T}\big\|\mathbf{Y}_m - \mathbf{Y}_m^0\big\|_F^2 &= \frac{1}{T}\sum_{t=1}^{T}\big\|\big(\mathbf{R}_m\mathbf{F}_t^{(m)}\mathbf{C}'_m - \widehat{\mathbf{R}}_m\widehat{\mathbf{F}}_t^{(m)}\widehat{\mathbf{C}}'_m\big) + \mathbf{M}_{p_m}\mathbf{E}_t^{(m)}\mathbf{M}_{q_m}\big\|_F^2 \\ &\lesssim_P p_m^{\delta_{r,1}^{(m)}} q_m^{\delta_{c,1}^{(m)}}\big(\Delta_{R,m} + \Delta_{C,m} + \Delta_{F,m}\big) + p_m q_m.\end{aligned} \tag{C.89}$$

For the second term in (C.88), for each $t \in [T]$,

$$\begin{aligned}&\Big\|\big(\mathbf{Q}_m\big(\mathbf{Q}'_{m_0}\mathbf{Q}_{m_0}\big)^{-1/2}\mathbf{H}_{Q,m_0}\big)\mathbf{H}'_{Q,m_0}\check{\mathbf{G}}_t\mathbf{H}_{J,m_0}\big(\mathbf{J}_m\big(\mathbf{J}'_{m_0}\mathbf{J}_{m_0}\big)^{-1/2}\mathbf{H}_{J,m_0}\big)' - \mathbf{Q}_m\mathbf{G}_t\mathbf{J}'_m\Big\|_F^2 \\ \lesssim_P\ &\big(\eta_{Q,m_0} + \eta_{J,m_0}\big)\big\|\mathbf{Q}_m\big\|^2\big\|\mathbf{J}_m\big\|^2\big\|\big(\mathbf{Q}'_{m_0}\mathbf{Q}_{m_0}\big)^{-1/2}\big\|^2\big\|\big(\mathbf{J}'_{m_0}\mathbf{J}_{m_0}\big)^{-1/2}\big\|^2\big\|\check{\mathbf{G}}_t\big\|_F^2 \\ =\ &O_P\left(\big(\eta_{Q,m_0} + \eta_{J,m_0}\big)\frac{p_m^{\delta_{q,1}^{(m)}} q_m^{\delta_{j,1}^{(m)}}}{p_{m_0}^{\delta_{q,k_r}^{(m_0)}} q_{m_0}^{\delta_{j,k_c}^{(m_0)}}}S_{m_0}\right),\end{aligned} \tag{C.90}$$

where the last line used Assumption (S1), the definition of S_{m_0} in Theorem 8, and $\|\check{\mathbf{G}}_t\|_F^2 = O_P(S_{m_0})$. Combining (C.88), (C.89), and (C.90), we have

$$\frac{1}{T}\big\|\mathbf{U}_m\big\|_F^2 \lesssim_P p_m^{\delta_{r,1}^{(m)}} q_m^{\delta_{c,1}^{(m)}}\big(\Delta_{R,m} + \Delta_{C,m} + \Delta_{F,m}\big) + p_m q_m + \big(\eta_{Q,m_0} + \eta_{J,m_0}\big)\frac{p_m^{\delta_{q,1}^{(m)}} q_m^{\delta_{j,1}^{(m)}}}{p_{m_0}^{\delta_{q,k_r}^{(m_0)}} q_{m_0}^{\delta_{j,k_c}^{(m_0)}}}S_{m_0}. \tag{C.91}$$

We next bound $\big\|T^{-1}\mathbf{U}_m\widehat{\mathbf{G}}'\big\|_F^2$. By definition, the (i,j) entry of $T^{-1}\mathbf{U}_m\widehat{\mathbf{G}}'$ is

$$\big(T^{-1}\mathbf{U}_m\widehat{\mathbf{G}}'\big)_{ij} = \frac{1}{T}\sum_{t=1}^{T}\big(\mathbf{U}_m\big)_{it}\big(\widehat{\mathbf{G}}\big)_{jt}, \tag{C.92}$$

where $\big(\mathbf{U}_m\big)_{it}$ denotes the (i,t) entry of $\mathbf{U}_m$ and $\big(\widehat{\mathbf{G}}\big)_{jt}$ denotes the (j,t) entry of $\widehat{\mathbf{G}}$. Recall the decompo-

sition $\mathbf{U}_m = (\mathbf{Y}_m - \mathbf{Y}_m^0) + (\mathbf{Y}_m^0 - \mathbf{K}_m^0\mathbf{G}^*)$, where $\mathbf{Y}_m^0$ is the matrix whose columns are $\mathbf{vec}(\mathbf{Q}_m\mathbf{G}_t\mathbf{J}_m')$, $t \in [T]$. Thus we have

$$\big(\mathbf{U}_m\big)_{it} = \Big[\mathbf{vec}\big(\check{\mathbf{Y}}_{G,t}^{(m)} - \mathbf{Q}_m\mathbf{G}_t\mathbf{J}_m'\big)\Big]_i + \Big[\mathbf{vec}(\mathbf{Q}_m\mathbf{G}_t\mathbf{J}_m') - \mathbf{K}_m^0\mathbf{G}_{\cdot t}^*\Big]_i,$$

where $\mathbf{G}_{\cdot t}^*$ denotes the t-th column of $\mathbf{G}^*$. By the Cauchy–Schwarz inequality,

$$\begin{aligned}\big(T^{-1}\mathbf{U}_m\widehat{\mathbf{G}}'\big)_{ij}^2 \le\ & 2\left[\frac{1}{T}\sum_{t=1}^{T}\Big[\mathbf{vec}\big(\check{\mathbf{Y}}_{G,t}^{(m)} - \mathbf{Q}_m\mathbf{G}_t\mathbf{J}_m'\big)\Big]_i\big(\widehat{\mathbf{G}}\big)_{jt}\right]^2 \\ & + 2\left[\frac{1}{T}\sum_{t=1}^{T}\Big[\mathbf{vec}(\mathbf{Q}_m\mathbf{G}_t\mathbf{J}_m') - \mathbf{K}_m^0\mathbf{G}_{\cdot t}^*\Big]_i\big(\widehat{\mathbf{G}}\big)_{jt}\right]^2. \end{aligned} \tag{C.93}$$

For the second term on the right-hand side of (C.93), by (C.90) and $\|\widehat{\mathbf{G}}\|_F^2 = O_P(TS_{m_0})$, we have

$$\begin{aligned}&\left(\frac{1}{T}\sum_{t=1}^{T}\Big[\mathbf{vec}(\mathbf{Q}_m\mathbf{G}_t\mathbf{J}_m') - \mathbf{K}_m^0\mathbf{G}_{\cdot t}^*\Big]_i\big(\widehat{\mathbf{G}}\big)_{jt}\right)^2 \\ \lesssim_P\ & \frac{1}{T}\frac{1}{p_m q_m}\big(\eta_{Q,m_0} + \eta_{J,m_0}\big)\frac{p_m^{\delta_{q,1}^{(m)}} q_m^{\delta_{j,1}^{(m)}}}{S_{m_0}}S_{m_0} = \frac{1}{Tp_m q_m}\big(\eta_{Q,m_0} + \eta_{J,m_0}\big)p_m^{\delta_{q,1}^{(m)}} q_m^{\delta_{j,1}^{(m)}}. \end{aligned} \tag{C.94}$$

For the first term on the right-hand side of (C.93), we further decompose

$$\Big[\mathbf{vec}\big(\check{\mathbf{Y}}_{G,t}^{(m)} - \mathbf{Q}_m\mathbf{G}_t\mathbf{J}_m'\big)\Big]_i = \Big[\mathbf{vec}\big(\mathbf{M}_{p_m}\mathbf{E}_t^{(m)}\mathbf{M}_{q_m}\big)\Big]_i + \Big[\mathbf{vec}\big(\mathbf{R}_m\mathbf{F}_t^{(m)}\mathbf{C}_m' - \widehat{\mathbf{R}}_m\widehat{\mathbf{F}}_t^{(m)}\widehat{\mathbf{C}}_m'\big)\Big]_i.$$

By Assumption (E2), the noise process $\{\mathbf{M}_{p_m}\mathbf{E}_t^{(m)}\mathbf{M}_{q_m}\}$ is independent of the factor process $\{\mathbf{G}_t\}$. By Lemma 1, the entries of $\{\mathbf{M}_{p_m}\mathbf{E}_t^{(m)}\mathbf{M}_{q_m}\}$ have summable autocovariance bounded by $C\rho(\cdot)$. By Assumptions (S1), (F1), and (L1), the entries of $\widehat{\mathbf{G}}$ have second moment of order S_{m_0}. Altogether, we obtain

$$\left(\frac{1}{T}\sum_{t=1}^{T}\Big[\mathbf{vec}\big(\mathbf{M}_{p_m}\mathbf{E}_t^{(m)}\mathbf{M}_{q_m}\big)\Big]_i\big(\widehat{\mathbf{G}}\big)_{jt}\right)^2 \lesssim_P \frac{S_{m_0}}{T}. \tag{C.95}$$

For $\mathbf{R}_m\mathbf{F}_t^{(m)}\mathbf{C}_m' - \widehat{\mathbf{R}}_m\widehat{\mathbf{F}}_t^{(m)}\widehat{\mathbf{C}}_m'$, by Assumption (IC2), the local factor $\mathbf{F}_t^{(m)}$ is uncorrelated with the global factor $\mathbf{G}_t$. Hence by Theorem 7, we have $\big\|\mathbf{R}_m\mathbf{F}_t^{(m)}\mathbf{C}_m' - \widehat{\mathbf{R}}_m\widehat{\mathbf{F}}_t^{(m)}\widehat{\mathbf{C}}_m'\big\|_F^2 \lesssim_P p_m^{\delta_{r,1}^{(m)}} q_m^{\delta_{c,1}^{(m)}}\big(\Delta_{R,m} + \Delta_{C,m} + \Delta_{F,m}\big)$, so that

$$\left(\frac{1}{T}\sum_{t=1}^{T}\Big[\mathbf{vec}\big(\mathbf{R}_m\mathbf{F}_t^{(m)}\mathbf{C}_m' - \widehat{\mathbf{R}}_m\widehat{\mathbf{F}}_t^{(m)}\widehat{\mathbf{C}}_m'\big)\Big]_i\big(\widehat{\mathbf{G}}\big)_{jt}\right)^2 \lesssim_P \frac{1}{Tp_m q_m}p_m^{\delta_{r,1}^{(m)}} q_m^{\delta_{c,1}^{(m)}}\big(\Delta_{R,m} + \Delta_{C,m} + \Delta_{F,m}\big)S_{m_0}. \tag{C.96}$$

Combining (C.93), (C.94), (C.95), and (C.96), summing over $i \in [p_m q_m]$ and $j \in [k_r k_c]$, we conclude

$$\left\| \frac{1}{T} \mathbf{U}_m \widehat{\mathbf{G}}' \right\|_F^2 \lesssim_P \frac{p_m q_m + p_m^{\delta_{r,1}^{(m)}} q_m^{\delta_{c,1}^{(m)}} \left(\Delta_{R,m} + \Delta_{C,m} + \Delta_{F,m} \right)}{T} S_{m_0}$$
$$+ \frac{1}{T} (\eta_{Q,m_0} + \eta_{J,m_0}) p_m^{\delta_{q,1}^{(m)}} q_m^{\delta_{j,1}^{(m)}} \lesssim \frac{p_m q_m + p_m^{\delta_{r,1}^{(m)}} q_m^{\delta_{c,1}^{(m)}} \left(\Delta_{R,m} + \Delta_{C,m} + \Delta_{F,m} \right)}{T} S_{m_0}. \quad \text{(C.97)}$$

With our notations above, by the construction in Step 3 of Section 3.3, we can write

$$\widehat{\mathbf{J}_m \otimes \mathbf{Q}_m} = \mathbf{Y}_m \widehat{\mathbf{G}}' \left(\widehat{\mathbf{G}} \widehat{\mathbf{G}}' \right)^{-1} = \left(\mathbf{K}_m^0 \mathbf{G}^* + \mathbf{U}_m \right) \widehat{\mathbf{G}}' \left(\widehat{\mathbf{G}} \widehat{\mathbf{G}}' \right)^{-1}$$
$$= \mathbf{K}_m^0 + \mathbf{K}_m^0 \left(\mathbf{G}^* \widehat{\mathbf{G}}' - \widehat{\mathbf{G}} \widehat{\mathbf{G}}' \right) \left(\widehat{\mathbf{G}} \widehat{\mathbf{G}}' \right)^{-1} + \mathbf{U}_m \widehat{\mathbf{G}}' \left(\widehat{\mathbf{G}} \widehat{\mathbf{G}}' \right)^{-1}. \quad \text{(C.98)}$$

We bound the two terms in (C.98). First, by Assumption (S1), the smallest eigenvalue of $T^{-1} \mathbf{G}^* \mathbf{G}^{*\prime}$ is of order S_{m_0}. Then we obtain

$$\left\| \left(\widehat{\mathbf{G}} \widehat{\mathbf{G}}' \right)^{-1} \right\| = O_P \left(T^{-1} S_{m_0}^{-1} \right). \quad \text{(C.99)}$$

Also, note that $\|\mathbf{G}^*\|_F^2 = O_P(T S_{m_0})$ and $\|\widehat{\mathbf{G}}\|_F^2 = O_P(T S_{m_0})$. Then for the first term in (C.98), we have

$$\left\| \mathbf{K}_m^0 \left(\mathbf{G}^* \widehat{\mathbf{G}}' - \widehat{\mathbf{G}} \widehat{\mathbf{G}}' \right) \left(\widehat{\mathbf{G}} \widehat{\mathbf{G}}' \right)^{-1} \right\|_F^2 \lesssim \|\mathbf{K}_m^0\|^2 \|\mathbf{G}^* - \widehat{\mathbf{G}}\|_F^2 \|\widehat{\mathbf{G}}\|^2 \left\| \left(\widehat{\mathbf{G}} \widehat{\mathbf{G}}' \right)^{-1} \right\|^2$$
$$= O_P \left(\|\mathbf{K}_m^0\|^2 \cdot T \Delta_G \cdot T S_{m_0} \cdot T^{-2} S_{m_0}^{-2} \right) = O_P \left(\|\mathbf{K}_m^0\|^2 \frac{\Delta_G}{S_{m_0}} \right). \quad \text{(C.100)}$$

For the second term in (C.98), by (C.97) and (C.99) we have

$$\left\| \mathbf{U}_m \widehat{\mathbf{G}}' \left(\widehat{\mathbf{G}} \widehat{\mathbf{G}}' \right)^{-1} \right\|_F^2 = \left\| \left(\frac{1}{T} \mathbf{U}_m \widehat{\mathbf{G}}' \right) \left(\frac{1}{T} \widehat{\mathbf{G}} \widehat{\mathbf{G}}' \right)^{-1} \right\|_F^2 \lesssim_P \left\| \frac{1}{T} \mathbf{U}_m \widehat{\mathbf{G}}' \right\|_F^2 \cdot \left\| \left(\frac{1}{T} \widehat{\mathbf{G}} \widehat{\mathbf{G}}' \right)^{-1} \right\|^2$$
$$= O_P \left(\frac{S_{m_0}}{T} \left\{ p_m q_m + p_m^{\delta_{r,1}^{(m)}} q_m^{\delta_{c,1}^{(m)}} \left(\Delta_{R,m} + \Delta_{C,m} + \Delta_{F,m} \right) \right\} \cdot S_{m_0}^{-2} \right)$$
$$= O_P \left(\frac{p_m q_m + p_m^{\delta_{r,1}^{(m)}} q_m^{\delta_{c,1}^{(m)}} \left(\Delta_{R,m} + \Delta_{C,m} + \Delta_{F,m} \right)}{T S_{m_0}} \right). \quad \text{(C.101)}$$

Combining (C.98), (C.100), and (C.101), we arrive at

$$\left\| \widehat{\mathbf{J}_m \otimes \mathbf{Q}_m} - \mathbf{K}_m^0 \right\|_F^2 \lesssim_P \|\mathbf{K}_m^0\|^2 \frac{\Delta_G}{S_{m_0}} + \frac{p_m q_m + p_m^{\delta_{r,1}^{(m)}} q_m^{\delta_{c,1}^{(m)}} \left(\Delta_{R,m} + \Delta_{C,m} + \Delta_{F,m} \right)}{T S_{m_0}} = \Delta_{K,m}, \quad \text{(C.102)}$$

where the last equality used the definition of $\Delta_{K,m}$ in the statement of Theorem 8, the fact that $\|\mathbf{K}_m^0\|_F^2 =$

$O_P\big(p_m^{\delta_{q,1}^{(m)}} q_m^{\delta_{j,1}^{(m)}}/S_{m_0}\big)$ which is direct from (C.87), and the condition from the statement of Theorem 8 that $\lambda_{\min}\big(\mathbf{Q}_{m_0}'\mathbf{Q}_{m_0}\big)\lambda_{\min}\big(\mathbf{J}_{m_0}'\mathbf{J}_{m_0}\big) = O\big\{\lambda_{\min}\big(\mathbf{Q}_m'\mathbf{Q}_m\big)\lambda_{\min}\big(\mathbf{J}_m'\mathbf{J}_m\big)\big\}$ and thus

$$\frac{p_m^{\delta_{r,1}^{(m)}} q_m^{\delta_{c,1}^{(m)}}\big(\Delta_{R,m}+\Delta_{C,m}+\Delta_{F,m}\big)}{TS_{m_0}} \lesssim \frac{\Delta_{G,m}}{TS_{m_0}} \lesssim \frac{\Delta_{G,m}}{S_{m_0}}\frac{1}{T}\frac{p_m^{\delta_{q,1}^{(m)}} q_m^{\delta_{j,1}^{(m)}}}{S_{m_0}} = o\left(\big\|\mathbf{K}_m^0\big\|^2\frac{\Delta_G}{S_{m_0}}\right).$$

The remaining arguments are similar to the previous ones in this proof on $\big\|\widehat{\mathbf{G}}_t - \mathbf{H}_{Q,m_0}'\breve{\mathbf{G}}_t\mathbf{H}_{J,m_0}\big\|_F^2$. In particular, by the construction in Step 3 of Section 3.3, the matrix $\mathcal{R}(\widehat{\mathbf{J}_m\otimes\mathbf{Q}}_m)$ has a rank-1 singular value decomposition

$$\mathcal{R}\big(\widehat{\mathbf{J}_m\otimes\mathbf{Q}}_m\big) = \widehat{\sigma}_{\Xi,m}\widehat{\mathbf{u}}_{\Xi,m}\widehat{\mathbf{v}}_{\Xi,m}' + \widehat{\Xi}_m^{(\mathrm{rem})}, \tag{C.103}$$

where the remainder $\widehat{\Xi}_m^{(\mathrm{rem})}$ includes the remaining singular components. Similarly, write the population version of the above as the rank-1 singular value decomposition and the remaining singular components (which are zero due to the following), such that

$$\mathcal{R}\big(\mathbf{K}_m^0\big) = \sigma_{*,m}\mathbf{u}_{*,m}\mathbf{v}_{*,m}' = \mathbf{vec}(\mathbf{Q}_m^*)\mathbf{vec}(\mathbf{J}_m^*)', \tag{C.104}$$

where $\sigma_{*,m}$ denotes the largest singular value of $\mathcal{R}(\mathbf{K}_m^0)$. By Assumptions (S1), (L1), and the asymptotic orthogonality of $\mathbf{H}_{Q,m_0}$ and $\mathbf{H}_{J,m_0}$ from Lemma 12, the leading singular value of $\mathcal{R}(\mathbf{K}_m^0)$ satisfies $\sigma_{*,m} \asymp_P 1$. Then by Weyl's inequality, (C.102), and recall that $\mathcal{R}(\cdot)$ preserves the Frobenius norm,

$$\big|\widehat{\sigma}_{\Xi,m} - \sigma_{*,m}\big| \lesssim_P \big\|\mathcal{R}(\widehat{\mathbf{J}_m\otimes\mathbf{Q}}_m) - \mathcal{R}(\mathbf{K}_m^0)\big\| \lesssim_P \Delta_{K,m}^{1/2}, \tag{C.105}$$

which then implies $\widehat{\sigma}_{\Xi,m} \asymp_P 1$. By the Davis–Kahan theorem (Yu et al., 2015, Theorem 2), we have

$$\big\|\widehat{\mathbf{u}}_{\Xi,m} - \mathbf{u}_{*,m}\big\|^2 \lesssim_P \frac{\big\|\mathcal{R}(\widehat{\mathbf{J}_m\otimes\mathbf{Q}}_m) - \mathcal{R}(\mathbf{K}_m^0)\big\|_F^2}{(\sigma_{*,m}-\sigma_{*,m}^{(2)})^2}, \quad \big\|\widehat{\mathbf{v}}_{\Xi,m} - \mathbf{v}_{*,m}\big\|^2 \lesssim_P \frac{\big\|\mathcal{R}(\widehat{\mathbf{J}_m\otimes\mathbf{Q}}_m) - \mathcal{R}(\mathbf{K}_m^0)\big\|_F^2}{(\sigma_{*,m}-\sigma_{*,m}^{(2)})^2},$$

where $\sigma_{*,m}^{(2)}$ denotes the second largest singular value of $\mathcal{R}(\mathbf{K}_m^0)$ such that $\sigma_{*,m}^{(2)} = 0$ due to (C.104), and thus $|\sigma_{*,m} - \sigma_{*,m}^{(2)}| \asymp_P 1$. Then with (C.105), we may conclude

$$\big\|\widehat{\mathbf{u}}_{\Xi,m} - \mathbf{u}_{*,m}\big\|^2 \lesssim_P \Delta_{K,m}, \quad \big\|\widehat{\mathbf{v}}_{\Xi,m} - \mathbf{v}_{*,m}\big\|^2 \lesssim_P \Delta_{K,m}. \tag{C.106}$$

Recall from Step 3 of Section 3.3 that $\mathbf{vec}(\widetilde{\mathbf{Q}}_m) = \widehat{\sigma}_{\Xi,m}^{1/2}\widehat{\mathbf{u}}_{\Xi,m}$ and $\mathbf{vec}(\widetilde{\mathbf{J}}_m) = \widehat{\sigma}_{\Xi,m}^{1/2}\widehat{\mathbf{v}}_{\Xi,m}$. Similarly, define their population versions $\mathbf{vec}(\mathbf{Q}_m^*) := \sigma_{*,m}^{1/2}\mathbf{u}_{*,m}$ and $\mathbf{vec}(\mathbf{J}_m^*) = \sigma_{*,m}^{1/2}\mathbf{v}_{*,m}$. Combining (C.105) and

(C.106), we obtain

$$\begin{aligned}\big\|\mathbf{vec}(\widetilde{\mathbf{Q}}_m) - \mathbf{vec}(\mathbf{Q}^*_m)\big\|^2 &= \big\|\widehat{\sigma}^{1/2}_{\Xi,m}\widehat{\mathbf{u}}_{\Xi,m} - \sigma^{1/2}_{*,m}\mathbf{u}_{*,m}\big\|^2 \\ \lesssim \big|\widehat{\sigma}^{1/2}_{\Xi,m} - \sigma^{1/2}_{*,m}\big|^2\big\|\widehat{\mathbf{u}}_{\Xi,m}\big\|^2 &+ \sigma_{*,m}\big\|\widehat{\mathbf{u}}_{\Xi,m} - \mathbf{u}_{*,m}\big\|^2 = O_P\big(\Delta_{K,m}\big),\end{aligned} \tag{C.107}$$

and analogously,

$$\big\|\mathbf{vec}(\widetilde{\mathbf{J}}_m) - \mathbf{vec}(\mathbf{J}^*_m)\big\|^2 = O_P\big(\Delta_{K,m}\big). \tag{C.108}$$

Thus finally, the re-estimated global loading matrices $\widetilde{\mathbf{Q}}_m$ and $\widetilde{\mathbf{J}}_m$ satisfy

$$\big\|\widetilde{\mathbf{Q}}_m - \mathbf{Q}^*_m\big\|^2_F = O_P\big(\Delta_{K,m}\big), \quad \big\|\widetilde{\mathbf{J}}_m - \mathbf{J}^*_m\big\|^2_F = O_P\big(\Delta_{K,m}\big), \tag{C.109}$$

where

$$\mathbf{Q}^*_m := \mathbf{Q}_m\big(\mathbf{Q}'_{m_0}\mathbf{Q}_{m_0}\big)^{-1/2}\mathbf{H}_{Q,m_0}, \quad \mathbf{J}^*_m := \mathbf{J}_m\big(\mathbf{J}'_{m_0}\mathbf{J}_{m_0}\big)^{-1/2}\mathbf{H}_{J,m_0}.$$

This completes the proof of the theorem. □

Proof of Theorem 9. For each $m \in [M]$ and any $c_m \in [k_r]$ and $d_m \in [k_c]$, we have

$$\begin{aligned}\widehat{\mathbf{Q}}'_{m,\cdot c_m}\check{\mathbf{Y}}^{(m)}_t\widehat{\mathbf{J}}_{m,\cdot d_m} &= (\widehat{\mathbf{J}}'_{m,\cdot d_m}\mathbf{J}_m \otimes \widehat{\mathbf{Q}}'_{m,\cdot c_m}\mathbf{Q}_m)\mathbf{vec}(\mathbf{G}_t) + (\widehat{\mathbf{J}}'_{m,\cdot d_m}\mathbf{C}_m \otimes \widehat{\mathbf{Q}}'_{m,\cdot c_m}\mathbf{R}_m)\mathbf{vec}(\mathbf{F}^{(m)}_t) \\ &+ \widehat{\mathbf{Q}}'_{m,\cdot c_m}\mathbf{M}_{p_m}\mathbf{E}^{(m)}_t\mathbf{M}_{q_m}\widehat{\mathbf{J}}_{m,\cdot d_m}.\end{aligned} \tag{C.110}$$

From the proof of Theorem 7, for some invertible constant order matrices $\mathbf{H}_q$ and $\mathbf{H}_j$, we have

$$\big\|\widehat{\mathbf{Q}}_m - \mathbf{Q}_m(\mathbf{Q}'_m\mathbf{Q}_m)^{-1/2}\mathbf{H}_{q,m}\big\| = O_P(\eta_{Q,m}) = o_P(1), \;\; \big\|\widehat{\mathbf{J}}_m - \mathbf{J}_m(\mathbf{J}'_m\mathbf{J}_m)^{-1/2}\mathbf{H}_{j,m}\big\| = O_P(\eta_{J,m}) = o_P(1). \tag{C.111}$$

Hence, defining $\mathbf{e}_{p,k}$ as the real unit vector of length p with 1 at its kth position, we have

$$\begin{aligned}\max_{c_m\in[k_r]}\big\|\widehat{\mathbf{Q}}'_{m,\cdot c_m}\mathbf{Q}_m\big\| = \max_{c_m\in[k_r]}\big\|\mathbf{e}'_{k_r,c_m}\widehat{\mathbf{Q}}'_m\mathbf{Q}_m\big\| &\asymp \big\|\widehat{\mathbf{Q}}'_m\mathbf{Q}_m\big\| \\ \geq \big\|(\mathbf{Q}'_m\mathbf{Q}_m)^{1/2}\mathbf{H}_{q,m}\big\| - \big\|\mathbf{Q}'_m(\widehat{\mathbf{Q}}_m - \mathbf{Q}_m(\mathbf{Q}'_m\mathbf{Q}_m)^{-1/2}\mathbf{H}_{q,m})\big\| &\asymp_P p_m^{\delta^{(m)}_{q,1}/2}.\end{aligned} \tag{C.112}$$

Similarly,

$$\max_{d_m\in[k_c]}\big\|\widehat{\mathbf{J}}'_{m,\cdot d_m}\mathbf{J}_m\big\| \asymp_P q_m^{\delta^{(m)}_{j,1}/2}. \tag{C.113}$$

Consider $m \in \mathcal{A}_G$. Then (C.110) implies that

$$T^{-1}\sum_{t=1}^{T}\widehat{\mathbf{Q}}'_{m,\cdot c_m}\check{\mathbf{Y}}_t^{(m)}\widehat{\mathbf{J}}_{m,\cdot d_m}\widehat{\mathbf{J}}'_{n,\cdot d_n}\check{\mathbf{Y}}_t^{(n)'}\widehat{\mathbf{Q}}_{m,\cdot c_n} = \sum_{j=0}^{9} I_j, \text{ where}$$

$$I_0 := (\widehat{\mathbf{J}}'_{m,\cdot d_m}\mathbf{J}_m \otimes \widehat{\mathbf{Q}}'_{m,\cdot c_m}\mathbf{Q}_m)T^{-1}\sum_{t=1}^{T}\mathbb{E}(\mathbf{vec}(\mathbf{G}_t))^{\otimes 2}(\widehat{\mathbf{J}}'_{n,\cdot d_n}\mathbf{J}_n \otimes \widehat{\mathbf{Q}}'_{n,\cdot c_n}\mathbf{Q}_n)',$$

$$I_1 := (\widehat{\mathbf{J}}'_{m,\cdot d_m}\mathbf{J}_m \otimes \widehat{\mathbf{Q}}'_{m,\cdot c_m}\mathbf{Q}_m)T^{-1}\sum_{t=1}^{T}(\mathbf{vec}(\mathbf{G}_t)^{\otimes 2} - \mathbb{E}(\mathbf{vec}(\mathbf{G}_t))^{\otimes 2})(\widehat{\mathbf{J}}'_{n,\cdot d_n}\mathbf{J}_n \otimes \widehat{\mathbf{Q}}'_{n,\cdot c_n}\mathbf{Q}_n)',$$

$$I_2 := (\widehat{\mathbf{J}}'_{m,\cdot d_m}\mathbf{J}_m \otimes \widehat{\mathbf{Q}}'_{m,\cdot c_m}\mathbf{Q}_m)T^{-1}\sum_{t=1}^{T}\mathbf{vec}(\mathbf{G}_t)\mathbf{vec}(\mathbf{F}_t^{(n)})'(\widehat{\mathbf{J}}'_{n,\cdot d_n}\mathbf{C}_n \otimes \widehat{\mathbf{Q}}'_{n,\cdot c_n}\mathbf{R}_n)',$$

$$I_3 := (\widehat{\mathbf{J}}'_{m,\cdot d_m}\mathbf{J}_m \otimes \widehat{\mathbf{Q}}'_{m,\cdot c_m}\mathbf{Q}_m)T^{-1}\sum_{t=1}^{T}\mathbf{vec}(\mathbf{G}_t)\widehat{\mathbf{J}}'_{n,\cdot d_n}\mathbf{M}_{q_n}\mathbf{E}_t^{(n)'}\mathbf{M}_{p_n}\widehat{\mathbf{Q}}_{n,\cdot c_n},$$

$$I_4 := (\widehat{\mathbf{J}}'_{m,\cdot d_m}\mathbf{C}_m \otimes \widehat{\mathbf{Q}}'_{m,\cdot c_m}\mathbf{R}_m)T^{-1}\sum_{t=1}^{T}\mathbf{vec}(\mathbf{F}_t^{(m)})\mathbf{vec}(\mathbf{G}_t)'(\widehat{\mathbf{J}}'_{n,\cdot d_n}\mathbf{J}_n \otimes \widehat{\mathbf{Q}}'_{n,\cdot c_n}\mathbf{Q}_n)',$$

$$I_5 := (\widehat{\mathbf{J}}'_{m,\cdot d_m}\mathbf{C}_m \otimes \widehat{\mathbf{Q}}'_{m,\cdot c_m}\mathbf{R}_m)T^{-1}\sum_{t=1}^{T}\mathbf{vec}(\mathbf{F}_t^{(m)})\mathbf{vec}(\mathbf{F}_t^{(n)})'(\widehat{\mathbf{J}}'_{n,\cdot d_n}\mathbf{C}_n \otimes \widehat{\mathbf{Q}}'_{n,\cdot c_n}\mathbf{R}_n)',$$

$$I_6 := (\widehat{\mathbf{J}}'_{m,\cdot d_m}\mathbf{C}_m \otimes \widehat{\mathbf{Q}}'_{m,\cdot c_m}\mathbf{R}_m)T^{-1}\sum_{t=1}^{T}\mathbf{vec}(\mathbf{F}_t^{(m)})\widehat{\mathbf{J}}'_{n,\cdot d_n}\mathbf{M}_{q_n}\mathbf{E}_t^{(n)'}\mathbf{M}_{p_n}\widehat{\mathbf{Q}}_{n,\cdot c_n},$$

$$I_7 := \widehat{\mathbf{Q}}'_{m,\cdot c_m}\mathbf{M}_{p_m}T^{-1}\sum_{t=1}^{T}\mathbf{E}_t^{(m)}\mathbf{M}_{q_m}\widehat{\mathbf{J}}_{m,\cdot d_m}\mathbf{vec}(\mathbf{G}_t)'(\widehat{\mathbf{J}}'_{n,\cdot d_n}\mathbf{J}_n \otimes \widehat{\mathbf{Q}}'_{n,\cdot c_n}\mathbf{Q}_n)',$$

$$I_8 := \widehat{\mathbf{Q}}'_{m,\cdot c_m}\mathbf{M}_{p_m}T^{-1}\sum_{t=1}^{T}\mathbf{E}_t^{(m)}\mathbf{M}_{q_m}\widehat{\mathbf{J}}_{m,\cdot d_m}\mathbf{vec}(\mathbf{F}_t^{(n)})'(\widehat{\mathbf{J}}'_{n,\cdot d_n}\mathbf{C}_n \otimes \widehat{\mathbf{Q}}'_{n,\cdot c_n}\mathbf{R}_n)',$$

$$I_9 := \widehat{\mathbf{Q}}'_{m,\cdot c_m}\mathbf{M}_{p_m}T^{-1}\sum_{t=1}^{T}\mathbf{E}_t^{(m)}\mathbf{M}_{q_m}\widehat{\mathbf{J}}_{m,\cdot d_m}\widehat{\mathbf{J}}'_{n,\cdot d_n}\mathbf{M}_{q_n}\mathbf{E}_t^{(n)'}\mathbf{M}_{p_n}\widehat{\mathbf{Q}}_{n,\cdot c_n},$$

with $I_0 = I_1 = I_4 = I_7 = 0$ if $n \notin \mathcal{A}_G$. By Assumption (S1) on $T^{-1}\sum_{t=1}^{T}\mathbb{E}\mathbf{vec}(\mathbf{G}_t)^{\otimes 2}$, and that the set of vectors $\{\mathbf{e}'_{k_c,d_n}\mathbf{H}'_{j,n}(\mathbf{J}'_n\mathbf{J}_n)^{1/2} \otimes \mathbf{e}'_{k_r,c_n}\mathbf{H}'_{q,n}(\mathbf{Q}'_n\mathbf{Q}_n)^{1/2}\}_{c_n\in[k_r],d_n\in[k_c]}$ is linearly independent by Assumption (L1), for each $n \neq m$ with $n \in \mathcal{A}_G$, there must exist $c_n \in [k_r], d_n \in [k_c]$ (through using (C.111), so that we can replace $\widehat{\mathbf{Q}}_{n,\cdot c_n}$ by $\mathbf{Q}_n(\mathbf{Q}'_n\mathbf{Q}_n)^{-1/2}\mathbf{H}_{q,n}\mathbf{e}_{k_r,c_n}$ and similarly $\widehat{\mathbf{J}}_{n,\cdot d_n}$ by $\mathbf{J}_n(\mathbf{J}'_n\mathbf{J}_n)^{-1/2}\mathbf{H}_{j,n}\mathbf{e}_{k_c,d_n}$) such that

$$\begin{aligned}\max_{c_m\in[k_r],d_m\in[k_c]} I_0^2 &\asymp_P \max_{c_m\in[k_r],d_m\in[k_c]} \left\|\mathbf{e}'_{k_c,d_m}\mathbf{H}'_{j,m}(\mathbf{J}'_m\mathbf{J}_m)^{1/2} \otimes \mathbf{e}'_{k_r,c_m}\mathbf{H}'_{q,m}(\mathbf{Q}'_m\mathbf{Q}_m)^{1/2}\right\|^2 \\ &\quad \cdot \left\|\mathbf{e}'_{k_c,d_n}\mathbf{H}'_{j,n}(\mathbf{J}'_n\mathbf{J}_n)^{1/2} \otimes \mathbf{e}'_{k_r,c_n}\mathbf{H}'_{q,n}(\mathbf{Q}'_n\mathbf{Q}_n)^{1/2}\right\|^2 \\ &\gtrsim_P p_m^{\delta_{q,1}^{(m)}} q_m^{\delta_{j,1}^{(m)}} \cdot p_n^{\delta_{q,k_r}^{(n)}} q_n^{\delta_{j,k_c}^{(n)}},\end{aligned} \tag{C.114}$$

where the last line used (C.112), (C.113) and Assumption (L1). By Lemma 2-2 and 2-3, we also have

$$\max_{c_m\in[k_r],d_m\in[k_c]} I_1^2 \lesssim_P p_m^{\delta_{q,1}^{(m)}} q_m^{\delta_{j,1}^{(m)}} \cdot p_n^{\delta_{q,1}^{(n)}} q_n^{\delta_{j,1}^{(n)}} R(T,d_{\text{fac},\star}), \quad \max_{c_m\in[k_r],d_m\in[k_c]} I_2^2 \lesssim_P p_m^{\delta_{q,1}^{(m)}} q_m^{\delta_{j,1}^{(m)}} \cdot p_n^{\delta_{r,1}^{(n)}} q_n^{\delta_{c,1}^{(n)}} R(T,d_{\text{fac},\star}),$$
$$\max_{c_m\in[k_r],d_m\in[k_c]} I_4^2 \lesssim_P p_m^{\delta_{r,1}^{(m)}} q_m^{\delta_{c,1}^{(m)}} \cdot p_n^{\delta_{q,1}^{(n)}} q_n^{\delta_{j,1}^{(n)}} R(T,d_{\text{fac},\star}), \quad \max_{c_m\in[k_r],d_m\in[k_c]} I_5^2 \lesssim_P p_m^{\delta_{r,1}^{(m)}} q_m^{\delta_{c,1}^{(m)}} \cdot p_n^{\delta_{r,1}^{(n)}} q_n^{\delta_{c,1}^{(n)}} R(T,d_{\text{fac},\star}).$$

Using similar proof for Lemma 3, and the fact that

$$\max_{i\in[p_m],j\in[k_r]} |(\widehat{\mathbf{Q}}_m - \mathbf{Q}_m(\mathbf{Q}_m'\mathbf{Q}_m)^{-1/2}\mathbf{H}_{q,m})_{ij}| = O_P(\|\widehat{\mathbf{Q}}_m - \mathbf{Q}_m(\mathbf{Q}_m'\mathbf{Q}_m)^{-1/2}\mathbf{H}_{q,m}\|) = o_P(\eta_{Q,m}) = o_P(1),$$

and similarly

$$\max_{i\in[q_m],j\in[k_c]} |(\widehat{\mathbf{J}}_m - \mathbf{J}_m(\mathbf{J}_m'\mathbf{J}_m)^{-1/2}\mathbf{H}_{j,m})_{ij}| = O_P(\|\widehat{\mathbf{J}}_m - \mathbf{J}_m(\mathbf{J}_m'\mathbf{J}_m)^{-1/2}\mathbf{H}_{j,m}\|) = o_P(\eta_{J,m}) = o_P(1),$$

we can prove the following in parallel to (D.33), (D.34) and (D.35)(details omitted):

$$\|T^{-1}\sum_{t=1}^T \mathbf{E}_t^{(m)}\widehat{\mathbf{v}}_m\mathbf{u}_t'\|_F^2 = O_P(T^{-1}p_m), \tag{C.115}$$
$$\|T^{-1}\sum_{t=1}^T \mathbf{E}_t^{(m)}\widehat{\mathbf{v}}_m\widehat{\mathbf{v}}_n'\mathbf{E}_t^{(n)'}\|_F^2 = O_P(T^{-1}p_mp_n), \tag{C.116}$$
$$\|T^{-1}\sum_{t=1}^T \mathbf{E}_t^{(m)}\widehat{\mathbf{v}}_m\widehat{\mathbf{v}}_m'\mathbf{E}_t^{(m)'}\|_F^2 = O_P(p_m + T^{-1}p_m^2), \tag{C.117}$$

where $\{\mathbf{u}_t\}$ is independent of $\{\mathbf{E}_t^{(m)}\}$ for each $m\in[M]$, and $\widehat{\mathbf{v}}_m, \widehat{\mathbf{v}}_n$ are unit vectors potentially dependent on $\{\mathbf{E}_t^{(m)}\}$ for some $m\in[M]$, but are consistent to some non-random unit vectors $\mathbf{v}_m$ and $\mathbf{v}_n$ respectively. With (C.115) and (C.116), we can show that

$$\max_{c_m\in[k_r],d_m\in[k_c]} I_3^2 \asymp_P p_m^{\delta_{q,1}^{(m)}} q_m^{\delta_{j,1}^{(m)}} \cdot T^{-1}p_n, \quad \max_{c_m\in[k_r],d_m\in[k_c]} I_6^2 \asymp_P p_m^{\delta_{r,1}^{(m)}} q_m^{\delta_{c,1}^{(m)}} \cdot T^{-1}p_n,$$
$$\max_{c_m\in[k_r],d_m\in[k_c]} I_7^2 \asymp_P T^{-1}p_m \cdot p_n^{\delta_{q,1}^{(n)}} q_n^{\delta_{j,1}^{(n)}}, \quad \max_{c_m\in[k_r],d_m\in[k_c]} I_8^2 \asymp_P T^{-1}p_m \cdot p_n^{\delta_{r,1}^{(n)}} q_n^{\delta_{c,1}^{(n)}},$$
$$\max_{c_m\in[k_r],d_m\in[k_c]} I_9^2 \asymp_P T^{-1}p_mp_n.$$

Using the rate assumption in the statement of the theorem, together with Assumption (R2), (C.114) and the rates for I_1^2 to I_9^2 result in

$$\sum_{n\neq m} \max_{c_m,c_m\in[k_r]} \max_{d_m,d_n\in[k_c]} \{T^{-1}\sum_{t=1}^T \widehat{\mathbf{Q}}_{m,\cdot c_m}'\check{\mathbf{Y}}_t^{(m)}\widehat{\mathbf{J}}_{m,\cdot d_m}\widehat{\mathbf{J}}_{n,\cdot d_n}'\check{\mathbf{Y}}_t^{(n)'}\widehat{\mathbf{Q}}_{n,\cdot c_n}\}^2 \gtrsim_P p_m^{\delta_{q,1}^{(m)}} q_m^{\delta_{j,1}^{(m)}} \max_{n\neq m, n\in\mathcal{A}_G} p_n^{\delta_{q,k_r}^{(n)}} q_n^{\delta_{j,k_c}^{(n)}}.$$

On the other hand, if $m = k \notin \mathcal{A}_G$, then $I_0 = I_1 = I_2 = I_3 = 0$, and the rates for I_4^2 to I_9^2 lead to

$$\sum_{n\neq k} \max_{c_k, c_n \in [k_r]} \max_{d_k, d_n \in [k_c]} \big\{ T^{-1} \sum_{t=1}^{T} \widehat{\mathbf{Q}}'_{k,\cdot c_k} \check{\mathbf{Y}}_t^{(k)} \widehat{\mathbf{J}}_{k,\cdot d_k} \widehat{\mathbf{J}}'_{n,\cdot d_n} \check{\mathbf{Y}}_t^{(n)'} \widehat{\mathbf{Q}}_{n,\cdot c_n} \big\}^2$$
$$\lesssim_P p_k^{\delta_{r,1}^{(k)}} q_k^{\delta_{c,1}^{(k)}} \max\big(\max_{n\in\mathcal{A}_G} p_n^{\delta_{q,1}^{(n)}} q_n^{\delta_{j,1}^{(n)}}, \max_{n\neq k, n\notin\mathcal{A}_G} p_n^{\delta_{r,1}^{(n)}} q_n^{\delta_{c,1}^{(n)}} \big) R(T, d_{\mathrm{fac},\star}).$$

Now we rewrite $\mathbf{Y}_t^{(m)}$ as

$$\check{\mathbf{Y}}_t^{(m)} = \mathbf{U}_m \mathbf{H}_t^{(m)} \mathbf{V}'_m + \mathbf{M}_{p_m} \mathbf{E}_t^{(m)} \mathbf{M}_{q_m},$$

where $\mathbf{U}_m := (\mathbf{Q}_m, \mathbf{R}_m)$, $\mathbf{V}_m := (\mathbf{J}_m, \mathbf{C}_m)$ and $\mathbf{H}_t^{(m)} := \mathrm{diag}(\mathbf{G}_t, \mathbf{F}_t^{(m)})$. We then have $\mathrm{rank}(\mathbf{U}_m) \leq k_r + k_r^{(m)}$ and $\mathrm{rank}(\mathbf{V}_m) \leq k_c + k_c^{(m)}$. Repeating our estimation methodology for just the local common components for individual groups using data $\{\check{\mathbf{Y}}_t^{(m)}\}$, we can use the same proof as Theorem 7 to arrive at $\big\|\widehat{\mathbf{U}}_m - \mathbf{U}_m(\mathbf{U}'_m\mathbf{U}_m)^{-1/2}\mathbf{H}_{u,m}\big\|_F^2 = o_P(1)$ and $\big\|\widehat{\mathbf{V}}_m - \mathbf{V}_m(\mathbf{V}'_m\mathbf{V}_m)^{-1/2}\mathbf{H}_{v,m}\big\|_F^2 = o_P(1)$, where $\mathbf{H}_{u,m}$ and $\mathbf{H}_{v,m}$ are two constant order invertible matrices. Then using (C.117) and Assumption (R2) and (S1), similar to the treatment before, we have for $m \in \mathcal{A}_G$,

$$\max_{c_m \in [k_r + k_r^{(m)}], d_m \in [k_c + k_c^{(m)}]} \big(T^{-1} \sum_{t=1}^{T} (\widehat{\mathbf{U}}'_{m,\cdot c_m} \mathbf{Q}_m \mathbf{G}_t \mathbf{J}'_m \widehat{\mathbf{V}}_{m,\cdot d_m} + \widehat{\mathbf{U}}'_{m,\cdot c_m} \mathbf{R}_m \mathbf{F}_t^{(m)} \mathbf{C}'_m \widehat{\mathbf{V}}_{m,\cdot d_m}$$
$$+ \widehat{\mathbf{U}}'_{m,\cdot c_m} \mathbf{M}_{p_m} \mathbf{E}_t^{(m)} \mathbf{M}_{q_m} \widehat{\mathbf{V}}_{m,\cdot d_m})^2 \big)^2$$
$$\gtrsim_P p_m^{2\delta_{q,1}^{(m)}} q_m^{2\delta_{j,1}^{(m)}} - p_m^{2\delta_{r,1}^{(m)}} q_m^{2\delta_{c,1}^{(m)}} - p_m - T^{-1} p_m^2 \asymp_P p_m^{2\delta_{q,1}^{(m)}} q_m^{2\delta_{j,1}^{(m)}}.$$

At the same time, if $m \notin \mathcal{A}_G$,

$$\max_{c_m \in [k_r + k_r^{(m)}], d_m \in [k_c + k_c^{(m)}]} \big(T^{-1} \sum_{t=1}^{T} (\widehat{\mathbf{U}}'_{m,\cdot c_m} \mathbf{R}_m \mathbf{F}_t^{(m)} \mathbf{C}'_m \widehat{\mathbf{V}}_{m,\cdot d_m} + \widehat{\mathbf{U}}'_{m,\cdot c_m} \mathbf{M}_{p_m} \mathbf{E}_t^{(m)} \mathbf{M}_{q_m} \widehat{\mathbf{V}}_{m,\cdot d_m})^2 \big)^2$$
$$\gtrsim_P p_m^{2\delta_{r,1}^{(m)}} q_m^{2\delta_{c,1}^{(m)}} - p_m - T^{-1} p_m^2 \asymp_P p_m^{2\delta_{r,1}^{(m)}} q_m^{2\delta_{c,1}^{(m)}}.$$

This completes the proof of the theorem. □

Proof of Theorem 10. Consider $\widehat{r}_1$. We omit the details for $\widehat{\ell}_1$ since its proof is parallel to that for $\widehat{r}_1$. It is not difficult to see that, through the finiteness of M,

$$\max_{n\in[M]} \max_{n\neq m} \sigma_{r_1}^2 \Big\{ T^{-1} \sum_{t=1}^{T} \widehat{\mathbf{a}}_t^{(m)} \widehat{\mathbf{a}}_t^{(n)'} \Big\} \lesssim \max_{m\in[M]} \lambda_i(\widehat{\mathbf{\Omega}}_{\mathbf{A}_1,m}) \lesssim \max_{m\in[M]} \max_{n\neq m} \sigma_1^2 \Big\{ T^{-1} \sum_{t=1}^{T} \widehat{\mathbf{a}}_t^{(m)} \widehat{\mathbf{a}}_t^{(n)'} \Big\}, \quad i \in [r_1], \tag{C.118}$$

where $\sigma_i(\mathbf{A})$ denotes the ith largest singular value of the matrix $\mathbf{A}$. At the same time, for $i \in [r_1]$,

$$\sigma_i\Big\{\mathbf{A}_{1m}\big(T^{-1}\sum_{t=1}^{T}\boldsymbol{\alpha}_t^{\otimes 2}\big)\mathbf{A}_{1n}'\Big\} - \sum_{j=2}^{5} I_j \leq \sigma_i\Big\{T^{-1}\sum_{t=1}^{T}\widehat{\mathbf{a}}_t^{(m)}\widehat{\mathbf{a}}_t^{(n)'}\Big\} \leq \sigma_1\Big\{\mathbf{A}_{1m}\big(T^{-1}\sum_{t=1}^{T}\boldsymbol{\alpha}_t^{\otimes 2}\big)\mathbf{A}_{1n}'\Big\} + \sum_{j=2}^{5} I_j, \tag{C.119}$$

where

$$I_2 := \sigma_1\Big\{\mathbf{A}_{2m}\big(T^{-1}\sum_{t=1}^{T}\boldsymbol{\alpha}_t^{(m)}\boldsymbol{\alpha}_t^{(n)'}\big)\mathbf{A}_{2n}'\Big\}, \quad I_3 := \sigma_1\Big\{T^{-1}\sum_{t=1}^{T} q_m^{-1}\mathbf{M}_{p_m}\mathbf{E}_t^{(m)}\mathbf{1}_{q_m}\big(\mathbf{A}_{1n}\boldsymbol{\alpha}_t + \mathbf{A}_{2n}\boldsymbol{\alpha}_t^{(n)}\big)'\Big\},$$
$$I_4 := \sigma_1\Big\{T^{-1}\sum_{t=1}^{T}\big(\mathbf{A}_{1m}\boldsymbol{\alpha}_t + \mathbf{A}_{2m}\boldsymbol{\alpha}_t^{(m)}\big)q_n^{-1}\mathbf{1}_{q_n}'\mathbf{E}_t^{(n)'}\mathbf{M}_{p_n}\Big\},$$
$$I_5 := \sigma_1\big\{T^{-1}q_m^{-1}q_n^{-1}\sum_{t=1}^{T}\mathbf{M}_{p_m}\mathbf{E}_t^{(m)}\mathbf{1}_{q_m}\mathbf{1}_{q_n}'\mathbf{E}_t^{(n)'}\mathbf{M}_{p_n}\big\}.$$

Firstly, for $i \in [r_1]$, using (D.14) and Assumption (S1),

$$\sigma_i\Big\{\mathbf{A}_{1m}\big(T^{-1}\sum_{t=1}^{T}\boldsymbol{\alpha}_t^{\otimes 2}\big)\mathbf{A}_{1n}'\Big\} \geq \lambda_i^{1/2}(\mathbf{A}_{1m}'\mathbf{A}_{1m})\lambda_{r_1}^{1/2}(\mathbf{A}_{1n}'\mathbf{A}_{1n})\lambda_{r_1}(T^{-1}\sum_{t=1}^{T}\boldsymbol{\alpha}_t^{\otimes 2})$$
$$\gtrsim_P p_m^{\gamma_r/2}p_n^{\gamma_r/2}(1 + R(T, d_{r,\star})^{1/2}).$$

Using (ME1), and (D.15) on I_2, (D.33) on I_3 and I_4 and (D.34) on I_5,

$$I_2 := O_P(p_m^{\gamma_r^{(m)}/2}p_n^{\gamma_r^{(n)}/2}R(T, d_{r,\star})^{1/2}) = o_P(p_m^{\gamma_r/2}p_n^{\gamma_r/2}),$$
$$I_3 := O_P(T^{-1/2}p_m^{1/2}q_m^{-1/2}\cdot(p_n^{\gamma_r/2} + p_n^{\gamma_r^{(n)}/2})) = o_P(p_m^{\gamma_r/2}p_n^{\gamma_r/2}),$$
$$I_4 := O_P(T^{-1/2}p_n^{1/2}q_n^{-1/2}\cdot(p_m^{\gamma_r/2} + p_m^{\gamma_r^{(m)}/2})) = o_P(p_m^{\gamma_r/2}p_n^{\gamma_r/2}),$$
$$I_5 := O_P(T^{-1/2}p_m^{1/2}p_n^{1/2}q_m^{-1/2}q_n^{-1/2}) = o_P(p_m^{\gamma_r/2}p_n^{\gamma_r/2}),$$

where we use (R1) as well for concluding that all the rates above are $o_P(p_m^{\gamma_r/2}p_n^{\gamma_r/2})$. Hence (C.119) implies that for $i \in [r_1]$,

$$\max_{m,n\in[M],n\neq m}\sigma_i^2\Big\{T^{-1}\sum_{t=1}^{T}\widehat{\mathbf{a}}_t^{(m)}\widehat{\mathbf{a}}_t^{(n)'}\Big\} \asymp_P \max_{m,n\in[M],n\neq m} p_m^{\gamma_r}p_n^{\gamma_r},$$

so that by (C.118), for $i \in [r_1]$,

$$\max_{m\in[M]}\lambda_i(\widehat{\boldsymbol{\Omega}}_{\mathbf{A}_1,m}) \asymp_P \max_{m,n\in[M],n\neq m} p_m^{\gamma_r}p_n^{\gamma_r}. \tag{C.120}$$

On the other hand, we can write

$$\begin{aligned}\widehat{\mathbf{\Omega}}_{\mathbf{A}_1,m} =: (M-1)^{-1}\sum_{n\neq m}\Big(\mathbf{A}_{1m}\big(T^{-1}\sum_{t=1}^{T}\boldsymbol{\alpha}_t^{\otimes 2}\big)\mathbf{A}_{1n}'+R_{m,n}\Big)^{\otimes 2}\\ =(M-1)^{-1}\sum_{n\neq m}\Big(\mathbf{A}_{1m}\big(T^{-1}\sum_{t=1}^{T}\boldsymbol{\alpha}_t^{\otimes 2}\big)\mathbf{A}_{1n}'\mathbf{A}_{1n}\big(T^{-1}\sum_{t=1}^{T}\boldsymbol{\alpha}_t^{\otimes 2}\big)\mathbf{A}_{1m}'+\mathbf{A}_{1m}\big(T^{-1}\sum_{t=1}^{T}\boldsymbol{\alpha}_t^{\otimes 2}\big)\mathbf{A}_{1n}'R_{m,n}'\\ +R_{m,n}\mathbf{A}_{1n}\big(T^{-1}\sum_{t=1}^{T}\boldsymbol{\alpha}_t^{\otimes 2}\big)\mathbf{A}_{1m}'+R_{m,n}R_{m,n}'\Big),\end{aligned}$$

so that for $i\in\{r_1+1,\dots,p_m\}$,

$$\begin{aligned}\lambda_i(\widehat{\mathbf{\Omega}}_{\mathbf{A}_1,m})\leq\lambda_1\Big\{(M-1)^{-1}\sum_{n\neq m}R_{m,n}R_{m,n}'\Big\}\leq\max_{n\neq m}\sigma_1^2(R_{m,n})\\ \lesssim\max_{n\neq m}\sum_{j=2}^{5}I_j^2=O_P(\max_{n\neq m}p_m^{\gamma_r^{(m)}}p_n^{\gamma_r^{(n)}}R(T,d_{r,\star})),\end{aligned}\tag{C.121}$$

where the last equality used Assumption (R1). Hence combining (C.120) and (C.121), we have

$$\frac{\max_{m\in[M]}\lambda_{i+1}(\widehat{\mathbf{\Omega}}_{\mathbf{A}_1,m})}{\max_{m\in[M]}\lambda_i(\widehat{\mathbf{\Omega}}_{\mathbf{A}_1,m})}\begin{cases}\asymp_P 1, & i\in[r_1-1];\\ \lesssim_P \frac{\max_{m\in[M],n\neq m}p_m^{\gamma_r^{(m)}}p_n^{\gamma_r^{(n)}}R(T,d_{r,\star})}{\max_{m\in[M],n\neq m}p_m^{\gamma_r}p_n^{\gamma_r}}=o_P(1), & i=r_1,\end{cases}$$

where the equation for the case $i=r_1$ used Assumption (R1). This implies $P(\widehat{r}_1\geq r_1)\to 1$. It is also clear that using (C.120) and (C.121), the ratio

$$\frac{\max_{m\in[M]}\lambda_{i+1}(\widehat{\mathbf{\Omega}}_{\mathbf{A}_1,m})+\max_{m\neq n\in[M]}Cp_m^{\gamma_r^{(m)}}p_n^{\gamma_r^{(n)}}R(T,d_{r,\star})}{\max_{m\in[M]}\lambda_i(\widehat{\mathbf{\Omega}}_{\mathbf{A}_1,m})+\max_{m\neq n\in[M]}Cp_m^{\gamma_r^{(m)}}p_n^{\gamma_r^{(n)}}R(T,d_{r,\star})}\begin{cases}\asymp_P 1, & i\in[r_1-1];\\ =o_P(1), & i=r_1,\\ \asymp_P 1, & i=r_1+1,\dots,p_m,\end{cases}$$

so that $P(\widehat{r}_1=r_1)\to 1$, which aligns with the statement of adding $\max_{m\neq n\in[M]}CT^{-1}p_mp_n$ to the ratio when all factors are strong and all processes have short memory.

For $\widehat{r}_2^{(m)}$ (We omit $\widehat{\ell}_2^{(m)}$ since the proof is parallel), given $i\in[r_2^{(m)}]$, consider

$$J_1-\sum_{i=2}^{6}J_i\lesssim\sigma_i\Big(T^{-1}\sum_{t=1}^{T}(\widehat{\mathbf{a}}_t^{(m)})^{\otimes 2}\widehat{\mathbf{A}}_{1m}^{\perp}\Big)\lesssim J_1+\sum_{i=2}^{6}J_i,\quad\text{where}\tag{C.122}$$

$$J_1:=\sigma_i\Big(\mathbf{A}_{2m}\big(T^{-1}\sum_{t=1}^{T}(\boldsymbol{\alpha}_t^{(m)})^{\otimes 2}\big)\mathbf{A}_{2m}'\mathbf{A}_{1m}^{\perp}\Big),$$

$$J_2:=\sigma_1\Big(\mathbf{A}_{1m}\big(T^{-1}\sum_{t=1}^{T}\boldsymbol{\alpha}_t^{\otimes 2}\big)(\mathbf{A}_{1m}'\mathbf{A}_{1m})^{1/2}(\mathbf{H}_{\mathbf{A}_1,m}^{-1})'(\check{\mathbf{A}}_{1m}\mathbf{H}_{A_1,m}-\widehat{\mathbf{A}}_{1m})'\widehat{\mathbf{A}}_{1m}^{\perp}\Big),$$

$$J_3 := \sigma_i\Big(\mathbf{A}_{2m}\big(T^{-1}\sum_{t=1}^{T}(\boldsymbol{\alpha}_t^{(m)})^{\otimes 2}\big)\mathbf{A}_{2m}'(\widehat{\mathbf{A}}_{1m}^{\perp}-\mathbf{A}_{1m}^{\perp})\Big),$$

$$J_4 := \sigma_1\Big(T^{-1}\sum_{t=1}^{T}q_m^{-1}\mathbf{M}_{p_m}\mathbf{E}_t^{(m)}\mathbf{1}_{q_m}(\mathbf{A}_{1m}\boldsymbol{\alpha}_t+\mathbf{A}_{2m}\boldsymbol{\alpha}_t^{(m)})'\widehat{\mathbf{A}}_{1m}^{\perp}\Big),$$

$$J_5 := \sigma_1\Big(T^{-1}\sum_{t=1}^{T}(\mathbf{A}_{1m}\boldsymbol{\alpha}_t+\mathbf{A}_{2m}\boldsymbol{\alpha}_t^{(m)})\mathbf{1}_{q_m}'\mathbf{E}_t^{(m)'}\mathbf{M}_{p_m}q_m^{-1}\widehat{\mathbf{A}}_{1m}^{\perp}\Big),$$

$$J_6 := \sigma_1\Big(T^{-1}q_m^{-2}\sum_{t=1}^{T}\mathbf{M}_{p_m}\mathbf{E}_t^{(m)}\mathbf{1}_{q_m}\mathbf{1}_{q_m}'\mathbf{E}_t^{(m)'}\mathbf{M}_{p_m}\widehat{\mathbf{A}}_{1m}^{\perp}\Big).$$

For $i \in [r_2^{(m)}]$, using (D.14) and Assumption (S1) and (L1),

$$J_1 = \sigma_i\Big(\big(T^{-1}\sum_{t=1}^{T}(\boldsymbol{\alpha}_t^{(m)})^{\otimes 2}\big)(\mathbf{A}_{2m}'\mathbf{A}_{1m}^{\perp})^{\otimes 2}\Big) \geq \sigma_i^2(\mathbf{A}_{2m}'\mathbf{A}_{1m}^{\perp})\sigma_{r_2^{(m)}}\Big(T^{-1}\sum_{t=1}^{T}(\boldsymbol{\alpha}_t^{(m)})^{\otimes 2}\Big) \asymp_P p_m^{\gamma_r^{(m)}},$$

Also,

$$\begin{aligned}
J_2 &= O_P(p_m^{\gamma_r}\eta_{A_{1m}}^{1/2}) = o_P(p_m^{\gamma_r^{(m)}}),\\
J_3 &= O_P(p_m^{\gamma_r^{(m)}}\big\|\widehat{\mathbf{A}}_{1m}^{\perp}-\mathbf{A}_{1m}^{\perp}\big\|) = O_P(p_m^{\gamma_r^{(m)}}\big\|\widehat{\mathbf{A}}_{1m}-\breve{\mathbf{A}}_{1m}\mathbf{H}_{A_1,m}\big\|) = o_P(p_m^{\gamma_r^{(m)}}),\\
J_4, J_5 &= O_P(T^{-1/2}p_m^{1/2}q_m^{-1/2}\cdot(p_m^{\gamma_r/2}+p_m^{\gamma_r^{(m)}/2})) = o_P(p_m^{\gamma_r^{(m)}}),\\
J_6 &= O_P(p_m^{1/2}q_m^{-1}+T^{-1/2}p_m q_m^{-1}) = o_P(p_m^{\gamma_r^{(m)}}),
\end{aligned}$$

where $\eta_{A_{1m}}$ is the rate defined in Theorem 3, and the equation for J_2 used the assumption in the statement of the theorem. We use (D.33) for J_4 and J_5 and that $\eta_{A_{1m}} = o(\nu_{r,m}^2)$ and (R1) to conclude that they are $o_P(p_m^{\gamma_r^{(m)}})$, while J_6 uses (D.35).

On the other hand, for $i \in \{r_2^{(m)+1}, \ldots, p_m\}$,

$$\sigma_i\Big(T^{-1}\sum_{t=1}^{T}(\widehat{\mathbf{a}}_t^{(m)})^{\otimes 2}\widehat{\mathbf{A}}_{1m}^{\perp}\Big) \lesssim_P \sum_{i=2}^{6}J_i = o_P(p_m^{\gamma_r^{(m)}}). \tag{C.123}$$

The rates for J_1 to J_6, combined with (C.122) and (C.123), allow us to conclude that

$$\frac{\lambda_{j+1}(\widehat{\boldsymbol{\Omega}}_{A_2,m})}{\lambda_j(\widehat{\boldsymbol{\Omega}}_{A_2,m})}\begin{cases}\asymp_P 1, & j \in [r_2^{(m)}-1];\\ = o_P(1), & j = r_2^{(m)},\end{cases}$$

so that $P(\widehat{r}_2^{(m)} \geq r_2^{(m)}) \to 1$. It is also clear that when all factors are strong and all processes have short

memory, $\eta_{A_{1m}} = T^{-1}$, so that the rate in (C.123) is $O_P(T^{-1/2}p_m)$, and the ratio

$$\frac{\lambda_{j+1}(\widehat{\mathbf{\Omega}}_{A_2,m}) + T^{-1}p_m^2}{\lambda_j(\widehat{\mathbf{\Omega}}_{A_2,m}) + T^{-1}p_m^2} \begin{cases} \asymp_P 1, & j \in [r_2^{(m)} - 1]; \\ = o_P(1), & j = r_2^{(m)}; \\ \asymp_P 1 & j = r_2^{(m)} + 1, \ldots, p_m, \end{cases}$$

so that $P(\widehat{r}_2^{(m)} = r_2^{(m)}) \to 1$.

Now for $\widehat{k}_r$ (we omit the details for $\widehat{k}_c$ since it involves parallel arguments), consider for $i \in [k_r]$,

$$\max_{m\in[M]} \max_{n\neq m} \sigma_i^2 \Big\{ T^{-1} \sum_{t=1}^{T} \breve{\mathbf{Y}}_t^{(m)} \mathbf{v}_b^{(m)} \mathbf{v}_b^{(n)'} \breve{\mathbf{Y}}_t^{(n)'} \Big\} \lesssim \max_{m\in[M]} \lambda_i(\mathbf{X}_{Q,m})$$
$$\lesssim \max_{m\in[M]} \max_{n\neq m} \sigma_1^2 \Big\{ T^{-1} \sum_{t=1}^{T} \breve{\mathbf{Y}}_t^{(m)} \mathbf{v}_b^{(m)} \mathbf{v}_b^{(n)'} \breve{\mathbf{Y}}_t^{(n)'} \Big\}. \quad \text{(C.124)}$$

At the same time, for $i \in [k_r]$,

$$K_{1,i} - \sum_{j=2}^{9} K_j \leq \sigma_i \Big\{ T^{-1} \sum_{t=1}^{T} \breve{\mathbf{Y}}_t^{(m)} \mathbf{v}_b^{(m)} \mathbf{v}_b^{(n)'} \breve{\mathbf{Y}}_t^{(n)'} \Big\} \leq K_{1,i} + \sum_{j=2}^{9} K_j, \quad \text{where}$$

$$K_{1,i} := \sigma_i \Big((\mathbf{v}_b^{(m)'} \mathbf{J}_m \otimes \mathbf{Q}_m) T^{-1} \sum_{t=1}^{T} \mathbf{vec}(\mathbf{G}_t)^{\otimes 2} (\mathbf{v}_b^{(n)'} \mathbf{J}_n \otimes \mathbf{Q}_n)' \Big),$$

$$K_2 := \sigma_1 \Big((\mathbf{v}_b^{(m)'} \mathbf{J}_m \otimes \mathbf{Q}_m) T^{-1} \sum_{t=1}^{T} \mathbf{vec}(\mathbf{G}_t) \mathbf{vec}(\mathbf{F}_t^{(n)})' (\mathbf{v}_b^{(n)'} \mathbf{C}_n \otimes \mathbf{R}_n)' \Big),$$

$$K_3 := \sigma_1 \Big((\mathbf{v}_b^{(m)'} \mathbf{J}_m \otimes \mathbf{Q}_m) T^{-1} \sum_{t=1}^{T} \mathbf{vec}(\mathbf{G}_t) \mathbf{v}_b^{(n)'} \mathbf{M}_{q_n} \mathbf{E}_t^{(n)'} \mathbf{M}_{p_n} \Big),$$

$$K_4 := \sigma_1 \Big((\mathbf{v}_b^{(m)'} \mathbf{C}_m \otimes \mathbf{R}_m) T^{-1} \sum_{t=1}^{T} \mathbf{vec}(\mathbf{F}_t^{(m)}) \mathbf{vec}(\mathbf{G}_t)' (\mathbf{v}_b^{(n)'} \mathbf{J}_n \otimes \mathbf{Q}_n)' \Big),$$

$$K_5 := \sigma_1 \Big((\mathbf{v}_b^{(m)'} \mathbf{C}_m \otimes \mathbf{R}_m) T^{-1} \sum_{t=1}^{T} \mathbf{vec}(\mathbf{F}_t^{(m)}) \mathbf{vec}(\mathbf{F}_t^{(n)})' (\mathbf{v}_b^{(n)'} \mathbf{C}_n \otimes \mathbf{R}_n)' \Big),$$

$$K_6 := \sigma_1 \Big((\mathbf{v}_b^{(m)'} \mathbf{C}_m \otimes \mathbf{R}_m) T^{-1} \sum_{t=1}^{T} \mathbf{vec}(\mathbf{F}_t^{(m)}) \mathbf{v}_b^{(n)'} \mathbf{M}_{q_n} \mathbf{E}_t^{(n)'} \mathbf{M}_{p_n} \Big),$$

$$K_7 := \sigma_1 \Big(\mathbf{M}_{p_m} \cdot T^{-1} \sum_{t=1}^{T} \mathbf{E}_t^{(m)} \mathbf{M}_{q_m} \mathbf{v}_b^{(m)} \mathbf{vec}(\mathbf{G}_t)' (\mathbf{v}_b^{(n)'} \mathbf{J}_n \otimes \mathbf{Q}_n)' \Big),$$

$$K_8 := \sigma_1 \Big(\mathbf{M}_{p_m} \cdot T^{-1} \sum_{t=1}^{T} \mathbf{E}_t^{(m)} \mathbf{M}_{q_m} \mathbf{v}_b^{(m)} \mathbf{vec}(\mathbf{F}_t^{(m)})' (\mathbf{v}_b^{(n)'} \mathbf{C}_n \otimes \mathbf{R}_n)' \Big),$$

$$K_9 := \sigma_1 \Big(\mathbf{M}_{p_m} \cdot T^{-1} \sum_{t=1}^{T} \mathbf{E}_t^{(m)} \mathbf{M}_{q_m} \mathbf{v}_b^{(m)} \mathbf{v}_b^{(n)'} \mathbf{M}_{q_n} \mathbf{E}_t^{(n)'} \cdot \mathbf{M}_{p_n} \Big).$$

To bound the above terms, firstly, we have $\|\mathbf{v}_b^{(m)'} \mathbf{J}_m\| \asymp q_m^{\delta_{j,1}^{(m)}/2}$ and $\|\mathbf{v}_a^{(m)'} \mathbf{Q}_m\| \asymp p_m^{\delta_{q,1}/2}$ by construction,

and hence using (D.14) and Assumption (S1), for $i \in [k_r]$,

$$K_{1,i} \gtrsim_P \lambda_1(\mathbf{J}_n' \mathbf{v}_b^{(n)} \mathbf{v}_b^{(m)} \mathbf{J}_m') \lambda_i(\mathbf{Q}_n' \mathbf{Q}_m) \lambda_{k_r+k_c}\Big(T^{-1}\sum_{t=1}^{T} \mathbf{vec}(\mathbf{G}_t)^{\otimes 2}\Big)$$
$$\gtrsim_P q_m^{\delta_{j,1}^{(m)}/2} q_n^{\delta_{j,1}^{(n)}/2} p_m^{\delta_{q,i}^{(m)}/2} p_n^{\delta_{q,k_r}^{(m)}/2}.$$

Also, using (D.15), (C.115) and (C.116),

$$K_2 = O_P(R(T, d_{\text{fac},\star})^{1/2} q_m^{\delta_{j,1}^{(m)}/2} q_n^{\delta_{c,1}^{(n)}/2} p_m^{\delta_{q,1}^{(m)}/2} p_n^{\delta_{r,1}^{(n)}/2}), \quad K_3 = O_P(T^{-1/2} p_n^{1/2} q_n^{1/2} \cdot q_m^{\delta_{j,1}^{(m)}/2} p_m^{\delta_{q,1}^{(m)}/2}),$$
$$K_4 = O_P(R(T, d_{\text{fac},\star})^{1/2} q_m^{\delta_{c,1}^{(m)}/2} q_n^{\delta_{j,1}^{(n)}/2} p_m^{\delta_{r,1}^{(m)}/2} p_n^{\delta_{q,1}^{(n)}/2}), \quad K_5 = O_P(R(T, d_{\text{fac},\star})^{1/2} q_m^{\delta_{c,1}^{(m)}/2} q_n^{\delta_{c,1}^{(n)}/2} p_m^{\delta_{r,1}^{(m)}/2} p_n^{\delta_{r,1}^{(n)}/2}),$$
$$K_6 = O_P(T^{-1/2} p_n^{1/2} q_n^{1/2} \cdot q_m^{\delta_{c,1}^{(m)}/2} p_m^{\delta_{r,1}^{(m)}/2}), \quad K_7 = O_P(T^{-1/2} p_m^{1/2} q_m^{1/2} \cdot q_n^{\delta_{j,1}^{(n)}/2} p_n^{\delta_{q,1}^{(n)}/2}),$$
$$K_8 = O_P(T^{-1/2} p_m^{1/2} q_m^{1/2} \cdot q_n^{\delta_{c,1}^{(n)}/2} p_n^{\delta_{r,1}^{(n)}/2}), \quad K_9 = O_P(T^{-1/2} p_m^{1/2} p_n^{1/2} q_m^{1/2} q_n^{1/2}).$$

By the assumption in the statement of the theorem, we also have for $i \in [k_r]$,

$$\max_{m,n\in[M],n\neq m,j=2,\ldots,9} K_j^2 = o_P(\max_{m,n\in[M],n\neq m} q_m^{\delta_{j,1}^{(m)}} q_n^{\delta_{j,1}^{(n)}} p_m^{\delta_{q,i}^{(m)}} p_n^{\delta_{q,k_r}^{(m)}}) \lesssim_P \max_{m,n\in[M],n\neq m} K_{1,i}^2 \asymp_P \max_{m\in[M]} \lambda_i(\mathbf{X}_{Q,m}),$$

after using (C.124). On the other hand, for $i \in \{k_r+1, \ldots, p_m\}$,

$$\max_{m\in[M]} \lambda_i(\mathbf{X}_{Q,m}) \lesssim_P \max_{m,n\in[M],n\neq m,j=2,\ldots,9} K_j^2$$
$$= O_P\Big(\max_{m,n\in[M],n\neq m} \big(T^{-1} p_m p_n q_m q_n + R(T, d_{\text{fac},\star}) p_m^{\delta_{q,1}^{(m)}} q_m^{\delta_{j,1}^{(m)}} p_n^{\delta_{q,1}^{(n)}} q_n^{\delta_{j,1}^{(n)}}\big)\Big).$$

With these intermediate results, we can conclude that for $i \in [k_r - 1]$,

$$\frac{\max_{m\in[M]} \lambda_{i+1}(\mathbf{X}_{Q,m})}{\max_{m\in[M]} \lambda_i(\mathbf{X}_{Q,m})} \gtrsim_P \frac{\max_{m,n\in[M],n\neq m} q_m^{\delta_{j,1}^{(m)}} q_n^{\delta_{j,1}^{(n)}} p_m^{\delta_{q,i+1}^{(m)}} p_n^{\delta_{q,k_r}^{(m)}}}{\max_{m,n\in[M],n\neq m} q_m^{\delta_{j,1}^{(m)}} q_n^{\delta_{j,1}^{(n)}} p_m^{\delta_{q,1}^{(m)}} p_n^{\delta_{q,1}^{(m)}}}$$
$$\succ_P \frac{\max_{m,n\in[M],n\neq m} \big(T^{-1} p_m p_n q_m q_n + R(T, d_{\text{fac},\star}) p_m^{\delta_{q,1}^{(m)}} q_m^{\delta_{j,1}^{(m)}} p_n^{\delta_{q,1}^{(n)}} q_n^{\delta_{j,1}^{(n)}}\big)}{\max_{m,n\in[M],n\neq m} q_m^{\delta_{j,1}^{(m)}} q_n^{\delta_{j,1}^{(n)}} p_m^{\delta_{q,k_r}^{(m)}} p_n^{\delta_{q,k_r}^{(m)}}} \quad (= o_P(1))$$
$$\gtrsim_P \frac{\max_{m\in[M]} \lambda_{k_r+1}(\mathbf{X}_{Q,m})}{\max_{m\in[M]} \lambda_{k_r}(\mathbf{X}_{Q,m})}, \tag{C.125}$$

where the second line is by the rate assumption in the statement of the theorem. The above implies $P(\widehat{k}_r \geq k_r) \to 1$. When all factors are strong and all processes have short memory, then for $i \in \{k_r + 1, \ldots, p_m\}$,

$$\max_{m\in[M]} \lambda_i(\mathbf{X}_{Q,m}) = O_P(\max_{m,n\in[M],n\neq m} T^{-1} p_m p_n q_m q_n),$$

so we can easily see that

$$\frac{\max_{m\in[M]}\lambda_{i+1}(\mathbf{X}_{Q,m})+\max_{m,n\in[M],n\neq m}CT^{-1}p_mp_nq_mq_n}{\max_{m\in[M]}\lambda_{i}(\mathbf{X}_{Q,m})+\max_{m,n\in[M],n\neq m}CT^{-1}p_mp_nq_mq_n}\asymp_P 1$$

for $i\in\{k_r+1,\ldots,p_m\}$. This, together with (C.125), proves $P(\widehat{k}_r=k_r)\to 1$.

For $\widehat{k}_c^{(m)}$ (the proof for $\widehat{k}_r^{(m)}$ runs parallel and is omitted), consider the decomposition of $\mathbf{X}_{C,m}$ (C.45), where we can see from the rate in (C.50) that for $i\in[k_c^{(m)}]$,

$$\lambda_i(\mathbf{X}_{C,m})\asymp_P q_m^{\delta_{c,i}^{(m)}}p_m^{\delta_{r,1}^{(m)}},$$

while for $i\in\{k_c^{(m)}+1,\ldots,q_m\}$, the rates $\left\|\mathcal{J}_5\right\|_F^2$ and $\left\|\mathcal{J}_6\right\|_F^2$ from (C.52) and (C.54) respectively dominated, and hence we have

$$\lambda_i(\mathbf{X}_{C,m})=O_P(q_m^{1-\delta_{c,1}^{(m)}}+T^{-1/2}q_mp_m^{1/2}+q_m^{(1+\delta_{j,1}^{(m)})/2}p_m^{(1+\delta_{q,1}^{(m)})/2}\eta_{Q,m}^{1/2}).$$

With these two intermediate results, we can conclude that for $i\in[k_c^{(m)}-1]$,

$$\begin{aligned}\frac{\lambda_{i+1}(\mathbf{X}_{C,m})}{\lambda_i(\mathbf{X}_{C,m})}&\gtrsim_P\frac{q_m^{\delta_{c,i+1}^{(m)}}p_m^{\delta_{r,1}^{(m)}}}{q_m^{\delta_{c,1}^{(m)}}p_m^{\delta_{r,1}^{(m)}}}\succ_P\frac{q_m^{1-\delta_{c,1}^{(m)}}+T^{-1/2}q_mp_m^{1/2}+q_m^{(1+\delta_{j,1}^{(m)})/2}p_m^{(1+\delta_{q,1}^{(m)})/2}\eta_{Q,m}^{1/2}}{q_m^{\delta_{c,k_c^{(m)}}^{(m)}}p_m^{\delta_{r,1}^{(m)}}}\quad(=o_P(1))\\&\gtrsim_P\frac{\lambda_{k_c^{(m)}+1}(\mathbf{X}_{C,m})}{\lambda_{k_c^{(m)}}(\mathbf{X}_{C,m})},\end{aligned}\tag{C.126}$$

where the second line is by the rate assumption in the statement of the theorem. The above implies $P(\hat{k}_c^{(m)}\geq k_c^{(m)})\to 1$. When all factors are strong and all processes have short memory, $\eta_{Q,m}=T^{-1}$, and hence for $i\in\{k_c^{(m)}+1,\ldots,q_m\}$,

$$\lambda_i(\mathbf{X}_{C,m})=O_P(T^{-1/2}p_mq_m).$$

Hence, we can see that for $i\in\{k_c^{(m)}+1,\ldots,q_m\}$,

$$\frac{\lambda_{i+1}(\mathbf{X}_{C,m})+CT^{-1/2}p_mq_m}{\lambda_i(\mathbf{X}_{C,m})+CT^{-1/2}p_mq_m}\asymp_P 1.$$

This, together with (C.126), proves $P(\hat{k}_c^{(m)}=k_c^{(m)})\to 1$. □

D Auxiliary Results and Proofs

Proposition 1 *(Contraction properties of orthogonal projectors) Let* $\mathbf{A} \in \mathbb{R}^{a\times b}$ *be any matrix, and let* $\mathbf{P} \in \mathbb{R}^{a\times a}$ *and* $\mathbf{Q} \in \mathbb{R}^{b\times b}$ *be orthogonal projectors. Then,*

$$\|\mathbf{PAQ}\| \le \|\mathbf{A}\|, \qquad \|\mathbf{PAQ}\|_F \le \|\mathbf{A}\|_F. \tag{D.1}$$

In particular, for any positive integer n*, define* $\mathbf{M}_n := \mathbf{I}_n - n^{-1}\mathbf{1}_n\mathbf{1}_n'$*. Then,* $\mathbf{M}_n$ *is the orthogonal projector onto* $\mathbf{1}_n^{\perp}$*, and*

$$\mathbf{M}_n' = \mathbf{M}_n, \qquad \mathbf{M}_n^2 = \mathbf{M}_n, \qquad \mathbf{M}_n\mathbf{1}_n = \mathbf{0}, \qquad \|\mathbf{M}_n\| \le 1. \tag{D.2}$$

Moreover,

$$\|\mathbf{M}_n\|_{\max} \le 1, \qquad \|\mathbf{M}_n\|_1 = \|\mathbf{M}_n\|_\infty = 2(1-n^{-1}) \le 2. \tag{D.3}$$

Proof of Proposition 1. Since $\mathbf{P}$ and $\mathbf{Q}$ are orthogonal projectors, $\|\mathbf{P}\| \le 1$ and $\|\mathbf{Q}\| \le 1$. Hence,

$$\|\mathbf{PAQ}\| \le \|\mathbf{P}\|\|\mathbf{A}\|\|\mathbf{Q}\| \le \|\mathbf{A}\|,$$
$$\|\mathbf{PAQ}\|_F \le \|\mathbf{P}\|\|\mathbf{A}\|_F\|\mathbf{Q}\| \le \|\mathbf{A}\|_F.$$

By direct calculation, $\mathbf{M}_n' = \mathbf{M}_n$, $\mathbf{M}_n^2 = \mathbf{M}_n$, and $\mathbf{M}_n\mathbf{1}_n = \mathbf{0}$. Thus, $\mathbf{M}_n$ is the orthogonal projector onto $\mathbf{1}_n^{\perp}$. Since the eigenvalues of an orthogonal projector belong to $\{0,1\}$, we have $\|\mathbf{M}_n\| \le 1$. Finally, its diagonal entries are $1-1/n$ and its off-diagonal entries are $-1/n$. Therefore,

$$\|\mathbf{M}_n\|_{\max} \le 1, \qquad \|\mathbf{M}_n\|_1 = \|\mathbf{M}_n\|_\infty = (1-n^{-1}) + (n-1)n^{-1} = 2(1-n^{-1}) \le 2.$$

This completes the proof. □

Lemma 1 *(Covariance bounds for* $\mathbf{E}_t^{(m)}$*) Let Assumptions (E1) and (E2) hold. Then there exist a non-negative sequence* $\{\rho(h) : h \in \mathbb{Z}\}$ *and a constant* $C < \infty$*, such that* $\sum_{h\in\mathbb{Z}} \rho(h) < \infty$*,* $\sup_{h\in\mathbb{Z}} \rho(h) \le 1$*, and, uniformly over* $h \in \mathbb{Z}$ *and all* t *such that* $t, t+h \in [T]$*,*

$$\max_{m\in[M]} \sup_{i\in[p_m]} \sup_{j\in[q_m]} \sum_{k=1}^{p_m}\sum_{l=1}^{q_m} \left|\mathbb{E}\big(E_{t+h,ij}^{(m)} E_{t,kl}^{(m)}\big)\right| \le C\rho(h). \tag{D.4}$$

In particular, uniformly over all t such that $t, t+h \in [T]$,

$$\max_{m\in[M]} \sup_{h\in\mathbb{Z}} \sup_{i\in[p_m]} \sup_{j\in[q_m]} \sum_{k=1}^{p_m}\sum_{l=1}^{q_m} \left|\mathbb{E}\big(E^{(m)}_{t+h,ij} E^{(m)}_{t,kl}\big)\right| \le C. \tag{D.5}$$

Moreover, uniformly over $h \in \mathbb{Z}$, all t such that $t, t+h \in [T]$, the covariance of the error products satisfies

$$\max_{m,n\in[M]} \sup_{i\in[p_m]} \sup_{j\in[q_m]} \sup_{k\in[p_n]} \sup_{l\in[q_n]} \sum_{a=1}^{q_m}\sum_{b=1}^{q_n} \left|\mathrm{Cov}\left(E^{(m)}_{t+h,ij} E^{(n)}_{t+h,kl}, E^{(m)}_{t,ia} E^{(n)}_{t,kb}\right)\right| \le C\rho(h). \tag{D.6}$$

Proof of Lemma 1. For $h \in \mathbb{Z}$, let $\gamma_e(h) := \sum_{w\ge 0} |a_{e,w} a_{e,w+|h|}|$, and for each $m \in [M]$, define analogously $\gamma_e^{(m)}(h) := \sum_{w\ge 0} |a^{(m)}_{e,w} a^{(m)}_{e,w+|h|}|$ and $\gamma_\epsilon(h)^{(m)} := \sum_{w\ge 0} |a^{(m)}_{\epsilon,w} a^{(m)}_{\epsilon,w+|h|}|$. By (E2),

$$\begin{aligned}\sum_{h\in\mathbb{Z}} \gamma_e(h) &= \sum_{w\ge 0} |a_{e,w}|^2 + 2\sum_{h=1}^{\infty}\sum_{w\ge 0} |a_{e,w}||a_{e,w+h}| \\ &= \sum_{w\ge 0} |a_{e,w}|^2 + 2\sum_{0\le w<r<\infty} |a_{e,w}||a_{e,r}| = \Big(\sum_{w\ge 0} |a_{e,w}|\Big)^2 \le c^2 < \infty,\end{aligned}$$

and analogously, for every $m \in [M]$, $\sum_{h\in\mathbb{Z}} \gamma_e^{(m)}(h) \le c^2$ and $\sum_{h\in\mathbb{Z}} \gamma_\epsilon^{(m)}(h) \le c^2$, where c is the constant in (E2). Moreover, by the Cauchy–Schwarz inequality and (E2),

$$\gamma_e(h) \le \big(\sum_{w\ge 0} a^2_{e,w}\big)^{1/2} \big(\sum_{w\ge 0} a^2_{e,w+|h|}\big)^{1/2} \le 1, \tag{D.7}$$

so that $\sup_h \gamma_e(h) \le 1$. The same argument gives $\sup_h \gamma_e^{(m)}(h) \le 1$ and $\sup_h \gamma_\epsilon^{(m)}(h) \le 1$. Since M is fixed, define

$$\rho(h) := \frac{1}{3}\left\{\gamma_e(h) + \max_{m\in[M]} \gamma_e^{(m)}(h) + \max_{m\in[M]} \gamma_\epsilon^{(m)}(h)\right\}.$$

Then $\rho(h) \ge 0$, $\sup_{h\in\mathbb{Z}} \rho(h) \le 1$, and

$$\sum_{h\in\mathbb{Z}} \rho(h) \le \frac{1}{3}\left\{\sum_{h\in\mathbb{Z}} \gamma_e(h) + \sum_{m=1}^{M}\sum_{h\in\mathbb{Z}} \gamma_e^{(m)}(h) + \sum_{m=1}^{M}\sum_{h\in\mathbb{Z}} \gamma_\epsilon^{(m)}(h)\right\} \le \frac{1+2M}{3} c^2 < \infty.$$

We next prove (D.4). Fix $m \in [M]$. By (E1), write

$$E^{(m)}_{t,ij} = (\mathbf{Q}_{e,m}\mathbf{G}_{e,t}\mathbf{J}'_{e,m})_{ij} + (\mathbf{R}_{e,m}\mathbf{F}^{(m)}_{e,t}\mathbf{C}'_{e,m})_{ij} + (\mathbf{\Sigma}_{\epsilon,m} \circ \boldsymbol{\epsilon}^{(m)}_t)_{ij} =: U^{(m)}_{t,ij} + V^{(m)}_{t,ij} + W^{(m)}_{t,ij}. \tag{D.8}$$

Since the three component processes are mutually independent and have mean zero under (E1) and (E2),

we have

$$\mathbb{E}(E^{(m)}_{t+h,ij}E^{(m)}_{t,kl}) = \mathbb{E}(U^{(m)}_{t+h,ij}U^{(m)}_{t,kl}) + \mathbb{E}(V^{(m)}_{t+h,ij}V^{(m)}_{t,kl}) + \mathbb{E}(W^{(m)}_{t+h,ij}W^{(m)}_{t,kl}). \tag{D.9}$$

We bound the three terms on the r.h.s. separately. First,

$$U^{(m)}_{t,ij} = \sum_{a=1}^{k_{e,r}}\sum_{b=1}^{k_{e,c}} (\mathbf{Q}_{e,m})_{ia}(\mathbf{G}_{e,t})_{ab}(\mathbf{J}_{e,m})_{jb}.$$

Hence

$$\mathbb{E}(U^{(m)}_{t+h,ij}U^{(m)}_{t,kl}) = \sum_{a=1}^{k_{e,r}}\sum_{b=1}^{k_{e,c}}\sum_{c=1}^{k_{e,r}}\sum_{d=1}^{k_{e,c}} (\mathbf{Q}_{e,m})_{ia}(\mathbf{Q}_{e,m})_{kc}(\mathbf{J}_{e,m})_{jb}(\mathbf{J}_{e,m})_{ld}\mathbb{E}\{(\mathbf{G}_{e,t+h})_{ab}(\mathbf{G}_{e,t})_{cd}\}.$$

By (E2), we have $\mathbb{E}\{(\mathbf{X}_{e,r})_{ab}(\mathbf{X}_{e,s})_{cd}\} = \mathbb{1}\{r=s\}\mathbb{1}\{a=c, b=d\}$, so that

$$\begin{aligned}
\mathbb{E}\{(\mathbf{G}_{e,t+h})_{ab}(\mathbf{G}_{e,t})_{cd}\} &= \sum_{u\geq 0}\sum_{v\geq 0} a_{e,u}a_{e,v}\mathbb{E}\{(\mathbf{X}_{e,t+h-u})_{ab}(\mathbf{X}_{e,t-v})_{cd}\} \\
&= \mathbb{1}\{a=c, b=d\}\sum_{u\geq 0}\sum_{v\geq 0} a_{e,u}a_{e,v}\mathbb{1}\{t+h-u = t-v\} \\
&= \mathbb{1}\{a=c, b=d\}\sum_{w\geq 0} a_{e,w+|h|}a_{e,w},
\end{aligned}$$

where the last equality follows by setting $w = v$ when $h \geq 0$ and $w = u$ when $h < 0$. Hence,

$$\big|\mathbb{E}\{(\mathbf{G}_{e,t+h})_{ab}(\mathbf{G}_{e,t})_{cd}\}\big| \leq \gamma_e(h)\mathbb{1}\{a=c, b=d\}.$$

Therefore,

$$\big|\mathbb{E}(U^{(m)}_{t+h,ij}U^{(m)}_{t,kl})\big| \leq \gamma_e(h)\sum_{a=1}^{k_{e,r}}\sum_{b=1}^{k_{e,c}} |(\mathbf{Q}_{e,m})_{ia}||(\mathbf{Q}_{e,m})_{ka}||(\mathbf{J}_{e,m})_{jb}||(\mathbf{J}_{e,m})_{lb}|.$$

Taking absolute values and summing over k and l, we obtain

$$\begin{aligned}
&\sum_{k=1}^{p_m}\sum_{l=1}^{q_m}\big|\mathbb{E}(U^{(m)}_{t+h,ij}U^{(m)}_{t,kl})\big| \\
&\leq \gamma_e(h)\sum_{a=1}^{k_{e,r}}\sum_{b=1}^{k_{e,c}} |(\mathbf{Q}_{e,m})_{ia}||(\mathbf{J}_{e,m})_{jb}|\Big(\sum_{k=1}^{p_m}|(\mathbf{Q}_{e,m})_{ka}|\Big)\Big(\sum_{l=1}^{q_m}|(\mathbf{J}_{e,m})_{lb}|\Big) \\
&\leq \gamma_e(h)\Big(\sum_{a=1}^{k_{e,r}}|(\mathbf{Q}_{e,m})_{ia}|\Big)\Big(\max_a\sum_{k=1}^{p_m}|(\mathbf{Q}_{e,m})_{ka}|\Big)\Big(\sum_{b=1}^{k_{e,c}}|(\mathbf{J}_{e,m})_{jb}|\Big)\Big(\max_b\sum_{l=1}^{q_m}|(\mathbf{J}_{e,m})_{lb}|\Big) \\
&\leq \gamma_e(h)\big(k_{e,r}\|\mathbf{Q}_{e,m}\|_1\big)\|\mathbf{Q}_{e,m}\|_1\big(k_{e,c}\|\mathbf{J}_{e,m}\|_1\big)\|\mathbf{J}_{e,m}\|_1 = O(\gamma_e(h)),
\end{aligned}$$

where we use $\sum_k |(\mathbf{Q}_{e,m})_{ka}| \le \|\mathbf{Q}_{e,m}\|_1$ and $\sum_l |(\mathbf{J}_{e,m})_{lb}| \le \|\mathbf{J}_{e,m}\|_1$ in the first inequality, $\sum_a |(\mathbf{Q}_{e,m})_{ia}| \le k_{e,r}\|\mathbf{Q}_{e,m}\|_1$ and $\sum_b |(\mathbf{J}_{e,m})_{jb}| \le k_{e,c}\|\mathbf{J}_{e,m}\|_1$ in the second inequality, and Assumption (E1) in the last equality. By the same argument,

$$\sum_{k=1}^{p_m}\sum_{l=1}^{q_m} |\mathbb{E}(V^{(m)}_{t+h,ij}V^{(m)}_{t,kl})| = O(\gamma_e^{(m)}(h)).$$

Finally, (E2) gives

$$|\mathbb{E}(\epsilon^{(m)}_{t+h,ij}\epsilon^{(m)}_{t,kl})| \le \gamma_\epsilon^{(m)}(h)\mathbb{1}\{(i,j)=(k,l)\}.$$

Therefore,

$$\begin{aligned}\sum_{k=1}^{p_m}\sum_{l=1}^{q_m} |\mathbb{E}(W^{(m)}_{t+h,ij}W^{(m)}_{t,kl})| &\le \sum_{k=1}^{p_m}\sum_{l=1}^{q_m} |(\mathbf{\Sigma}_{\epsilon,m})_{ij}(\mathbf{\Sigma}_{\epsilon,m})_{kl}|\gamma_\epsilon^{(m)}(h)\mathbb{1}\{(i,j)=(k,l)\} \\ &= \gamma_\epsilon^{(m)}(h)(\mathbf{\Sigma}_{\epsilon,m})^2_{ij} = O(\gamma_\epsilon^{(m)}(h)),\end{aligned}$$

by the uniform boundedness of $\mathbf{\Sigma}_{\epsilon,m}$ in (E1). Combining the above bounds, there exists an absolute constant C such that

$$\begin{aligned}\sum_{k=1}^{p_m}\sum_{l=1}^{q_m} |\mathbb{E}(U^{(m)}_{t+h,ij}U^{(m)}_{t,kl})| &\le \frac{C}{3}\gamma_e(h), \\ \sum_{k=1}^{p_m}\sum_{l=1}^{q_m} |\mathbb{E}(V^{(m)}_{t+h,ij}V^{(m)}_{t,kl})| &\le \frac{C}{3}\gamma_e^{(m)}(h), \\ \sum_{k=1}^{p_m}\sum_{l=1}^{q_m} |\mathbb{E}(W^{(m)}_{t+h,ij}W^{(m)}_{t,kl})| &\le \frac{C}{3}\gamma_\epsilon^{(m)}(h),\end{aligned}$$

uniformly over $m \in [M]$, $h \in \mathbb{Z}$, $i \in [p_m]$, and $j \in [q_m]$. By (D.9) and the triangle inequality,

$$\begin{aligned}\sum_{k=1}^{p_m}\sum_{l=1}^{q_m} |\mathbb{E}(E^{(m)}_{t+h,ij}E^{(m)}_{t,kl})| &\le \frac{C}{3}\left(\gamma_e(h) + \gamma_e^{(m)}(h) + \gamma_\epsilon^{(m)}(h)\right) \\ &\le \frac{C}{3}\left(\gamma_e(h) + \max_{m\in[M]}\gamma_e^{(m)}(h) + \max_{m\in[M]}\gamma_\epsilon^{(m)}(h)\right) = C\rho(h),\end{aligned}$$

which concludes (D.4). Since $\sup_{h\in\mathbb{Z}}\rho(h) \le 1$, (D.5) follows immediately.

We next show (D.6). For $h \in \mathbb{Z}$, define $\kappa_e(h) := \sum_{w\ge 0} a^2_{e,w}a^2_{e,w+|h|}$, and for each $m \in [M]$, define $\kappa_e^{(m)}(h) := \sum_{w\ge0}\left(a^{(m)}_{e,w}\right)^2\left(a^{(m)}_{e,w+|h|}\right)^2$ and $\kappa_\epsilon^{(m)}(h) := \sum_{w\ge0}\left(a^{(m)}_{\epsilon,w}\right)^2\left(a^{(m)}_{\epsilon,w+|h|}\right)^2$. Since

$$\kappa_e(h) \le \gamma_e(h)^2 \le \gamma_e(h) \le 3\rho(h),$$

and analogously $\kappa_e^{(m)}(h) \le 3\rho(h)$ and $\kappa_\epsilon^{(m)}(h) \le 3\rho(h)$ for every $m \in [M]$, we have

$$\kappa_e(h) + \max_{m\in[M]} \kappa_e^{(m)}(h) + \max_{m\in[M]} \kappa_\epsilon^{(m)}(h) \le 9\rho(h).$$

Moreover, for any two sequences γ_1 and γ_2 chosen from $\{\gamma_e, \gamma_e^{(m)}, \gamma_\epsilon^{(m)} : m \in [M]\}$, we have

$$\gamma_1(h)\gamma_2(h) \le 9\rho(h)^2 \le 9\rho(h),$$

where the second inequality used $\rho(h) \le 1$.

Recall the decomposition in (D.8). Expanding each of the four error entries in the covariance yields finitely many terms. We first consider

$$S_U(h) := \sum_{a=1}^{q_m}\sum_{b=1}^{q_n} \left|\mathrm{Cov}\left(U_{t+h,ij}^{(m)} U_{t+h,kl}^{(n)}, U_{t,ia}^{(m)} U_{t,kb}^{(n)}\right)\right|.$$

By the definition of $U_{t,ij}^{(m)}$,

$$U_{t,ij}^{(m)} = \sum_{c=1}^{k_{e,r}}\sum_{d=1}^{k_{e,c}} (\mathbf{Q}_{e,m})_{ic}(\mathbf{J}_{e,m})_{jd}(\mathbf{G}_{e,t})_{cd}.$$

It follows that

$$\begin{aligned}
&\mathrm{Cov}\left(U_{t+h,ij}^{(m)} U_{t+h,kl}^{(n)}, U_{t,ia}^{(m)} U_{t,kb}^{(n)}\right) \\
&= \sum_{c_1=1}^{k_{e,r}}\sum_{d_1=1}^{k_{e,c}}\sum_{c_2=1}^{k_{e,r}}\sum_{d_2=1}^{k_{e,c}}\sum_{c_3=1}^{k_{e,r}}\sum_{d_3=1}^{k_{e,c}}\sum_{c_4=1}^{k_{e,r}}\sum_{d_4=1}^{k_{e,c}} (\mathbf{Q}_{e,m})_{ic_1}(\mathbf{J}_{e,m})_{jd_1}(\mathbf{Q}_{e,n})_{kc_2}(\mathbf{J}_{e,n})_{ld_2} \\
&\quad \cdot (\mathbf{Q}_{e,m})_{ic_3}(\mathbf{J}_{e,m})_{ad_3}(\mathbf{Q}_{e,n})_{kc_4}(\mathbf{J}_{e,n})_{bd_4} \\
&\quad \cdot \mathrm{Cov}\left((\mathbf{G}_{e,t+h})_{c_1d_1}(\mathbf{G}_{e,t+h})_{c_2d_2}, (\mathbf{G}_{e,t})_{c_3d_3}(\mathbf{G}_{e,t})_{c_4d_4}\right).
\end{aligned}$$

By (E2), we have

$$\begin{aligned}
&\left|\mathrm{Cov}\left((\mathbf{G}_{e,t+h})_{c_1d_1}(\mathbf{G}_{e,t+h})_{c_2d_2}, (\mathbf{G}_{e,t})_{c_3d_3}(\mathbf{G}_{e,t})_{c_4d_4}\right)\right| \\
&\le C\gamma_e(h)^2 \mathbb{1}\{(c_1,d_1) = (c_3,d_3),\ (c_2,d_2) = (c_4,d_4)\} \\
&\quad + C\gamma_e(h)^2 \mathbb{1}\{(c_1,d_1) = (c_4,d_4),\ (c_2,d_2) = (c_3,d_3)\} \\
&\quad + C\kappa_e(h) \mathbb{1}\{(c_1,d_1) = (c_2,d_2) = (c_3,d_3) = (c_4,d_4)\}.
\end{aligned}$$

Substituting the above into the preceding expansion and using the triangle inequality, we obtain

$$
\begin{aligned}
S_U(h) \le{}& C\gamma_e(h)^2 \sum_{c_1=1}^{k_{e,r}} \sum_{d_1=1}^{k_{e,c}} \sum_{c_2=1}^{k_{e,r}} \sum_{d_2=1}^{k_{e,c}} |(\mathbf{Q}_{e,m})_{ic_1}(\mathbf{J}_{e,m})_{jd_1}(\mathbf{Q}_{e,n})_{kc_2}(\mathbf{J}_{e,n})_{ld_2}| \\
&\cdot \sum_{a=1}^{q_m} |(\mathbf{Q}_{e,m})_{ic_1}(\mathbf{J}_{e,m})_{ad_1}| \sum_{b=1}^{q_n} |(\mathbf{Q}_{e,n})_{kc_2}(\mathbf{J}_{e,n})_{bd_2}| \\
&+ C\gamma_e(h)^2 \sum_{c_1=1}^{k_{e,r}} \sum_{d_1=1}^{k_{e,c}} \sum_{c_2=1}^{k_{e,r}} \sum_{d_2=1}^{k_{e,c}} |(\mathbf{Q}_{e,m})_{ic_1}(\mathbf{J}_{e,m})_{jd_1}(\mathbf{Q}_{e,n})_{kc_2}(\mathbf{J}_{e,n})_{ld_2}| \\
&\cdot \sum_{a=1}^{q_m} |(\mathbf{Q}_{e,m})_{ic_2}(\mathbf{J}_{e,m})_{ad_2}| \sum_{b=1}^{q_n} |(\mathbf{Q}_{e,n})_{kc_1}(\mathbf{J}_{e,n})_{bd_1}| \\
&+ C\kappa_e(h) \sum_{c=1}^{k_{e,r}} \sum_{d=1}^{k_{e,c}} |(\mathbf{Q}_{e,m})_{ic}(\mathbf{J}_{e,m})_{jd}(\mathbf{Q}_{e,n})_{kc}(\mathbf{J}_{e,n})_{ld}| \\
&\cdot \sum_{a=1}^{q_m} |(\mathbf{Q}_{e,m})_{ic}(\mathbf{J}_{e,m})_{ad}| \sum_{b=1}^{q_n} |(\mathbf{Q}_{e,n})_{kc}(\mathbf{J}_{e,n})_{bd}|.
\end{aligned}
$$

Since $k_{e,r}$ and $k_{e,c}$ are fixed, and since $\|\mathbf{Q}_{e,m}\|_1 = \|\mathbf{J}_{e,m}\|_1 = O(1)$ and $\|\mathbf{Q}_{e,n}\|_1 = \|\mathbf{J}_{e,n}\|_1 = O(1)$ for all $m, n \in [M]$, we have, uniformly over $i \in [p_m]$, $j \in [q_m]$, $k \in [p_n]$, and $l \in [q_n]$,

$$
S_U(h) \le C\gamma_e(h)^2 + C\kappa_e(h) \le C\rho(h). \tag{D.10}
$$

Next, define

$$
S_V(h) := \sum_{a=1}^{q_m} \sum_{b=1}^{q_n} \left| \mathrm{Cov}\left(V_{t+h,ij}^{(m)} V_{t+h,kl}^{(n)}, V_{t,ia}^{(m)} V_{t,kb}^{(n)} \right) \right|.
$$

When $m = n$, the above arguments apply to the local error process. When $m \neq n$, the cross-group independence of the local error innovation processes in (E2) yields, uniformly over $i \in [p_m]$, $j \in [q_m]$, $k \in [p_n]$, and $l \in [q_n]$,

$$
S_V(h) \le C\gamma_e^{(m)}(h)\gamma_e^{(n)}(h) + C\mathbb{1}\{m = n\}\kappa_e^{(m)}(h) \le C\rho(h). \tag{D.11}
$$

Similarly, define

$$
S_W(h) := \sum_{a=1}^{q_m} \sum_{b=1}^{q_n} \left| \mathrm{Cov}\left(W_{t+h,ij}^{(m)} W_{t+h,kl}^{(n)}, W_{t,ia}^{(m)} W_{t,kb}^{(n)} \right) \right|.
$$

Using the independence and uniformly bounded fourth moments of the idiosyncratic innovations in (E2),

together with the uniform boundedness of the entries of $\mathbf{\Sigma}_{\epsilon,m}$ in (E1), we obtain

$$S_W(h) \le C\gamma_\epsilon^{(m)}(h)\gamma_\epsilon^{(n)}(h) + C\mathbb{1}\{m=n\}\kappa_\epsilon^{(m)}(h) \le C\rho(h). \tag{D.12}$$

It remains to bound the terms involving at least two components among $U_{t,ij}^{(m)}$, $V_{t,ij}^{(m)}$, and $W_{t,ij}^{(m)}$. These remaining terms fall into the following two cases.

(i) At least one driving innovation process appears only once in the covariance. Such terms have zero covariance. For example,

$$\begin{aligned}
&\mathrm{Cov}\left(U_{t+h,ij}^{(m)}U_{t+h,kl}^{(n)}, U_{t,ia}^{(m)}V_{t,kb}^{(n)}\right)\\
&= \mathbb{E}\left(U_{t+h,ij}^{(m)}U_{t+h,kl}^{(n)}U_{t,ia}^{(m)}V_{t,kb}^{(n)}\right) - \mathbb{E}\left(U_{t+h,ij}^{(m)}U_{t+h,kl}^{(n)}\right)\mathbb{E}\left(U_{t,ia}^{(m)}V_{t,kb}^{(n)}\right)\\
&= \mathbb{E}\left(U_{t+h,ij}^{(m)}U_{t+h,kl}^{(n)}U_{t,ia}^{(m)}\right)\mathbb{E}\left(V_{t,kb}^{(n)}\right) - \mathbb{E}\left(U_{t+h,ij}^{(m)}U_{t+h,kl}^{(n)}\right)\mathbb{E}\left(U_{t,ia}^{(m)}\right)\mathbb{E}\left(V_{t,kb}^{(n)}\right) = 0.
\end{aligned}$$

The same argument applies whenever one of the driving innovation processes occurs only once. For example,

$$\mathrm{Cov}\left(U_{t+h,ij}^{(m)}V_{t+h,kl}^{(n)}, U_{t,ia}^{(m)}W_{t,kb}^{(n)}\right) = 0.$$

(ii) Two independent driving innovation processes each appear twice. There are two possible arrangements.

(a) *The two occurrences of each process lie on the same side of the covariance.* In this case, the two products are independent, so their covariance is zero. For example,

$$\mathrm{Cov}\left(U_{t+h,ij}^{(m)}U_{t+h,kl}^{(n)}, V_{t,ia}^{(m)}V_{t,kb}^{(n)}\right) = 0.$$

(b) *Each process contributes once on each side of the covariance.* We first consider the arrangements in which each innovation process contributes one factor at time $t+h$ and one factor at time t. As a representative case, write

$$\begin{aligned}
S_{UV}(h) &:= \sum_{a=1}^{q_m}\sum_{b=1}^{q_n}\left|\mathrm{Cov}\left(U_{t+h,ij}^{(m)}V_{t+h,kl}^{(n)}, U_{t,ia}^{(m)}V_{t,kb}^{(n)}\right)\right|\\
&= \sum_{a=1}^{q_m}\sum_{b=1}^{q_n}\left|\mathbb{E}\left(U_{t+h,ij}^{(m)}U_{t,ia}^{(m)}\right)\mathbb{E}\left(V_{t+h,kl}^{(n)}V_{t,kb}^{(n)}\right)\right|\\
&\le \left\{\sum_{a=1}^{q_m}\left|\mathbb{E}\left(U_{t+h,ij}^{(m)}U_{t,ia}^{(m)}\right)\right|\right\}\left\{\sum_{b=1}^{q_n}\left|\mathbb{E}\left(V_{t+h,kl}^{(n)}V_{t,kb}^{(n)}\right)\right|\right\}\\
&\le C\gamma_e(h)\gamma_e^{(n)}(h) \le C\rho(h).
\end{aligned}$$

We finally consider the arrangement in which the two component types exchange their group assignments across the two sides of the covariance. For example,

$$\mathrm{Cov}\left(U^{(m)}_{t+h,ij}V^{(n)}_{t+h,kl}, V^{(m)}_{t,ia}U^{(n)}_{t,kb}\right).$$

If $m \neq n$, then the covariance is zero by case (i). If $m = n$, uniformly over $m, n \in [M]$, independence of the global and local innovation processes gives

$$\begin{aligned}
&\sum_{a=1}^{q_m}\sum_{b=1}^{q_n}\left|\mathrm{Cov}\left(U^{(m)}_{t+h,ij}V^{(n)}_{t+h,kl}, V^{(m)}_{t,ia}U^{(n)}_{t,kb}\right)\right| \\
&= \sum_{a=1}^{q_m}\sum_{b=1}^{q_n}\left|\mathbb{E}\left(U^{(m)}_{t+h,ij}U^{(m)}_{t,kb}\right)\mathbb{E}\left(V^{(m)}_{t+h,kl}V^{(m)}_{t,ia}\right)\right| \\
&\leq C\gamma_e(h)\gamma_e^{(m)}(h) \leq C\rho(h).
\end{aligned}$$

Cases (i) and (ii) cover all terms involving at least two of $U^{(m)}_{t,ij}$, $V^{(m)}_{t,ij}$, and $W^{(m)}_{t,ij}$, where these terms are not necessarily corresponding to the same group m. Combining these bounds with (D.10), (D.11) and (D.12), we obtain, uniformly over $h \in \mathbb{Z}$ and all t such that $t, t+h \in [T]$:

$$\max_{m,n\in[M]}\sup_{i\in[p_m]}\sup_{j\in[q_m]}\sup_{k\in[p_n]}\sup_{l\in[q_n]}\sum_{a=1}^{q_m}\sum_{b=1}^{q_n}\left|\mathrm{Cov}\left(E^{(m)}_{t+h,ij}E^{(n)}_{t+h,kl}, E^{(m)}_{t,ia}E^{(n)}_{t,kb}\right)\right| \leq C\rho(h).$$

This proves (D.6) and completes the proof. □

Lemma 2 *(Basic properties of $\mathcal{G}_T$ processes) Let $\{\mathbf{u}_t\}_{t\in[T]} \in \mathcal{G}_T(\{d_{u,1},\ldots,d_{u,m_u}\}|\{\mathbf{x}_{u,i}, i \in \mathbb{Z}\})$ be a p_u-dimensional process, where p_u and m_u are bounded. Define $d_u := \max_{1\leq j\leq m_u} d_{u,j}$. Then:*

1. *There exists a constant $C < \infty$ such that the second moments are uniformly bounded as*

$$\sup_{t\in[T]}\mathbb{E}\|\mathbf{u}_t\|^2 \leq C. \tag{D.13}$$

2. *For any $h \in \{0,\ldots,T-1\}$, the centered sample autocovariance satisfies*

$$\left\|T^{-1}\sum_{t=1}^{T-h}\left(\mathbf{u}_t\mathbf{u}'_{t+h} - \mathbb{E}(\mathbf{u}_t\mathbf{u}'_{t+h})\right)\right\|_F^2 = O_P(R(T,d_u)). \tag{D.14}$$

3. *Let $\{\mathbf{v}_t\}_{t\in[T]} \in \mathcal{G}_T(\{d_{v,1},\ldots,d_{v,m_v}\}|\{\mathbf{x}_{v,i}, i \in \mathbb{Z}\})$ be another p_v-dimensional process, independent of $\{\mathbf{u}_t\}_{t\in[T]}$, where p_v and m_v are bounded. Define $d_v := \max_{1\leq j\leq m_v} d_{v,j}$. For $d_{uv} := (d_u + d_v)/2$, the*

sample cross-moment satisfies

$$\left\| T^{-1} \sum_{t=1}^{T} \mathbf{u}_t \mathbf{v}_t' \right\|_F^2 = O_P(R(T, d_{uv})). \tag{D.15}$$

Proof of Lemma 2. We first show (D.13). Fix $t \in [T]$, and suppose that t belongs to segment $a \in [m_u]$ of $\{\mathbf{u}_t\}$. By the definition of $\mathcal{G}_T(\cdot)$,

$$\mathbf{u}_t = \sum_{w \geq 0} \psi_{u,a}(w) \mathbf{x}_{u,t-w},$$

where $\psi_{u,a}(0) = 1$, and for $w \geq 1$,

$$\psi_{u,a}(w) = w^{d_{u,a}-1} l_{u,a}(w), \qquad d_{u,a} < 1/2.$$

Since $l_{u,a}(w)$ converges to a finite constant and m_u is bounded,

$$|\psi_{u,a}(w)| \lesssim w^{d_{u,a}-1}, \qquad w \geq 1.$$

Since the entries of the innovation sequence $\{\mathbf{x}_{u,t}\}$ are independent across coordinates and over time and have zero means and uniformly bounded second moments,

$$\begin{aligned} \mathbb{E}\|\mathbf{u}_t\|^2 &= \sum_{w \geq 0} \psi_{u,a}(w)^2 \mathbb{E}\left\|\mathbf{x}_{u,t-w}\right\|^2 \\ &\lesssim p_u \sum_{w \geq 0} \psi_{u,a}(w)^2 \lesssim p_u \left(1 + \sum_{w \geq 1} w^{2d_{u,a}-2} \right) \lesssim 1, \end{aligned}$$

because $d_{u,a} < 1/2$, while p_u and m_u are bounded. Taking the supremum over $t \in [T]$ gives (D.13).

We next show (D.14). Fix $h \in \{0, \ldots, T-1\}$, and define

$$\widetilde{\mathbf{\Gamma}}_u(h) := T^{-1} \sum_{t=1}^{T-h} \left(\mathbf{u}_t \mathbf{u}_{t+h}' - \mathbb{E}(\mathbf{u}_t \mathbf{u}_{t+h}') \right).$$

Since p_u is bounded, it is enough to bound the variance of each entry. The proof proceeds in three steps. We first control the covariance between observations associated with arbitrary segment pairs. We then bound the covariance between the corresponding products. Finally, we group the pairs of time indices according to their lag.

First, fix an entry (i,j). For $t,s \in [T-h]$, let

$$k = J_u(t), \qquad r = J_u(t+h), \qquad k' = J_u(s), \qquad r' = J_u(s+h),$$

where $J_u(t) \in [m_u]$ denotes the segment containing time t, and write $\tau = t - s$. Then

$$\mathrm{Var}\left(\widetilde{\Gamma}_{u,ij}(h)\right) = T^{-2} \sum_{t=1}^{T-h} \sum_{s=1}^{T-h} \mathrm{Cov}\left(u_{t,i}u_{t+h,j}, u_{s,i}u_{s+h,j}\right). \tag{D.16}$$

Because all coordinates use the same filter within each segment and the innovation coordinates are identically distributed, the covariance between two observations from the same coordinate index depends only on their segment labels and time lag. For $k,k' \in [m_u]$ and $\tau \geq 0$, define

$$\gamma_{k,k'}(\tau) := \sigma_u^2 \sum_{w \geq 0} \psi_{u,k}(w+\tau)\psi_{u,k'}(w), \tag{D.17}$$

where $\sigma_u^2 := \mathbb{E}(x_{u,0,1}^2) < \infty$, and, for $\tau < 0$, set $\gamma_{k,k'}(\tau) := \gamma_{k',k}(-\tau)$. By the temporal independence of the innovations, whenever $J_u(t) = k$, $J_u(s) = k'$, and $t - s = \tau$, it holds that $\mathrm{Cov}(u_{t,i}, u_{s,i}) = \gamma_{k,k'}(\tau)$ for every $i \in [p_u]$. For each segment $a \in [m_u]$, since each $l_{u,a}(w)$ converges to a finite constant, we have

$$|\psi_{u,a}(w)| \lesssim w^{d_{u,a}-1}, \qquad w \geq 1. \tag{D.18}$$

Moreover, by the Cauchy–Schwarz inequality and the square summability of the filters,

$$\max_{k,k' \in [m_u]} |\gamma_{k,k'}(0)| \lesssim \max_{k,k' \in [m_u]} \left\{ \sum_{w \geq 0} \psi_{u,k}(w)^2 \right\}^{1/2} \left\{ \sum_{w \geq 0} \psi_{u,k'}(w)^2 \right\}^{1/2} \lesssim 1.$$

Suppose first that $d_u > 0$. For $\tau \geq 1$, since $d_{u,a} \leq d_u$ for every $a \in [m_u]$,

$$\begin{aligned} |\gamma_{k,k'}(\tau)| &\lesssim \tau^{d_u-1} + \sum_{w \geq 1} (w+\tau)^{d_u-1} w^{d_u-1} \\ &\lesssim \tau^{d_u-1} + \tau^{d_u-1} \sum_{w=1}^{\tau} w^{d_u-1} + \sum_{w > \tau} w^{2d_u-2}. \end{aligned}$$

Since $0 < d_u < 1/2$, integral comparison gives

$$\sum_{w=1}^{\tau} w^{d_u-1} \leq 1 + \int_1^{\tau} x^{d_u-1} \mathrm{d}x = 1 + \frac{\tau^{d_u} - 1}{d_u} \lesssim \tau^{d_u},$$

and

$$\sum_{w>\tau} w^{2d_u-2} \le \int_{\tau}^{\infty} x^{2d_u-2}\mathrm{d}x \lesssim \tau^{2d_u-1}.$$

Combining these bounds yields $|\gamma_{k,k'}(\tau)| \lesssim \tau^{2d_u-1}$ for $\tau \ge 1$. Hence,

$$\max_{k,k'\in[m_u]} |\gamma_{k,k'}(\tau)|^2 \lesssim (1+|\tau|)^{4d_u-2}, \qquad \tau \in \mathbb{Z}. \tag{D.19}$$

Therefore, by the standard partial-sum bounds for p-series,

$$\sum_{\tau=-2(T-1)}^{2(T-1)} \max_{k,k'\in[m_u]} |\gamma_{k,k'}(\tau)|^2 \lesssim 1 + \sum_{\tau=1}^{2(T-1)} \tau^{4d_u-2} \lesssim \begin{cases} 1, & 0 < d_u < 1/4, \\ \log T, & d_u = 1/4, \\ T^{4d_u-1}, & 1/4 < d_u < 1/2. \end{cases} \tag{D.20}$$

If $d_u = 0$, then (D.18), together with $\psi_{u,a}(0) = 1$, gives

$$|\psi_{u,a}(w)| \lesssim (1+w)^{-1}, \qquad a \in [m_u], \quad w \ge 0.$$

Hence, for $\tau \ge 1$, splitting the sum in (D.17) at $w = \tau$ gives

$$\begin{aligned} |\gamma_{k,k'}(\tau)| &\lesssim \sum_{w=0}^{\tau} \frac{1}{(1+w)(1+w+\tau)} + \sum_{w>\tau} \frac{1}{(1+w)(1+w+\tau)} \\ &\lesssim (1+\tau)^{-1} \sum_{w=0}^{\tau} (1+w)^{-1} + \sum_{w>\tau} (1+w)^{-2} \\ &\lesssim (1+\tau)^{-1} \log(2+\tau). \end{aligned}$$

Together with the bound at $\tau = 0$ and the symmetry for negative lags, we have

$$\max_{k,k'\in[m_u]} |\gamma_{k,k'}(\tau)| \lesssim (1+|\tau|)^{-1} \log(2+|\tau|), \qquad \tau \in \mathbb{Z}, \tag{D.21}$$

so that

$$\sum_{\tau\in\mathbb{Z}} \max_{k,k'\in[m_u]} |\gamma_{k,k'}(\tau)|^2 < \infty. \tag{D.22}$$

If $d_u < 0$, all segment filters are absolutely summable. Extending each filter by zero on negative indices,

we have

$$\sum_{\tau\in\mathbb{Z}}|\gamma_{k,k'}(\tau)|\lesssim\sum_{\tau\in\mathbb{Z}}\sum_{w\in\mathbb{Z}}|\psi_{u,k}(w+\tau)||\psi_{u,k'}(w)|=\left\{\sum_{v\in\mathbb{Z}}|\psi_{u,k}(v)|\right\}\left\{\sum_{w\in\mathbb{Z}}|\psi_{u,k'}(w)|\right\}<\infty.$$

Hence, it follows that

$$\sum_{\tau\in\mathbb{Z}}\max_{k,k'\in[m_u]}|\gamma_{k,k'}(\tau)|<\infty,\qquad\sum_{\tau\in\mathbb{Z}}\max_{k,k'\in[m_u]}|\gamma_{k,k'}(\tau)|^2<\infty. \tag{D.23}$$

Combining (D.20), (D.22), and (D.23), we have

$$B_u(T):=\sum_{\tau=-2(T-1)}^{2(T-1)}\max_{k,k'\in[m_u]}|\gamma_{k,k'}(\tau)|^2\lesssim\begin{cases}1, & d_u<1/4,\\ \log T, & d_u=1/4,\\ T^{4d_u-1}, & d_u>1/4,\end{cases} \tag{D.24}$$

In particular, $T^{-1}B_u(T)=O(R(T,d_u))$.

We next bound the covariance between the products. Recall that $k=J_u(t)$, $r=J_u(t+h)$, $k'=J_u(s)$, $r'=J_u(s+h)$, and $\tau=t-s$. If $i\neq j$, independence across coordinates and the zero-mean property give

$$\mathrm{Cov}(u_{t,i}u_{t+h,j},u_{s,i}u_{s+h,j})=\gamma_{k,k'}(\tau)\gamma_{r,r'}(\tau).$$

If $i=j$, the fourth-moment expansion gives

$$\mathrm{Cov}(u_{t,i}u_{t+h,i},u_{s,i}u_{s+h,i})=\gamma_{k,k'}(\tau)\gamma_{r,r'}(\tau)+\gamma_{k,r'}(\tau-h)\gamma_{r,k'}(\tau+h)+\mathcal{K}_{k,r,k',r'}(\tau,h), \tag{D.25}$$

where $\mathcal{K}_{k,r,k',r'}(\tau,h)$ denotes the fourth-cumulant contribution, viz. after extending each filter by zero on negative indices,

$$\mathcal{K}_{k,r,k',r'}(\tau,h):=\kappa_{4,u}\sum_{\ell\in\mathbb{Z}}\psi_{u,k}(\ell)\psi_{u,r}(\ell+h)\psi_{u,k'}(\ell-\tau)\psi_{u,r'}(\ell+h-\tau),$$

where $\kappa_{4,u}:=\mathbb{E}(x_{u,0,1}^4)-3\{\mathbb{E}(x_{u,0,1}^2)\}^2$ is finite by the bounded fourth moment assumption. Define

$$a_q:=\psi_{u,k}(q)\psi_{u,r}(q+h),\qquad b_q:=\psi_{u,k'}(q)\psi_{u,r'}(q+h).$$

Then, $|\mathcal{K}_{k,r,k',r'}(\tau,h)|\lesssim\sum_{q\in\mathbb{Z}}|a_q|\,|b_{q-\tau}|$. Summing over τ and applying the Cauchy–Schwarz inequality

gives

$$\sum_{\tau\in\mathbb{Z}}|\mathcal{K}_{k,r,k',r'}(\tau,h)| \lesssim \left(\sum_{q\in\mathbb{Z}}|a_q|\right)\left(\sum_{q\in\mathbb{Z}}|b_q|\right)$$
$$\lesssim \left\{\sum_{q\in\mathbb{Z}}|\psi_{u,k}(q)|^2\right\}^{1/2}\left\{\sum_{q\in\mathbb{Z}}|\psi_{u,r}(q+h)|^2\right\}^{1/2}\left\{\sum_{q\in\mathbb{Z}}|\psi_{u,k'}(q)|^2\right\}^{1/2}\left\{\sum_{q\in\mathbb{Z}}|\psi_{u,r'}(q+h)|^2\right\}^{1/2} \lesssim 1.$$

Therefore, we have

$$\sum_{\tau\in\mathbb{Z}}\max_{k,r,k',r'\in[m_u]}|\mathcal{K}_{k,r,k',r'}(\tau,h)| \lesssim 1. \tag{D.26}$$

For the two covariance-product terms in (D.25), we have

$$\sum_{\tau=-(T-1)}^{T-1}\max_{k,r,k',r'\in[m_u]}|\gamma_{k,k'}(\tau)\gamma_{r,r'}(\tau)| \le \sum_{\tau=-(T-1)}^{T-1}\left\{\max_{a,b\in[m_u]}|\gamma_{a,b}(\tau)|\right\}^2 \le B_u(T), \tag{D.27}$$

and, by the Cauchy–Schwarz inequality,

$$\sum_{\tau=-(T-1)}^{T-1}\max_{k,r,k',r'\in[m_u]}|\gamma_{k,r'}(\tau-h)\gamma_{r,k'}(\tau+h)|$$
$$\le \left\{\sum_{\tau=-(T-1)}^{T-1}\max_{k,r'\in[m_u]}|\gamma_{k,r'}(\tau-h)|^2\right\}^{1/2}\left\{\sum_{\tau=-(T-1)}^{T-1}\max_{r,k'\in[m_u]}|\gamma_{r,k'}(\tau+h)|^2\right\}^{1/2} \le B_u(T), \tag{D.28}$$

where the last inequality follows because both shifted ranges $\{\tau-h:|\tau|\le T-1\}$ and $\{\tau+h:|\tau|\le T-1\}$ are contained in $\{-2(T-1),\dots,2(T-1)\}$. For fixed i,j and h, define

$$D_{u,ij,h}(\tau) := \max_{\substack{t,s\in[T-h]\\ t-s=\tau}}\left|\mathrm{Cov}\left(u_{t,i}u_{t+h,j},u_{s,i}u_{s+h,j}\right)\right|,$$

where the maximum over an empty set is taken to be zero by convention. Combining (D.26), (D.27), and (D.28), we obtain

$$\sum_{\tau=-(T-1)}^{T-1}D_{u,ij,h}(\tau) \lesssim B_u(T)+1. \tag{D.29}$$

For each τ, the number of pairs $(t,s)\in[T-h]^2$ satisfying $t-s=\tau$ is $\max\{T-h-|\tau|,0\}$. Therefore, by

(D.16),

$$\begin{aligned}\mathrm{Var}\left(\widetilde{\Gamma}_{u,ij}(h)\right) &\le T^{-2}\sum_{\tau=-(T-1)}^{T-1}\max\{T-h-|\tau|,0\}D_{u,ij,h}(\tau)\\ &\le T^{-1}\sum_{\tau=-(T-1)}^{T-1}D_{u,ij,h}(\tau)\\ &\lesssim T^{-1}(B_u(T)+1).\end{aligned}$$

Since $T^{-1}B_u(T) = O(R(T,d_u))$ and $T^{-1} = O(R(T,d_u))$, it follows that

$$\mathrm{Var}\left(\widetilde{\Gamma}_{u,ij}(h)\right) = O(R(T,d_u)).$$

Since each entry of $\widetilde{\mathbf{\Gamma}}_u(h)$ is centered and p_u is bounded,

$$\mathbb{E}\left\|\widetilde{\mathbf{\Gamma}}_u(h)\right\|_F^2 = \sum_{i=1}^{p_u}\sum_{j=1}^{p_u}\mathbb{E}\left(\widetilde{\Gamma}_{u,ij}(h)^2\right) = \sum_{i=1}^{p_u}\sum_{j=1}^{p_u}\mathrm{Var}\left(\widetilde{\Gamma}_{u,ij}(h)\right) = O(R(T,d_u)).$$

Markov's inequality then gives

$$\left\|\widetilde{\mathbf{\Gamma}}_u(h)\right\|_F^2 = O_P(R(T,d_u)),$$

which proves (D.14).

To prove (D.15), define

$$\widetilde{\mathbf{\Sigma}}_{uv} := T^{-1}\sum_{t=1}^{T}\mathbf{u}_t\mathbf{v}_t'.$$

Fix an entry $(i,j) \in [p_u]\times[p_v]$. Since the two processes $\{\mathbf{u}_t\}$ and $\{\mathbf{v}_t\}$ are independent and have mean zero, we have $\mathbb{E}(u_{t,i}v_{t,j}) = 0$ for every $t \in [T]$. Hence,

$$\mathrm{Var}\left(\widetilde{\Sigma}_{uv,ij}\right) = T^{-2}\sum_{t=1}^{T}\sum_{s=1}^{T}\mathrm{Cov}(u_{t,i}v_{t,j},u_{s,i}v_{s,j}) = T^{-2}\sum_{t=1}^{T}\sum_{s=1}^{T}\mathrm{Cov}(u_{t,i},u_{s,i})\mathrm{Cov}(v_{t,j},v_{s,j}), \tag{D.30}$$

where the second equality follows from the independence of $\{\mathbf{u}_t\}$ and $\{\mathbf{v}_t\}$.

Let $J_u(t)$ and $J_v(t)$ denote the segment indices of $\mathbf{u}_t$ and $\mathbf{v}_t$, respectively. For $a,a' \in [m_u]$ and $\tau \ge 0$, define

$$\gamma_{u,a,a'}(\tau) := \sigma_u^2\sum_{w\ge 0}\psi_{u,a}(w+\tau)\psi_{u,a'}(w),$$

where $\sigma_u^2 := \mathbb{E}(x_{u,0,1}^2) < \infty$, and set $\gamma_{u,a,a'}(\tau) := \gamma_{u,a',a}(-\tau)$ when $\tau < 0$. Define $\gamma_{v,b,b'}(\tau)$ analogously for $b, b' \in [m_v]$. By the temporal independence of the innovations, whenever $J_u(t) = a$, $J_u(s) = a'$, and $t - s = \tau$, we have $\mathrm{Cov}(u_{t,i}, u_{s,i}) = \gamma_{u,a,a'}(\tau)$, and the analogous equality holds for $\{\mathbf{v}_t\}$. In particular,

$$\max_{\substack{a,a' \in [m_u] \\ b,b' \in [m_v]}} |\gamma_{u,a,a'}(\tau)\gamma_{v,b,b'}(\tau)| = \left\{ \max_{a,a' \in [m_u]} |\gamma_{u,a,a'}(\tau)| \right\} \left\{ \max_{b,b' \in [m_v]} |\gamma_{v,b,b'}(\tau)| \right\}.$$

We first consider the case $d_u > 0$ and $d_v > 0$. Taking square roots in (D.19), applied separately to $\{\mathbf{u}_t\}$ and $\{\mathbf{v}_t\}$, gives

$$\max_{\substack{a,a' \in [m_u] \\ b,b' \in [m_v]}} |\gamma_{u,a,a'}(\tau)\gamma_{v,b,b'}(\tau)| \lesssim (1 + |\tau|)^{2d_u - 1}(1 + |\tau|)^{2d_v - 1} = (1 + |\tau|)^{4d_{uv} - 2}, \tag{D.31}$$

where $d_{uv} := (d_u + d_v)/2$.

Next, consider the case where $d_u \leq 0$ or $d_v \leq 0$. We divide it into two disjoint subcases. First, suppose that $\min\{d_u, d_v\} < 0$. Without loss of generality, let $d_u < 0$. The argument in (D.23) gives

$$\sum_{\tau \in \mathbb{Z}} \max_{a,a' \in [m_u]} |\gamma_{u,a,a'}(\tau)| < \infty.$$

Moreover, by the Cauchy–Schwarz inequality and (D.13),

$$\sup_{\tau \in \mathbb{Z}} \max_{b,b' \in [m_v]} |\gamma_{v,b,b'}(\tau)| \leq \sigma_v^2 \max_{b,b' \in [m_v]} \left\{ \sum_{w \geq 0} \psi_{v,b}(w)^2 \right\}^{1/2} \left\{ \sum_{w \geq 0} \psi_{v,b'}(w)^2 \right\}^{1/2} \lesssim 1,$$

where $\sigma_v^2 := \mathbb{E}(x_{v,0,1}^2)$. It follows that

$$\sum_{\tau \in \mathbb{Z}} \max_{\substack{a,a' \in [m_u] \\ b,b' \in [m_v]}} |\gamma_{u,a,a'}(\tau)\gamma_{v,b,b'}(\tau)| < \infty.$$

It remains to consider the case $\min\{d_u, d_v\} = 0$. Without loss of generality, let $d_u = 0$ and $d_v \geq 0$. The bound in (D.21) gives

$$\max_{a,a' \in [m_u]} |\gamma_{u,a,a'}(\tau)| \lesssim (1 + |\tau|)^{-1} \log(2 + |\tau|).$$

If $d_v > 0$, then

$$\max_{\substack{a,a' \in [m_u] \\ b,b' \in [m_v]}} |\gamma_{u,a,a'}(\tau)\gamma_{v,b,b'}(\tau)| \lesssim (1 + |\tau|)^{2d_v - 2} \log(2 + |\tau|),$$

which is summable because $2d_v - 2 < -1$. If $d_v = 0$, the same bound in (D.21) gives

$$\max_{\substack{a,a' \in [m_u] \\ b,b' \in [m_v]}} |\gamma_{u,a,a'}(\tau)\gamma_{v,b,b'}(\tau)| \lesssim (1+|\tau|)^{-2} \log^2(2+|\tau|),$$

which is also summable. Hence, whenever $d_u \le 0$ or $d_v \le 0$, we have

$$\sum_{\tau \in \mathbb{Z}} \max_{\substack{a,a' \in [m_u] \\ b,b' \in [m_v]}} |\gamma_{u,a,a'}(\tau)\gamma_{v,b,b'}(\tau)| < \infty. \tag{D.32}$$

In this case, since $d_u \le 0$ or $d_v \le 0$, while both $d_u, d_v < 1/2$, we have $d_{uv} < 1/4$.

Define

$$B_{uv}(T) := \sum_{\tau=-(T-1)}^{T-1} \max_{\substack{a,a' \in [m_u] \\ b,b' \in [m_v]}} |\gamma_{u,a,a'}(\tau)\gamma_{v,b,b'}(\tau)|.$$

Combining the two cases (D.31) and (D.32), we obtain

$$B_{uv}(T) \lesssim \begin{cases} 1, & d_{uv} < 1/4, \\ \log T, & d_{uv} = 1/4, \\ T^{4d_{uv}-1}, & d_{uv} > 1/4. \end{cases}$$

Consequently, $T^{-1}B_{uv}(T) = O(R(T, d_{uv}))$. For each $\tau \in \{-(T-1), \dots, T-1\}$, the number of pairs $(t,s) \in [T]^2$ satisfying $t - s = \tau$ is $T - |\tau|$. Therefore, grouping the pairs according to their lag gives

$$\begin{aligned} \mathrm{Var}\left(\widetilde{\Sigma}_{uv,ij}\right) &\le T^{-2} \sum_{\tau=-(T-1)}^{T-1} (T-|\tau|) \max_{\substack{a,a' \in [m_u] \\ b,b' \in [m_v]}} |\gamma_{u,a,a'}(\tau)\gamma_{v,b,b'}(\tau)| \\ &\le T^{-1} B_{uv}(T) = O(R(T, d_{uv})). \end{aligned}$$

Since p_u and p_v are bounded and $\mathbb{E}(\widetilde{\Sigma}_{uv,ij}) = 0$,

$$\mathbb{E}\left\|\widetilde{\mathbf{\Sigma}}_{uv}\right\|_F^2 = \sum_{i=1}^{p_u} \sum_{j=1}^{p_v} \mathrm{Var}\left(\widetilde{\Sigma}_{uv,ij}\right) = O(R(T, d_{uv})).$$

Markov's inequality then gives

$$\left\|\widetilde{\mathbf{\Sigma}}_{uv}\right\|_F^2 = O_P(R(T, d_{uv})),$$

which proves (D.15) and completes the proof. □

Lemma 3 *(Auxiliary bounds for error interactions) Let Assumptions (E1) and (E2) hold. Let* $\{\mathbf{u}_t\}_{t\in[T]} \in \mathcal{G}_T(\{d_{u,1},\dots,d_{u,m_u}\}|\{\mathbf{x}_{u,i}, i \in \mathbb{Z}\})$ *be a* p_u*-dimensional piecewise-stationary time series, where* p_u *and* m_u *are bounded, and assume that* $\{\mathbf{u}_t\}_{t\in[T]}$ *is independent of* $\{\mathbf{E}_t^{(m)} : t \in [T], m \in [M]\}$*. Let* $d_u := \max_{1\le j\le m_u} d_{u,j}$*. For each* $m, n \in [M]$ *with* $m \neq n$*, the following bounds hold:*

$$\big\|T^{-1}\sum_{t=1}^{T}\mathbf{E}_t^{(m)}\mathbf{1}_{q_m}\mathbf{u}_t'\big\|_F^2 = O_P(T^{-1}p_m q_m), \tag{D.33}$$

$$\big\|T^{-1}\sum_{t=1}^{T}\mathbf{E}_t^{(m)}\mathbf{1}_{q_m}\mathbf{1}_{q_n}'\mathbf{E}_t^{(n)'}\big\|_F^2 = O_P(T^{-1}p_m p_n q_m q_n), \tag{D.34}$$

$$\big\|T^{-1}\sum_{t=1}^{T}\mathbf{E}_t^{(m)}\mathbf{1}_{q_m}\mathbf{1}_{q_m}'\mathbf{E}_t^{(m)'}\big\|_F^2 = O_P(p_m q_m^2 + T^{-1}p_m^2 q_m^2). \tag{D.35}$$

Moreover, for any deterministic vector $\mathbf{w} = (w_1,\dots,w_T)'$ *and for each* $m \in [M]$*,*

$$\Big\|T^{-1}\sum_{t=1}^{T} w_t \mathbf{M}_{p_m}\mathbf{E}_t^{(m)}\mathbf{1}_{q_m}\Big\|^2 = O_P\big(T^{-2}p_m q_m\|\mathbf{w}\|^2\big). \tag{D.36}$$

Proof of Lemma 3. We first prove (D.33).

$$\begin{aligned}\mathbb{E}\Big\|T^{-1}\sum_{t=1}^{T}\mathbf{E}_t^{(m)}\mathbf{1}_{q_m}\mathbf{u}_t'\Big\|_F^2 &= T^{-2}\sum_{t=1}^{T}\sum_{s=1}^{T}\mathbb{E}\left(\mathrm{tr}(\mathbf{E}_t^{(m)}\mathbf{1}_{q_m}\mathbf{u}_t'\mathbf{u}_s\mathbf{1}_{q_m}'\mathbf{E}_s^{(m)'})\right)\\ &= T^{-2}\sum_{t=1}^{T}\sum_{s=1}^{T}\mathbb{E}(\mathbf{1}_{q_m}'\mathbf{E}_s^{(m)'}\mathbf{E}_t^{(m)}\mathbf{1}_{q_m})\mathbb{E}(\mathbf{u}_t'\mathbf{u}_s).\end{aligned}$$

The second equality follows because the whole process $\{\mathbf{u}_t\}_{t\in[T]}$ is independent of $\{\mathbf{E}_t^{(m)}\}_{t\in[T]}$. Define the admissible index set $I_T(h) := \{t \in [T] : t + h \in [T]\}$. Re-indexing by lag $h := s - t$ gives

$$\mathbb{E}\Big\|T^{-1}\sum_{t=1}^{T}\mathbf{E}_t^{(m)}\mathbf{1}_{q_m}\mathbf{u}_t'\Big\|_F^2 = T^{-2}\sum_{h=-(T-1)}^{T-1}\sum_{t\in I_T(h)}\mathbb{E}(\mathbf{1}_{q_m}'\mathbf{E}_{t+h}^{(m)'}\mathbf{E}_t^{(m)}\mathbf{1}_{q_m})\mathbb{E}(\mathbf{u}_t'\mathbf{u}_{t+h}).$$

We bound the two expectations on the r.h.s. separately. First, expand

$$\big|\mathbb{E}(\mathbf{1}_{q_m}'\mathbf{E}_{t+h}^{(m)'}\mathbf{E}_t^{(m)}\mathbf{1}_{q_m})\big| = \Big|\mathbb{E}\big(\sum_{i=1}^{p_m}\sum_{j=1}^{q_m}\sum_{l=1}^{q_m}E_{t+h,ij}^{(m)}E_{t,il}^{(m)}\big)\Big| \le \sum_{i=1}^{p_m}\sum_{j=1}^{q_m}\sum_{l=1}^{q_m}\big|\mathbb{E}(E_{t+h,ij}^{(m)}E_{t,il}^{(m)})\big|.$$

By Lemma 1, for each fixed (i,j),

$$\sum_{l=1}^{q_m}\big|\mathbb{E}(E_{t+h,ij}^{(m)}E_{t,il}^{(m)})\big| \le \sum_{k=1}^{p_m}\sum_{l=1}^{q_m}\big|\mathbb{E}(E_{t+h,ij}^{(m)}E_{t,kl}^{(m)})\big| \le C\rho(h),$$

uniformly over t, h, m, i, j. Summing over i and j, we obtain

$$\left|\mathbb{E}(\mathbf{1}'_{q_m}\mathbf{E}_{t+h}^{(m)'}\mathbf{E}_t^{(m)}\mathbf{1}_{q_m})\right| \le Cp_m q_m \rho(h). \tag{D.37}$$

Moreover, by (D.13) in Lemma 2 and the Cauchy–Schwarz inequality,

$$\sup_{t\in I_T(h)} |\mathbb{E}(\mathbf{u}'_t\mathbf{u}_{t+h})| \le \sup_{t,s\in[T]} \left(\mathbb{E}\|\mathbf{u}_t\|^2\right)^{1/2} \left(\mathbb{E}\|\mathbf{u}_s\|^2\right)^{1/2} \le C. \tag{D.38}$$

Combining (D.37) and (D.38),

$$\begin{aligned}
\mathbb{E}\left\|T^{-1}\sum_{t=1}^{T}\mathbf{E}_t^{(m)}\mathbf{1}_{q_m}\mathbf{u}'_t\right\|_F^2 &\le CT^{-2}p_m q_m \sum_{h=-(T-1)}^{T-1}\sum_{t\in I_T(h)}\rho(h) \\
&\le CT^{-1}p_m q_m \sum_{h\in\mathbb{Z}}\rho(h) \\
&= O(T^{-1}p_m q_m),
\end{aligned}$$

and Markov's inequality proves (D.33).

We next prove (D.34). First, for each group $m \in [M]$, define $\boldsymbol{\zeta}_t^{(m)} := q_m^{-1}\mathbf{E}_t^{(m)}\mathbf{1}_{q_m}$. Then

$$\mathbb{E}\left\|T^{-1}\sum_{t=1}^{T}\mathbf{E}_t^{(m)}\mathbf{1}_{q_m}\mathbf{1}'_{q_n}\mathbf{E}_t^{(n)'}\right\|_F^2 = q_m^2 q_n^2 \sum_{i=1}^{p_m}\sum_{k=1}^{p_n}\mathbb{E}\left(T^{-1}\sum_{t=1}^{T}\zeta_{t,i}^{(m)}\zeta_{t,k}^{(n)}\right)^2.$$

We first show that, for $m \neq n$ and $t, s \in [T]$,

$$\mathbb{E}\left(\zeta_{t,i}^{(m)}\zeta_{s,k}^{(n)}\right) = 0. \tag{D.39}$$

Write $\zeta_{t,i}^{(m)} = \zeta_{G,t,i}^{(m)} + \zeta_{F,t,i}^{(m)} + \zeta_{\epsilon,t,i}^{(m)}$, where the three terms are the corresponding row-averaged components induced respectively by $\mathbf{Q}_{e,m}\mathbf{G}_{e,t}\mathbf{J}'_{e,m}$, $\mathbf{R}_{e,m}\mathbf{F}_{e,t}^{(m)}\mathbf{C}'_{e,m}$, and $\boldsymbol{\Sigma}_{\epsilon,m} \circ \boldsymbol{\epsilon}_t^{(m)}$ in (E1). By the mean-zero assumptions and the mutual independence of the global error, local error, and idiosyncratic error innovation processes in (E1) and (E2), all cross terms vanish, and hence

$$\mathbb{E}\left(\zeta_{t,i}^{(m)}\zeta_{s,k}^{(n)}\right) = \mathbb{E}\left(\zeta_{G,t,i}^{(m)}\zeta_{G,s,k}^{(n)}\right) + \mathbb{E}\left(\zeta_{F,t,i}^{(m)}\zeta_{F,s,k}^{(n)}\right) + \mathbb{E}\left(\zeta_{\epsilon,t,i}^{(m)}\zeta_{\epsilon,s,k}^{(n)}\right).$$

The last two terms are zero by the cross-group independence of the group-specific local error processes and the idiosyncratic error processes in (E2). It remains to consider the global-global term. By (E2),

$$\mathbb{E}\left((\mathbf{G}_{e,t})_{ab}(\mathbf{G}_{e,s})_{cd}\right) = \mathbb{1}\{a=c, b=d\}\sum_{u,v\ge 0} a_{e,u}a_{e,v}\,\mathbb{1}\{t-u=s-v\}.$$

Therefore,

$$\begin{aligned}\mathbb{E}\{\zeta^{(m)}_{G,t,i}\zeta^{(n)}_{G,s,k}\} &= (q_m q_n)^{-1}\sum_{j=1}^{q_m}\sum_{l=1}^{q_n}\mathbb{E}\left((\mathbf{Q}_{e,m}\mathbf{G}_{e,t}\mathbf{J}'_{e,m})_{ij}(\mathbf{Q}_{e,n}\mathbf{G}_{e,s}\mathbf{J}'_{e,n})_{kl}\right)\\ &= (q_m q_n)^{-1}\left(\sum_{u,v\geq 0} a_{e,u}a_{e,v}\mathbb{1}\{t-u=s-v\}\right)(\mathbf{Q}_{e,m}\mathbf{Q}'_{e,n})_{ik}\left(\mathbf{1}'_{q_m}\mathbf{J}_{e,m}\mathbf{J}'_{e,n}\mathbf{1}_{q_n}\right)\\ &= 0,\end{aligned}$$

where the last equality follows from the cross-group orthogonality $\mathbf{Q}_{e,m}\mathbf{Q}'_{e,n}=\mathbf{0}$ or $\mathbf{J}_{e,m}\mathbf{J}'_{e,n}=\mathbf{0}$ in (E1). This proves (D.39).

By (D.39), for each fixed $i\in[p_m]$ and $k\in[p_n]$,

$$\begin{aligned}&\mathbb{E}\left(T^{-1}\sum_{t=1}^{T}\zeta^{(m)}_{t,i}\zeta^{(n)}_{t,k}\right)^2 = T^{-2}\sum_{h=-(T-1)}^{T-1}\sum_{t\in I_T(h)}\mathrm{Cov}\left(\zeta^{(m)}_{t+h,i}\zeta^{(n)}_{t+h,k},\zeta^{(m)}_{t,i}\zeta^{(n)}_{t,k}\right)\\ &\leq T^{-2}q_m^{-2}q_n^{-2}\sum_{h=-(T-1)}^{T-1}\sum_{t\in I_T(h)}\sum_{j=1}^{q_m}\sum_{l=1}^{q_n}\sum_{a=1}^{q_m}\sum_{b=1}^{q_n}\left|\mathrm{Cov}\left(E^{(m)}_{t+h,ij}E^{(n)}_{t+h,kl},E^{(m)}_{t,ia}E^{(n)}_{t,kb}\right)\right|.\end{aligned}$$

By (D.6) in Lemma 1,

$$\sum_{a=1}^{q_m}\sum_{b=1}^{q_n}\left|\mathrm{Cov}\left(E^{(m)}_{t+h,ij}E^{(n)}_{t+h,kl},E^{(m)}_{t,ia}E^{(n)}_{t,kb}\right)\right|\leq C\rho(h),$$

uniformly over all admissible indices. Therefore,

$$\begin{aligned}\mathbb{E}\left(T^{-1}\sum_{t=1}^{T}\zeta^{(m)}_{t,i}\zeta^{(n)}_{t,k}\right)^2 &\leq CT^{-2}q_m^{-2}q_n^{-2}\sum_{h=-(T-1)}^{T-1}\sum_{t\in I_T(h)}\sum_{j=1}^{q_m}\sum_{l=1}^{q_n}\rho(h)\\ &\leq CT^{-1}q_m^{-1}q_n^{-1}\sum_{h\in\mathbb{Z}}\rho(h)\\ &= O(T^{-1}q_m^{-1}q_n^{-1}).\end{aligned}$$

Consequently,

$$\begin{aligned}\mathbb{E}\left\|T^{-1}\sum_{t=1}^{T}\mathbf{E}^{(m)}_t\mathbf{1}_{q_m}\mathbf{1}'_{q_n}\mathbf{E}^{(n)'}_t\right\|_F^2 &\leq Cq_m^2q_n^2\sum_{i=1}^{p_m}\sum_{k=1}^{p_n}T^{-1}q_m^{-1}q_n^{-1}\\ &= O(T^{-1}p_mp_nq_mq_n).\end{aligned}$$

Markov's inequality proves (D.34).

We next prove (D.35). Using the same notation $\boldsymbol{\zeta}_t^{(m)} = q_m^{-1}\mathbf{E}_t^{(m)}\mathbf{1}_{q_m}$, we have

$$T^{-1}\sum_{t=1}^{T}\mathbf{E}_t^{(m)}\mathbf{1}_{q_m}\mathbf{1}_{q_m}'\mathbf{E}_t^{(m)'} = T^{-1}q_m^2\sum_{t=1}^{T}\boldsymbol{\zeta}_t^{(m)}\boldsymbol{\zeta}_t^{(m)'}.$$

Hence,

$$\begin{aligned}
&\mathbb{E}\Big\|T^{-1}\sum_{t=1}^{T}\mathbf{E}_t^{(m)}\mathbf{1}_{q_m}\mathbf{1}_{q_m}'\mathbf{E}_t^{(m)'}\Big\|_F^2 \\
&= q_m^4\mathbb{E}\Big\|T^{-1}\sum_{t=1}^{T}\boldsymbol{\zeta}_t^{(m)}\boldsymbol{\zeta}_t^{(m)'}\Big\|_F^2 \\
&\le 2q_m^4\Big\|T^{-1}\sum_{t=1}^{T}\mathbb{E}\left(\boldsymbol{\zeta}_t^{(m)}\boldsymbol{\zeta}_t^{(m)'}\right)\Big\|_F^2 + 2q_m^4\mathbb{E}\Big\|T^{-1}\sum_{t=1}^{T}\left[\boldsymbol{\zeta}_t^{(m)}\boldsymbol{\zeta}_t^{(m)'} - \mathbb{E}\left(\boldsymbol{\zeta}_t^{(m)}\boldsymbol{\zeta}_t^{(m)'}\right)\right]\Big\|_F^2.
\end{aligned}$$

We first bound the deterministic part. Let $\boldsymbol{\Sigma}_{\zeta,t}^{(m)} := \mathbb{E}\left(\boldsymbol{\zeta}_t^{(m)}\boldsymbol{\zeta}_t^{(m)'}\right)$. For each fixed $t \in [T]$ and $i \in [p_m]$, Lemma 1 with $h = 0$ gives

$$\begin{aligned}
\sum_{k=1}^{p_m}\left|(\boldsymbol{\Sigma}_{\zeta,t}^{(m)})_{ik}\right| &= q_m^{-2}\sum_{k=1}^{p_m}\left|\sum_{j=1}^{q_m}\sum_{l=1}^{q_m}\mathbb{E}(E_{t,ij}^{(m)}E_{t,kl}^{(m)})\right| \\
&\le q_m^{-2}\sum_{j=1}^{q_m}\sum_{k=1}^{p_m}\sum_{l=1}^{q_m}\left|\mathbb{E}(E_{t,ij}^{(m)}E_{t,kl}^{(m)})\right| \\
&\le Cq_m^{-1}.
\end{aligned}$$

Therefore,

$$\left\|\boldsymbol{\Sigma}_{\zeta,t}^{(m)}\right\|_F^2 = \sum_{i=1}^{p_m}\sum_{k=1}^{p_m}\left|(\boldsymbol{\Sigma}_{\zeta,t}^{(m)})_{ik}\right|^2 \le \sum_{i=1}^{p_m}\left\{\sum_{k=1}^{p_m}\left|(\boldsymbol{\Sigma}_{\zeta,t}^{(m)})_{ik}\right|\right\}^2 = O(p_m q_m^{-2}),$$

uniformly over $t \in [T]$. By Jensen's inequality,

$$\Big\|T^{-1}\sum_{t=1}^{T}\mathbb{E}\left(\boldsymbol{\zeta}_t^{(m)}\boldsymbol{\zeta}_t^{(m)'}\right)\Big\|_F^2 \le T^{-1}\sum_{t=1}^{T}\left\|\boldsymbol{\Sigma}_{\zeta,t}^{(m)}\right\|_F^2 = O(p_m q_m^{-2}).$$

We next bound the centered stochastic part. Write

$$\begin{aligned}
&\mathbb{E}\Big\|T^{-1}\sum_{t=1}^{T}\left[\boldsymbol{\zeta}_t^{(m)}\boldsymbol{\zeta}_t^{(m)'} - \mathbb{E}\left(\boldsymbol{\zeta}_t^{(m)}\boldsymbol{\zeta}_t^{(m)'}\right)\right]\Big\|_F^2 \\
&= T^{-2}\sum_{i=1}^{p_m}\sum_{k=1}^{p_m}\sum_{h=-(T-1)}^{T-1}\sum_{t\in I_T(h)}\mathrm{Cov}\left(\zeta_{t+h,i}^{(m)}\zeta_{t+h,k}^{(m)}, \zeta_{t,i}^{(m)}\zeta_{t,k}^{(m)}\right).
\end{aligned}$$

Taking absolute values and using $\boldsymbol{\zeta}_t^{(m)} = q_m^{-1}\mathbf{E}_t^{(m)}\mathbf{1}_{q_m}$, we obtain

$$\begin{aligned}
&\mathbb{E}\left\|T^{-1}\sum_{t=1}^{T}\left[\boldsymbol{\zeta}_t^{(m)}\boldsymbol{\zeta}_t^{(m)\prime} - \mathbb{E}\left(\boldsymbol{\zeta}_t^{(m)}\boldsymbol{\zeta}_t^{(m)\prime}\right)\right]\right\|_F^2 \\
&\leq T^{-2}q_m^{-4}\sum_{i=1}^{p_m}\sum_{k=1}^{p_m}\sum_{h=-(T-1)}^{T-1}\sum_{t\in I_T(h)}\sum_{j=1}^{q_m}\sum_{l=1}^{q_m}\sum_{a=1}^{q_m}\sum_{b=1}^{q_m}\left|\mathrm{Cov}\left(E_{t+h,ij}^{(m)}E_{t+h,kl}^{(m)}, E_{t,ia}^{(m)}E_{t,kb}^{(m)}\right)\right|.
\end{aligned}$$

By (D.6) in Lemma 1, with the two group indices both equal to m,

$$\sum_{a=1}^{q_m}\sum_{b=1}^{q_m}\left|\mathrm{Cov}\left(E_{t+h,ij}^{(m)}E_{t+h,kl}^{(m)}, E_{t,ia}^{(m)}E_{t,kb}^{(m)}\right)\right| \leq C\rho(h),$$

uniformly over all admissible indices. Hence,

$$\begin{aligned}
&\mathbb{E}\left\|T^{-1}\sum_{t=1}^{T}\left[\boldsymbol{\zeta}_t^{(m)}\boldsymbol{\zeta}_t^{(m)\prime} - \mathbb{E}\left(\boldsymbol{\zeta}_t^{(m)}\boldsymbol{\zeta}_t^{(m)\prime}\right)\right]\right\|_F^2 \\
&\leq CT^{-2}q_m^{-4}\sum_{i=1}^{p_m}\sum_{k=1}^{p_m}\sum_{h=-(T-1)}^{T-1}\sum_{t\in I_T(h)}\sum_{j=1}^{q_m}\sum_{l=1}^{q_m}\rho(h) \\
&\leq CT^{-1}p_m^2q_m^{-2}\sum_{h\in\mathbb{Z}}\rho(h) \\
&= O(T^{-1}p_m^2q_m^{-2}).
\end{aligned}$$

Combining the deterministic and centered stochastic parts, we obtain

$$\mathbb{E}\left\|T^{-1}\sum_{t=1}^{T}\mathbf{E}_t^{(m)}\mathbf{1}_{q_m}\mathbf{1}_{q_m}'\mathbf{E}_t^{(m)\prime}\right\|_F^2 = O(p_mq_m^2 + T^{-1}p_m^2q_m^2).$$

Markov's inequality proves (D.35).

To prove (D.36), retain the notation $\boldsymbol{\zeta}_t^{(m)} = q_m^{-1}\mathbf{E}_t^{(m)}\mathbf{1}_{q_m}$. Then,

$$\begin{aligned}
\mathbb{E}\left\|T^{-1}\sum_{t=1}^{T}w_t\mathbf{M}_{p_m}\mathbf{E}_t^{(m)}\mathbf{1}_{q_m}\right\|^2 &= T^{-2}q_m^2\sum_{t=1}^{T}\sum_{s=1}^{T}w_tw_s\mathbb{E}\left(\boldsymbol{\zeta}_t^{(m)\prime}\mathbf{M}_{p_m}\boldsymbol{\zeta}_s^{(m)}\right) \\
&\leq T^{-2}q_m^2\sum_{t=1}^{T}\sum_{s=1}^{T}|w_tw_s|\left|\mathbb{E}\left(\boldsymbol{\zeta}_t^{(m)\prime}\mathbf{M}_{p_m}\boldsymbol{\zeta}_s^{(m)}\right)\right|.
\end{aligned}$$

For any $h\in\mathbb{Z}$ and $t\in I_T(h)$, write

$$\begin{aligned}
\left|\mathbb{E}\left(\boldsymbol{\zeta}_{t+h}^{(m)\prime}\mathbf{M}_{p_m}\boldsymbol{\zeta}_t^{(m)}\right)\right| &= q_m^{-2}\left|\sum_{i=1}^{p_m}\sum_{k=1}^{p_m}(\mathbf{M}_{p_m})_{ik}\sum_{j=1}^{q_m}\sum_{l=1}^{q_m}\mathbb{E}\left(E_{t+h,ij}^{(m)}E_{t,kl}^{(m)}\right)\right| \\
&\leq q_m^{-2}\sum_{i=1}^{p_m}\sum_{j=1}^{q_m}\sum_{k=1}^{p_m}\sum_{l=1}^{q_m}\left|\mathbb{E}\left(E_{t+h,ij}^{(m)}E_{t,kl}^{(m)}\right)\right|
\end{aligned}$$

$$\leq Cq_m^{-2}\sum_{i=1}^{p_m}\sum_{j=1}^{q_m}\rho(h)$$
$$= Cp_m q_m^{-1}\rho(h),$$

where the first inequality uses $|(\mathbf{M}_{p_m})_{ik}| \leq 1$, and the last inequality follows from (D.4) in Lemma 1. Therefore,

$$\mathbb{E}\Big\|T^{-1}\sum_{t=1}^{T} w_t \mathbf{M}_{p_m}\mathbf{E}_t^{(m)}\mathbf{1}_{q_m}\Big\|^2 \leq CT^{-2}p_m q_m \sum_{h=-(T-1)}^{T-1}\sum_{t\in I_T(h)}|w_t w_{t+h}|\rho(h).$$

Recall the $p=1, q=2, r=2$ case of the discrete Young inequality: if $\sum_{h\in\mathbb{Z}}|a_h|<\infty$ and $\sum_{t\in\mathbb{Z}} b_t^2<\infty$, then

$$\sum_{s\in\mathbb{Z}}\left(\sum_{t\in\mathbb{Z}}|a_{s-t}||b_t|\right)^2 \leq \left(\sum_{h\in\mathbb{Z}}|a_h|\right)^2\sum_{t\in\mathbb{Z}} b_t^2.$$

Extending w_t by zero outside $[T]$, and taking $a_h=\rho(h)$ and $b_t=w_t$ in the preceding inequality, we have

$$\sum_{h=-(T-1)}^{T-1}\sum_{t\in I_T(h)}|w_t w_{t+h}|\rho(h) = \sum_{s\in\mathbb{Z}}|w_s|\sum_{t\in\mathbb{Z}}\rho(s-t)|w_t|$$
$$\leq \left(\sum_{s\in\mathbb{Z}} w_s^2\right)^{1/2}\left\{\sum_{s\in\mathbb{Z}}\left(\sum_{t\in\mathbb{Z}}\rho(s-t)|w_t|\right)^2\right\}^{1/2} \leq \left(\sum_{h\in\mathbb{Z}}\rho(h)\right)\sum_{t\in\mathbb{Z}} w_t^2 \leq C\|\mathbf{w}\|^2,$$

where the second inequality follows from the discrete Young inequality, and the last inequality follows from $\sum_{h\in\mathbb{Z}}\rho(h)<\infty$ in Lemma 1. Consequently,

$$\mathbb{E}\Big\|T^{-1}\sum_{t=1}^{T} w_t \mathbf{M}_{p_m}\mathbf{E}_t^{(m)}\mathbf{1}_{q_m}\Big\|^2 \leq CT^{-2}p_m q_m\|\mathbf{w}\|^2.$$

Markov's inequality proves (D.36). This completes the proof. □

Lemma 4 *(Asymptotic orthogonality of projected rotation) Let* $\mathbf{U}, \widehat{\mathbf{U}} \in \mathbb{R}^{d\times r}$ *satisfy* $\mathbf{U}'\mathbf{U} = \mathbf{I}_r$ *and* $\widehat{\mathbf{U}}'\widehat{\mathbf{U}} = \mathbf{I}_r$. *Define* $\mathbf{P}_U := \mathbf{U}\mathbf{U}'$, $\widehat{\mathbf{P}}_U := \widehat{\mathbf{U}}\widehat{\mathbf{U}}'$, *and* $\mathbf{H}_U := \mathbf{U}'\widehat{\mathbf{U}}$. *Then,*

$$\widehat{\mathbf{U}} - \mathbf{U}\mathbf{H}_U = (\mathbf{I}_d - \mathbf{P}_U)\widehat{\mathbf{U}}. \tag{D.40}$$

Moreover, if $\big\|\widehat{\mathbf{U}} - \mathbf{U}\mathbf{H}_U\big\|_F = o_P(1)$, *then*

$$\mathbf{H}_U'\mathbf{H}_U = \mathbf{I}_r + o_P(1), \qquad \mathbf{H}_U\mathbf{H}_U' = \mathbf{I}_r + o_P(1), \tag{D.41}$$

and $\mathbf{H}_U$ *is invertible with probability tending to one. In addition,*

$$\left\|\widehat{\mathbf{P}}_U - \mathbf{P}_U\right\|_F^2 = 2\left\|\widehat{\mathbf{U}} - \mathbf{U}\mathbf{H}_U\right\|_F^2. \tag{D.42}$$

Proof of Lemma 4. By the definition of $\mathbf{H}_U$, $\mathbf{U}\mathbf{H}_U = \mathbf{U}\mathbf{U}'\widehat{\mathbf{U}} = \mathbf{P}_U\widehat{\mathbf{U}}$. Therefore,

$$\widehat{\mathbf{U}} - \mathbf{U}\mathbf{H}_U = \widehat{\mathbf{U}} - \mathbf{P}_U\widehat{\mathbf{U}} = (\mathbf{I}_d - \mathbf{P}_U)\widehat{\mathbf{U}}.$$

Let $\mathbf{E}_U := \widehat{\mathbf{U}} - \mathbf{U}\mathbf{H}_U$. Since $\mathbf{U}'\mathbf{E}_U = 0$, we have

$$\mathbf{I}_r = \widehat{\mathbf{U}}'\widehat{\mathbf{U}} = (\mathbf{U}\mathbf{H}_U + \mathbf{E}_U)'(\mathbf{U}\mathbf{H}_U + \mathbf{E}_U) = \mathbf{H}_U'\mathbf{H}_U + \mathbf{E}_U'\mathbf{E}_U.$$

Hence

$$\left\|\mathbf{H}_U'\mathbf{H}_U - \mathbf{I}_r\right\| = \left\|\mathbf{E}_U\right\|^2 \le \left\|\mathbf{E}_U\right\|_F^2 = o_P(1).$$

Since $\mathbf{H}_U$ is square, $\mathbf{H}_U'\mathbf{H}_U$ and $\mathbf{H}_U\mathbf{H}_U'$ have the same eigenvalues, given by the squared singular values of $\mathbf{H}_U$. Hence

$$\left\|\mathbf{H}_U\mathbf{H}_U' - \mathbf{I}_r\right\| = \max_{1\le j\le r}|\sigma_j^2(\mathbf{H}_U) - 1| = \left\|\mathbf{H}_U'\mathbf{H}_U - \mathbf{I}_r\right\| = o_P(1).$$

Also, since $\lambda_{\min}(\mathbf{H}_U'\mathbf{H}_U) = 1 + o_P(1)$, $\mathbf{H}_U$ is invertible with probability tending to one.

Finally, write

$$\begin{aligned}\left\|\widehat{\mathbf{P}}_U - \mathbf{P}_U\right\|_F^2 &= \mathrm{tr}(\widehat{\mathbf{P}}_U) + \mathrm{tr}(\mathbf{P}_U) - 2\mathrm{tr}(\widehat{\mathbf{P}}_U\mathbf{P}_U)\\ &= 2r - 2\mathrm{tr}(\widehat{\mathbf{U}}'\mathbf{U}\mathbf{U}'\widehat{\mathbf{U}})\\ &= 2r - 2\mathrm{tr}(\mathbf{H}_U'\mathbf{H}_U).\end{aligned}$$

On the other hand, from $\mathbf{I}_r = \mathbf{H}_U'\mathbf{H}_U + \mathbf{E}_U'\mathbf{E}_U$, we have

$$\left\|\mathbf{E}_U\right\|_F^2 = \mathrm{tr}(\mathbf{E}_U'\mathbf{E}_U) = r - \mathrm{tr}(\mathbf{H}_U'\mathbf{H}_U).$$

Therefore,

$$\left\|\widehat{\mathbf{P}}_U - \mathbf{P}_U\right\|_F^2 = 2\left\|\mathbf{E}_U\right\|_F^2 = 2\left\|\widehat{\mathbf{U}} - \mathbf{U}\mathbf{H}_U\right\|_F^2.$$

This completes the proof.

Lemma 5 *(Bounds for $\mathbf{S}_{A_1,mn}$ and $\mathbf{\Delta}_{A_1,mn}$) For any $m,n \in [M]$ with $m \neq n$, let $\mathbf{S}_{A_1,mn}$ and $\widehat{\mathbf{S}}_{A_1,mn}$ be defined in (C.9) and (C.10), respectively, and define $\mathbf{\Delta}_{A_1,mn} := \widehat{\mathbf{S}}_{A_1,mn} - \mathbf{S}_{A_1,mn}$. Under the assumptions of Theorem 3,*

$$\left\|\mathbf{S}_{A_1,mn}\right\|_F^2 = O(p_m^{\gamma_r} p_n^{\gamma_r}), \tag{D.43}$$

$$\left\|\mathbf{\Delta}_{A_1,mn}\right\|_F^2 = o_P(p_m^{\gamma_r} p_n^{\gamma_r}). \tag{D.44}$$

Furthermore, defining $\mathbf{P}_{A_1,m} := \breve{\mathbf{A}}_{1m}\breve{\mathbf{A}}_{1m}'$, we have

$$\left\|(\mathbf{I}_{p_m} - \mathbf{P}_{A_1,m})\mathbf{\Delta}_{A_1,mn}\right\|_F^2 = O_P\Big(R(T,d_{r,\star})p_m^{\gamma_r^{(m)}} p_n^{\gamma_r^{(n)}} + T^{-1}p_m p_n^{\gamma_{r,\max}^{(n)}} q_m^{-1}\Big). \tag{D.45}$$

Proof of Lemma 5. Recall from (C.9) and (C.10) that

$$\mathbf{S}_{A_1,mn} := T^{-1}q_m^{-1}q_n^{-1}\mathbf{M}_{p_m}\sum_{t=1}^{T}\mathbb{E}\left(\mathbf{Y}_t^{(m)}\mathbf{1}_{q_m}\mathbf{1}_{q_n}'\mathbf{Y}_t^{(n)'}\right)\mathbf{M}_{p_n},$$

$$\widehat{\mathbf{S}}_{A_1,mn} := T^{-1}q_m^{-1}q_n^{-1}\mathbf{M}_{p_m}\sum_{t=1}^{T}\left(\mathbf{Y}_t^{(m)}\mathbf{1}_{q_m}\mathbf{1}_{q_n}'\mathbf{Y}_t^{(n)'}\right)\mathbf{M}_{p_n}.$$

We first prove (D.43). By (D.13) in Lemma 2,

$$\left\|T^{-1}\sum_{t=1}^{T}\mathbb{E}(\boldsymbol{\alpha}_t\boldsymbol{\alpha}_t')\right\|_F \leq T^{-1}\sum_{t=1}^{T}\mathbb{E}\|\boldsymbol{\alpha}_t\boldsymbol{\alpha}_t'\|_F = T^{-1}\sum_{t=1}^{T}\mathbb{E}\|\boldsymbol{\alpha}_t\|^2 = O(1).$$

By (C.2), we have $\mathbf{S}_{A_1,mn} = \mathbf{A}_{1m}\left(T^{-1}\sum_{t=1}^{T}\mathbb{E}(\boldsymbol{\alpha}_t\boldsymbol{\alpha}_t')\right)\mathbf{A}_{1n}'$. Hence, by (ME1),

$$\left\|\mathbf{S}_{A_1,mn}\right\|_F^2 \leq \left\|\mathbf{A}_{1m}\right\|_F^2\left\|T^{-1}\sum_{t=1}^{T}\mathbb{E}(\boldsymbol{\alpha}_t\boldsymbol{\alpha}_t')\right\|_F^2\left\|\mathbf{A}_{1n}\right\|_F^2 = O\big(p_m^{\gamma_r}p_n^{\gamma_r}\big),$$

which proves (D.43).

We next prove (D.44) and (D.45). By (IC1), for each $m \in [M]$,

$$q_m^{-1}\mathbf{M}_{p_m}\mathbf{Y}_t^{(m)}\mathbf{1}_{q_m} = \mathbf{a}_t^{(m)} + q_m^{-1}\mathbf{M}_{p_m}\mathbf{E}_t^{(m)}\mathbf{1}_{q_m}.$$

Substituting these identities into the definitions of $\mathbf{S}_{A_1,mn}$ and $\widehat{\mathbf{S}}_{A_1,mn}$ gives

$$\mathbf{\Delta}_{A_1,mn} = \widehat{\mathbf{S}}_{A_1,mn} - \mathbf{S}_{A_1,mn} = \mathcal{I}_{A_1,1} + \mathcal{I}_{A_1,2} + \mathcal{I}_{A_1,3} + \mathcal{I}_{A_1,4},$$

where

$$\mathcal{I}_{A_1,1} := T^{-1}\sum_{t=1}^{T}\mathbf{a}_t^{(m)}\mathbf{a}_t^{(n)'} - \mathbf{S}_{A_1,mn}, \qquad \mathcal{I}_{A_1,2} := (Tq_n)^{-1}\sum_{t=1}^{T}\mathbf{a}_t^{(m)}\mathbf{1}_{q_n}'\mathbf{E}_t^{(n)'}\mathbf{M}_{p_n},$$

$$\mathcal{I}_{A_1,3} := (Tq_m)^{-1}\sum_{t=1}^{T}\mathbf{M}_{p_m}\mathbf{E}_t^{(m)}\mathbf{1}_{q_m}\mathbf{a}_t^{(n)'}, \qquad \mathcal{I}_{A_1,4} := (Tq_mq_n)^{-1}\sum_{t=1}^{T}\mathbf{M}_{p_m}\mathbf{E}_t^{(m)}\mathbf{1}_{q_m}\mathbf{1}_{q_n}'\mathbf{E}_t^{(n)'}\mathbf{M}_{p_n}.$$

We bound the four components separately. Throughout the remainder of the proof, we repeatedly use the contraction properties (D.1) in Proposition 1 to remove orthogonal projectors from upper bounds without further mention. By (C.2), since $\mathbf{S}_{A_1,mn} = T^{-1}\sum_{t=1}^{T}\mathbb{E}\left(\mathbf{a}_t^{(m)}\mathbf{a}_t^{(n)'}\right)$, we have

$$\mathcal{I}_{A_1,1} = T^{-1}\sum_{t=1}^{T}\left\{\mathbf{a}_t^{(m)}\mathbf{a}_t^{(n)'} - \mathbb{E}\left(\mathbf{a}_t^{(m)}\mathbf{a}_t^{(n)'}\right)\right\}.$$

Using $\mathbf{a}_t^{(m)} = \mathbf{A}_{1m}\boldsymbol{\alpha}_t + \mathbf{A}_{2m}\boldsymbol{\alpha}_t^{(m)}$ and $\mathbf{a}_t^{(n)} = \mathbf{A}_{1n}\boldsymbol{\alpha}_t + \mathbf{A}_{2n}\boldsymbol{\alpha}_t^{(n)}$, together with the sample orthogonality and cross-group uncorrelatedness conditions in (IC2), we obtain

$$\mathcal{I}_{A_1,1} = \mathbf{A}_{1m}\left\{T^{-1}\sum_{t=1}^{T}\left(\boldsymbol{\alpha}_t\boldsymbol{\alpha}_t' - \mathbb{E}(\boldsymbol{\alpha}_t\boldsymbol{\alpha}_t')\right)\right\}\mathbf{A}_{1n}' + \mathbf{A}_{2m}\left(T^{-1}\sum_{t=1}^{T}\boldsymbol{\alpha}_t^{(m)}\boldsymbol{\alpha}_t^{(n)'}\right)\mathbf{A}_{2n}'.$$

It follows from (D.14) and (D.15) in Lemma 2, together with (ME1) and (ME2), that

$$\left\|\mathcal{I}_{A_1,1}\right\|_F^2 = O_P\left(R(T,d_{r,\star})\left(p_m^{\gamma_r}p_n^{\gamma_r} + p_m^{\gamma_r^{(m)}}p_n^{\gamma_r^{(n)}}\right)\right). \tag{D.46}$$

Moreover, since $(\mathbf{I}_{p_m} - \mathbf{P}_{A_1,m})\mathbf{A}_{1m} = \mathbf{0}$,

$$(\mathbf{I}_{p_m} - \mathbf{P}_{A_1,m})\mathcal{I}_{A_1,1} = (\mathbf{I}_{p_m} - \mathbf{P}_{A_1,m})\mathbf{A}_{2m}\left(T^{-1}\sum_{t=1}^{T}\boldsymbol{\alpha}_t^{(m)}\boldsymbol{\alpha}_t^{(n)'}\right)\mathbf{A}_{2n}'.$$

Thus,

$$\left\|(\mathbf{I}_{p_m} - \mathbf{P}_{A_1,m})\mathcal{I}_{A_1,1}\right\|_F^2 = O_P\left(R(T,d_{r,\star})p_m^{\gamma_r^{(m)}}p_n^{\gamma_r^{(n)}}\right). \tag{D.47}$$

For $\mathcal{I}_{A_1,2}$, write

$$\mathcal{I}_{A_1,2} = \mathbf{A}_{1m}\left\{(Tq_n)^{-1}\sum_{t=1}^{T}\boldsymbol{\alpha}_t\mathbf{1}_{q_n}'\mathbf{E}_t^{(n)'}\mathbf{M}_{p_n}\right\} + \mathbf{A}_{2m}\left\{(Tq_n)^{-1}\sum_{t=1}^{T}\boldsymbol{\alpha}_t^{(m)}\mathbf{1}_{q_n}'\mathbf{E}_t^{(n)'}\mathbf{M}_{p_n}\right\}.$$

Applying (D.33) in Lemma 3 to $\{\boldsymbol{\alpha}_t\}$ and $\{\boldsymbol{\alpha}_t^{(m)}\}$, together with (ME1),

$$\left\|\mathcal{I}_{A_1,2}\right\|_F^2 = O_P\left(T^{-1}\left(p_m^{\gamma_r} + p_m^{\gamma_r^{(m)}}\right)p_n q_n^{-1}\right). \tag{D.48}$$

Since left-multiplication by $(\mathbf{I}_{p_m} - \mathbf{P}_{A_1,m})$ eliminates the term involving $\mathbf{A}_{1m}$,

$$(\mathbf{I}_{p_m} - \mathbf{P}_{A_1,m})\mathcal{I}_{A_1,2} = (\mathbf{I}_{p_m} - \mathbf{P}_{A_1,m})\mathbf{A}_{2m}\left\{(Tq_n)^{-1}\sum_{t=1}^{T}\boldsymbol{\alpha}_t^{(m)}\mathbf{1}_{q_n}'\mathbf{E}_t^{(n)'}\mathbf{M}_{p_n}\right\}.$$

Consequently,

$$\left\|(\mathbf{I}_{p_m} - \mathbf{P}_{A_1,m})\mathcal{I}_{A_1,2}\right\|_F^2 = O_P\left(T^{-1}p_m^{\gamma_r^{(m)}}p_n q_n^{-1}\right). \tag{D.49}$$

For $\mathcal{I}_{A_1,3}$, we have

$$\mathcal{I}_{A_1,3} = q_m^{-1}\left(T^{-1}\sum_{t=1}^{T}\mathbf{M}_{p_m}\mathbf{E}_t^{(m)}\mathbf{1}_{q_m}\boldsymbol{\alpha}_t'\right)\mathbf{A}_{1n}' + q_m^{-1}\left(T^{-1}\sum_{t=1}^{T}\mathbf{M}_{p_m}\mathbf{E}_t^{(m)}\mathbf{1}_{q_m}\boldsymbol{\alpha}_t^{(n)'}\right)\mathbf{A}_{2n}'.$$

Applying (D.33) in Lemma 3 and (ME1) gives

$$\left\|\mathcal{I}_{A_1,3}\right\|_F^2 = O_P\left(T^{-1}p_m q_m^{-1}\left(p_n^{\gamma_r} + p_n^{\gamma_r^{(n)}}\right)\right) = O_P\left(T^{-1}p_m p_n^{\gamma_{r,\max}^{(n)}}q_m^{-1}\right). \tag{D.50}$$

Consequently,

$$\left\|(\mathbf{I}_{p_m} - \mathbf{P}_{A_1,m})\mathcal{I}_{A_1,3}\right\|_F^2 \le \left\|\mathcal{I}_{A_1,3}\right\|_F^2 = O_P\left(T^{-1}p_m p_n^{\gamma_{r,\max}^{(n)}}q_m^{-1}\right). \tag{D.51}$$

For $\mathcal{I}_{A_1,4}$, by (D.34) in Lemma 3,

$$\left\|\mathcal{I}_{A_1,4}\right\|_F^2 = O_P\left(T^{-1}p_m p_n q_m^{-1}q_n^{-1}\right), \tag{D.52}$$

$$\left\|(\mathbf{I}_{p_m} - \mathbf{P}_{A_1,m})\mathcal{I}_{A_1,4}\right\|_F^2 = O_P\left(T^{-1}p_m p_n q_m^{-1}q_n^{-1}\right). \tag{D.53}$$

Using $\left\|\sum_{j=1}^4 \mathcal{I}_{A_1,j}\right\|_F^2 \lesssim \sum_{j=1}^4 \left\|\mathcal{I}_{A_1,j}\right\|_F^2$, together with (D.46), (D.48), (D.50), and (D.52), we obtain

$$\begin{aligned}\left\|\boldsymbol{\Delta}_{A_1,mn}\right\|_F^2 = O_P\Bigg(&R(T,d_{r,\star})\left(p_m^{\gamma_r}p_n^{\gamma_r} + p_m^{\gamma_r^{(m)}}p_n^{\gamma_r^{(n)}}\right) + T^{-1}p_m^{\gamma_{r,\max}^{(m)}}p_n q_n^{-1} \\ &+ T^{-1}p_m p_n^{\gamma_{r,\max}^{(n)}}q_m^{-1} + T^{-1}p_m p_n q_m^{-1}q_n^{-1}\Bigg).\end{aligned} \tag{D.54}$$

To deduce (D.44), first note that $T^{-1} = O(R(T, d_{r,\star}))$. Moreover, by (R1), $p_s^{1-\gamma_r} q_s^{-1}$ and $p_s^{1-\gamma_r^{(s)}} q_s^{-1}$ are both $o(1)$ uniformly in $s \in [M]$. Since $p_m^{\gamma_{r,\max}^{(m)}} \le p_m^{\gamma_r} + p_m^{\gamma_r^{(m)}}$, we have

$$\begin{aligned} T^{-1} p_m^{\gamma_{r,\max}^{(m)}} p_n q_n^{-1} &\lesssim R(T, d_{r,\star}) \Big\{ p_m^{\gamma_r} p_n^{\gamma_r} p_n^{1-\gamma_r} q_n^{-1} + p_m^{\gamma_r^{(m)}} p_n^{\gamma_r^{(n)}} p_n^{1-\gamma_r^{(n)}} q_n^{-1} \Big\} \\ &= o\left(R(T, d_{r,\star}) \{ p_m^{\gamma_r} p_n^{\gamma_r} + p_m^{\gamma_r^{(m)}} p_n^{\gamma_r^{(n)}} \} \right). \end{aligned}$$

The term $T^{-1} p_m p_n^{\gamma_{r,\max}^{(n)}} q_m^{-1}$ follows symmetrically, while the term $T^{-1} p_m p_n q_m^{-1} q_n^{-1}$ is of still smaller order since $p_m^{1-\gamma_{r,\max}^{(m)}} q_m^{-1} = o(1)$. Hence, (D.54) reduces to

$$\left\| \boldsymbol{\Delta}_{A_1,mn} \right\|_F^2 = O_P\left(R(T, d_{r,\star}) \{ p_m^{\gamma_r} p_n^{\gamma_r} + p_m^{\gamma_r^{(m)}} p_n^{\gamma_r^{(n)}} \} \right).$$

After division by $p_m^{\gamma_r} p_n^{\gamma_r}$, the r.h.s. is $O_P\left(R(T, d_{r,\star}) \{ 1 + \nu_{r,m} \nu_{r,n} \} \right)$. For $m \neq n$,

$$1 + \nu_{r,m} \nu_{r,n} \le 2(\nu_{r,m} \vee \nu_{r,m}^{-1})(1 \vee \nu_{r,n}),$$

so (R1) gives $R(T, d_{r,\star}) \{ 1 + \nu_{r,m} \nu_{r,n} \} = o(1)$ uniformly over $m \neq n$. This proves (D.44).

Finally, by $T^{-1} = O(R(T, d_{r,\star}))$ and (R1), we have $T^{-1} p_m^{\gamma_r^{(m)}} p_n q_n^{-1} = o\left(R(T, d_{r,\star}) p_m^{\gamma_r^{(m)}} p_n^{\gamma_r^{(n)}} \right)$, while $T^{-1} p_m p_n q_m^{-1} q_n^{-1} = o\left(T^{-1} p_m p_n^{\gamma_{r,\max}^{(n)}} q_m^{-1} \right)$. Combining (D.47), (D.49), (D.51), and (D.53), we obtain

$$\left\| (\mathbf{I}_{p_m} - \mathbf{P}_{A_1,m}) \boldsymbol{\Delta}_{A_1,mn} \right\|_F^2 = O_P\Big(R(T, d_{r,\star}) p_m^{\gamma_r^{(m)}} p_n^{\gamma_r^{(n)}} + T^{-1} p_m p_n^{\gamma_{r,\max}^{(n)}} q_m^{-1} \Big),$$

which proves (D.45). This completes the proof. □

Lemma 6 *(Bounding $\widehat{\mathbf{D}}_{A_1,m}^{-1}$) Under the assumptions of Theorem 3, define $\omega_{A_1,m} := p_m^{\gamma_r} \sum_{n \neq m} p_n^{\gamma_r}$, and let $\widehat{\mathbf{D}}_{A_1,m}$ be defined in (C.13). Then, for each $m \in [M]$,*

$$\lambda_{r_1}(\boldsymbol{\Omega}_{A_1,m}) \asymp \omega_{A_1,m}, \qquad \lambda_{r_1}(\widehat{\boldsymbol{\Omega}}_{A_1,m}) \asymp_P \omega_{A_1,m}, \qquad \left\| \widehat{\mathbf{D}}_{A_1,m}^{-1} \right\|_F^2 = O_P(\omega_{A_1,m}^{-2}). \tag{D.55}$$

Proof of Lemma 6. By (ME1), we have $\lambda_{r_1}(\mathbf{A}_{1m}' \mathbf{A}_{1m}) \asymp \lambda_1(\mathbf{A}_{1m}' \mathbf{A}_{1m}) \asymp p_m^{\gamma_r}$. Moreover, since $\mathbf{A}_{1n}' \mathbf{A}_{1n} \succeq 0$ for every $n \neq m$,

$$\begin{aligned} \lambda_{r_1}\left(\sum_{n \neq m} \mathbf{A}_{1n}' \mathbf{A}_{1n} \right) &\ge \sum_{n \neq m} \lambda_{r_1}(\mathbf{A}_{1n}' \mathbf{A}_{1n}) \asymp \sum_{n \neq m} p_n^{\gamma_r}, \\ \lambda_1\left(\sum_{n \neq m} \mathbf{A}_{1n}' \mathbf{A}_{1n} \right) &\le \sum_{n \neq m} \lambda_1(\mathbf{A}_{1n}' \mathbf{A}_{1n}) \asymp \sum_{n \neq m} p_n^{\gamma_r}. \end{aligned}$$

By (S1), $\lambda_{r_1}\left(T^{-1}\sum_{t=1}^{T}\mathbb{E}(\boldsymbol{\alpha}_t\boldsymbol{\alpha}_t')\right) \asymp \lambda_1\left(T^{-1}\sum_{t=1}^{T}\mathbb{E}(\boldsymbol{\alpha}_t\boldsymbol{\alpha}_t')\right) \asymp 1$. Together with the preceding bounds and (ME1), we have

$$\lambda_{r_1}(\boldsymbol{\Omega}_{A_1,m}) \asymp \lambda_1(\boldsymbol{\Omega}_{A_1,m}) \asymp p_m^{\gamma_r}\sum_{n\neq m} p_n^{\gamma_r} = \omega_{A_1,m}.$$

By (C.14), Lemma 5,

$$\begin{aligned}\big\|\widehat{\boldsymbol{\Omega}}_{A_1,m} - \boldsymbol{\Omega}_{A_1,m}\big\|_F &\lesssim \sum_{n\neq m}\Big\{\big\|\boldsymbol{\Delta}_{A_1,mn}\big\|_F\big\|\mathbf{S}_{A_1,mn}\big\|_F + \big\|\boldsymbol{\Delta}_{A_1,mn}\big\|_F^2\Big\}\\ &= o_P\left(p_m^{\gamma_r}\sum_{n\neq m} p_n^{\gamma_r}\right) = o_P(\omega_{A_1,m}).\end{aligned}$$

Therefore, Weyl's inequality gives

$$\big|\lambda_{r_1}(\widehat{\boldsymbol{\Omega}}_{A_1,m}) - \lambda_{r_1}(\boldsymbol{\Omega}_{A_1,m})\big| \le \big\|\widehat{\boldsymbol{\Omega}}_{A_1,m} - \boldsymbol{\Omega}_{A_1,m}\big\| \le \big\|\widehat{\boldsymbol{\Omega}}_{A_1,m} - \boldsymbol{\Omega}_{A_1,m}\big\|_F = o_P(\omega_{A_1,m}),$$

and hence $\lambda_{r_1}(\widehat{\boldsymbol{\Omega}}_{A_1,m}) \asymp_P \omega_{A_1,m}$. Thus, $\widehat{\mathbf{D}}_{A_1,m}$ is invertible with probability tending to one and, by (C.13),

$$\big\|\widehat{\mathbf{D}}_{A_1,m}^{-1}\big\|_F^2 = \sum_{i=1}^{r_1}\lambda_i^{-2}(\widehat{\boldsymbol{\Omega}}_{A_1,m}) \lesssim \lambda_{r_1}^{-2}(\widehat{\boldsymbol{\Omega}}_{A_1,m}) = O_P(\omega_{A_1,m}^{-2}),$$

which completes the proof. □

Lemma 7 *(Bounds for $\mathbf{S}_{A_2,m}$ and $\boldsymbol{\Delta}_{A_2,m}$) For any $m \in [M]$, let $\mathbf{S}_{A_2,m}$ and $\widehat{\mathbf{S}}_{A_2,m}$ be defined in (C.19), and define $\boldsymbol{\Delta}_{A_2,m} := \widehat{\mathbf{S}}_{A_2,m} - \mathbf{S}_{A_2,m}$. Under the assumptions of Theorem 4,*

$$\big\|\mathbf{S}_{A_2,m}\big\|_F^2 = O_P(p_m^{2\gamma_r^{(m)}}), \tag{D.56}$$

$$\big\|\boldsymbol{\Delta}_{A_2,m}\big\|_F^2 = o_P(p_m^{2\gamma_r^{(m)}}). \tag{D.57}$$

Furthermore, defining $\mathbf{P}_{A_2,m} := \breve{\mathbf{A}}_{2m}\breve{\mathbf{A}}_{2m}'$, we have

$$\begin{aligned}\big\|(\mathbf{I}_{p_m} - \mathbf{P}_{A_2,m})\boldsymbol{\Delta}_{A_2,m}\big\|_F^2 = O_P\Big(&R(T,d_{r,\star})p_m^{\gamma_r+\gamma_r^{(m)}}\max_{n\in[M]\setminus\{m\}} p_n^{\gamma_r^{(n)}-\gamma_r}\\ &+ T^{-1}p_m^{\gamma_r+1}q_m^{-1}\max_{n\in[M]} p_n^{\gamma_{r,\max}^{(n)}-\gamma_r} + p_m q_m^{-2}\Big).\end{aligned} \tag{D.58}$$

Proof of Lemma 7. Recall from (C.19) that

$$\mathbf{S}_{A_2,m} := T^{-1}\sum_{t=1}^{T}\mathbf{a}_t^{(m)}\mathbf{a}_t^{(m)'}\mathbf{P}_{A_1,m}^{\perp}, \qquad \widehat{\mathbf{S}}_{A_2,m} := T^{-1}\sum_{t=1}^{T}\widehat{\mathbf{a}}_t^{(m)}\widehat{\mathbf{a}}_t^{(m)'}\widehat{\mathbf{P}}_{A_1,m}^{\perp},$$

where $\mathbf{P}_{A_1,m}^{\perp} = \mathbf{I}_{p_m} - \mathbf{P}_{A_1,m}$ and $\widehat{\mathbf{P}}_{A_1,m}^{\perp} = \mathbf{I}_{p_m} - \widehat{\mathbf{P}}_{A_1,m}$. Consequently,

$$\boldsymbol{\Delta}_{A_2,m} = T^{-1}\sum_{t=1}^{T}\mathbf{a}_t^{(m)}\mathbf{a}_t^{(m)'}\left(\widehat{\mathbf{P}}_{A_1,m}^{\perp} - \mathbf{P}_{A_1,m}^{\perp}\right) + T^{-1}\sum_{t=1}^{T}\left(\widehat{\mathbf{a}}_t^{(m)}\widehat{\mathbf{a}}_t^{(m)'} - \mathbf{a}_t^{(m)}\mathbf{a}_t^{(m)'}\right)\widehat{\mathbf{P}}_{A_1,m}^{\perp}. \tag{D.59}$$

We first prove (D.56). By (D.13) and (D.14) in Lemma 2, and (IC2), it follows that

$$T^{-1}\sum_{t=1}^{T}\mathbf{a}_t^{(m)}\mathbf{a}_t^{(m)'} = \mathbf{A}_{1m}\left(T^{-1}\sum_{t=1}^{T}\boldsymbol{\alpha}_t\boldsymbol{\alpha}_t'\right)\mathbf{A}_{1m}' + \mathbf{A}_{2m}\left(T^{-1}\sum_{t=1}^{T}\boldsymbol{\alpha}_t^{(m)}\boldsymbol{\alpha}_t^{(m)'}\right)\mathbf{A}_{2m}'. \tag{D.60}$$

Since $\mathbf{A}_{1m}'\mathbf{P}_{A_1,m}^{\perp} = \mathbf{0}$,

$$\mathbf{S}_{A_2,m} = \mathbf{A}_{2m}\left(T^{-1}\sum_{t=1}^{T}\boldsymbol{\alpha}_t^{(m)}\boldsymbol{\alpha}_t^{(m)'}\right)\mathbf{A}_{2m}'\mathbf{P}_{A_1,m}^{\perp}. \tag{D.61}$$

Therefore, by (ME1), Proposition 1, and Lemma 2,

$$\left\|\mathbf{S}_{A_2,m}\right\|_F^2 \le \left\|\mathbf{A}_{2m}\right\|_F^4\left\|T^{-1}\sum_{t=1}^{T}\boldsymbol{\alpha}_t^{(m)}\boldsymbol{\alpha}_t^{(m)'}\right\|_F^2 = O_P(p_m^{2\gamma_r^{(m)}}),$$

which proves (D.56).

We next prove (D.57) and (D.58). By (IC1), $\widehat{\mathbf{a}}_t^{(m)} - \mathbf{a}_t^{(m)} = q_m^{-1}\mathbf{M}_{p_m}\mathbf{E}_t^{(m)}\mathbf{1}_{q_m}$. Using (D.59), write

$$\boldsymbol{\Delta}_{A_2,m} = \mathcal{I}_{A_2,1} + \mathcal{I}_{A_2,2} + \mathcal{I}_{A_2,3} + \mathcal{I}_{A_2,4},$$

where

$$\mathcal{I}_{A_2,1} := T^{-1}\sum_{t=1}^{T}\mathbf{a}_t^{(m)}\mathbf{a}_t^{(m)'}\left(\widehat{\mathbf{P}}_{A_1,m}^{\perp} - \mathbf{P}_{A_1,m}^{\perp}\right), \quad \mathcal{I}_{A_2,2} := (Tq_m)^{-1}\sum_{t=1}^{T}\mathbf{a}_t^{(m)}\mathbf{1}_{q_m}'\mathbf{E}_t^{(m)'}\mathbf{M}_{p_m}\widehat{\mathbf{P}}_{A_1,m}^{\perp},$$
$$\mathcal{I}_{A_2,3} := (Tq_m)^{-1}\sum_{t=1}^{T}\mathbf{M}_{p_m}\mathbf{E}_t^{(m)}\mathbf{1}_{q_m}\mathbf{a}_t^{(m)'}\widehat{\mathbf{P}}_{A_1,m}^{\perp}, \ \mathcal{I}_{A_2,4} := (Tq_m^2)^{-1}\sum_{t=1}^{T}\mathbf{M}_{p_m}\mathbf{E}_t^{(m)}\mathbf{1}_{q_m}\mathbf{1}_{q_m}'\mathbf{E}_t^{(m)'}\mathbf{M}_{p_m}\widehat{\mathbf{P}}_{A_1,m}^{\perp}.$$

We bound these four terms separately.

By Theorem 3 and (C.18),

$$\left\|\widehat{\mathbf{P}}_{A_1,m} - \mathbf{P}_{A_1,m}\right\|_F^2 = O_P\Big(R(T,d_{r,\star})p_m^{\gamma_r^{(m)}-\gamma_r}\max_{n\in[M]\setminus\{m\}}p_n^{\gamma_r^{(n)}-\gamma_r} + T^{-1}p_m^{1-\gamma_r}q_m^{-1}\max_{n\in[M]\setminus\{m\}}p_n^{\gamma_{r,\max}^{(n)}-\gamma_r}\Big). \tag{D.62}$$

For $\mathcal{I}_{A_2,1}$, since $\widehat{\mathbf{P}}^{\perp}_{A_1,m} - \mathbf{P}^{\perp}_{A_1,m} = \mathbf{P}_{A_1,m} - \widehat{\mathbf{P}}_{A_1,m}$, together with (D.60) and (ME1), we have

$$\left\|\mathcal{I}_{A_2,1}\right\|_F^2 = O_P\left((p_m^{2\gamma_r} + p_m^{2\gamma_r^{(m)}})\left\|\widehat{\mathbf{P}}_{A_1,m} - \mathbf{P}_{A_1,m}\right\|_F^2\right). \tag{D.63}$$

Furthermore, since $(\mathbf{I}_{p_m} - \mathbf{P}_{A_2,m})\mathbf{A}_{2m} = \mathbf{0}$,

$$(\mathbf{I}_{p_m} - \mathbf{P}_{A_2,m})\mathcal{I}_{A_2,1} = (\mathbf{I}_{p_m} - \mathbf{P}_{A_2,m})\mathbf{A}_{1m}\left(T^{-1}\sum_{t=1}^{T}\boldsymbol{\alpha}_t\boldsymbol{\alpha}_t'\right)\mathbf{A}_{1m}'(\mathbf{P}_{A_1,m} - \widehat{\mathbf{P}}_{A_1,m}).$$

Consequently,

$$\left\|(\mathbf{I}_{p_m} - \mathbf{P}_{A_2,m})\mathcal{I}_{A_2,1}\right\|_F^2 = O_P\left(p_m^{2\gamma_r}\left\|\widehat{\mathbf{P}}_{A_1,m} - \mathbf{P}_{A_1,m}\right\|_F^2\right). \tag{D.64}$$

For $\mathcal{I}_{A_2,2}$, write

$$\begin{aligned}\mathcal{I}_{A_2,2} = {} & \mathbf{A}_{1m}\left\{(Tq_m)^{-1}\sum_{t=1}^{T}\boldsymbol{\alpha}_t\mathbf{1}_{q_m}'\mathbf{E}_t^{(m)'}\mathbf{M}_{p_m}\right\}\widehat{\mathbf{P}}^{\perp}_{A_1,m} \\ & + \mathbf{A}_{2m}\left\{(Tq_m)^{-1}\sum_{t=1}^{T}\boldsymbol{\alpha}_t^{(m)}\mathbf{1}_{q_m}'\mathbf{E}_t^{(m)'}\mathbf{M}_{p_m}\right\}\widehat{\mathbf{P}}^{\perp}_{A_1,m}.\end{aligned}$$

Applying (D.33) in Lemma 3, together with (ME1) and Proposition 1, gives

$$\left\|\mathcal{I}_{A_2,2}\right\|_F^2 = O_P\left(T^{-1}p_m q_m^{-1}(p_m^{\gamma_r} + p_m^{\gamma_r^{(m)}})\right). \tag{D.65}$$

Since left-multiplication by $\mathbf{I}_{p_m} - \mathbf{P}_{A_2,m}$ eliminates the term involving $\mathbf{A}_{2m}$,

$$\left\|(\mathbf{I}_{p_m} - \mathbf{P}_{A_2,m})\mathcal{I}_{A_2,2}\right\|_F^2 = O_P\left(T^{-1}p_m^{\gamma_r+1}q_m^{-1}\right). \tag{D.66}$$

For $\mathcal{I}_{A_2,3}$, we have

$$\begin{aligned}\mathcal{I}_{A_2,3} = {} & q_m^{-1}\left\{T^{-1}\sum_{t=1}^{T}\mathbf{M}_{p_m}\mathbf{E}_t^{(m)}\mathbf{1}_{q_m}\boldsymbol{\alpha}_t'\right\}\mathbf{A}_{1m}'\widehat{\mathbf{P}}^{\perp}_{A_1,m} \\ & + q_m^{-1}\left\{T^{-1}\sum_{t=1}^{T}\mathbf{M}_{p_m}\mathbf{E}_t^{(m)}\mathbf{1}_{q_m}\boldsymbol{\alpha}_t^{(m)'}\right\}\mathbf{A}_{2m}'\widehat{\mathbf{P}}^{\perp}_{A_1,m}.\end{aligned}$$

By (D.33) in Lemma 3, (ME1), and Proposition 1,

$$\left\|\mathcal{I}_{A_2,3}\right\|_F^2 = O_P\left(T^{-1}p_m q_m^{-1}(p_m^{\gamma_r} + p_m^{\gamma_r^{(m)}})\right) = O_P\left(T^{-1}p_m^{\gamma_{r,\max}^{(m)}+1}q_m^{-1}\right). \tag{D.67}$$

The same bound remains valid after left-multiplication by $\mathbf{I}_{p_m} - \mathbf{P}_{A_2,m}$, and hence

$$\left\|(\mathbf{I}_{p_m} - \mathbf{P}_{A_2,m})\mathcal{I}_{A_2,3}\right\|_F^2 = O_P\left(T^{-1}p_m^{\gamma_{r,\max}^{(m)}+1}q_m^{-1}\right). \tag{D.68}$$

For $\mathcal{I}_{A_2,4}$, by (D.35) in Lemma 3 and Proposition 1,

$$\left\|\mathcal{I}_{A_2,4}\right\|_F^2 = O_P\left(p_m q_m^{-2} + T^{-1}p_m^2 q_m^{-2}\right), \tag{D.69}$$

$$\left\|(\mathbf{I}_{p_m} - \mathbf{P}_{A_2,m})\mathcal{I}_{A_2,4}\right\|_F^2 = O_P\left(p_m q_m^{-2} + T^{-1}p_m^2 q_m^{-2}\right). \tag{D.70}$$

Using $\left\|\sum_{j=1}^4 \mathcal{I}_{A_2,j}\right\|_F^2 \lesssim \sum_{j=1}^4 \left\|\mathcal{I}_{A_2,j}\right\|_F^2$, together with (D.63), (D.65), (D.67), and (D.69), we obtain

$$\begin{aligned}\left\|\mathbf{\Delta}_{A_2,m}\right\|_F^2 = O_P\Big((p_m^{2\gamma_r} + p_m^{2\gamma_r^{(m)}})\left\|\widehat{\mathbf{P}}_{A_1,m} - \mathbf{P}_{A_1,m}\right\|_F^2 \\ + T^{-1}p_m^{\gamma_{r,\max}^{(m)}+1}q_m^{-1} + p_m q_m^{-2} + T^{-1}p_m^2 q_m^{-2}\Big).\end{aligned} \tag{D.71}$$

To deduce (D.57), Theorem 3 and (C.18) give

$$\begin{aligned}\left(1 + \nu_{r,m}^{-2}\right)\left\|\widehat{\mathbf{P}}_{A_1,m} - \mathbf{P}_{A_1,m}\right\|_F^2 = O_P\Bigg(R(T, d_{r,\star})(\nu_{r,m} + \nu_{r,m}^{-1}) \max_{n\in[M]\setminus\{m\}} \nu_{r,n} \\ + \frac{1}{T}\left\{\frac{p_m^{1-\gamma_r}}{q_m} + \frac{p_m^{1-\gamma_r^{(m)}}}{q_m}\nu_{r,m}^{-1}\right\}\left(1 \vee \max_{n\in[M]\setminus\{m\}} \nu_{r,n}\right)\Bigg) = o_P(1),\end{aligned}$$

where the last equality follows from (R1) and $T^{-1} = O(R(T, d_{r,\star}))$. Moreover,

$$T^{-1}p_m^{\gamma_{r,\max}^{(m)}+1-2\gamma_r^{(m)}}q_m^{-1} = T^{-1}p_m^{1-\gamma_r^{(m)}}q_m^{-1}(1 \vee \nu_{r,m}^{-1}) = o(1),$$

while the remaining two terms in (D.71), after division by $p_m^{2\gamma_r^{(m)}}$, are also $o(1)$ by (R1). Hence (D.57) follows.

Finally, combining (D.64), (D.66), (D.68), and (D.70) with (D.62), we obtain

$$\begin{aligned}\left\|(\mathbf{I}_{p_m} - \mathbf{P}_{A_2,m})\mathbf{\Delta}_{A_2,m}\right\|_F^2 = O_P\Big(R(T, d_{r,\star})p_m^{\gamma_r+\gamma_r^{(m)}} \max_{n\in[M]\setminus\{m\}} p_n^{\gamma_r^{(n)}-\gamma_r} \\ + T^{-1}p_m^{\gamma_r+1}q_m^{-1} \max_{n\in[M]} p_n^{\gamma_{r,\max}^{(n)}-\gamma_r} + p_m q_m^{-2}\Big),\end{aligned}$$

where $T^{-1}p_m^2 q_m^{-2}$ is absorbed into the second term under (R1). This proves (D.58), and completes the proof. □

Lemma 8 *(Bounding $\widehat{\mathbf{D}}_{A_2,m}^{-1}$) Under the assumptions of Theorem 4, define $\omega_{A_2,m} := p_m^{2\gamma_r^{(m)}}$, and let $\widehat{\mathbf{D}}_{A_2,m}$*

be defined in (C.23). *Then, for each* $m \in [M]$,

$$\lambda_{r_2^{(m)}}(\mathbf{\Omega}_{A_2,m}) \asymp_P \omega_{A_2,m}, \qquad \lambda_{r_2^{(m)}}(\widehat{\mathbf{\Omega}}_{A_2,m}) \asymp_P \omega_{A_2,m}, \qquad \left\|\widehat{\mathbf{D}}^{-1}_{A_2,m}\right\|_F^2 = O_P(\omega^{-2}_{A_2,m}). \tag{D.72}$$

Proof of Lemma 8. By Lemma 2, (S1), and Weyl's inequality,

$$\lambda_{r_2^{(m)}}\left(T^{-1}\sum_{t=1}^{T}\boldsymbol{\alpha}_t^{(m)}\boldsymbol{\alpha}_t^{(m)'}\right) \asymp_P \lambda_1\left(T^{-1}\sum_{t=1}^{T}\boldsymbol{\alpha}_t^{(m)}\boldsymbol{\alpha}_t^{(m)'}\right) \asymp_P 1.$$

Since $\mathbf{P}^{\perp}_{A_1,m} = \mathbf{A}^{\perp}_{1m}\mathbf{A}^{\perp'}_{1m}$,

$$\begin{aligned}&(\mathbf{A}'_{2m}\mathbf{A}_{2m})^{-1/2}\mathbf{A}'_{2m}\mathbf{P}^{\perp}_{A_1,m}\mathbf{A}_{2m}(\mathbf{A}'_{2m}\mathbf{A}_{2m})^{-1/2}\\ &= \left\{\mathbf{A}^{\perp'}_{1m}\mathbf{A}_{2m}(\mathbf{A}'_{2m}\mathbf{A}_{2m})^{-1/2}\right\}'\left\{\mathbf{A}^{\perp'}_{1m}\mathbf{A}_{2m}(\mathbf{A}'_{2m}\mathbf{A}_{2m})^{-1/2}\right\}.\end{aligned}$$

By (S1), (ME1), and Proposition 1, this gives

$$\lambda_{r_2^{(m)}}(\mathbf{A}'_{2m}\mathbf{P}^{\perp}_{A_1,m}\mathbf{A}_{2m}) \asymp \lambda_1(\mathbf{A}'_{2m}\mathbf{P}^{\perp}_{A_1,m}\mathbf{A}_{2m}) \asymp p_m^{\gamma_r^{(m)}}.$$

Therefore, (ME1) gives

$$\lambda_{r_2^{(m)}}(\mathbf{\Omega}_{A_2,m}) \asymp_P \lambda_1(\mathbf{\Omega}_{A_2,m}) \asymp_P p_m^{2\gamma_r^{(m)}} = \omega_{A_2,m}.$$

Let $\mathbf{\Delta}_{A_2,m} := \widehat{\mathbf{S}}_{A_2,m} - \mathbf{S}_{A_2,m}$. Since $\widehat{\mathbf{\Omega}}_{A_2,m} = \widehat{\mathbf{S}}_{A_2,m}\widehat{\mathbf{S}}'_{A_2,m}$ and $\mathbf{\Omega}_{A_2,m} = \mathbf{S}_{A_2,m}\mathbf{S}'_{A_2,m}$,

$$\widehat{\mathbf{\Omega}}_{A_2,m} - \mathbf{\Omega}_{A_2,m} = \mathbf{\Delta}_{A_2,m}\mathbf{S}'_{A_2,m} + \mathbf{S}_{A_2,m}\mathbf{\Delta}'_{A_2,m} + \mathbf{\Delta}_{A_2,m}\mathbf{\Delta}'_{A_2,m}.$$

Lemma 7 gives $\left\|\mathbf{S}_{A_2,m}\right\|_F = O_P(p_m^{\gamma_r^{(m)}})$ and $\left\|\mathbf{\Delta}_{A_2,m}\right\|_F = o_P(p_m^{\gamma_r^{(m)}})$. Therefore,

$$\left\|\widehat{\mathbf{\Omega}}_{A_2,m} - \mathbf{\Omega}_{A_2,m}\right\|_F \lesssim \left\|\mathbf{\Delta}_{A_2,m}\right\|_F\left\|\mathbf{S}_{A_2,m}\right\|_F + \left\|\mathbf{\Delta}_{A_2,m}\right\|_F^2 = o_P(p_m^{2\gamma_r^{(m)}}) = o_P(\omega_{A_2,m}).$$

By Weyl's inequality,

$$\left|\lambda_{r_2^{(m)}}(\widehat{\mathbf{\Omega}}_{A_2,m}) - \lambda_{r_2^{(m)}}(\mathbf{\Omega}_{A_2,m})\right| \le \left\|\widehat{\mathbf{\Omega}}_{A_2,m} - \mathbf{\Omega}_{A_2,m}\right\|_F = o_P(\omega_{A_2,m}),$$

so that $\lambda_{r_2^{(m)}}(\widehat{\mathbf{\Omega}}_{A_2,m}) \asymp_P \omega_{A_2,m}$. Thus, $\widehat{\mathbf{D}}_{A_2,m}$ is invertible with probability tending to one and, by (C.23),

$$\left\|\widehat{\mathbf{D}}_{A_2,m}^{-1}\right\|_F^2 = \sum_{i=1}^{r_2^{(m)}} \lambda_i^{-2}(\widehat{\mathbf{\Omega}}_{A_2,m}) \leq r_2^{(m)} \lambda_{r_2^{(m)}}^{-2}(\widehat{\mathbf{\Omega}}_{A_2,m}) = O_P(\omega_{A_2,m}^{-2}),$$

which completes the proof. □

Lemma 9 *(Properties of $\mathbf{\Omega}_{Q,m}$) For any $m, n \in [M]$ with $n \neq m$, define*

$$\mathbf{\Omega}_{Q,m} := \frac{1}{M-1}\sum_{n=1;n\neq m}^{M} \mathbf{S}_{Q,mn}^{\otimes 2}, \quad \mathbf{S}_{Q,mn} := \frac{1}{T}\sum_{t=1}^{T}\mathbb{E}\left(\mathbf{Q}_m\mathbf{G}_t\mathbf{J}_m'\mathbf{v}_b^{(m)}\mathbf{v}_b^{(n)'}\mathbf{J}_n\mathbf{G}_t'\mathbf{Q}_n'\right)$$
$$\widehat{\mathbf{S}}_{Q,mn} := \frac{1}{T}\sum_{t=1}^{T}\breve{\mathbf{Y}}_t^{(m)}\mathbf{v}_b^{(m)}\mathbf{v}_b^{(n)'}\breve{\mathbf{Y}}_t^{(n)'}, \quad \mathbf{\Delta}_{Q,mn} := \widehat{\mathbf{S}}_{Q,mn} - \mathbf{S}_{Q,mn}.$$

Then under Assumptions (IC1), (E1), (E2), (F1), and (L1), we have

$$\begin{aligned}\|\mathbf{\Delta}_{Q,mn}\|_F^2 \lesssim_P\ & p_m^{\delta_{q,1}^{(m)}} p_n^{\delta_{q,1}^{(n)}} s_{J,m}^2 s_{J,n}^2 R(T, d_{fac,\star}) + p_m^{\delta_{r,1}^{(m)}} p_n^{\delta_{q,1}^{(n)}} s_{J,n}^2 q_m^{\delta_{c,1}^{(m)}} R(T, d_{fac,\star}) \\ & + p_m^{\delta_{q,1}^{(m)}} p_n^{\delta_{r,1}^{(n)}} s_{J,m}^2 q_n^{\delta_{c,1}^{(n)}} R(T, d_{fac,\star}) + p_m^{\delta_{r,1}^{(m)}} p_n^{\delta_{r,1}^{(n)}} q_m^{\delta_{c,1}^{(m)}} q_n^{\delta_{c,1}^{(n)}} R(T, d_{fac,\star}) + T^{-1} p_m p_n^{\delta_{q,1}^{(n)}} s_{J,n}^2 \\ & + T^{-1} p_m p_n^{\delta_{r,1}^{(n)}} q_n^{\delta_{c,1}^{(n)}} + T^{-1} p_m^{\delta_{q,1}^{(m)}} s_{J,m}^2 p_n + T^{-1} p_m^{\delta_{r,1}^{(m)}} q_m^{\delta_{c,1}^{(m)}} p_n + T^{-1} p_m p_n.\end{aligned}$$

Further define $\breve{\mathbf{Q}}_m := \mathbf{Q}_m(\mathbf{Q}_m'\mathbf{Q}_m)^{-1/2}$. Then under Assumptions (L2) and (R2), there exists an asymptotically orthogonal matrix $\mathbf{H}_{Q,m}$ with $\mathbf{H}_{Q,m}\mathbf{H}_{Q,m}' = \mathbf{I} + o_P(1)$, such that

$$\left\|\widehat{\mathbf{Q}}_m - \breve{\mathbf{Q}}_m\mathbf{H}_{Q,m}\right\|^2 = O_P\left\{R(T, d_{fac,\star})\ p_m^{2(\delta_{q,1}^{(m)} - \delta_{q,k_r}^{(m)})}\left(\frac{\max_{n\in[M]\setminus\{m\}} s_{J,n}^2 p_n^{\delta_{q,1}^{(n)}}}{\max_{n\in[M]\setminus\{m\}} s_{J,n}^2 p_n^{\delta_{q,k_r}^{(n)}}}\right)^2\right\}.$$

Analogous results hold for $\widehat{\mathbf{J}}_m$, with asymptotically orthogonal matrix $\mathbf{H}_{J,m}$.

Proof of Lemma 9. By Assumption (F1), we note that $T^{-1}\sum_{t=1}^{T}\mathbb{E}[\mathbf{z}_{G,t}^{\otimes 2}]$ and $T^{-1}\sum_{t=1}^{T}\mathbb{E}[(\mathbf{z}_{F,t}^{(m)})^{\otimes 2}]$ are diagonal due to the independence between cross-sectional entries. Without loss of generality, we further normalize these long-run covariance matrices to be the identity matrix. Namely, we will assume throughout the proof that

$$\frac{1}{T}\sum_{t=1}^{T}\mathbb{E}[\mathbf{z}_{G,t}^{\otimes 2}] = \mathbf{I}_{k_r k_c}, \quad \frac{1}{T}\sum_{t=1}^{T}\mathbb{E}[(\mathbf{z}_{F,t}^{(m)})^{\otimes 2}] = \mathbf{I}_{k_r^{(m)} k_c^{(m)}}. \tag{D.73}$$

This is equivalent to absorbing the entry-wise variance into $\mathbf{\Sigma}_{Q,m}$, $\mathbf{\Sigma}_{J,m}$, $\mathbf{\Sigma}_{R,m}$ and $\mathbf{\Sigma}_{C,m}$ (from Assumption (L1)) which are arbitrary positive-definite matrices with eigenvalues uniformly bounded away from 0 and infinity. Then for any group $m \in [M]$, by definitions we have

$$\begin{aligned}&\breve{\mathbf{Y}}_t^{(m)} = \mathbf{Y}_t^{(m)} - \widehat{c}_t^{(m)}\mathbf{1}_{p_m}\mathbf{1}_{q_m}' - \widehat{\mathbf{a}}_t^{(m)}\mathbf{1}_{q_m}' - \mathbf{1}_{p_m}\widehat{\mathbf{b}}_t^{(m)'} \\ &= \mathbf{Y}_t^{(m)} - (p_m q_m)^{-1}\mathbf{1}_{p_m}\mathbf{1}_{p_m}'\mathbf{Y}_t^{(m)}\mathbf{1}_{q_m}\mathbf{1}_{q_m}' - q_m^{-1}\mathbf{M}_{p_m}\mathbf{Y}_t^{(m)}\mathbf{1}_{q_m}\mathbf{1}_{q_m}' - p_m^{-1}\mathbf{1}_{p_m}\mathbf{1}_{p_m}'\mathbf{Y}_t^{(m)}\mathbf{M}_{q_m} \\ &= \mathbf{M}_{p_m}\mathbf{Y}_t^{(m)}\mathbf{M}_{q_m} + p_m^{-1}\mathbf{1}_{p_m}\mathbf{1}_{p_m}'\mathbf{Y}_t^{(m)}\mathbf{M}_{q_m} + q_m^{-1}\mathbf{Y}_t^{(m)}\mathbf{1}_{q_m}\mathbf{1}_{q_m}' - (p_m q_m)^{-1}\mathbf{1}_{p_m}\mathbf{1}_{p_m}'\mathbf{Y}_t^{(m)}\mathbf{1}_{q_m}\mathbf{1}_{q_m}' \\ &\quad - q_m^{-1}\mathbf{M}_{p_m}\mathbf{Y}_t^{(m)}\mathbf{1}_{q_m}\mathbf{1}_{q_m}' - p_m^{-1}\mathbf{1}_{p_m}\mathbf{1}_{p_m}'\mathbf{Y}_t^{(m)}\mathbf{M}_{q_m} = \mathbf{M}_{p_m}\mathbf{Y}_t^{(m)}\mathbf{M}_{q_m}.\end{aligned}$$

Together with our model (3.1), Assumption (IC1), and the fact that $\mathbf{M}_a \mathbf{1}_a = \mathbf{0}$ for any positive integer a, we further arrive at

$$\begin{aligned}\check{\mathbf{Y}}_t^{(m)} &= \mathbf{M}_{p_m} \mathbf{Y}_t^{(m)} \mathbf{M}_{q_m} = \mathbf{M}_{p_m} \mathbf{Q}_m \mathbf{G}_t \mathbf{J}_m' \mathbf{M}_{q_m} + \mathbf{M}_{p_m} \mathbf{R}_m \mathbf{F}_t^{(m)} \mathbf{C}_m' \mathbf{M}_{q_m} + \mathbf{M}_{p_m} \mathbf{E}_t^{(m)} \mathbf{M}_{q_m} \\ &= \mathbf{Q}_m \mathbf{G}_t \mathbf{J}_m' + \mathbf{R}_m \mathbf{F}_t^{(m)} \mathbf{C}_m' + \mathbf{M}_{p_m} \mathbf{E}_t^{(m)} \mathbf{M}_{q_m}. \end{aligned} \tag{D.74}$$

Then it holds that

$$\begin{aligned}&\widehat{\mathbf{S}}_{Q,mn} \\ &= \frac{1}{T} \sum_{t=1}^{T} \left(\mathbf{Q}_m \mathbf{G}_t \mathbf{J}_m' \mathbf{v}_b^{(m)} + \mathbf{R}_m \mathbf{F}_t^{(m)} \mathbf{C}_m' \mathbf{v}_b^{(m)} + \mathbf{M}_{p_m} \mathbf{E}_t^{(m)} \mathbf{M}_{q_m} \mathbf{v}_b^{(m)} \right) \\ &\quad \cdot \left(\mathbf{v}_b^{(n)'} \mathbf{J}_n \mathbf{G}_t' \mathbf{Q}_n' + \mathbf{v}_b^{(n)'} \mathbf{C}_n \mathbf{F}_t^{(n)'} \mathbf{R}_n' + \mathbf{v}_b^{(n)'} \mathbf{M}_{q_n} \mathbf{E}_t^{(n)'} \mathbf{M}_{p_n} \right) \\ &= \frac{1}{T} \sum_{t=1}^{T} \mathbf{Q}_m \mathbf{G}_t \mathbf{J}_m' \mathbf{v}_b^{(m)} \mathbf{v}_b^{(n)'} \mathbf{J}_n \mathbf{G}_t' \mathbf{Q}_n' + \frac{1}{T} \sum_{t=1}^{T} \mathbf{R}_m \mathbf{F}_t^{(m)} \mathbf{C}_m' \mathbf{v}_b^{(m)} \mathbf{v}_b^{(n)'} \mathbf{J}_n \mathbf{G}_t' \mathbf{Q}_n' \\ &\quad + \frac{1}{T} \sum_{t=1}^{T} \mathbf{M}_{p_m} \mathbf{E}_t^{(m)} \mathbf{M}_{q_m} \mathbf{v}_b^{(m)} \mathbf{v}_b^{(n)'} \mathbf{J}_n \mathbf{G}_t' \mathbf{Q}_n' + \frac{1}{T} \sum_{t=1}^{T} \mathbf{Q}_m \mathbf{G}_t \mathbf{J}_m' \mathbf{v}_b^{(m)} \mathbf{v}_b^{(n)'} \mathbf{C}_n \mathbf{F}_t^{(n)'} \mathbf{R}_n' \\ &\quad + \frac{1}{T} \sum_{t=1}^{T} \mathbf{R}_m \mathbf{F}_t^{(m)} \mathbf{C}_m' \mathbf{v}_b^{(m)} \mathbf{v}_b^{(n)'} \mathbf{C}_n \mathbf{F}_t^{(n)'} \mathbf{R}_n' + \frac{1}{T} \sum_{t=1}^{T} \mathbf{M}_{p_m} \mathbf{E}_t^{(m)} \mathbf{M}_{q_m} \mathbf{v}_b^{(m)} \mathbf{v}_b^{(n)'} \mathbf{C}_n \mathbf{F}_t^{(n)'} \mathbf{R}_n' \\ &\quad + \frac{1}{T} \sum_{t=1}^{T} \mathbf{Q}_m \mathbf{G}_t \mathbf{J}_m' \mathbf{v}_b^{(m)} \mathbf{v}_b^{(n)'} \mathbf{M}_{q_n} \mathbf{E}_t^{(n)'} \mathbf{M}_{p_n} + \frac{1}{T} \sum_{t=1}^{T} \mathbf{R}_m \mathbf{F}_t^{(m)} \mathbf{C}_m' \mathbf{v}_b^{(m)} \mathbf{v}_b^{(n)'} \mathbf{M}_{q_n} \mathbf{E}_t^{(n)'} \mathbf{M}_{p_n} \\ &\quad + \frac{1}{T} \sum_{t=1}^{T} \mathbf{M}_{p_m} \mathbf{E}_t^{(m)} \mathbf{M}_{q_m} \mathbf{v}_b^{(m)} \mathbf{v}_b^{(n)'} \mathbf{M}_{q_n} \mathbf{E}_t^{(n)'} \mathbf{M}_{p_n} \\ &=: \mathcal{I}_{Q,mn} + \mathcal{II}_{Q,mn} + \mathcal{III}_{Q,mn} + \mathcal{IV}_{Q,mn} + \mathcal{V}_{Q,mn} + \mathcal{VI}_{Q,mn} + \mathcal{VII}_{Q,mn} + \mathcal{VIII}_{Q,mn} + \mathcal{IX}_{Q,mn}. \end{aligned} \tag{D.75}$$

Consider first $\mathcal{I}_{Q,mn} - \mathbf{S}_{Q,mn}$. Define $\mathbf{z}_{G,t} := \mathbf{vec}(\mathbf{G}_t)$ for any $t \in [T]$. As the dimensions of $\mathbf{G}_t$ are bounded, Lemma 2 implies that

$$\left\| \frac{1}{T} \sum_{t=1}^{T} \left\{ \mathbf{z}_{G,t} \mathbf{z}_{G,t}' - \mathbb{E}\left[\mathbf{z}_{G,t} \mathbf{z}_{G,t}' \right] \right\} \right\|_F^2 = O_P(R(T, d_{\text{fac},\star})), \tag{D.76}$$

where $R(\cdot, \cdot)$ is defined in (4.1) and $d_{\text{fac},\star}$ is the largest long-memory exponent among all processes in $\{\mathbf{G}_t\}$ and $\{\mathbf{F}_t^{(m)}\}$ for $m \in [M]$. Then from Assumption (L1), by the triangle inequality we have

$$\left\| \mathcal{I}_{Q,mn} - \mathbf{S}_{Q,mn} \right\|_F^2 = \left\| \mathbf{Q}_m \frac{1}{T} \sum_{t=1}^{T} \left(\mathbf{G}_t \mathbf{J}_m' \mathbf{v}_b^{(m)} \mathbf{v}_b^{(n)'} \mathbf{J}_n \mathbf{G}_t' - \mathbb{E}\left[\mathbf{G}_t \mathbf{J}_m' \mathbf{v}_b^{(m)} \mathbf{v}_b^{(n)'} \mathbf{J}_n \mathbf{G}_t' \right] \right) \mathbf{Q}_n' \right\|_F^2$$

$$
\begin{aligned}
&\lesssim \frac{1}{T^2}\cdot p_m^{\delta_{q,1}^{(m)}}\cdot p_n^{\delta_{q,1}^{(n)}}\cdot \left\|\sum_{t=1}^{T}\left(\mathbf{G}_t\mathbf{J}_m'\mathbf{v}_b^{(m)}\mathbf{v}_b^{(n)'}\mathbf{J}_n\mathbf{G}_t' - \mathbb{E}\left[\mathbf{G}_t\mathbf{J}_m'\mathbf{v}_b^{(m)}\mathbf{v}_b^{(n)'}\mathbf{J}_n\mathbf{G}_t'\right]\right)\right\|_F^2 \\
&= \frac{1}{T^2}\cdot p_m^{\delta_{q,1}^{(m)}}\cdot p_n^{\delta_{q,1}^{(n)}}\cdot \left\|\sum_{t=1}^{T}\left\{(\mathbf{G}_t\otimes\mathbf{G}_t)\mathbf{vec}(\mathbf{J}_m'\mathbf{v}_b^{(m)}\mathbf{v}_b^{(n)'}\mathbf{J}_n) - \mathbb{E}\left[(\mathbf{G}_t\otimes\mathbf{G}_t)\mathbf{vec}(\mathbf{J}_m'\mathbf{v}_b^{(m)}\mathbf{v}_b^{(n)'}\mathbf{J}_n)\right]\right\}\right\|_F^2 \\
&\le \frac{1}{T^2}\cdot p_m^{\delta_{q,1}^{(m)}}\cdot p_n^{\delta_{q,1}^{(n)}}\cdot \left\|\sum_{t=1}^{T}\{\mathbf{G}_t\otimes\mathbf{G}_t - \mathbb{E}[\mathbf{G}_t\otimes\mathbf{G}_t]\}\right\|_F^2\cdot\left\|\mathbf{J}_m'\mathbf{v}_b^{(m)}\mathbf{v}_b^{(n)'}\mathbf{J}_n\right\|_F^2 \\
&\asymp p_m^{\delta_{q,1}^{(m)}}\cdot p_n^{\delta_{q,1}^{(n)}}\cdot \left\|\frac{1}{T}\sum_{t=1}^{T}\left\{\mathbf{z}_{G,t}\mathbf{z}_{G,t}' - \mathbb{E}\left[\mathbf{z}_{G,t}\mathbf{z}_{G,t}'\right]\right\}\right\|_F^2\cdot\left\|\mathbf{J}_m'\mathbf{v}_b^{(m)}\mathbf{v}_b^{(n)'}\mathbf{J}_n\right\|_F^2 \\
&\lesssim_P p_m^{\delta_{q,1}^{(m)}}\cdot p_n^{\delta_{q,1}^{(n)}}\cdot R(T,d_{\mathrm{fac},\star})\cdot\left\|\mathbf{J}_m'\mathbf{v}_b^{(m)}\mathbf{v}_b^{(n)'}\mathbf{J}_n\right\|_F^2 \\
&\lesssim_P p_m^{\delta_{q,1}^{(m)}}\cdot p_n^{\delta_{q,1}^{(n)}}\cdot s_{J,m}^2\cdot s_{J,n}^2\cdot R(T,d_{\mathrm{fac},\star}),
\end{aligned}
\tag{D.77}
$$

where the first equivalence follows from the fact that the Frobenius norm of $\mathbf{G}_t\otimes\mathbf{G}_t$ equals that of $\mathbf{z}_{G,t}\mathbf{z}_{G,t}'$ under a fixed permutation of their entries, the second last inequality used (D.76), and the last used the definitions of $s_{J,m}$ and $s_{J,n}$.

By Assumption (L1), analogous arguments in (D.77), and the fact $\mathbf{v}_b^{(m)}$ and $\mathbf{v}_b^{(n)}$ are unit vectors by construction, we may also obtain

$$
\|\mathcal{II}_{Q,mn}\|_F^2 \lesssim_P p_m^{\delta_{r,1}^{(m)}}\cdot p_n^{\delta_{q,1}^{(n)}}\cdot s_{J,n}^2\cdot q_m^{\delta_{c,1}^{(m)}}\cdot R(T,d_{\mathrm{fac},\star}), \tag{D.78}
$$

$$
\|\mathcal{IV}_{Q,mn}\|_F^2 \lesssim_P p_m^{\delta_{q,1}^{(m)}}\cdot p_n^{\delta_{r,1}^{(n)}}\cdot s_{J,m}^2\cdot q_n^{\delta_{c,1}^{(n)}}\cdot R(T,d_{\mathrm{fac},\star}), \tag{D.79}
$$

$$
\|\mathcal{V}_{Q,mn}\|_F^2 \lesssim_P p_m^{\delta_{r,1}^{(m)}}\cdot p_n^{\delta_{r,1}^{(n)}}\cdot q_m^{\delta_{c,1}^{(m)}}\cdot q_n^{\delta_{c,1}^{(n)}}\cdot R(T,d_{\mathrm{fac},\star}). \tag{D.80}
$$

We next bound the noise-involved terms in (D.75), i.e., $\mathcal{III}_{Q,mn}$, $\mathcal{VI}_{Q,mn}$, $\mathcal{VII}_{Q,mn}$, $\mathcal{VIII}_{Q,mn}$, and $\mathcal{IX}_{Q,mn}$. For each $m\in[M]$, define $a_m := \mathbf{M}_{q_m}\mathbf{v}_b^{(m)}$ and $b_m := \mathbf{J}_m'\mathbf{v}_b^{(m)}$. By construction, $\|a_m\|\le 1$ and $\|b_m\| = s_{J,m}$. We first present a simple but useful bound. For any deterministic vector $a\in\mathbb{R}^{q_m}$ with $\|a\|\le 1$, Lemma 1 implies that, uniformly in t,

$$
\mathbb{E}\left\|\mathbf{M}_{p_m}\mathbf{E}_t^{(m)}a\right\|^2 = O(p_m), \tag{D.81}
$$

which used that for each row $i\in[p_m]$, the covariance matrix of the row vector $(\mathbf{E}_t^{(m)})_{i\cdot}$ has row-wise ℓ_1-norms bounded by a constant under Lemma 1, and as the covariance matrix is symmetric and positive semidefinite, its spectral norm is bounded by the same constant, and thus the above bound follows by summing over $i\in[p_m]$. Also, by Assumptions (F1) and (L1), we have

$$
\mathbb{E}\left\|\mathbf{Q}_n\mathbf{G}_t b_n\right\|^2 \lesssim p_n^{\delta_{q,1}^{(n)}}s_{J,n}^2, \qquad \mathbb{E}\left\|\mathbf{R}_n\mathbf{F}_t^{(n)}\mathbf{C}_n'\mathbf{v}_b^{(n)}\right\|^2 \lesssim p_n^{\delta_{r,1}^{(n)}}q_n^{\delta_{c,1}^{(n)}}. \tag{D.82}
$$

Consider $\mathcal{III}_{Q,mn}$. For neatness, define $u_{n,t} := \mathbf{Q}_n \mathbf{G}_t b_n$ and $e_{m,t} := \mathbf{M}_{p_m} \mathbf{E}_t^{(m)} a_m$. Then we can write $\mathcal{III}_{Q,mn} = T^{-1} \sum_{t=1}^{T} e_{m,t} u'_{n,t}$. By Assumption (E2), the processes $\{e_{m,t}\}_{t\in[T]}$ and $\{u_{n,t}\}_{t\in[T]}$ are independent. Hence we have

$$\mathbb{E}\big\|\mathcal{III}_{Q,mn}\big\|_F^2 = \frac{1}{T^2} \sum_{t=1}^{T} \sum_{s=1}^{T} \mathbb{E}\Big(u'_{n,t} u_{n,s}\Big) \mathbb{E}\Big(e'_{m,t} e_{m,s}\Big). \tag{D.83}$$

We bound the two expectations on the right-hand side separately. First, by the Cauchy–Schwarz inequality and (D.82), we have

$$\Big|\mathbb{E}\Big(u'_{n,t} u_{n,s}\Big)\Big| \leq \Big(\mathbb{E}\big\|u_{n,t}\big\|^2\Big)^{1/2} \Big(\mathbb{E}\big\|u_{n,s}\big\|^2\Big)^{1/2} \lesssim p_n^{\delta_{q,1}^{(n)}} s_{J,n}^2. \tag{D.84}$$

Second, by the lagged version of (D.81), for any $t, s \in [T]$,

$$\Big|\mathbb{E}\Big(e'_{m,t} e_{m,s}\Big)\Big| \lesssim p_m \rho(|t-s|), \tag{D.85}$$

where $\rho(\cdot)$ is the summable sequence from Lemma 1. Substituting (D.84) and (D.85) into (D.83) gives

$$\begin{aligned}\mathbb{E}\big\|\mathcal{III}_{Q,mn}\big\|_F^2 &\lesssim T^{-2} p_m p_n^{\delta_{q,1}^{(n)}} s_{J,n}^2 \sum_{t=1}^{T} \sum_{s=1}^{T} \rho(|t-s|) \\ &\lesssim T^{-2} \cdot p_m p_n^{\delta_{q,1}^{(n)}} s_{J,n}^2 \cdot T = T^{-1} p_m p_n^{\delta_{q,1}^{(n)}} s_{J,n}^2,\end{aligned} \tag{D.86}$$

where the second line follows from evaluating the double sum, viz. setting $h = t - s$ gives

$$\sum_{t=1}^{T} \sum_{s=1}^{T} \rho(|t-s|) = \sum_{h=-(T-1)}^{T-1} (T - |h|) \rho(|h|) \leq T \sum_{h=-\infty}^{\infty} \rho(|h|) = O(T),$$

where the final equality follows from the summability of $\rho(\cdot)$ as in Lemma 1. We mark that (D.86) carries T^{-1} rather than $R(T, d_{\text{fac},\star})$ as in e.g. (D.77), since here we consider the cross-moment between the factor and noise processes, whereas previously, say (D.77), the sample covariance of factor processes is considered.

The same arguments can be applied to $\mathcal{VI}_{Q,mn}$, $\mathcal{VII}_{Q,mn}$, $\mathcal{VIII}_{Q,mn}$ and $\mathcal{IX}_{Q,mn}$, yielding

$$\big\|\mathcal{VI}_{Q,mn}\big\|_F^2 = O_P\Big(T^{-1} p_m p_n^{\delta_{r,1}^{(n)}} q_n^{\delta_{c,1}^{(n)}}\Big), \tag{D.87}$$

$$\big\|\mathcal{VII}_{Q,mn}\big\|_F^2 = O_P\Big(T^{-1} p_m^{\delta_{q,1}^{(m)}} s_{J,m}^2 p_n\Big), \tag{D.88}$$

$$\big\|\mathcal{VIII}_{Q,mn}\big\|_F^2 = O_P\Big(T^{-1} p_m^{\delta_{r,1}^{(m)}} q_m^{\delta_{c,1}^{(m)}} p_n\Big), \tag{D.89}$$

$$\big\|\mathcal{IX}_{Q,mn}\big\|_F^2 = O_P\Big(T^{-1} p_m p_n\Big), \tag{D.90}$$

where in each case, the bound follows by separating the independent vector processes, applying (D.81) to the error part, (D.82) to the factor part, and using the short-memory property of the error process to sum over t, s. Lastly, combining (D.75), (D.77), (D.78), (D.86), (D.79), (D.80), (D.87), (D.88), (D.89), and (D.90), together with the definitions of $\mathbf{\Delta}_{Q,mn}$, we may conclude that

$$
\begin{aligned}
&\|\mathbf{\Delta}_{Q,mn}\|_F^2 = \|\widehat{\mathbf{S}}_{Q,mn} - \mathbf{S}_{Q,mn}\|_F^2 \\
&\lesssim \left\|\mathcal{I}_{Q,mn} - \mathbf{S}_{Q,mn}\right\|_F^2 + \left\|\mathcal{II}_{Q,mn}\right\|_F^2 + \left\|\mathcal{III}_{Q,mn}\right\|_F^2 + \left\|\mathcal{IV}_{Q,mn}\right\|_F^2 + \left\|\mathcal{V}_{Q,mn}\right\|_F^2 \\
&\quad + \left\|\mathcal{VI}_{Q,mn}\right\|_F^2 + \left\|\mathcal{VII}_{Q,mn}\right\|_F^2 + \left\|\mathcal{VIII}_{Q,mn}\right\|_F^2 + \left\|\mathcal{IX}_{Q,mn}\right\|_F^2 \\
&\lesssim_P p_m^{\delta_{q,1}^{(m)}} p_n^{\delta_{q,1}^{(n)}} s_{J,m}^2 s_{J,n}^2 R(T, d_{\text{fac},\star}) + p_m^{\delta_{r,1}^{(m)}} p_n^{\delta_{q,1}^{(n)}} s_{J,n}^2 q_m^{\delta_{c,1}^{(m)}} R(T, d_{\text{fac},\star}) + p_m^{\delta_{q,1}^{(m)}} p_n^{\delta_{r,1}^{(n)}} s_{J,m}^2 q_n^{\delta_{c,1}^{(n)}} R(T, d_{\text{fac},\star}) \\
&\quad + p_m^{\delta_{r,1}^{(m)}} p_n^{\delta_{r,1}^{(n)}} q_m^{\delta_{c,1}^{(m)}} q_n^{\delta_{c,1}^{(n)}} R(T, d_{\text{fac},\star}) + T^{-1} p_m p_n^{\delta_{q,1}^{(n)}} s_{J,n}^2 + T^{-1} p_m p_n^{\delta_{r,1}^{(n)}} q_n^{\delta_{c,1}^{(n)}} \\
&\quad + T^{-1} p_m^{\delta_{q,1}^{(m)}} s_{J,m}^2 p_n + T^{-1} p_m^{\delta_{r,1}^{(m)}} q_m^{\delta_{c,1}^{(m)}} p_n + T^{-1} p_m p_n.
\end{aligned}
$$

This completes the proof on $\|\mathbf{\Delta}_{Q,mn}\|_F^2$.

Next, we focus on the k_r-th eigenvalue of $\mathbf{\Omega}_{Q,m}$. First, by Assumption (F1), for any $m, n \in [M]$, we may simplify and write

$$
\begin{aligned}
\mathbf{S}_{Q,mn} &= \mathbf{Q}_m \mathbb{E}\left(\mathbf{G}_t \mathbf{J}_m' \mathbf{v}_b^{(m)} \mathbf{v}_b^{(n)'} \mathbf{J}_n \mathbf{G}_t'\right) \mathbf{Q}_n' \\
&= \operatorname{tr}\left(\mathbf{J}_m' \mathbf{v}_b^{(m)} \mathbf{v}_b^{(n)'} \mathbf{J}_n\right) \mathbf{Q}_m \mathbf{Q}_n' = \left(\mathbf{v}_b^{(n)'} \mathbf{J}_n \mathbf{J}_m' \mathbf{v}_b^{(m)}\right) \mathbf{Q}_m \mathbf{Q}_n' =: w_{mn} \mathbf{Q}_m \mathbf{Q}_n',
\end{aligned} \tag{D.91}
$$

where by Assumption (L2), the selected projection directions satisfy

$$
|w_{mn}| \asymp s_{J,m} s_{J,n} \tag{D.92}
$$

for all $m, n \in [M]$ with $n \neq m$. Hence we have $\mathbf{S}_{Q,mn}$ of rank k_r, and

$$
\|\mathbf{S}_{Q,mn}\| \asymp_P \left(s_{J,m} s_{J,n} p_m^{\delta_{q,1}^{(m)}/2} p_n^{\delta_{q,1}^{(n)}/2}\right). \tag{D.93}
$$

Now plugging (D.91) in $\mathbf{\Omega}_{Q,m}$, we have

$$
\begin{aligned}
\mathbf{\Omega}_{Q,m} &= \frac{1}{M-1} \sum_{n=1; n\neq m}^{M} w_{mn}^2 \mathbf{Q}_m \mathbf{Q}_n' \mathbf{Q}_n \mathbf{Q}_m' \\
&= \mathbf{Q}_m \left\{ \frac{1}{M-1} \sum_{n=1; n\neq m}^{M} w_{mn}^2 \mathbf{Q}_n' \mathbf{Q}_n \right\} \mathbf{Q}_m' =: \mathbf{Q}_m \mathbf{G}_{Q,m} \mathbf{Q}_m'.
\end{aligned} \tag{D.94}
$$

We next derive the order of the k_r-th eigenvalue of $\mathbf{\Omega}_{Q,m}$. For any nonzero vector $\mathbf{x} \in \mathbb{R}^{k_r}$,

$$\begin{aligned}\mathbf{x}'\mathbf{G}_{Q,m}\mathbf{x} &= \frac{1}{M-1}\sum_{n=1;n\neq m}^{M} w_{mn}^2 \mathbf{x}'\mathbf{Q}_n'\mathbf{Q}_n\mathbf{x} = \frac{1}{M-1}\sum_{n=1;n\neq m}^{M} w_{mn}^2 \|\mathbf{Q}_n\mathbf{x}\|^2 \\ &\geq \frac{1}{M-1}\sum_{n=1;n\neq m}^{M} w_{mn}^2 \lambda_{\min}(\mathbf{Q}_n'\mathbf{Q}_n)\|\mathbf{x}\|^2. \end{aligned} \tag{D.95}$$

By Assumption (L1), $\lambda_{\min}(\mathbf{Q}_n'\mathbf{Q}_n) \asymp p_n^{\delta_{q,k_r}^{(n)}}$ for each $n \in [M]$. Combining this with (D.92) and (D.95) gives

$$\lambda_{\min}(\mathbf{G}_{Q,m}) \gtrsim \frac{s_{J,m}^2}{M-1}\sum_{n=1;n\neq m}^{M} s_{J,n}^2 p_n^{\delta_{q,k_r}^{(n)}} \asymp s_{J,m}^2 \max_{n\in[M]\setminus\{m\}} s_{J,n}^2 p_n^{\delta_{q,k_r}^{(n)}}, \tag{D.96}$$

where also used that M is fixed. As the right-hand side of (D.96) is bounded away from zero, the matrix $\mathbf{G}_{Q,m}$ is positive definite with all eigenvalues bounded away from zero. Then from (D.94), the nonzero eigenvalues of $\mathbf{\Omega}_{Q,m}$ are the same as those of $\mathbf{G}_{Q,m}^{1/2}\mathbf{Q}_m'\mathbf{Q}_m\mathbf{G}_{Q,m}^{1/2}$. Therefore, we may conclude that

$$\lambda_{k_r}(\mathbf{\Omega}_{Q,m}) \geq \lambda_{\min}(\mathbf{G}_{Q,m})\lambda_{\min}(\mathbf{Q}_m'\mathbf{Q}_m) \asymp \lambda_{\min}(\mathbf{G}_{Q,m}) p_m^{\delta_{q,k_r}^{(m)}} \gtrsim p_m^{\delta_{q,k_r}^{(m)}} s_{J,m}^2 \max_{n\in[M]\setminus\{m\}} s_{J,n}^2 p_n^{\delta_{q,k_r}^{(n)}}, \tag{D.97}$$

by Assumption (L1) and (D.96), which particularly implies $\mathbf{\Omega}_{Q,m}$ has rank k_r.

Next, note that by Assumption (R2), we may further simplify $\|\mathbf{\Delta}_{Q,mn}\|_F^2$ as

$$\|\mathbf{\Delta}_{Q,mn}\|_F^2 = O_P\Big(R(T,d_{\mathrm{fac},\star}) p_m^{\delta_{q,1}^{(m)}} p_n^{\delta_{q,1}^{(n)}} s_{J,m}^2 s_{J,n}^2\Big) = O_P\Big(R(T,d_{\mathrm{fac},\star})\,\|\mathbf{S}_{Q,mn}\|_F^2\Big),$$

where the last equality used (D.93). Hence by (3.8) and the triangle inequality,

$$\begin{aligned}
&\Big\|\mathbf{X}_{Q,m} - \mathbf{\Omega}_{Q,m}\Big\| = \left\|\frac{1}{T^2(M-1)}\sum_{n=1;n\neq m}^{M}\left\{\sum_{t=1}^{T}\breve{\mathbf{Y}}_t^{(m)}\mathbf{v}_b^{(m)}\mathbf{v}_b^{(n)\prime}\breve{\mathbf{Y}}_t^{(n)\prime}\right\}^{\otimes 2} - \mathbf{\Omega}_{Q,m}\right\| \\
&= \left\|\left(\frac{1}{M-1}\sum_{n=1;n\neq m}^{M}\widehat{\mathbf{S}}_{Q,mn}^{\otimes 2}\right) - \mathbf{\Omega}_{Q,m}\right\| \leq \frac{1}{M-1}\sum_{n=1;n\neq m}^{M}\Big\|\widehat{\mathbf{S}}_{Q,mn}\widehat{\mathbf{S}}_{Q,mn}' - \mathbf{S}_{Q,mn}\mathbf{S}_{Q,mn}'\Big\| \\
&\leq \frac{1}{M-1}\sum_{n=1;n\neq m}^{M}\Big\|\mathbf{\Delta}_{Q,mn}\widehat{\mathbf{S}}_{Q,mn}' + \mathbf{S}_{Q,mn}\mathbf{\Delta}_{Q,mn}'\Big\| \\
&= \frac{1}{M-1}\sum_{n=1;n\neq m}^{M}\Big\|\mathbf{\Delta}_{Q,mn}\mathbf{\Delta}_{Q,mn}' + \mathbf{\Delta}_{Q,mn}\mathbf{S}_{Q,mn}' + \mathbf{S}_{Q,mn}\mathbf{\Delta}_{Q,mn}'\Big\| \\
&\lesssim \max_{n\in[M]\setminus\{m\}}\big(\|\mathbf{\Delta}_{Q,mn}\|^2 + \|\mathbf{S}_{Q,mn}\|\|\mathbf{\Delta}_{Q,mn}\|\big) \\
&\lesssim R(T,d_{\mathrm{fac},\star})^{1/2}\max_{n\in[M]\setminus\{m\}}\|\mathbf{S}_{Q,mn}\|_F^2 \asymp R(T,d_{\mathrm{fac},\star})^{1/2} p_m^{\delta_{q,1}^{(m)}} s_{J,m}^2 \max_{n\in[M]\setminus\{m\}} s_{J,n}^2 p_n^{\delta_{q,1}^{(n)}},
\end{aligned}$$

which, combined with (D.97), applying the Davis–Kahan theorem (Yu et al., 2015, Theorem 2) gives the

desired result on $\widehat{\mathbf{Q}}_m$, and thus concludes this proof. □

Lemma 10 *Let all assumptions in Theorem 7 hold. Fix any $m \in [M]$. Recall $\check{\mathbf{Q}}_m = \mathbf{Q}_m(\mathbf{Q}_m'\mathbf{Q}_m)^{-1/2}$ from Lemma 9, and similarly let $\check{\mathbf{J}}_m := \mathbf{J}_m(\mathbf{J}_m'\mathbf{J}_m)^{-1/2}$. Also, recall $\mathbf{H}_{Q,m}$ and $\mathbf{H}_{J,m}$ from Lemma 9. For each $m \in [M]$, with $\check{\mathbf{Y}}_{G,t}^{(m)} = \check{\mathbf{Y}}_t^{(m)} - \widehat{\mathbf{R}}_m\widehat{\mathbf{F}}_t^{(m)}\widehat{\mathbf{C}}_m'$ from Step 3 of Section 3.3, define*

$$\check{\mathbf{G}}_{t,m} := \check{\mathbf{Q}}_m'\mathbf{Q}_m\mathbf{G}_t\mathbf{J}_m'\check{\mathbf{J}}_m, \quad \widehat{\mathbf{G}}_{t,m} := \widehat{\mathbf{Q}}_m'\check{\mathbf{Y}}_{G,t}^{(m)}\widehat{\mathbf{J}}_m.$$

Then for every $t \in [T]$, $\widehat{\mathbf{G}}_{t,m}$ satisfies that

$$\begin{aligned}&\left\|\widehat{\mathbf{G}}_{t,m} - \mathbf{H}_{Q,m}'\check{\mathbf{G}}_{t,m}\mathbf{H}_{J,m}\right\|_F^2 \\ &= O_P\bigg(\min(p_m,q_m) + p_m^{\delta_{r,1}^{(m)}}q_m^{\delta_{c,1}^{(m)}}\big(\Delta_{R,m}+\Delta_{C,m}+\Delta_{F,m}\big) + \big(\eta_{Q,m}+\eta_{J,m}\big)p_mq_m\bigg),\end{aligned}$$

where $\Delta_{R,m}$, $\Delta_{C,m}$, $\Delta_{F,m}$, $\eta_{Q,m}$, and $\eta_{J,m}$ are defined in the statement of Theorem 7.

Proof of Lemma 10. By construction, we may decompose

$$\widehat{\mathbf{G}}_{t,m} = \mathbf{H}_{Q,m}'\check{\mathbf{Q}}_m'\mathbf{Q}_m\mathbf{G}_t\mathbf{J}_m'\check{\mathbf{J}}_m\mathbf{H}_{J,m} + \mathcal{R}_{Q,m} + \mathcal{R}_{J,m} + \mathcal{R}_{E,m} + \mathcal{R}_{\text{local},m}, \tag{D.98}$$

where

$$\begin{aligned}\mathcal{R}_{Q,m} &:= \big(\widehat{\mathbf{Q}}_m - \check{\mathbf{Q}}_m\mathbf{H}_{Q,m}\big)'\big(\mathbf{Q}_m\mathbf{G}_t\mathbf{J}_m' + \mathbf{M}_{p_m}\mathbf{E}_t^{(m)}\mathbf{M}_{q_m} + \mathbf{R}_m\mathbf{F}_t^{(m)}\mathbf{C}_m' - \widehat{\mathbf{R}}_m\widehat{\mathbf{F}}_t^{(m)}\widehat{\mathbf{C}}_m'\big)\widehat{\mathbf{J}}_m,\\ \mathcal{R}_{J,m} &:= \mathbf{H}_{Q,m}'\check{\mathbf{Q}}_m'\big(\mathbf{Q}_m\mathbf{G}_t\mathbf{J}_m' + \mathbf{M}_{p_m}\mathbf{E}_t^{(m)}\mathbf{M}_{q_m} + \mathbf{R}_m\mathbf{F}_t^{(m)}\mathbf{C}_m' - \widehat{\mathbf{R}}_m\widehat{\mathbf{F}}_t^{(m)}\widehat{\mathbf{C}}_m'\big)\big(\widehat{\mathbf{J}}_m - \check{\mathbf{J}}_m\mathbf{H}_{J,m}\big),\\ \mathcal{R}_{E,m} &:= \mathbf{H}_{Q,m}'\check{\mathbf{Q}}_m'\mathbf{M}_{p_m}\mathbf{E}_t^{(m)}\mathbf{M}_{q_m}\check{\mathbf{J}}_m\mathbf{H}_{J,m},\\ \mathcal{R}_{\text{local},m} &:= \mathbf{H}_{Q,m}'\check{\mathbf{Q}}_m'\big(\mathbf{R}_m\mathbf{F}_t^{(m)}\mathbf{C}_m' - \widehat{\mathbf{R}}_m\widehat{\mathbf{F}}_t^{(m)}\widehat{\mathbf{C}}_m'\big)\check{\mathbf{J}}_m\mathbf{H}_{J,m}.\end{aligned}$$

Note that $\check{\mathbf{Q}}_m$ and $\check{\mathbf{J}}_m$ have orthonormal columns, $\mathbf{H}_{Q,m}$ and $\mathbf{H}_{J,m}$ are asymptotically orthogonal, and $\widehat{\mathbf{J}}_m$ has orthonormal columns. Hence we have

$$\left\|\check{\mathbf{Q}}_m\right\| = \left\|\check{\mathbf{J}}_m\right\| = \left\|\widehat{\mathbf{J}}_m\right\| = 1, \quad \left\|\mathbf{H}_{Q,m}\right\| = \left\|\mathbf{H}_{J,m}\right\| = O_P(1). \tag{D.99}$$

Then consider $\mathcal{R}_{E,m}$. As the columns of $\check{\mathbf{J}}_m\mathbf{H}_{J,m}$ are asymptotically orthonormal, by (D.81), it holds that

$$\left\|\mathcal{R}_{E,m}\right\|_F^2 \le \left\|\mathbf{H}_{Q,m}'\check{\mathbf{Q}}_m'\right\|^2 \cdot \left\|\mathbf{M}_{p_m}\mathbf{E}_t^{(m)}\mathbf{M}_{q_m}\check{\mathbf{J}}_m\mathbf{H}_{J,m}\right\|_F^2 \lesssim_P \min(p_m,q_m), \tag{D.100}$$

where the equality follows from (D.99) and the fact that $\left\|\mathbf{M}_{p_m}\mathbf{E}_t^{(m)}\mathbf{M}_{q_m}\check{\mathbf{J}}_m\mathbf{H}_{J,m}\right\|_F^2 = O_P(p_m)$ as in the

proof of Theorem 7, and a symmetrical argument which also gives the rate in $O_P(q_m)$.

Next, consider $\mathcal{R}_{\text{local},m}$. By Theorem 7, we have

$$
\begin{aligned}
&\big\|\mathbf{R}_m\mathbf{F}_t^{(m)}\mathbf{C}_m' - \widehat{\mathbf{R}}_m\widehat{\mathbf{F}}_t^{(m)}\widehat{\mathbf{C}}_m'\big\|_F^2 \\
\lesssim\ & \big\|\mathbf{R}_m\mathbf{F}_t^{(m)}\mathbf{C}_m' - \widehat{\mathbf{R}}_m\mathbf{H}_{R,m}^{-1}\check{\mathbf{F}}_t^{(m)}\mathbf{H}_{C,m}\widehat{\mathbf{C}}_m'\big\|_F^2 + \big\|\widehat{\mathbf{R}}_m\big(\widehat{\mathbf{F}}_t^{(m)} - \mathbf{H}_{R,m}^{-1}\check{\mathbf{F}}_t^{(m)}\mathbf{H}_{C,m}\big)\widehat{\mathbf{C}}_m'\big\|_F^2 \\
=\ & \big\|\mathbf{R}_m\mathbf{F}_t^{(m)}\mathbf{C}_m' - \widehat{\mathbf{R}}_m\mathbf{H}_{R,m}^{-1}\big(\mathbf{R}_m'\mathbf{R}_m\big)^{1/2}\mathbf{F}_t^{(m)}\big(\mathbf{C}_m'\mathbf{C}_m\big)^{1/2}\mathbf{H}_{C,m}\widehat{\mathbf{C}}_m'\big\|_F^2 \\
&+ \big\|\widehat{\mathbf{R}}_m\big(\widehat{\mathbf{F}}_t^{(m)} - \mathbf{H}_{R,m}^{-1}\check{\mathbf{F}}_t^{(m)}\mathbf{H}_{C,m}\big)\widehat{\mathbf{C}}_m'\big\|_F^2 \\
\lesssim_P\ & \big\|\mathbf{R}_m\mathbf{F}_t^{(m)}\mathbf{C}_m' - \mathbf{R}_m(\mathbf{R}_m'\mathbf{R}_m)^{-1/2}\big(\mathbf{R}_m'\mathbf{R}_m\big)^{1/2}\mathbf{F}_t^{(m)}\big(\mathbf{C}_m'\mathbf{C}_m\big)^{1/2}\big(\mathbf{C}_m'\mathbf{C}_m\big)^{-1/2}\mathbf{C}_m'\big\|_F^2 \\
&+ \big\|\widehat{\mathbf{R}}_m\big(\widehat{\mathbf{F}}_t^{(m)} - \mathbf{H}_{R,m}^{-1}\check{\mathbf{F}}_t^{(m)}\mathbf{H}_{C,m}\big)\widehat{\mathbf{C}}_m'\big\|_F^2 \\
\lesssim_P\ & \big\|\widehat{\mathbf{R}}_m\big(\widehat{\mathbf{F}}_t^{(m)} - \mathbf{H}_{R,m}^{-1}\check{\mathbf{F}}_t^{(m)}\mathbf{H}_{C,m}\big)\widehat{\mathbf{C}}_m'\big\|_F^2 \lesssim_P p_m^{\delta_{r,1}^{(m)}}\, q_m^{\delta_{c,1}^{(m)}}\big(\Delta_{R,m} + \Delta_{C,m} + \Delta_{F,m}\big),
\end{aligned}
\tag{D.101}
$$

where $\Delta_{R,m}$, $\Delta_{C,m}$ and $\Delta_{F,m}$ are defined in the statement of Theorem 7. Combining (D.99) and (D.101), we thus have

$$
\begin{aligned}
\big\|\mathcal{R}_{\text{local},m}\big\|_F^2 &\le \big\|\mathbf{H}_{Q,m}'\check{\mathbf{Q}}_m'\big\|^2 \cdot \big\|\mathbf{R}_m\mathbf{F}_t^{(m)}\mathbf{C}_m' - \widehat{\mathbf{R}}_m\widehat{\mathbf{F}}_t^{(m)}\widehat{\mathbf{C}}_m'\big\|_F^2 \cdot \big\|\check{\mathbf{J}}_m\mathbf{H}_{J,m}\big\|^2 \\
&\lesssim_P p_m^{\delta_{r,1}^{(m)}}\, q_m^{\delta_{c,1}^{(m)}}\big(\Delta_{R,m} + \Delta_{C,m} + \Delta_{F,m}\big).
\end{aligned}
\tag{D.102}
$$

It remains to consider $\mathcal{R}_{Q,m}$ and $\mathcal{R}_{J,m}$. For $\mathcal{R}_{Q,m}$, by Assumptions (F1), (L1), (D.35), and (D.101), we have

$$
\begin{aligned}
&\big\|\mathbf{Q}_m\mathbf{G}_t\mathbf{J}_m' + \mathbf{M}_{p_m}\mathbf{E}_t^{(m)}\mathbf{M}_{q_m} + \mathbf{R}_m\mathbf{F}_t^{(m)}\mathbf{C}_m' - \widehat{\mathbf{R}}_m\widehat{\mathbf{F}}_t^{(m)}\widehat{\mathbf{C}}_m'\big\|_F^2 \\
\lesssim_P\ & p_m^{\delta_{q,1}^{(m)}}\, q_m^{\delta_{j,1}^{(m)}} + p_m q_m + p_m^{\delta_{r,1}^{(m)}}\, q_m^{\delta_{c,1}^{(m)}}\big(\Delta_{R,m} + \Delta_{C,m} + \Delta_{F,m}\big),
\end{aligned}
$$

which, combined with Lemma 9 and (D.99), give

$$
\big\|\mathcal{R}_{Q,m}\big\|_F^2 \lesssim_P \eta_{Q,m}\Big\{p_m^{\delta_{q,1}^{(m)}}\, q_m^{\delta_{j,1}^{(m)}} + p_m q_m + p_m^{\delta_{r,1}^{(m)}}\, q_m^{\delta_{c,1}^{(m)}}\big(\Delta_{R,m} + \Delta_{C,m} + \Delta_{F,m}\big)\Big\}.
\tag{D.103}
$$

Similarly, $\mathcal{R}_{J,m}$ satisfies that

$$
\big\|\mathcal{R}_{J,m}\big\|_F^2 \lesssim_P \eta_{J,m}\Big\{p_m^{\delta_{q,1}^{(m)}}\, q_m^{\delta_{j,1}^{(m)}} + p_m q_m + p_m^{\delta_{r,1}^{(m)}}\, q_m^{\delta_{c,1}^{(m)}}\big(\Delta_{R,m} + \Delta_{C,m} + \Delta_{F,m}\big)\Big\}.
\tag{D.104}
$$

Finally, combining (D.98), (D.100), (D.102), (D.103), and (D.104), we conclude that

$$
\Big\|\widehat{\mathbf{G}}_{t,m} - \mathbf{H}_{Q,m}'\check{\mathbf{Q}}_m'\mathbf{Q}_m\mathbf{G}_t\mathbf{J}_m'\check{\mathbf{J}}_m\mathbf{H}_{J,m}\Big\|_F^2
$$

$$
\begin{aligned}
&\lesssim_P \min(p_m, q_m) + p_m^{\delta_{r,1}^{(m)}} q_m^{\delta_{c,1}^{(m)}} \big(\Delta_{R,m} + \Delta_{C,m} + \Delta_{F,m}\big) \\
&\quad + \big(\eta_{Q,m} + \eta_{J,m}\big)\Big\{ p_m^{\delta_{q,1}^{(m)}} q_m^{\delta_{j,1}^{(m)}} + p_m q_m + p_m^{\delta_{r,1}^{(m)}} q_m^{\delta_{c,1}^{(m)}} \big(\Delta_{R,m} + \Delta_{C,m} + \Delta_{F,m}\big)\Big\} \\
&\lesssim_P \min(p_m, q_m) + p_m^{\delta_{r,1}^{(m)}} q_m^{\delta_{c,1}^{(m)}} \big(\Delta_{R,m} + \Delta_{C,m} + \Delta_{F,m}\big) + \big(\eta_{Q,m} + \eta_{J,m}\big) p_m q_m,
\end{aligned}
$$

which completes the proof of Lemma 10. □

Lemma 11 *By construction in Step 3 of Section 3.3, it holds that*

$$
\mathcal{R}\big(\widehat{\mathbf{\Xi}}_{J,m} \otimes \widehat{\mathbf{\Xi}}_{Q,m}\big) = \widehat{\sigma}_{\Xi,m} \widehat{\mathbf{u}}_{\Xi,m} \widehat{\mathbf{v}}_{\Xi,m}'.
$$

Proof of Lemma 11. Recall from Step 3 of Section 3.3 the definition of the notation $\mathcal{R}(\mathbf{W})$ for a block matrix $\mathbf{W} = (\mathbf{W}_{ij})_{i\in[k_c], j\in[k_c]}$, with each sub-matrix $\mathbf{W}_{ij}$ of size $k_r \times k_r$, such that

$$
\mathcal{R}(\mathbf{W}) = (\mathbf{vec}(\mathbf{W}_{11}), \ldots, \mathbf{vec}(\mathbf{W}_{k_c 1}), \ldots, \mathbf{vec}(\mathbf{W}_{1 k_c}), \ldots, \mathbf{vec}(\mathbf{W}_{k_c k_c})).
$$

Then note that each sub-matrix in $\widehat{\mathbf{\Xi}}_{J,m} \otimes \widehat{\mathbf{\Xi}}_{Q,m}$ is exactly the corresponding element in $\widehat{\mathbf{\Xi}}_{J,m}$ multiplied by the matrix $\widehat{\mathbf{\Xi}}_{Q,m}$; hence the vectorization of such sub-matrix would be the corresponding element in $\widehat{\mathbf{\Xi}}_{J,m}$ multiplied by $\mathbf{vec}(\widehat{\mathbf{\Xi}}_{Q,m}) = \widehat{\sigma}_{\Xi,m}^{1/2} \widehat{\mathbf{u}}_{\Xi,m}$ by construction. Together with the construction of $\widehat{\mathbf{\Xi}}_{J,m}$, we complete the proof of this lemma. □

Lemma 12 *Let all assumptions in Lemma 9 hold, and recall the matrices $\mathbf{H}_{Q,m}$ and $\mathbf{H}_{J,m}$ within. Then*

$$
\mathbf{H}_{Q,m}'\mathbf{H}_{Q,m} = \mathbf{I}_{k_r} + O_P\big(\eta_{Q,m}^{1/2}\big), \quad \mathbf{H}_{J,m}'\mathbf{H}_{J,m} = \mathbf{I}_{k_c} + O_P\big(\eta_{J,m}^{1/2}\big),
$$

where $\eta_{Q,m}$ and $\eta_{J,m}$ are defined in Theorem 7.

Proof of Lemma 12. For $\mathbf{H}_{Q,m}$, note that

$$
\begin{aligned}
\mathbf{I}_{k_r} &= \widehat{\mathbf{Q}}_m'\widehat{\mathbf{Q}}_m = (\widehat{\mathbf{Q}}_m - \check{\mathbf{Q}}_m\mathbf{H}_{Q,m})'\widehat{\mathbf{Q}}_m + \mathbf{H}_{Q,m}'\check{\mathbf{Q}}_m'\widehat{\mathbf{Q}}_m \\
&= (\widehat{\mathbf{Q}}_m - \check{\mathbf{Q}}_m\mathbf{H}_{Q,m})'\widehat{\mathbf{Q}}_m + \mathbf{H}_{Q,m}'\check{\mathbf{Q}}_m'(\widehat{\mathbf{Q}}_m - \check{\mathbf{Q}}_m\mathbf{H}_{Q,m}) + \mathbf{H}_{Q,m}'\check{\mathbf{Q}}_m'\check{\mathbf{Q}}_m\mathbf{H}_{Q,m} \\
&= (\widehat{\mathbf{Q}}_m - \check{\mathbf{Q}}_m\mathbf{H}_{Q,m})'\widehat{\mathbf{Q}}_m + \mathbf{H}_{Q,m}'\check{\mathbf{Q}}_m'(\widehat{\mathbf{Q}}_m - \check{\mathbf{Q}}_m\mathbf{H}_{Q,m}) \\
&\quad + \mathbf{H}_{Q,m}'(\mathbf{Q}_m'\mathbf{Q}_m)^{-1/2}\mathbf{Q}_m'\mathbf{Q}_m(\mathbf{Q}_m'\mathbf{Q}_m)^{-1/2}\mathbf{H}_{Q,m} \\
&= \mathbf{H}_{Q,m}'\mathbf{H}_{Q,m} + O_P\left\{ R(T, d_{\mathrm{fac},\star})^{1/2}\, p_m^{(\delta_{q,1}^{(m)} - \delta_{q,k_r}^{(m)})} \left(\frac{\max_{n\in[M]\setminus\{m\}} s_{J,n}^2 p_n^{\delta_{q,1}^{(n)}}}{\max_{n\in[M]\setminus\{m\}} s_{J,n}^2 p_n^{\delta_{q,k_r}^{(n)}}} \right) \right\} \\
&= \mathbf{H}_{Q,m}'\mathbf{H}_{Q,m} + O_P\big(\eta_{Q,m}^{1/2}\big),
\end{aligned}
$$

where the last equality used Lemma 9, and the $O_P(\cdot)$ bound is in terms of the Frobenius norm. Analogous arguments hold for $\mathbf{H}_{J,m}$. This thus completes the proof of this lemma. □

E Additional Numerical Results

E.1 Simulation

E.1.1 Heavy-tailed robustness

We repeat the main effect settings (Ia)–(Id) and common component settings (IIa)–(IId) under heavy-tailed innovations. The DGP is identical to that in Section 6.1, except that the Gaussian innovations in the latent time-series processes are replaced by Student $t(6)$ innovations. Each setting is crossed with $T \in \{100, 400\}$, with 500 Monte Carlo replications.

The main effect results in Table E.1 remain close to their Gaussian counterparts. The direct moment estimators are essentially unchanged, while the loading-space and reconstructed-component errors retain the same improvement with T; heavy tails mainly increase finite-sample dispersion in some entries.

Table E.1: Main effect estimation accuracy. All entries are Monte Carlo means multiplied by 10^2, with standard deviations in parentheses. The latent-process innovations follow a Student $t(6)$ distribution.

Settings	Ia		Ib		Ic		Id	
	$T = 100$	$T = 400$	$T = 100$	$T = 400$	$T = 100$	$T = 400$	$T = 100$	$T = 400$
$\mathrm{MSE}(\widehat{c}_t^{(m)})$	0.05(0.01)	0.05(0.00)	0.04(0.01)	0.04(0.00)	0.05(0.01)	0.05(0.00)	0.02(0.00)	0.02(0.00)
$\mathrm{MSE}(\widehat{\mathbf{a}}_t^{(m)})$	0.60(0.08)	0.58(0.06)	0.49(0.07)	0.47(0.05)	1.19(0.17)	1.14(0.12)	0.30(0.04)	0.29(0.03)
$\mathrm{MSE}(\widehat{\mathbf{b}}_t^{(m)})$	0.60(0.08)	0.58(0.06)	0.49(0.07)	0.47(0.05)	1.19(0.17)	1.16(0.12)	0.60(0.07)	0.58(0.04)
$\mathrm{MSE}(\widehat{\mu}_t)$	0.03(0.01)	0.03(0.00)	0.02(0.01)	0.02(0.00)	0.03(0.01)	0.03(0.00)	0.01(0.00)	0.01(0.00)
$\mathrm{MSE}(\widehat{\mu}_t^{(m)})$	0.08(0.02)	0.08(0.01)	0.06(0.01)	0.06(0.01)	0.08(0.02)	0.08(0.01)	0.04(0.01)	0.04(0.00)
$\mathcal{D}(\mathbf{A}_{1m}, \widehat{\mathbf{A}}_{1m})$	7.04(5.98)	2.80(1.13)	9.81(10.06)	3.22(1.73)	7.37(5.11)	2.97(1.01)	7.38(6.06)	2.70(1.15)
$\mathcal{D}(\mathbf{A}_{1m}, \widetilde{\mathbf{A}}_{1m})$	7.03(6.01)	2.79(1.13)	9.80(10.07)	3.21(1.73)	7.34(5.10)	2.96(1.00)	7.37(6.06)	2.69(1.15)
$\mathcal{D}(\mathbf{A}_{2m}, \widehat{\mathbf{A}}_{2m})$	6.33(4.15)	2.74(0.92)	8.23(6.34)	3.15(1.16)	6.68(2.95)	2.94(0.82)	6.41(3.80)	2.58(0.87)
$\mathcal{D}(\mathbf{B}_{1m}, \widehat{\mathbf{B}}_{1m})$	6.95(5.47)	2.74(1.10)	9.76(10.80)	3.26(1.57)	7.73(5.95)	2.89(1.04)	5.12(3.82)	2.03(0.65)
$\mathcal{D}(\mathbf{B}_{1m}, \widetilde{\mathbf{B}}_{1m})$	6.93(5.47)	2.73(1.10)	9.74(10.80)	3.26(1.57)	7.70(5.94)	2.87(1.04)	5.11(3.82)	2.02(0.64)
$\mathcal{D}(\mathbf{B}_{2m}, \widehat{\mathbf{B}}_{2m})$	6.19(3.70)	2.70(0.96)	8.25(7.19)	3.19(1.23)	6.90(3.66)	2.89(0.88)	4.57(2.53)	1.99(0.58)
$\mathrm{MSE}(\widetilde{\mathbf{A}}_{1m}\widehat{\boldsymbol{\alpha}}_t)$	0.79(5.71)	0.10(0.07)	1.58(6.01)	0.12(0.15)	0.62(1.22)	0.14(0.05)	0.73(3.14)	0.08(0.06)
$\mathrm{MSE}(\widehat{\mathbf{A}}_{2m}\widehat{\boldsymbol{\alpha}}_t^{(m)})$	0.83(2.89)	0.19(0.08)	1.56(4.20)	0.20(0.13)	0.84(1.12)	0.28(0.07)	0.78(2.17)	0.13(0.07)
$\mathrm{MSE}(\widetilde{\mathbf{B}}_{1m}\widehat{\boldsymbol{\beta}}_t)$	0.65(2.98)	0.10(0.06)	1.60(5.63)	0.12(0.10)	0.83(3.69)	0.13(0.06)	0.29(0.94)	0.05(0.02)
$\mathrm{MSE}(\widehat{\mathbf{B}}_{2m}\widehat{\boldsymbol{\beta}}_t^{(m)})$	0.77(2.69)	0.18(0.08)	1.75(5.34)	0.21(0.13)	0.98(2.72)	0.28(0.08)	0.38(1.30)	0.09(0.03)

The common component results in Table E.2 show the same robustness. Refitting remains effective and the estimation errors continue to decrease with T, while some small-sample entries exhibit greater Monte Carlo variation.

We also repeat the $S_{G,m}$ behavior experiment in Section 6.1.4 under the baseline design (IIIa), using 400 Monte Carlo replications and replacing the Gaussian innovations by Student $t(6)$ innovations. Figure E.1 shows that the active and inactive group averages remain clearly separated at $T = 200$, with a wider gap as more groups become active. Figure E.2 shows the same pattern over T: the inactive averages decrease, while the active averages remain comparatively stable. These results closely match the Gaussian baseline.

Table E.2: Common component estimation accuracy. All entries are Monte Carlo means multiplied by 10^2, with standard deviations in parentheses. The latent-process innovations follow a Student $t(6)$ distribution.

Settings	IIa		IIb		IIc		IId	
	$T = 100$	$T = 400$	$T = 100$	$T = 400$	$T = 100$	$T = 400$	$T = 100$	$T = 400$
$\mathcal{D}(\mathbf{Q}_m, \widehat{\mathbf{Q}}_m)$	8.14(3.45)	3.72(1.32)	9.66(3.98)	4.69(1.89)	9.28(3.41)	4.45(1.44)	5.85(2.47)	2.75(1.01)
$\mathcal{D}(\mathbf{Q}_m, \widetilde{\mathbf{Q}}_m)$	1.84(0.46)	0.79(0.14)	1.97(1.54)	0.81(0.21)	3.22(0.64)	1.45(0.19)	1.16(0.20)	0.50(0.07)
$\mathcal{D}(\mathbf{J}_m, \widehat{\mathbf{J}}_m)$	8.41(3.41)	3.85(1.45)	9.92(4.38)	4.61(1.76)	9.43(3.51)	4.33(1.41)	7.92(3.03)	3.87(1.43)
$\mathcal{D}(\mathbf{J}_m, \widetilde{\mathbf{J}}_m)$	1.60(0.23)	0.71(0.08)	1.67(1.42)	0.67(0.08)	3.08(0.71)	1.40(0.17)	1.48(0.16)	0.67(0.06)
$\mathcal{D}(\mathbf{R}_m, \widehat{\mathbf{R}}_m)$	2.60(0.75)	1.01(0.17)	3.23(1.89)	1.04(0.22)	4.44(1.87)	2.43(0.57)	2.20(0.70)	0.78(0.15)
$\mathcal{D}(\mathbf{C}_m, \widehat{\mathbf{C}}_m)$	2.57(0.71)	1.00(0.17)	3.14(1.78)	1.05(0.21)	4.33(0.96)	2.41(0.57)	2.13(0.51)	0.85(0.11)
$\text{MSE}(\widetilde{\mathbf{Q}}_m \widehat{\mathbf{G}}_t \widetilde{\mathbf{J}}'_m)$	0.07(0.03)	0.04(0.01)	0.19(2.72)	0.03(0.01)	0.26(0.31)	0.16(0.02)	0.04(0.01)	0.02(0.00)
$\text{MSE}(\widehat{\mathbf{R}}_m \widehat{\mathbf{F}}_t^{(m)} \widehat{\mathbf{C}}'_m)$	0.27(0.15)	0.13(0.03)	0.42(2.05)	0.12(0.03)	0.76(0.60)	0.48(0.12)	0.14(0.06)	0.06(0.01)

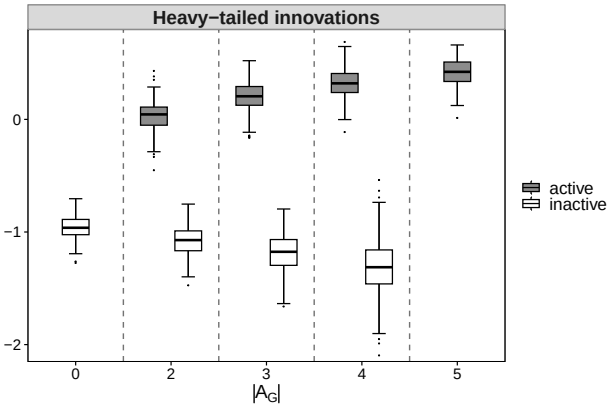


Figure E.1: Within-replication averages of $S_{G,m}$ over the active and inactive groups at $T = 200$ under the baseline design with Student $t(6)$ innovations. Each boxplot is formed across Monte Carlo replications. The y-axis reports the base-10 logarithm of the group average.

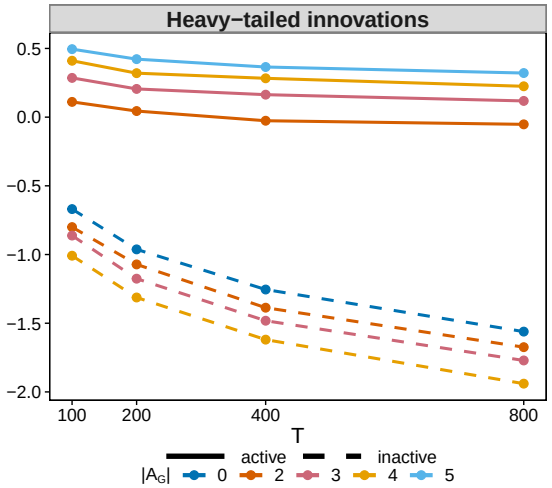


Figure E.2: Monte Carlo medians of the within-replication averages of $S_{G,m}$ over the active and inactive groups, respectively, over $T \in \{100, 200, 400, 800\}$ under Student $t(6)$ innovations. The y-axis reports the statistics on the $\log_{10}$ scale. Solid and dashed lines correspond to active and inactive groups, respectively.

E.1.2 Order-statistic calibration for detection and screening

We further examine the choice of order statistic in the bootstrap procedures of Section 5. For existence testing, we compare tests based on $S_{G,(1)}$ and $S_{G,(2)}$, each calibrated by its corresponding bootstrap distribution. For screening, we compare two procedures indexed by $j \in \{1, 2\}$. The $S_{G,(j)}$-based procedure sets both $\tau_1^{(j)}$ and $\tau_2^{(j)}$ equal to the empirical 95% quantile of $S^*_{G,(j)}$ and applies the corresponding existence gate and groupwise screening rule. Since $\tau_1^{(j)} = \tau_2^{(j)}$, the existence gate is automatically satisfied whenever at least two groups exceed the screening cutoff. All results below use the baseline setting (IIIa), 400 Monte Carlo replications, and $B = 199$ bootstrap replications.

Table E.3 shows that the two existence tests have very similar finite-sample performance. Both maintain size close to the nominal 5% level and have nearly identical power across the alternatives.

Table E.3: Bootstrap existence tests based on $S_{G,(1)}$ and $S_{G,(2)}$ in the baseline setting (IIIa). Each statistic is calibrated by its corresponding bootstrap distribution. Entries are empirical rejection rates, with $|\mathcal{A}_G| = 0$ representing size and $|\mathcal{A}_G| \geq 2$ representing power.

T	Statistic	$\lvert\mathcal{A}_G\rvert = 0$	$\lvert\mathcal{A}_G\rvert = 2$	$\lvert\mathcal{A}_G\rvert = 3$	$\lvert\mathcal{A}_G\rvert = 4$	$\lvert\mathcal{A}_G\rvert = 5$
100	$S_{G,(1)}$	0.048	0.800	0.993	1.000	1.000
	$S_{G,(2)}$	0.043	0.828	0.993	1.000	1.000
200	$S_{G,(1)}$	0.058	0.998	1.000	1.000	1.000
	$S_{G,(2)}$	0.058	0.995	1.000	1.000	1.000
400	$S_{G,(1)}$	0.058	0.998	1.000	1.000	1.000
	$S_{G,(2)}$	0.050	0.998	1.000	1.000	1.000

The difference is more pronounced for screening. As shown in Table E.4, the $S_{G,(1)}$-based procedure is substantially more conservative: it further reduces false positives under the null but yields noticeably lower true positive and exact-selection rates, especially when T is small or few groups are active. The gap narrows as T and the number of active groups increase. For $|\mathcal{A}_G| = 3$ and 4, false positives are zero under both procedures in the baseline design, so the differences in exact recovery are driven by missed active groups.

Taken together, the results support using the second-largest statistic throughout the proposed procedure. It performs similarly to the largest statistic for existence testing, while providing substantially higher screening recovery with low false positive rates.

Table E.4: Bootstrap screening procedures based on $S_{G,(1)}$ and $S_{G,(2)}$ in the baseline setting (IIIa). For each procedure, the same bootstrap order statistic is used to calibrate both the existence and screening cutoffs.

		$T = 100$		$T = 200$		$T = 400$	
$\lvert\mathcal{A}_G\rvert$	Metric	$S_{G,(1)}$	$S_{G,(2)}$	$S_{G,(1)}$	$S_{G,(2)}$	$S_{G,(1)}$	$S_{G,(2)}$
	TPR	–	–	–	–	–	–
0	FPR	0.001	0.019	0.000	0.026	0.002	0.021
	Exact	0.998	0.958	1.000	0.943	0.995	0.950
	TPR	0.255	0.826	0.755	0.995	0.973	0.998
2	FPR	0.000	0.001	0.000	0.002	0.000	0.000
	Exact	0.255	0.825	0.755	0.990	0.973	0.998
	TPR	0.729	0.951	0.947	0.998	0.994	1.000
3	FPR	0.000	0.000	0.000	0.000	0.000	0.000
	Exact	0.497	0.867	0.870	0.993	0.983	1.000
	TPR	0.860	0.971	0.959	0.996	0.998	0.999
4	FPR	0.000	0.000	0.000	0.000	0.000	0.000
	Exact	0.570	0.887	0.855	0.983	0.990	0.998
	TPR	0.896	0.985	0.971	0.995	0.999	1.000
5	FPR	–	–	–	–	–	–
	Exact	0.595	0.927	0.858	0.975	0.993	1.000

E.2 Real data

E.2.1 Data construction and preprocessing

Table E.5 summarizes the construction of the nine indicators used in the empirical analysis. The last indicator is a rolling version of the illiquidity measure of Amihud (2002), based on the ratio of absolute return to dollar trading volume. Rolling indicators are constructed stock by stock using November–December 2022 as warm-up history. ADF tests do not reveal systematic unit-root behavior in the constructed series, so no differencing is applied; we also do not winsorize the data.

For each indicator, the mean and standard deviation are first computed from all finite observed cells in the 2023–2024 primary sample, before any imputation, and used for standardization. The panel is then standardized using these values. Ordinary missing cells are imputed using the same-day sector median, with the same-day market median as a fallback; observations after structural exits are left missing. Finally, each stock-indicator series is demeaned using its 2023-2024 time mean.

Table E.5: Construction of the equity indicators.

Indicator	Construction	Window
Daily return	CRSP daily total return, $r_t = \texttt{DlyRet}_t$	Daily
20-day cumulative log return	$\sum_{\ell=0}^{19} \log(1 + r_{t-\ell})$	20 days
Log dollar volume	$\log(1 + \texttt{DlyPrcVol}_t)$	Daily
Log turnover	$\log\{\texttt{DlyPrcVol}_t/(1000\,\texttt{DlyCap}_t)\}$	Daily
20-day relative turnover	Current log turnover minus its mean over the previous 20 trading days	20 lagged days
20-day log realized volatility	Log sample standard deviation of daily returns over the latest 20 trading days	20 days
High-low range	$\log(\texttt{DlyHigh}_t/\texttt{DlyLow}_t)$	Daily
Log bid-ask spread	$\log[1 + 10^4\{2(\texttt{DlyAsk}_t - \texttt{DlyBid}_t)/(\texttt{DlyAsk}_t + \texttt{DlyBid}_t)\}]$	Daily
20-day log Amihud illiquidity	$\log\left[1 + \frac{1}{20}\sum_{\ell=0}^{19} \frac{10^6\lvert r_{t-\ell}\rvert}{\texttt{DlyPrcVol}_{t-\ell}}\right]$	20 days

E.2.2 Additional real-data results

More evidence of using MMEFM structure. Under (IC1), the four components, grand mean, row main effect, column main effect, and the rest interaction component are orthogonal, so for each sector m and date t, $\|\mathbf{Y}_t^{(m)}\|_F^2 = p_m q_m (\widehat{c}_t^{(m)})^2 + q_m \|\widehat{\mathbf{a}}_t^{(m)}\|^2 + p_m \|\widehat{\mathbf{b}}_t^{(m)}\|^2 + \|\check{\mathbf{Y}}_t^{(m)}\|_F^2$. After aggregation over sectors and dates, the additive components account for 43.23% of the total squared magnitude. Figure E.3 shows that the interaction component is the largest part in every sector, but the main effects remain substantial and vary across sectors. The grand-mean contribution is small throughout.

Since the fitted global and local components are not exactly orthogonal, we compare their squared

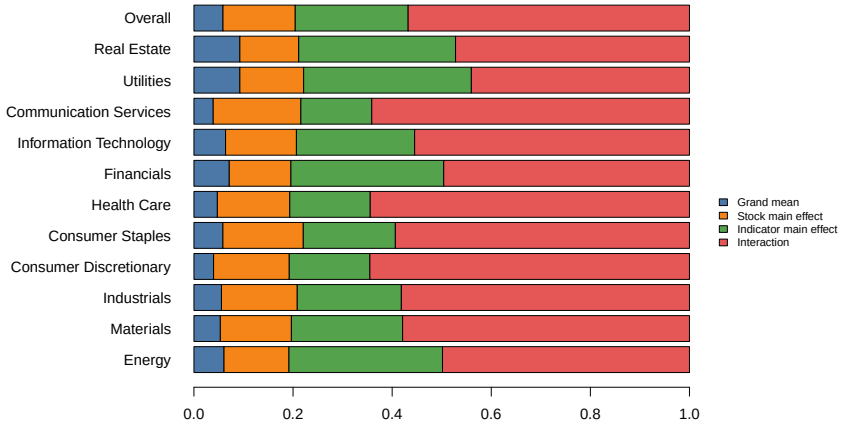


Figure E.3: Orthogonal decomposition of time-varying squared Frobenius norms across sectors.

magnitudes ratios. After aggregation over sectors and dates, the global magnitude ratios are

$$\text{Indicator main effect:} \quad \frac{\sum_{m=1}^{M}\sum_{t=1}^{T}\left\|\mathbf{1}_{p_m}(\widetilde{\mathbf{B}}_{1m}\widehat{\boldsymbol{\beta}}_t)'\right\|_F^2}{\sum_{m=1}^{M}\sum_{t=1}^{T}\left\{\left\|\mathbf{1}_{p_m}(\widetilde{\mathbf{B}}_{1m}\widehat{\boldsymbol{\beta}}_t)'\right\|_F^2+\left\|\mathbf{1}_{p_m}(\widehat{\mathbf{B}}_{2m}\widehat{\boldsymbol{\beta}}_t^{(m)})'\right\|_F^2\right\}}=42.81\%,$$

$$\text{Common component:} \quad \frac{\sum_{m=1}^{M}\sum_{t=1}^{T}\left\|\widetilde{\mathbf{Q}}_m\widehat{\mathbf{G}}_t\widetilde{\mathbf{J}}_m'\right\|_F^2}{\sum_{m=1}^{M}\sum_{t=1}^{T}\left\{\left\|\widetilde{\mathbf{Q}}_m\widehat{\mathbf{G}}_t\widetilde{\mathbf{J}}_m'\right\|_F^2+\left\|\widehat{\mathbf{R}}_m\widehat{\mathbf{F}}_t^{(m)}\widehat{\mathbf{C}}_m'\right\|_F^2\right\}}=26.63\%.$$

The sector-level ratios are computed in the same way by restricting the sums to each sector. Figure E.4 reports the corresponding sector-level ratios together with the screening statistics.

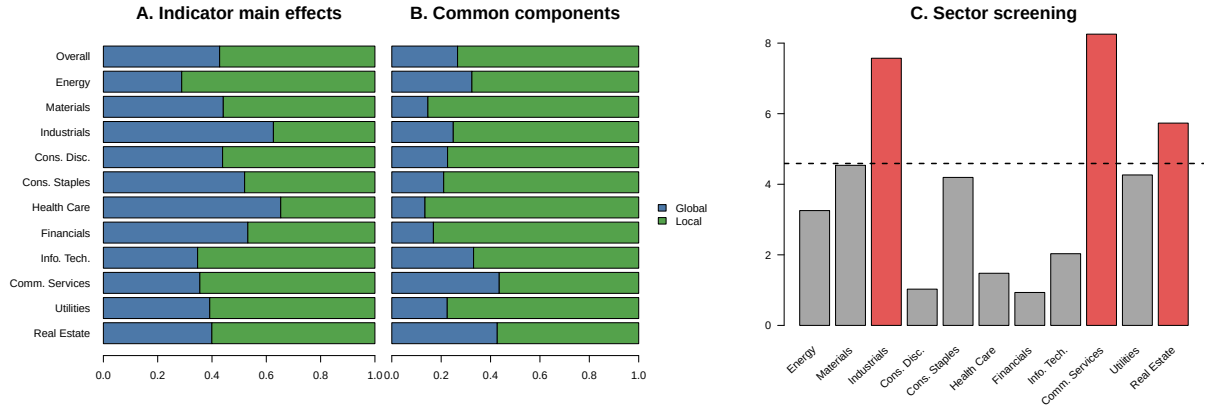


Figure E.4: Evidence for a multilevel structure. Panels A and B report global–local squared Frobenius magnitude ratios for the indicator main effects and common components, with Overall formed after aggregation across sectors. Panel C reports $S_{G,m}$; the dashed line is the 95% cutoff calibrated from $B = 199$ circular-shift bootstrap replications, and red bars denote screened sectors.

More results. Table E.6 reports the selected factor dimensions. The main text reports the global indicator main effect and illustrative local estimates for Financials and Information Technology. For completeness, Figure E.5 reports the global stock main effect, and Figures E.6–E.14 report local MMEFM estimates for the remaining nine sectors. In each sector-specific figure, Panels A–B show local stock-effect loadings and factor series, Panels C–D show local indicator-effect loadings and factor series, and Panels E–G show local common-component stock loadings, factor series, and indicator loadings.

Table E.6: Selected factor dimensions for the MMEFM: (r_1, ℓ_1) and $(r_2^{(m)}, \ell_2^{(m)})$ for the main effects, and (k_r, k_c) and $(k_r^{(m)}, k_c^{(m)})$ for the common component.

	Main effects	Common component
Global (all sectors)	$(1,1)$	$(1,1)$
Local ranks by sector		
Energy	$(1,2)$	$(2,1)$
Materials	$(1,2)$	$(3,1)$
Industrials	$(1,1)$	$(1,2)$
Consumer Discretionary	$(1,2)$	$(1,1)$
Consumer Staples	$(2,2)$	$(2,1)$
Health Care	$(1,1)$	$(2,1)$
Financials	$(1,1)$	$(1,2)$
Information Technology	$(1,2)$	$(1,1)$
Communication Services	$(1,2)$	$(2,1)$
Utilities	$(3,2)$	$(2,1)$
Real Estate	$(1,2)$	$(1,1)$

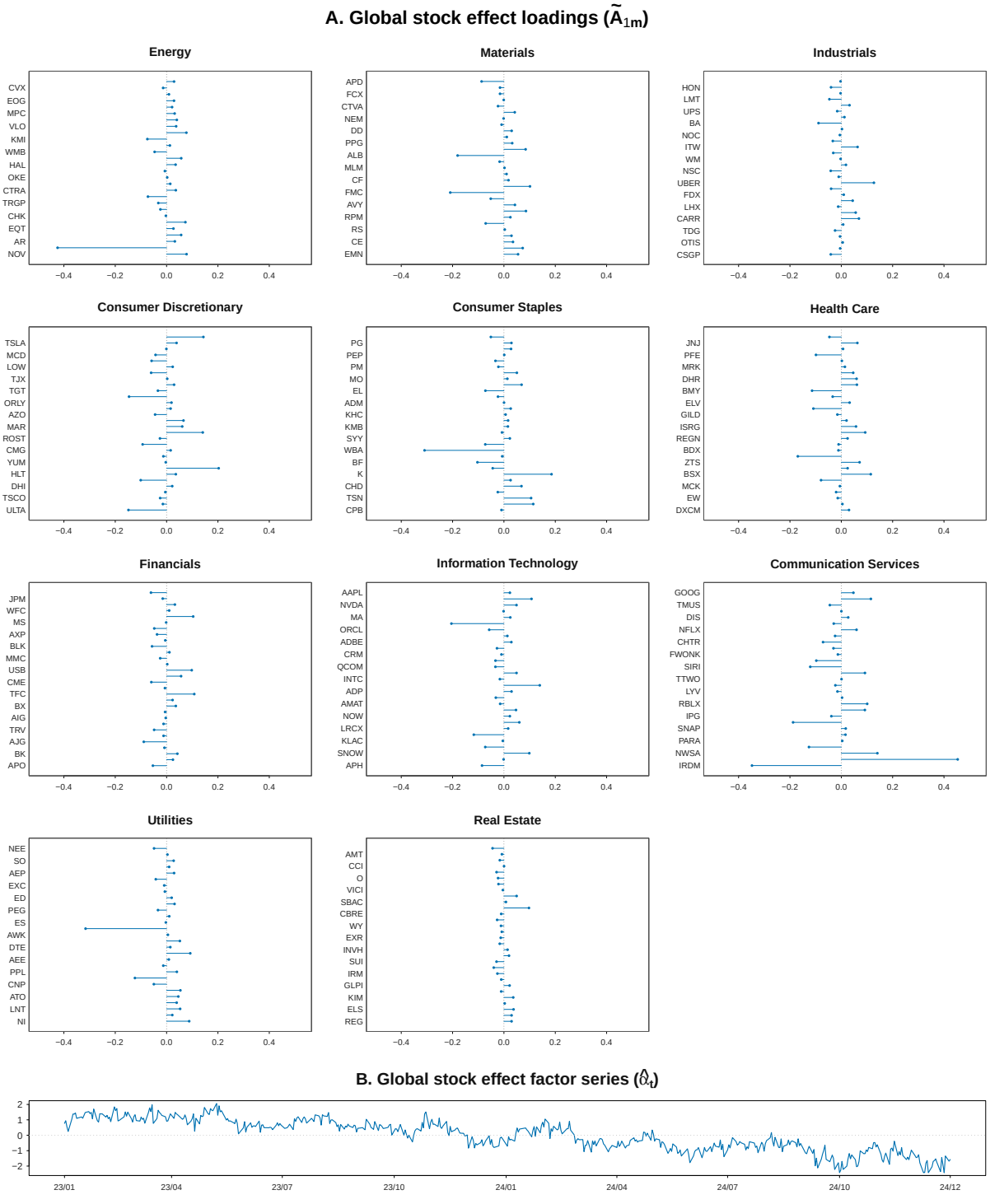


Figure E.5: Estimated global stock main effects.

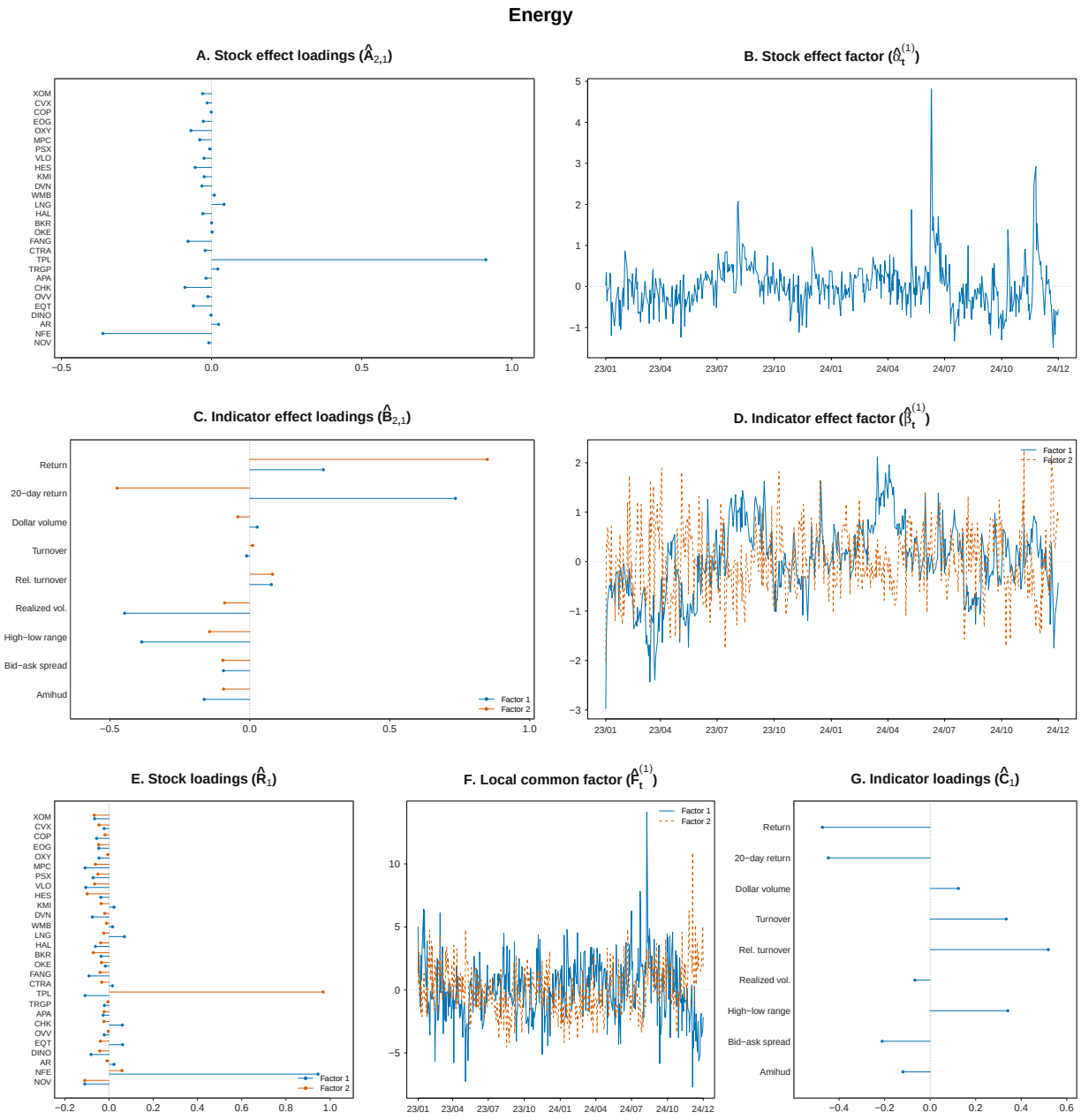


Figure E.6: Estimated local MMEFM components for Energy.

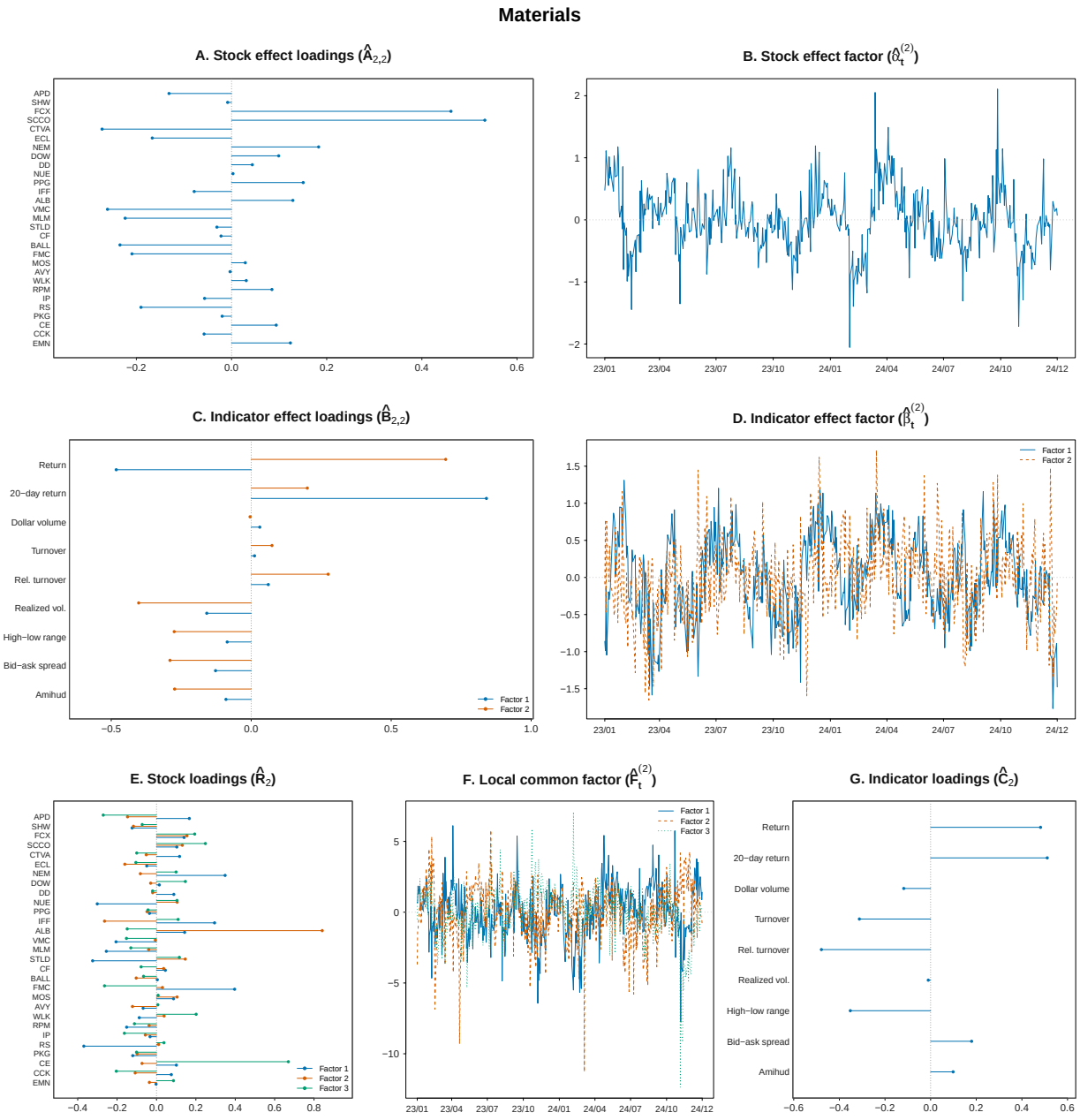


Figure E.7: Estimated local MMEFM components for Materials.

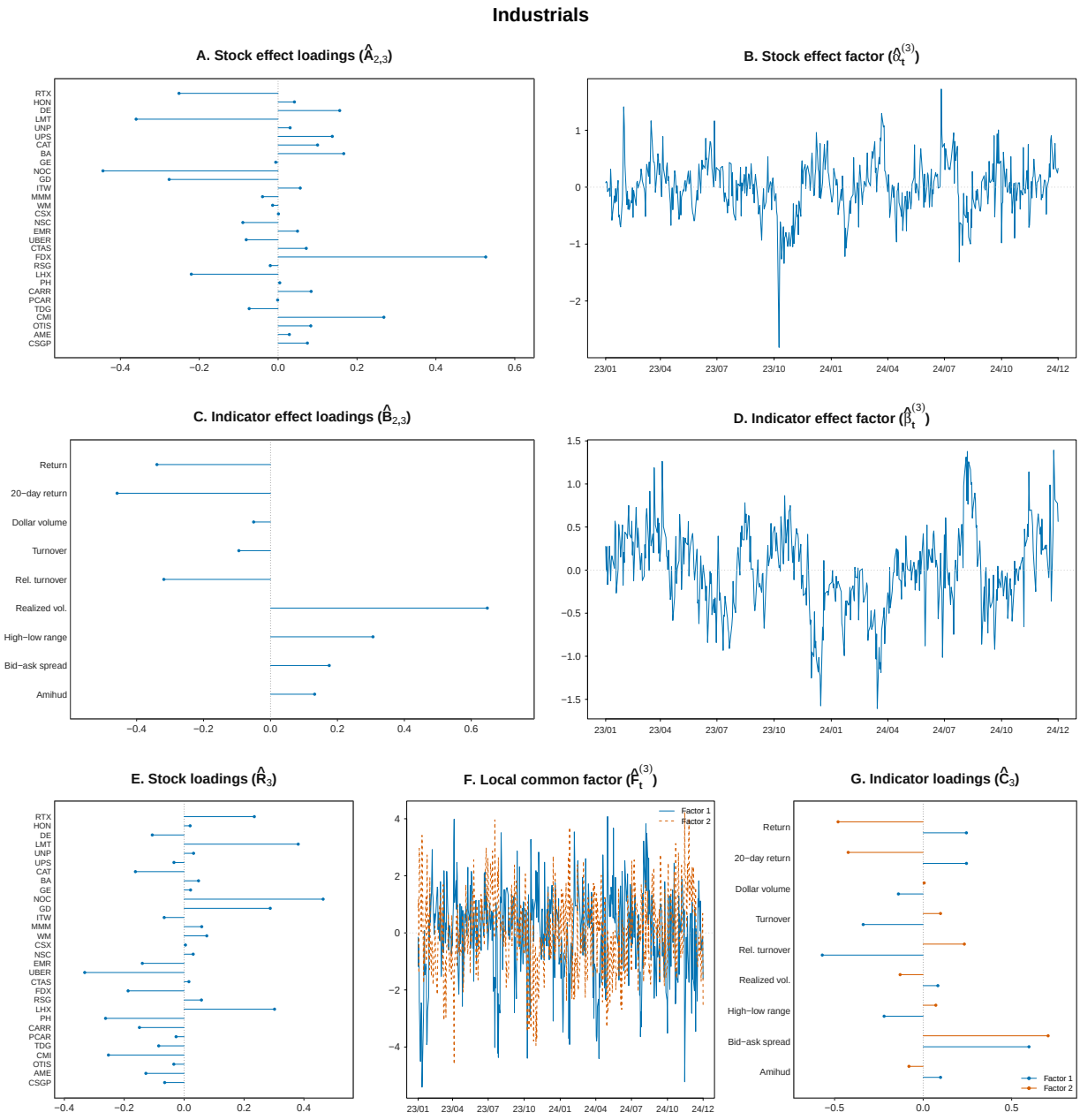


Figure E.8: Estimated local MMEFM components for Industrials.

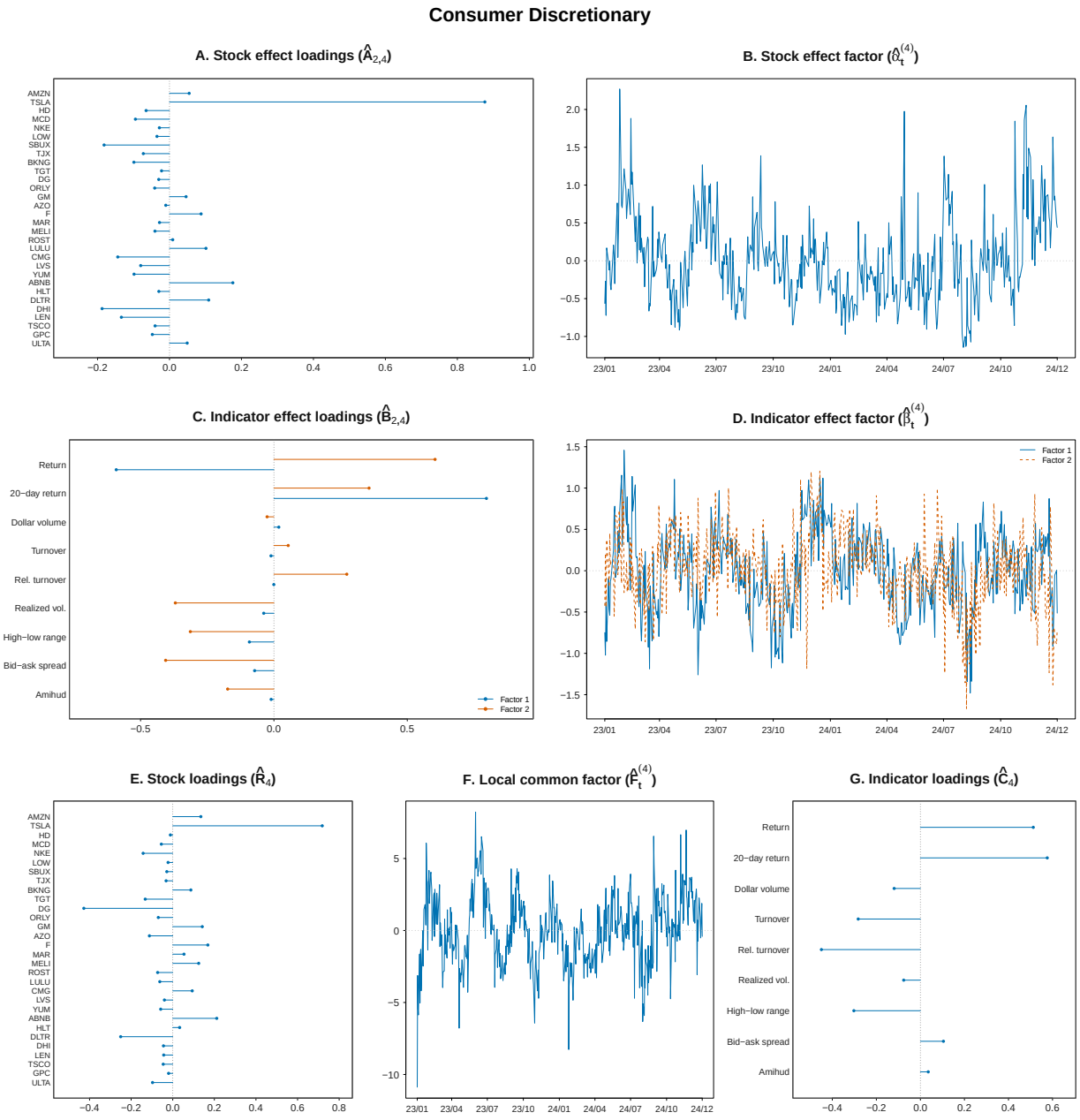


Figure E.9: Estimated local MMEFM components for Consumer Discretionary.

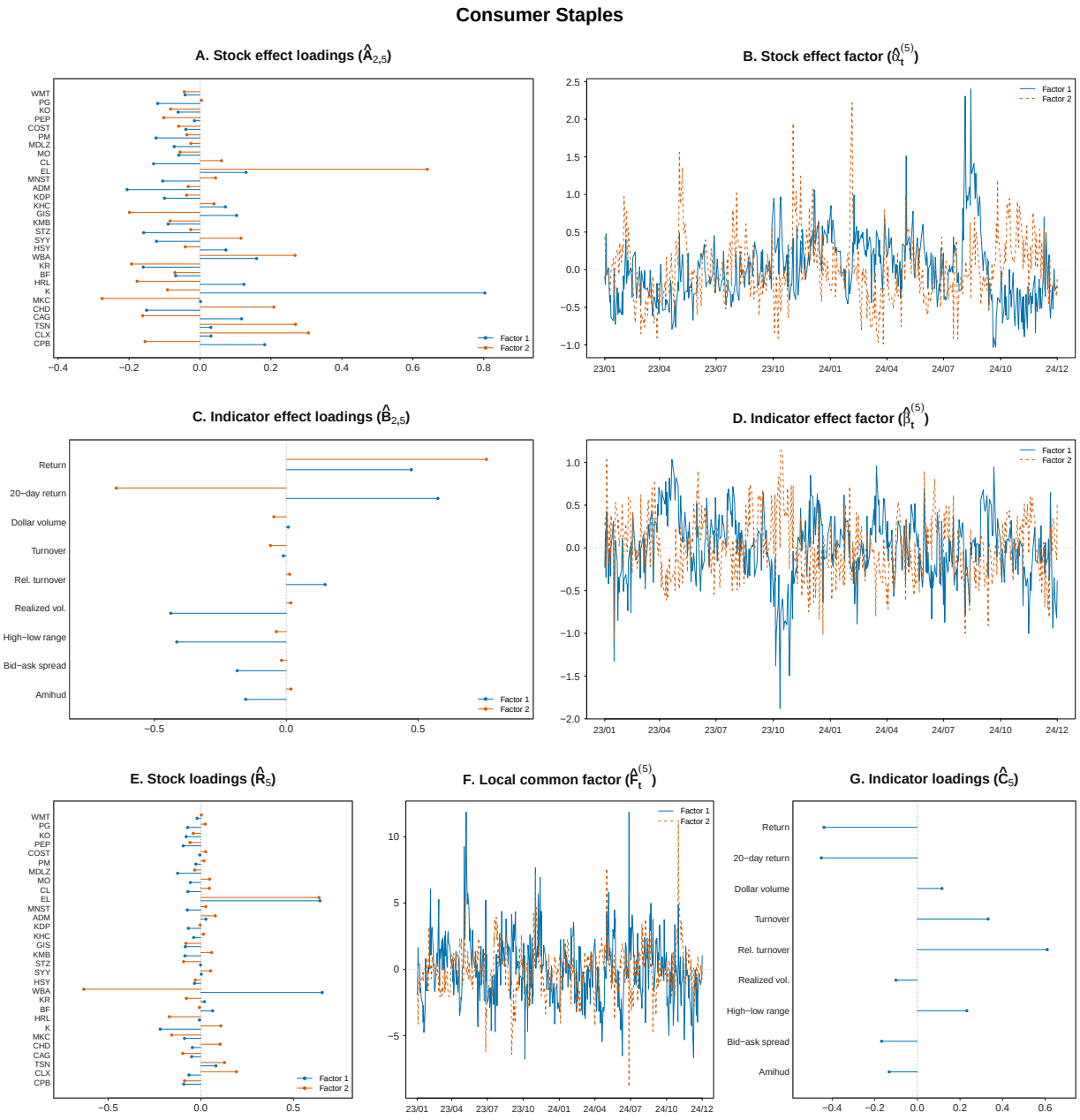


Figure E.10: Estimated local MMEFM components for Consumer Staples.

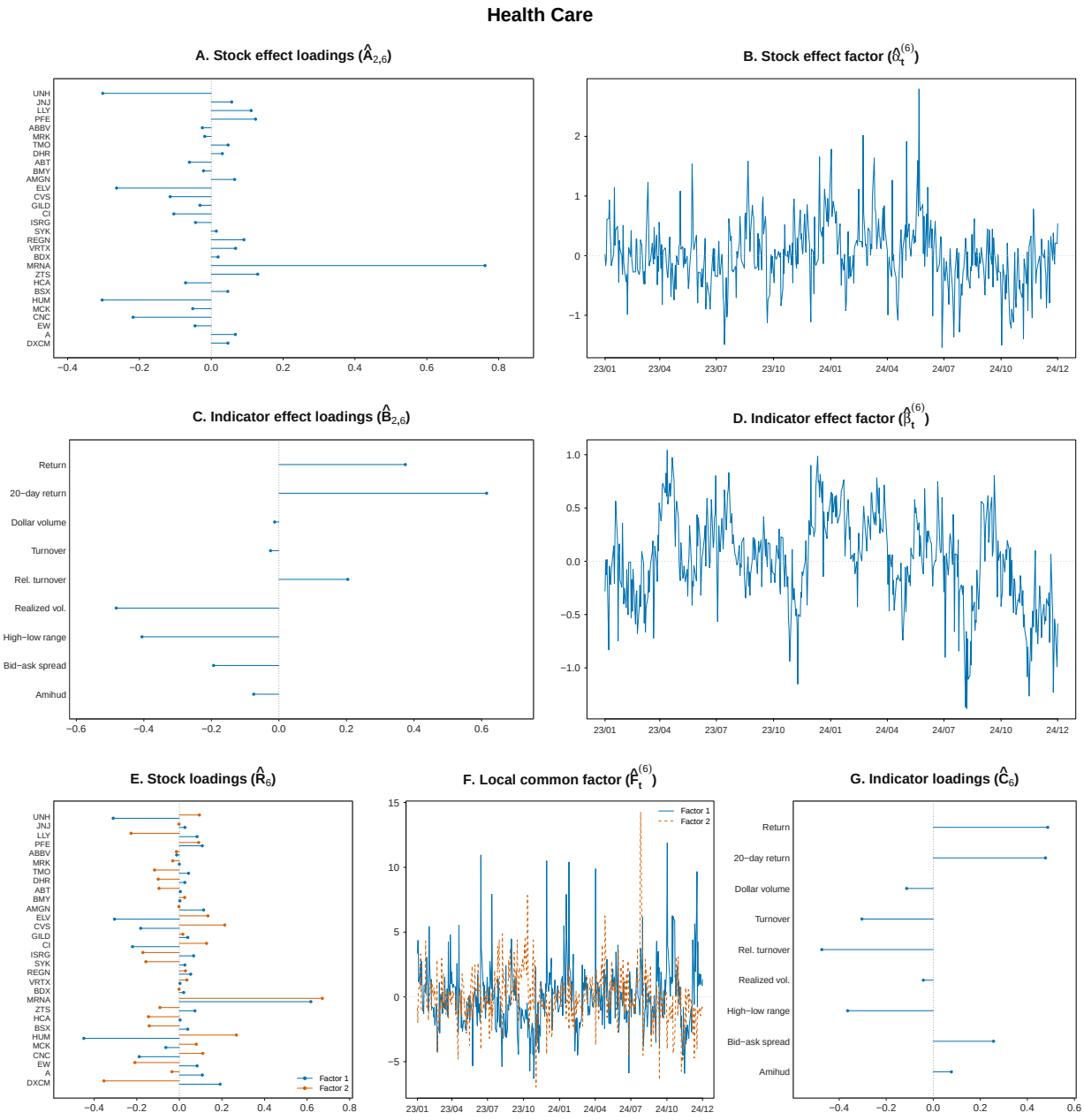


Figure E.11: Estimated local MMEFM components for Health Care.

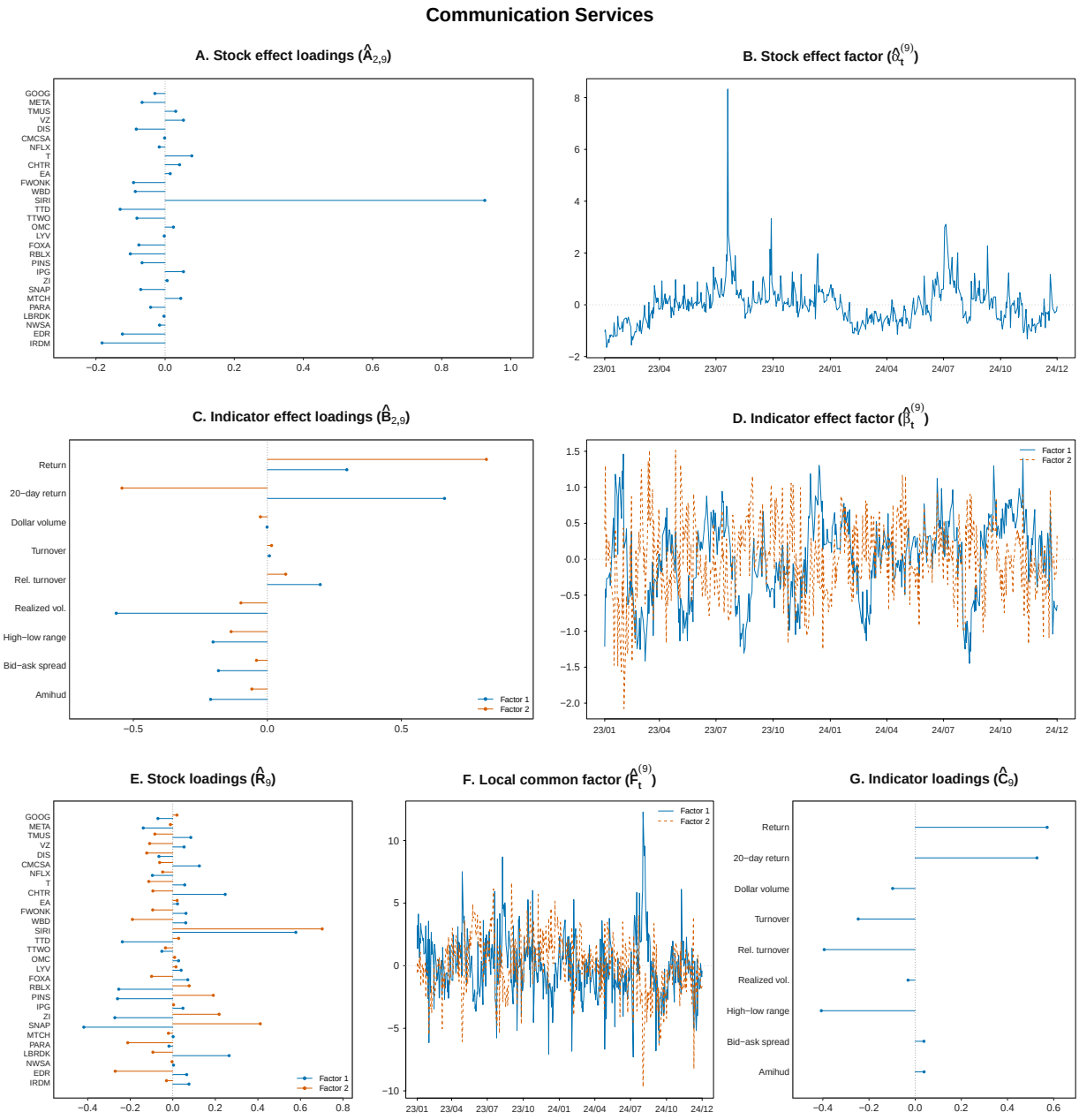


Figure E.12: Estimated local MMEFM components for Communication Services.

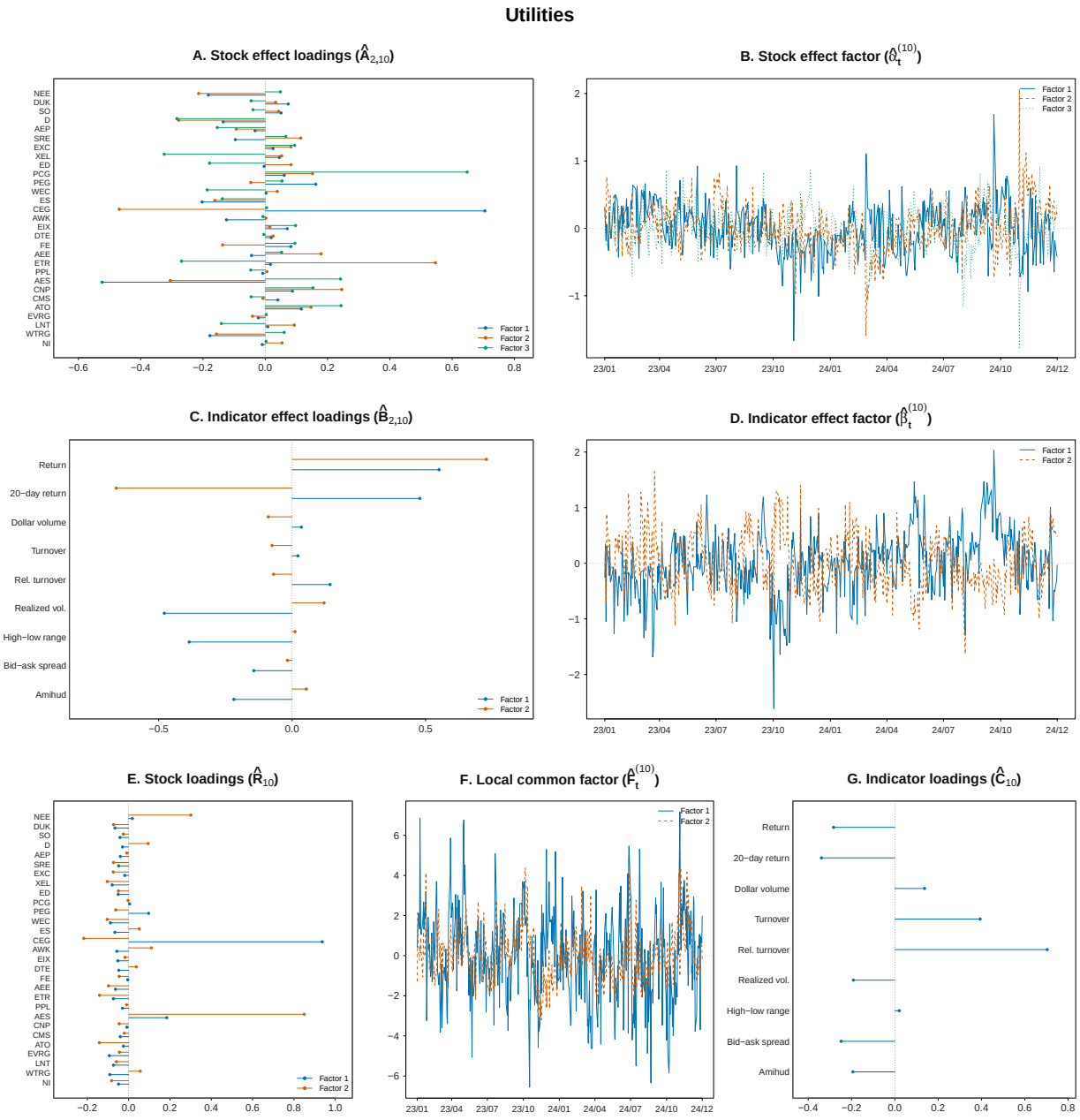


Figure E.13: Estimated local MMEFM components for Utilities.

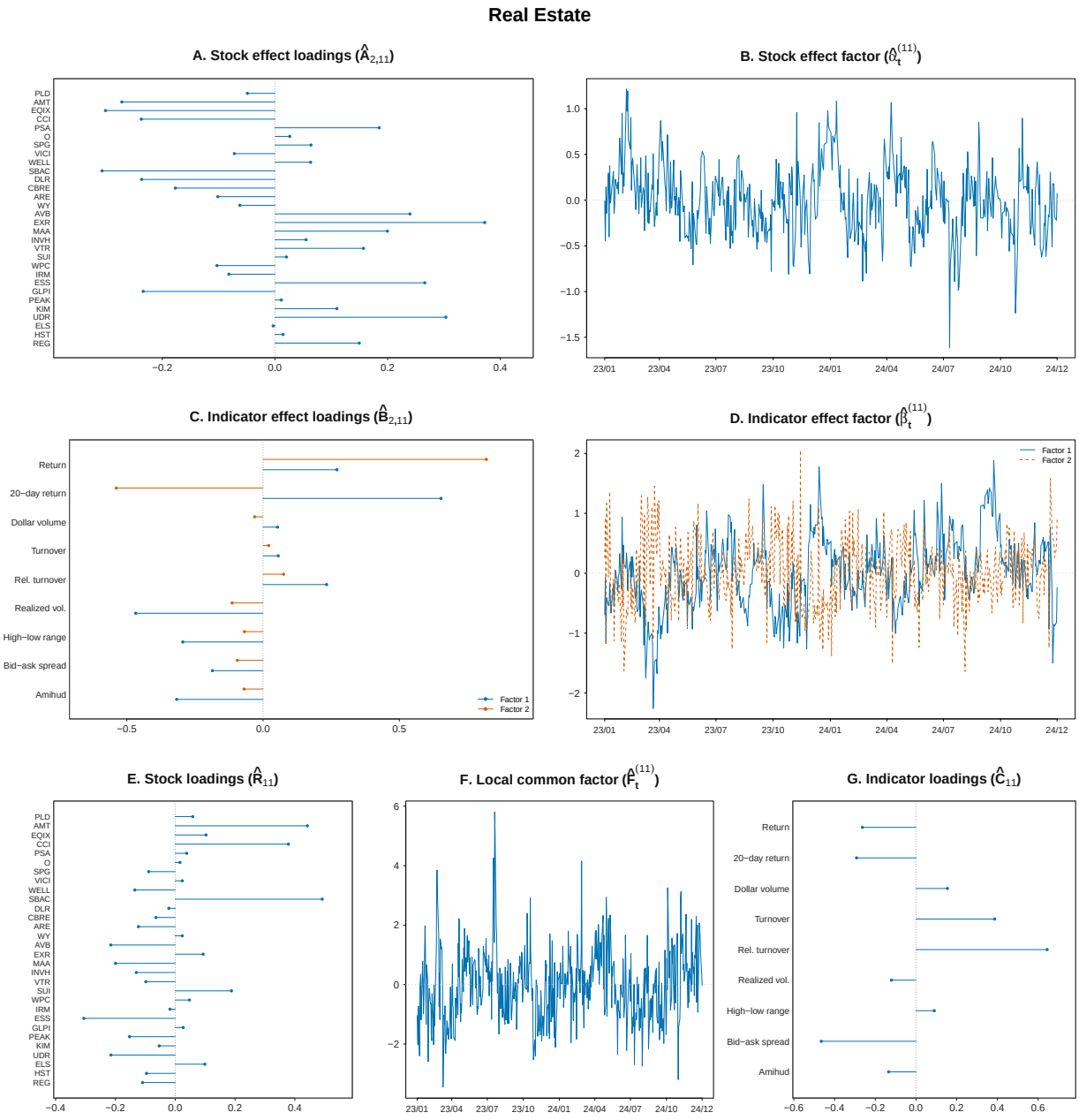


Figure E.14: Estimated local MMEFM components for Real Estate.

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